IB CHEMISTRY HL

Textbook Review Package

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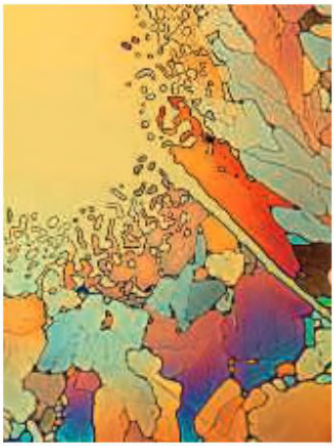
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▲ Figure 11 Polarized light micrograph of ammonium nitrate crystals. Ammonium nitrate contains two polyatomic ions: NH_4^+ and NO_3^-. Its uses include fertilizers and rocket propellants

Name	Formula
ammonium	NH_4^+
hydroxide	OH^-
nitrate	NO_3^-
hydrogencarbonate	HCO_3^-
carbonate	CO_3^{2-}
sulfate	SO_4^{2-}
phosphate	PO_4^{3-}

▲ Table 4 Common polyatomic ions

Name	Formula
potassium fluoride	KF
magnesium fluoride	MgF_2
calcium carbonate	$CaCO_3$
barium hydroxide	$Ba(OH)_2$
iron(III) oxide	Fe_2O_3
silver(I) sulfide	Ag_2S

▲ Table 5 Names and formulas of some ionic compounds

Activity

Determine whether the following pairs of elements are likely to bond ionically using the following two methods:

i. look at their positions in the periodic table

ii. refer to their electronegativity values in the data booklet.

a. Li and F
b. Rb and Ga
c. Ca and I
d. As and S
e. P and Cl
f. Ag and Br

It is often incorrectly said that only ionic bonds form when a metallic element and a non-metallic element bond together. There are substances, such as aluminium chloride, $AlCl_3$, that do not fit this description. Aluminium is a metal and chlorine is a non-metal, so you would expect them to bond ionically. But the compound has properties that are characteristic of covalent compounds, such as a low melting point and high volatility. The electronegativity difference between these two elements (1.6) suggests they do not bond ionically.

Polyatomic ions

Some ionic compounds contain more than two elements. For instance, NH_4Cl, which is made up of NH_4^+ cations and Cl^- anions. Ammonium ions, NH_4^+, are **polyatomic ions**. As their name suggests, polyatomic ions are ions that contain several atoms.

You are expected to know the names and formulas of the polyatomic ions shown in table 4.

ATL Self-management skills

You will need to spend some time memorizing the names and formulas of the polyatomic ions in table 4. Some students like to use flashcards, others make up mnemonics. What strategies will you use? How will you make sure you actively engage with them?

Naming ionic compounds

Consider the list of ionic compounds shown in table 5. Can you notice any patterns in their names?

You should notice that, in the names of ionic compounds:

- the cation name is given first and is followed by the anion
- cations adopt the name of the parent atom and the name remains unchanged
- monatomic anions adopt the first part of the name of the parent atom, followed by the suffix ***-ide***. If the anion is polyatomic, refer to table 4
- the name of the compound does not reflect the number of ions in the formula.

Practice questions

9. State the name of each of the following compounds:
   a. RbF
   b. Al_2S_3
   c. AlN
   d. $Sr(OH)_2$
   e. $BaCO_3$
   f. NH_4HCO_3

Anions are **conjugate bases** of common acids. The strength of acids and stability of their anions can be compared quantitatively using their dissociation constants, K_a, which will be introduced in *Reactivity 3.1*.

The formulas of ionic compounds

The name of an ionic compound tells you what elements it contains, but not the ratio of the ions in it. The basis for working out the formula of an ionic compound is remembering that the net charge of the compound is zero, so the positive charges and negative charges must cancel out. First, determine the charge of the anion and the cation, then work out how many of each ion you need to reach a total charge of zero.

Worked example 4

Deduce the formulas of the following ionic compounds:

a. calcium oxide b. calcium nitride c. sodium carbonate d. aluminium nitrate

Solution

a. To deduce the formula of calcium oxide, work through the following steps.

Step 1: Determine the charges of the cation and the anion

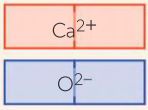
Calcium has an electron configuration of $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2$. Calcium atoms have two outer shell electrons, so they form ions with a 2+ charge.

Oxygen has an electron configuration of $1s^2\,2s^2\,2p^4$. Oxygen atoms have six outer shell electrons, so they form ions with a 2– charge.

Therefore, calcium ions = Ca^{2+} and oxide ions = O^{2-}.

Step 2: Determine how many of each ion are needed in order to achieve a net charge of zero

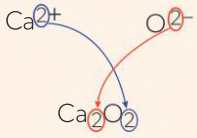
There are two methods you can use for this step. The first is the **bar diagram** method. Write out the ions as blocks equal to the number of charges on each individual ion:



The bar diagram contains one calcium ion and one oxide ion, so the ratio of calcium to oxide in the compound is 1:1.

$$Ca_1O_1$$

The second method is the **criss-cross rule**. Swap the charges and turn them into the other ion's subscript, ignoring the sign:



Then, simplify the ratio:

$$Ca_1O_1$$

Step 3: Check that the net charge is zero

You check your working by adding up the charges of each individual ion. If you did Step 2 correctly, the charges will add to zero:

$$\text{Total positive charge} = \frac{Ca^{2+}}{2+}$$

$$\text{Total negative charge} = \frac{O^{2-}}{2-}$$

Net charge = 2 – 2 = 0

Step 4: Write the formula

This is a straightforward example where the magnitude of charge is equal for the cation and anion, and hence the formula is CaO.

b. Calcium nitride is a more complex example as the ions have different charges. Work through the steps as before.

Step 1: Determine the charges of the cation and the anion

$$Ca: 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$$

Calcium atoms have two outer shell electrons, so they form ions with a 2+ charge.

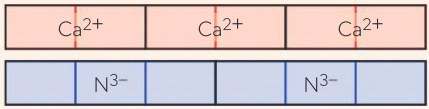
$$N: 1s^2\ 2s^2\ 2p^3$$

Nitrogen atoms have five outer shell electrons, so they form ions with a 3– charge.

Calcium ions = Ca^{2+} and nitride ions = N^{3-}

Step 2: Determine how many of each ion are needed in order to achieve a net charge of zero

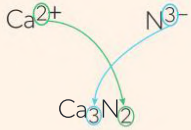
Bar diagram method



The bar diagram contains three calcium ions and two nitride ions, so the ratio of calcium to nitride ions in the compound is 3:2

$$Ca_3N_2$$

Criss-cross rule



There is no need to simplify the ratio here because it is already in its simplest form.

Step 3: Check that the net charge is zero

Ca^{2+}
Ca^{2+}
Ca^{2+}

N^{3-}
N^{3-}

Total positive charge = 6+ Total negative charge = 6–

Net charge = 6 – 6 = 0

Step 4: Write the formula

$$Ca_3N_2$$

c. Sodium carbonate contains a polyatomic ion. You must not split up or change the ratio of atoms in the polyatomic cluster. Treat it like an indivisible entity and draw brackets around it if the formula contains more than one such ion.

Step 1: Determine the charges of the cation and the anion

$Na: 1s^2\ 2s^2\ 2p^6\ 3s^1$

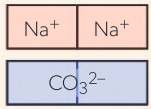
Sodium atoms have one outer shell electron, so they form ions with a 1+ charge.

Carbonate ions, CO_3^{2-}, have a 2– charge.

Sodium ions = Na^+ and carbonate ions = CO_3^{2-}

Step 2: Determine how many of each ion are needed in order to achieve a net charge of zero

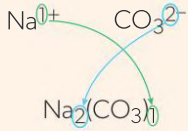
Bar diagram method



The bar diagram contains two sodium ions and one carbonate ion, so the ratio of sodium to carbonate in the compound is 2:1

$$Na_2CO_3$$

Criss-cross rule



Remember, you can draw brackets around the polyatomic ion to remind yourself that its formula does not change. Again, there is no need to simplify the ratio here.

Step 3: Check that the net charge is zero

Na^+
Na^+

CO_3^{2-}

Total positive charge = 2+ Total negative charge = 2–

Net charge = 2 – 2 = 0

Step 4: Write the formula

The formula is Na_2CO_3. Note that in the final answer there are no brackets around the polyatomic ion because the formula contains only one carbonate ion.

d. The final example, aluminium nitrate, also contains a polyatomic ion. Follow the same steps as before.

Step 1: Determine the charges of the cation and the anion

Al: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

Aluminium atoms have three outer shell electrons, so they form ions with a 3+ charge.

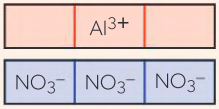
Nitrate ions, NO_3^-, have a 1– charge.

Al ions = Al^{3+} and nitrate ions = NO_3^-

> If you do not remember the charge on a polyatomic ion, revise table 4. Make sure that you learn these formulas and charges off by heart.

Step 2: Determine how many of each ion are needed in order to achieve a net charge of zero

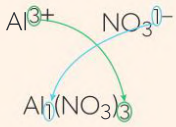
Bar diagram method



The bar diagram contains one aluminium ion and three nitrate ions, so the ratio of aluminium to nitrate in the compound is 1:3.

$$Al_1(NO_3)_3$$

Criss-cross rule



Again, there is no need to simplify the ratio here.

Step 3: Check that the net charge is zero

Al^{3+} Total positive charge = 3+

NO_3^-, NO_3^-, NO_3^- Total negative charge = 3–

Net charge = 3 – 3 = 0

Step 4: Write the formula

The formula is $Al(NO_3)_3$. Note that brackets are used to indicate the presence of more than one nitrate ion. The formula should not be written as AlN_3O_9.

Practice questions

10. Deduce the formula of each of the following compounds:
    a. magnesium oxide
    b. strontium chloride
    c. sodium sulfide
    d. lithium nitride
    e. lithium nitrate
    f. barium hydrogencarbonate
    g. ammonium phosphate.

Names of ionic compounds that contain transition elements have the oxidation number of the transition metal ion in brackets. This is covered in *Structure 3.1*.

Linking questions

Why is the formation of an ionic compound from its elements a redox reaction? (Reactivity 3.2)

AHL How is formal charge used to predict the preferred structure of sulfate? (Structure 2.2)

AHL Polyatomic anions are conjugate bases of common acids. What is the relationship between their stability and the conjugate acid's dissociation constant, K_a? (Reactivity 3.1)

Naming of oxyanions

Oxyanions are polyatomic anions that include oxygen atoms. When naming oxyanions, you should include the oxidation state of the non-oxygen atom as a Roman numeral at the end of the name. The Roman numerals are enclosed in brackets with no space between the ion name and the bracket.

For example, consider the anion MnO_4^-. The anion is named manganate(VII), as Mn has an oxidation state of +7. The neutral compound $KMnO_4$ is called potassium manganate(VII). This compound is also known under its traditional name, potassium permanganate.

While the Roman numerals are often used to refer to the oxidation state of a transition metal, the non-oxygen atom in an oxyanion can also be a non-metal. $KClO_4$, for example, is called potassium chlorate(VII).

In practice, the Roman numeral is omitted for some common oxyanions. These are shown in table 3.

Ionic formula	Common name	Systematic name
NO_2^-	nitrite	nitrate(III)
NO_3^-	nitrate	nitrate(V)
SO_3^{2-}	sulfite	sulfate(IV)
SO_4^{2-}	sulfate	sulfate(VI)

▲ Table 3 The common names and systematic names of some oxyanions. Both are acceptable in examinations

Practice questions

5. Write the chemical formulas for the following compounds:
   a. hydrogen nitrate
   b. sodium phosphate(V)
   c. magnesium bromate(III)
   d. potassium sulfite

Linking question

How can oxidation states be used to analyse redox reactions? (Reactivity 3.2)

ATL Communication skills

Scientists have used symbols to identify different elements for a long time (figure 14). Nowadays, we use the symbols in the periodic table, and the nomenclature conventions encouraged by the International Union of Pure and Applied Chemistry (IUPAC). Your study of chemistry will involve using symbols as they are stated in the data booklet, writing chemical formulas correctly (including subscripts and brackets where necessary) and naming substances according to IUPAC guidance.

Test your chemical communication skills by finding the error in each of the following and writing the correct answer:

1. HCL
2. NE
3. O3
4. MgOH2
5. cuprum(II) oxide



▲ Figure 14 The symbols used by alchemists for copper and iron, respectively

Structure 1.4 Counting particles by mass: the mole

How do we quantify matter on the atomic scale?

Atoms are extremely small, so any physical object contains a huge number of these particles. There are more atoms in a glass of water than glasses of water in all of the oceans combined. The unit of the amount of substance, the mole, enables chemists to deal comfortably with large numbers of very small particles. At the same time, the concepts of molar, relative atomic and relative molecular masses allow the use of small numbers for expressing masses of atomic species.

Understandings

Structure 1.4.1 — The mole (mol) is the SI unit of amount of substance. One mole contains exactly the number of elementary entities given by the Avogadro constant.

Structure 1.4.2 — Masses of atoms are compared on a scale relative to ^{12}C and are expressed as relative atomic mass (A_r) and relative formula mass (M_r).

Structure 1.4.3 — Molar mass (M) has the units g mol^{-1}.

Structure 1.4.4 — The empirical formula of a compound gives the simplest ratio of atoms of each element present in that compound. The molecular formula gives the actual number of atoms of each element present in a molecule.

Structure 1.4.5 — The molar concentration is determined by the amount of solute and the volume of solution.

Structure 1.4.6 — Avogadro's law states that equal volumes of all gases measured under the same conditions of temperature and pressure contain equal numbers of molecules.

The mole (*Structure 1.4.1*)

Atoms and molecules are so small that their masses cannot be measured directly. Even a million atoms of lead, Pb, the heaviest stable element, would have a mass of only 3.4×10^{-16} g. This is too small to be weighed even on the most sensitive analytical balance. At the same time, the number of Pb atoms in 1 g of lead is huge, about 2.9×10^{21}, which is hard to imagine, let alone count. Therefore, chemists need a unit that allows them to work comfortably with both very small masses and very large numbers of atoms. This unit, the **mole**, was devised in the 19th century and quickly became one of the most useful concepts in chemistry.

The mole (with the unit "mol") is the SI unit of amount of substance that contains $6.02214076 \times 10^{23}$ elementary entities of that substance. An elementary entity can be an atom, a molecule, an electron or any other species. In this book, we will use the rounded value of the mole: 1 mol = 6.02×10^{23}.



▲ Figure 1 One mole quantities of different substances (left to right): aluminium, water, copper, sucrose and sodium chloride

Prefix	Symbol	Factor
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}

▲ Table 1 Decimal prefixes

Avogadro's constant (N_A) is the conversion factor linking the number of particles and **amount of substance** in moles. It has the unit of mol^{-1}:

$N_A = 6.02 \times 10^{23}\ mol^{-1}$

In chemical calculations, Avogadro's constant is used in the same way as any other conversion factor (table 1). For example, to convert kilograms into grams, we need to multiply the mass in kg by 1,000. Similarly, to convert the amount of substance (n) into the number of atoms or any other structural units (N), we need to multiply that amount by N_A:

$N = n \times N_A$

In chemistry texts, the term "amount of substance" is often abbreviated to just "amount".

Worked example 1

Calculate the amount of lead (Pb), in mol and mmol, in a sample containing 2.9×10^{21} atoms of this element.

Solution

To find n, we can rearrange the equation $N = n \times N_A$ as follows:

$$n = \frac{N}{N_A}$$

Therefore, $n(\text{Pb}) = \frac{2.9 \times 10^{21}}{6.02 \times 10^{23}} \approx 4.8 \times 10^{-3}\ \text{mol}$

According to table 1, 1 mmol = 10^{-3} mol, so 4.8×10^{-3} mol = 4.8 mmol.
In this example, both answers (4.8×10^{-3} mol and 4.8 mmol) have been rounded to two significant figures, the same as in the least precise value used in the division (2.9×10^{21}).

The use of correct **significant figures** is discussed in the *Tools for chemistry* chapter.

Activity

Calculate:

a. the number of atoms in 2.5 mol of copper metal

b. the number of molecules in 0.25 mol of water

c. the number of atoms in 0.25 mol of water

ATL Research skills

The mole is a huge number, and it is useful for counting particles because they are so small. Measuring amounts of everyday objects in moles can help use to convey just how large this number is.

Choose one of the following and conduct the necessary research to reach an approximate answer.

- How many moles of grains of sand are in a desert of your choice?
- How many moles of water molecules are in a large sea or ocean of your choice?
- One mole of human cells represents roughly how many people?
- What is the age of the universe, in moles of seconds?
- How tall is a stack of one mole of sheets of paper?
- How many moles of air are in your school building?

Relative molecular mass and molar mass (*Structure 1.4.2 and 1.4.3*)

In *Structure 1.2*, we introduced the concept of **relative atomic mass,** A_r, which is the ratio of the mass of a certain atom to one-twelfth of the mass of a carbon-12 atom. Similarly, **relative molecular mass,** M_r, is the ratio of the mass of a molecule or other multiatomic species to one-twelfth of the mass of a carbon-12 atom. Both A_r and M_r are ratios, so they have no units.

To find the M_r of a molecule, we need to add together the A_r values for all atoms in that molecule.

Worked example 2

Calculate the M_r for a molecule of water.

Solution

Water, H_2O, is composed of two hydrogen atoms ($A_r = 1.01$) and one oxygen atom ($A_r = 16.00$). Therefore $M_r(H_2O) = 2 \times 1.01 + 16.00 = 18.02$.
You should always use the actual (not rounded) values of A_r, which are given in the data booklet and the periodic table at the end of this book. Similarly, keep all significant figures in calculated M_r values and never round them to the nearest integer number.

If a substance is composed of ions instead of molecules, the M_r for that substance is calculated using the smallest **formula unit**. For example, calcium chloride ($CaCl_2$) is an ionic compound that consists of many calcium cations (Ca^{2+}) and twice as many chloride anions (Cl^-). Its smallest formula unit contains one Ca^{2+} and two Cl^- ions. The ions have approximately the same masses as neutral atoms because the masses of electrons are negligible.

Therefore, $M_r(CaCl_2) = A_r(Ca) + 2 \times A_r(Cl) = 40.08 + (2 \times 35.45) = 110.98$.

Many ionic compounds form **hydrates**: compounds in which water molecules form coordination bonds (Structure 2.2) with the ions. One of the most common hydrates is copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$. Copper(II) sulfate pentahydrate forms large, clear, deep-blue crystals (figure 2). The **stoichiometric coefficient** "5" before "H_2O" means that one formula unit of copper(II) sulfate is bound with five molecules of water. Therefore, the M_r value for this hydrate can be calculated as follows:

$$M_r(CuSO_4 \cdot 5H_2O) = A_r(Cu) + A_r(S) + 4 \times A_r(O) + 5 \times M_r(H_2O)$$
$$= 63.55 + 32.07 + (4 \times 16.00) + (5 \times 18.02)$$
$$= 249.72$$

The composition and structure of ionic compounds will be discussed in *Structure 2.1*.

Activity

Calculate the M_r values for the following species:

a. ammonia, NH_3

b. sulfuric acid, H_2SO_4

c. sodium sulfate decahydrate, $Na_2SO_4 \cdot 10H_2O$



▲ Figure 2 Crystals of copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$

Molar mass, M, of a chemical substance is the mass of 1 mol of that substance. Molar mass is numerically equal to relative molecular mass (for substances with molecular and ionic structures) or relative atomic mass (for substances with atomic structure). For example, $M(Na) = 22.99\ g\ mol^{-1}$ and $M(H_2O) = 18.02\ g\ mol^{-1}$.

Science as a shared endeavour

A shared understanding of common terminology helps scientists to communicate effectively. This terminology is constantly being updated.

Historically, the mole was defined as the amount of substance that contained as many elementary entities (atoms, molecules, ions, electrons or other particles) as there were atoms in 0.012 kg (or 12 g) of carbon-12. However, the numerical value of the mole (approximately 6.02×10^{23}) had to be revised frequently, as the improvements in instrumentation allowed scientists to measure mass with greater precision.

On 16 November 2018, scientists from more than 60 countries met at the *General Conference on Weights and Measures* in Versailles, France. It was agreed here that all SI base units, including the mole, were defined in terms of physical constants instead of physical objects. Following these changes, one mole of a substance is now defined exactly as $6.02214076 \times 10^{23}$ elementary entities of that substance.

The 2018 redefinition of the mole means that the mass of 1 mol of carbon-12 no longer equals 12 g exactly. As a result, the numerical values of M (defined through two exact SI quantities, the kilogram and the mole) no longer match the numerical values of their respective A_r or M_r (defined through the experimentally determined mass of a carbon-12 atom). However, the differences between these numerical values are so small (approximately 4×10^{-8} %) that they can be ignored for all practical purposes.

Why are constants and values continuously being revised and updated?

How do scientists achieve a shared understanding of changes made to existing definitions?

The amount (n), mass (m) and molar mass (M) of any substance are related as follows:

$$n = \frac{m}{M}$$

This is probably the most common expression in chemistry, as it is used in almost all stoichiometric calculations. Although the base SI unit of mass is the kilogram, the masses of chemical substances are traditionally expressed in grams, and molar masses in g mol^{-1}.

Worked example 3

Table sugar is often sold in the form of cubes that are made almost entirely of sucrose. Sucrose is an organic compound with the molecular formula $C_{12}H_{22}O_{11}$. Calculate:

a. the molar mass of sucrose

b. the amount of sucrose in one cube (2.80 g) of sugar

c. the number of oxygen atoms in one cube of sugar

Solution

a. $M_r(C_{12}H_{22}O_{11}) = 12 \times 12.01 + 22 \times 1.01 + 11 \times 16.00 = 342.34$

$M(C_{12}H_{22}O_{11}) = 342.34\ \text{g mol}^{-1}$

b. $n = \frac{m}{M}$

$$n(C_{12}H_{22}O_{11}) = \frac{2.80\ \text{g}}{342.34\ \text{g mol}^{-1}} \approx 0.00818\ \text{mol}$$

c. One mole of sucrose contains 11 mol of oxygen atoms, so

$$n(O) = 11 \times n(C_{12}H_{22}O_{11})$$

$$= 11 \times 0.00818\ \text{mol} \approx 0.0900\ \text{mol}$$

$$N(O) = n(O) \times N_A = 0.0900\ \text{mol} \times 6.02 \times 10^{23}\ \text{mol}^{-1} \approx 5.42 \times 10^{22}$$

Activity

Calculate:

a. the molar mass of sulfuric acid, H_2SO_4

b. the amount of substance in 1.00 g of sulfuric acid

c. the number of hydrogen atoms in 1.00 g of sulfuric acid





▲ Figure 3 There are more oxygen atoms in one sugar cube than the estimated total insect population on Earth (10^{19}) and total grains of sand on Earth's beaches (10^{21})

Measurement

Atoms, molecules and ions are so small that counting them directly is virtually impossible. The concept of the mole is powerful because it relates number of particles to mass, which can be easily measured.

As with all measurements, mass has an uncertainty associated with it. Consider the mass of a sample of calcium carbonate, $CaCO_3$, is found to be 3.500 g ± 0.001 g. This means the mass measurement can be inaccurate by up to 0.001 g in either direction. This is clearly a minuscule mass. How many moles does it represent? How many particles does it represent? Do a quick calculation and find out. You will see that in moles the uncertainty is tiny, but in terms of particles, it is quite large.

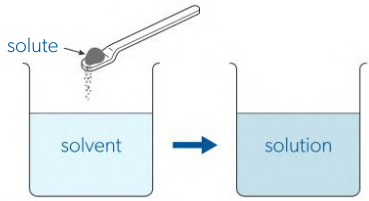
Is a measurement uncertainty ever negligible? If so, when? Think about these questions as you proceed through the DP chemistry course, particularly when doing experiments that involve making measurements.

Solutions and concentration (*Structure 1.4.5*)

Many chemical reactions are carried out in solutions. Solutions are easier to handle and mix than solids or gases. Sometimes a solvent is used because it can affect the properties of dissolved substances or participate in chemical reactions.

Solutions are homogeneous mixtures of two or more components. Each solution consists of a **solvent** and one or more **solutes**. The solvent is usually the major component of the solution, so the properties of the whole solution are similar to the properties of the solvent. The other components of the solution are called solutes. For example, a solution of sugar in water is more like water (clear colourless liquid) than sugar (white crystalline powder), so water is the solvent while sugar is the solute. In this topic, we will consider only **aqueous solutions** (from the Latin *aqua* meaning "water"), in which the solvent is water.

Homogeneous and heterogeneous mixtures are discussed in *Structure 1.1.*



▲ Figure 7 How a solution is formed

In some cases, the identity of the solvent is unclear: for example, if we mix ethanol and water, each of these liquids can be called a solvent. However, if water is present in the mixture, it is traditionally regarded as the solvent, even if it is not the major component. For example, we say "96% solution of ethanol in water" rather than "4% solution of water in ethanol".

Solutions are often classified according to the mass or mole ratio between the solute and solvent. A **concentrated solution** contains a large proportion of solute, and so has a high ratio of solute to solvent, while a **dilute solution** has a small proportion of solute, and so has a low ratio of solute to solvent. Generally, the term "concentrated" refers to solutions with much more than 10 g of the solute per 100 g of the solvent, and the term "dilute" refers to solutions with much less than 10 g of the solute per 100 g of the solvent.

TOK

Some words do not have precise definitions and their choice and interpretation is context dependent. The terms "concentrated" and "dilute" are not precisely defined and should be used with care. For example, most chemists would call a solution of 5 g of sulfuric acid (H_2SO_4) in 100 g of water "dilute", as much higher proportions of sulfuric acid to water are commonly used in laboratories. At the same time, a solution of 5 g of potassium permanganate ($KMnO_4$) in 100 g of water would be considered very concentrated by any medical worker, as typical concentrations of potassium permanganate in antiseptic solutions are less than 0.1 g per 100 g of water.

The concentrations in the examples above could be expressed numerically. To what extent does expressing a quantity numerically help or hinder the communication of knowledge?

Quantitatively, the composition of solutions is expressed in terms of concentration. **Molar concentration**, c, also known as **molarity**, is the ratio of the amount of a solute to the volume of the solution:

$$c_{\text{solute}} = \frac{n_{\text{solute}}}{V_{\text{solution}}}$$

The most common units for molar concentration are mol dm^{-3} (which is the same as mol L^{-1}). For very dilute solutions, smaller units (mmol dm^{-3} or μmol dm^{-3}) can also be used:

1 mmol dm^{-3} = 1×10^{-3} mol dm^{-3}

1 μmol dm^{-3} = 1×10^{-6} mol dm^{-3}

The units of molar concentrations are sometimes abbreviated as M (for mol dm^{-3}) or mM (for mmol dm^{-3}). For example, the expression "2.5 M NaOH" means that each dm^3 of the solution contains 2.5 mol of sodium hydroxide.

Note that the term "molar concentration" refers to a specific substance, not the whole solution. For example, it is incorrect to say that "the concentration of a sodium chloride solution is 1.0 mol dm^{-3}", as it is not clear whether we are talking about the concentration of sodium chloride or water. The correct statement would be "the concentration of sodium chloride in a solution is 1.0 mol dm^{-3}".

Molar concentration is often represented by square brackets around the solute formula. For example, the expression $[NH_3] = 0.5$ M refers to a 0.5 mol dm^{-3} solution of ammonia. Similarly, the expression $[Cl^-]$ refers to the molar concentration of chloride ions in a solution.

Worked example 7

Calculate the molar concentration of sodium chloride, in mol dm^{-3}, in a solution prepared by dissolving 3.60 g of NaCl(s) in water to make 25.0 cm^3 of the final solution.

Solution

First, calculate the molar mass of sodium chloride:

$$M(\text{NaCl}) = 22.99 + 35.45 = 58.44\ \text{g mol}^{-1}$$

Then use $n = \frac{m}{M}$ to calculate the amount of solute:

$$n(\text{NaCl}) = \frac{3.60\ \text{g}}{58.44\ \text{g mol}^{-1}} \approx 0.0616\ \text{mol}$$

Convert the volume to dm^3 by dividing by 1,000:

$$V(\text{solution}) = 25.0\ \text{cm}^3 = 0.0250\ \text{dm}^3$$

Use $c = \frac{n}{V}$ to calculate the concentration:

$$c(\text{NaCl}) = \frac{0.0616\ \text{mol}}{0.0250\ \text{dm}^3} \approx 2.46\ \text{mol dm}^{-3}.$$

Activity

Calculate the mass of sulfuric acid, H_2SO_4, in 50.0 cm^3 of a solution where $[H_2SO_4] = 1.50\ \text{mol dm}^{-3}$.

The composition of a solution is sometimes expressed as the **mass concentration**, ρ_{solute}, of the solute, which is the ratio of the mass of the solute to the volume of the solution:

$$\rho_{solute} = \frac{m_{solute}}{V_{solution}}$$

Worked example 8

Calculate the mass concentration of sodium chloride in the solution from worked example 7.

Solution

If we know the mass of the solute and the volume of the solution, we can calculate the mass concentration as follows:

$$\rho(\text{NaCl}) = \frac{3.60\ \text{g}}{0.0250\ \text{dm}^3} = 144\ \text{g dm}^{-3}$$

Alternatively, the mass concentration of NaCl can be found from its molar concentration and molar mass, using the relationship $\rho_{solute} = c_{solute} \times M_{solute}$:

$$\rho(\text{NaCl}) = c(\text{NaCl}) \times M(\text{NaCl}) = 2.46\ \text{mol dm}^{-3} \times 58.44\ \text{g mol}^{-1} \approx 144\ \text{g dm}^{-3}$$

Activity

Calculate the molar concentration of sulfuric acid, in mol dm^{-3} in a solution with $\rho(H_2SO_4) = 0.150\ \text{g cm}^{-3}$.

The most common units for mass concentration are g dm^{-3} and g cm^{-3}. Mass concentration and molar concentration of the same solute are related by molar mass, as follows:

$$c_{solute} = \frac{\rho_{solute}}{M_{solute}}$$

Worked example 9

A standard solution was prepared by dissolving 6.624 g of sodium carbonate, Na_2CO_3, in deionized water using a 250 cm^3 volumetric flask. An analytical pipette was used to transfer 10.0 cm^3 sample of this solution to a 100 cm^3 volumetric flask, and the flask was topped up to the graduation mark with deionized water. Calculate the concentration, in mol dm^{-3}, of sodium carbonate in the new solution.

Solution

First, we need to find the concentration of sodium carbonate in the standard solution:

$$M(Na_2CO_3) = 2 \times 22.99 + 12.01 + 3 \times 16.00 = 105.99\ g\ mol^{-1}$$

$$n(Na_2CO_3) = \frac{6.624\ g}{105.99\ g\ mol^{-1}} \approx 0.06250\ mol$$

$$V_{standard} = 250\ cm^3 = 0.250\ dm^3$$

Note that the accuracy of a typical volumetric flask is three significant figures.

$$c_{standard}(Na_2CO_3) = \frac{0.06250\ mol}{0.250\ dm^3} = 0.250\ mol\ dm^{-3}$$

Then we need to calculate the concentration of sodium carbonate in the new solution.

First, calculate the amount of Na_2CO_3 in the sample. Remember to convert all volumes to dm^3.

$$V_{sample} = 10.0\ cm^3 = 0.0100\ dm^3$$

$$c_{standard}(Na_2CO_3) = c_{sample}(Na_2CO_3) = 0.250\ mol\ dm^{-3}$$

$$n_{sample}(Na_2CO_3) = 0.250\ mol\ dm^{-3} \times 0.0100\ dm^3 = 0.00250\ mol$$

When the sample is diluted with deionized water to produce the new solution, the amount of solute does not change. Therefore

$$n_{sample}(Na_2CO_3) = n_{new}(Na_2CO_3) = 0.00250\ mol$$

Now you can work out the concentration of Na_2CO_3 in the new solution by dividing the amount of Na_2CO_3 by the volume of the new solution:

$$V_{new} = 100\ cm^3 = 0.100\ dm^3$$

$$c_{new}(Na_2CO_3) = \frac{0.00250\ mol}{0.100\ dm^3} = 0.0250\ mol\ dm^{-3}$$

It is a common practice to store chemicals in the form of concentrated solutions (so-called **stock solutions**) and dilute them to the required concentration when needed. Stock solutions with a known concentration of the solute are called **standard solutions**.

To determine the concentration of the standard solution in worked example 9, we did the following two calculations:

1. $n_{sample} = c_{sample} \times V_{sample}$
2. $c_{new} = \frac{n_{new}}{V_{new}}$

You know that $n_{sample} = n_{new}$, so you can substitute equation 1 into equation 2. This gives the following expression:

$$c_{new} = \frac{c_{sample} \times V_{sample}}{V_{new}}$$

Therefore, to calculate the concentration of a solute in a new solution, you just need to know the original concentration of the solute, the volume of the original solution, and the volume of the new solution. In summary:

$$c_1 \times V_1 = c_2 \times V_2$$

Practice question

3. A standard solution was prepared by dissolving 2.497 g of copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$, in deionized water using a 100 cm^3 volumetric flask. A 5.00 cm^3 sample of this solution was diluted to 250.0 cm^3. Calculate the concentration, in mol dm^{-3}, of copper(II) sulfate in the final solution.

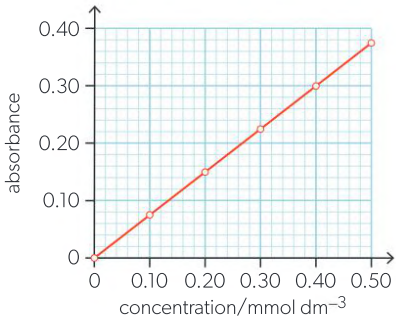
The process for preparing standard solutions is discussed in the *Tools for chemistry* chapter.

Case study: spectrophotometry and calibration curves

Spectrophotometry is an analytical technique based on the measurement of the intensity of visible, ultraviolet and near-infrared radiation. This technique is commonly used for determining concentrations of coloured substances in solutions.

A spectrophotometer produces light of a certain wavelength, which passes through a small sample of the studied solution. The photodetector measures the intensity of the transmitted light and converts it into the **absorbance**. Absorbance is a value describing the amount of light absorbed by the sample. Initially, several standard solutions of the studied substance are prepared by serial dilution (figure 9), and their absorbances are measured. These absorbances are plotted against concentrations, producing a **calibration curve** (figure 8). The calibration curve is then used for determining the unknown concentration of the coloured substance in the studied solution.

In the general case, a calibration curve relates a measurable property (such as absorbance, pH or electrical conductivity) of the solution to the concentration of the solute. The unknown concentration can be found by measuring that property and plotting the result of the measurement on the calibration curve.



▲ Figure 8 A typical calibration curve



▲ Figure 9 A series of standard solutions of potassium permanganate

Data-based question

The calibration curve in figure 8 was obtained using a series of standard solutions of potassium permanganate, $KMnO_4$. A solution with unknown concentration of $KMnO_4$ has an absorbance of 0.285. Determine the concentration of $KMnO_4$ in that solution.

Ideally, the calibration curve should be linear, pass through the origin and have a tilt of approximately 45°. If the curve does not meet any of these requirements, it should be constructed again using a slightly different wavelength of light and/or different set of standard solutions. Sometimes linearity can only be achieved within a narrow range of concentrations. In this case, the studied solution can be diluted, so the concentration of the studied substance falls within the range of calibration curve. In the last case, some additional calculations will be required to relate the concentrations of the studied substance in the diluted and original solutions.

Another technique, **colorimetry**, is based on the same principles as spectrophotometry but limited to visible light. The terms "colorimetry" and "spectrophotometry" are often used interchangeably, which is not entirely correct but very common.

Concentration uncertainty of a standard solution

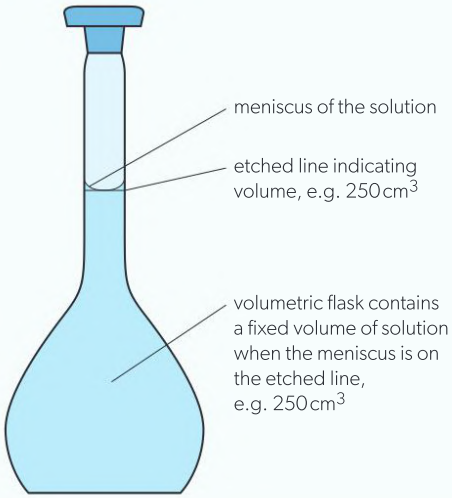
A standard solution is a solution of known concentration. In this activity, you will prepare two standard solutions of copper(II) sulfate, each by using different equipment. By propagating the measurement uncertainties, you will assess the precision of the concentration values. You will determine the actual concentration of your solutions using a colorimeter. This will then allow you to assess the accuracy of the concentrations.

Relevant skills

- Tool 1: Measuring volume and mass
- Tool 1: Standard solution preparation
- Tool 3: Calculate and interpret percentage error and percentage uncertainty
- Tool 3: Express quantities and uncertainties to an appropriate number of significant figures
- Tool 3: Record measurement uncertainties
- Tool 3: Propagate uncertainties
- Inquiry 2: Assess accuracy and precision

Safety

- Wear eye protection.
- Solid copper(II) sulfate is an irritant and toxic to the environment
- Dispose of all solutions appropriately.



Materials

- Wash bottle containing distilled water
- Weighing boats (2)
- 100 cm^3 beakers (2)
- Stirring rods (2)
- Funnels (2)
- Pipettes
- Spatula
- Reagent bottles (2)
- Blank labels
- Colorimeter
- Cuvettes
- Calibration curve relating concentration of copper(II) sulfate and absorbance
- Copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$

Additional equipment for solution 1:

- 100 cm^3 volumetric flask
- Milligram balance (three decimal places)

Additional equipment for solution 2:

- 100 cm^3 measuring cylinder
- Centigram balance (two decimal places)

Instructions

1. Use the equipment provided to prepare two copper(II) sulfate standard solutions, both with concentration 0.020 mol dm^{-3}. When preparing solution 1, you should use the volumetric flask and milligram balance. For solution 2, use the measuring cylinder and centigram balance.
2. Record the measurements you make along the way, including their uncertainties.
3. Following your teacher's instructions on how to use the colorimeter, measure the absorbance of your solutions.
4. Refer to the calibration curve to determine the actual concentration of your solutions.

Questions

1. Determine the uncertainty of the concentrations of solutions 1 and 2.
2. Calculate the percentage error of the concentrations of solutions 1 and 2.
3. Assess the precision and accuracy of the concentrations of solutions 1 and 2.

Reactivity 2.1 How much? The amount of chemical change

How are chemical equations used to calculate reacting ratios?

Chemical equations contain essential information about the nature of chemical changes, the participating species, the states of these species and the proportions in which they are consumed or produced. In chemical changes, atoms can be combined or rearranged, but never created or destroyed. So, the total number of atoms of each type remains constant when reactants are transformed into products. As a result, all species participating in a chemical reaction are used up or formed in fixed mole ratios, which are described by the stoichiometric coefficients before these species. For gases, the volumes of reacting species are also proportional to their stoichiometric coefficients.

It is important to understand that a chemical equation reflects only the changes in amounts or volumes of reacting species, but not the initial or final quantities of these species. In many cases, one of the reactants may be present in excess, or the reaction may not come to completion. Even in these cases, the mole ratios can be used to calculate any unknown quantity (amount, volume, mass and so on) of a reactant or product from given quantities of other species participating in the reaction.

Understandings

Reactivity 2.1.1—Chemical equations show the ratio of reactants and products in a reaction.

Reactivity 2.1.2—The mole ratio of an equation can be used to determine:

- the masses and/or volumes of reactants and products
- the concentrations of reactants and products for reactions occurring in solution.

Reactivity 2.1.3—The limiting reactant determines the theoretical yield.

Reactivity 2.1.4—The percentage yield is calculated from the ratio of experimental yield to theoretical yield.

Reactivity 2.1.5—The atom economy is a measure of efficiency in green chemistry.

Stoichiometry and the mole ratio (*Reactivity 2.1.1*)

Stoichiometry is the **quantitative** study of the reactants and products involved in a chemical reaction. If you know the reactants and products, you can write a balanced chemical equation by adding **stoichiometric coefficients** before each species to ensure that the number of each type of atom in the reactants and products is identical.

TOK

On the internet, find the Wikipedia article on *Stoichiometry*. Then, select a language that you are not familiar with from the menu on the left of the page. Even though you will not be able to read the text, you will probably recognize many of the diagrams, chemical equations and related mathematical calculations.

Chemical equations describe processes using a universal and internationally adopted set of symbols. To what extent does the use of universal "languages" help knowledge development?

Worked example 1

Ethane gas, C_2H_6, is completely combusted in oxygen, O_2, to form carbon dioxide, CO_2, and water, H_2O, under standard conditions. Write a balanced equation for this reaction.

Solution

First, write the unbalanced equation for the reaction, with the reactants on one side of the reaction arrow and the products on the other:

$$C_2H_6 + O_2 \rightarrow CO_2 + H_2O$$

Then, count the number of atoms of each type on both sides of the equation, adding stoichiometric coefficients where required:

- There are two carbon atoms in the reactants, and one in the products. Write the coefficient "2" before CO_2 to balance the carbon atoms.

  $C_2H_6 + O_2 \rightarrow 2CO_2 + H_2O$

- There are six hydrogen atoms in the reactants and two in the products. Write the coefficient "3" before H_2O to balance the hydrogen atoms.

  $C_2H_6 + O_2 \rightarrow 2CO_2 + 3H_2O$

- There are two oxygen atoms in the reactants and now seven in the products. Write the coefficient "3.5" before O_2 to balance the oxygen atoms.

  $C_2H_6 + 3.5O_2 \rightarrow 2CO_2 + 3H_2O$

- To get a whole-number ratio, multiply each coefficient by 2.

  $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$

- If you know the physical states of the species involved in the reaction, you should write the state symbols after each species:

  $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(l)$

Practice questions

1. The anaerobic fermentation of glucose, $C_6H_{12}O_6$, forms ethanol, C_2H_5OH, and carbon dioxide, CO_2. Write the balanced equation for this reaction.

State	Symbol
solid	s
liquid	l
gas	g
aqueous solution	aq

▲ Table 1 State symbols commonly used in chemical equations

The stoichiometric coefficients in a balanced chemical equation are proportional to the amounts of reacting species. The ratio of the stoichiometric coefficients is called the **mole ratio**. Reactants will always react in relative amounts equal to the mole ratio of the reactants, to produce relative amounts of the products equal to the mole ratio of the products.

For example, the equation for the photosynthesis of glucose, $C_6H_{12}O_6$, is given below:

$$6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$$

The mole ratio of the reactants is 1 : 1, and the mole ratio of the products is 1 : 6. This means that equal amounts of carbon dioxide and water will be consumed in the reaction to produce six times less glucose than oxygen.

Linking question

When is it useful to use half-equations? (Reactivity 3.2)

Limiting reactants (*Reactivity 2.1.2* and *Reactivity 2.1.3*)

The relative amounts of reactants in a reaction mixture might not be equal to the mole ratio, so some reactants could be left over even if all other reactants are consumed completely. The **limiting reactant** is the reactant used up completely, while other reactants are present in excess. It controls the amount of product formed in the reaction and therefore the **theoretical yield** of the reaction. The **excess reactant** is the reactant that is added to the reaction mixture in a greater proportion than that required by the mole ratio. Some of the excess reactant will be left over when the reaction reaches completion.

The reaction yield will be discussed in *Reactivity 2.1.4*.

For example, imagine that 1.0 mol of hydrogen is mixed with 3.0 mol of oxygen and the mixture is ignited. An explosion occurs, and hydrogen and oxygen react in a 2 : 1 mole ratio, producing water:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

The reaction stops when all the hydrogen is consumed: 1.0 mol of hydrogen will react with 0.5 mol of oxygen and 2.5 mol of oxygen remains unreacted. Therefore, the final mixture will contain 1.0 mol of water and 2.5 mol of oxygen.

It is helpful to record the amounts of all substances as follows:

	$2H_2(g) + O_2(g) \rightarrow$		$2H_2O(l)$
$n_{initial}$ / mol	1.0	3.0	0
Δn / mol	−1.0	−0.5	+1.0
n_{final} / mol	0	2.5	1.0

- The first row under the equation represents the initial mixture, in which the substances can be present in any proportions. We can take any amounts of reactants and mix them together, regardless of the stoichiometric coefficients.
- The second row shows how the amount of each substance changes in the course of the reaction. The sign before each amount shows whether it decreases (for reactants) or increases (for products). These changes must be proportional to the mole ratio shown by the stoichiometric coefficients.
- The last row represents the final mixture, where the amounts of all substances are calculated as $n_{final} = n_{initial} + \Delta n$. As with the initial mixture, the composition of the final mixture is not related to the stoichiometric coefficients.

In this example, the amount of the reaction product (water) is limited by the amount of hydrogen, which is consumed completely in the reaction. Therefore, hydrogen is the limiting reactant and oxygen is in excess.

Practice questions

2. Household lighters often contain pressurized butane, C_4H_{10}, which is combusted upon release:

   $$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

   Determine the limiting reactant in the following mixtures:

   a. 20 molecules of C_4H_{10} and 100 molecules of O_2
   b. 0.20 mol of C_4H_{10} and 2.6 mol of O_2
   c. 8.72 g of C_4H_{10} and 28.8 g of O_2

You can find the amount of a substance, n, from its mass, m, using $n = \frac{m}{M}$, where M is the **molar mass** in g mol^{-1}. Molar mass can be calculated using the A_r values given in the data booklet. This was covered in *Structure 1.4*.

The concept of the limiting reactant allows us to determine the **extent of reaction**. Consider the mixture of 1.0 mol of hydrogen and 3.0 mol of oxygen again. If we double the amount of hydrogen, the amount of oxygen that reacts doubles, and the amount of water formed will also double:

	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$		
$n_{initial}$ / mol	1.0	3.0	0
Δn / mol	−1.0	−0.5	+1.0
n_{final} / mol	0	2.5	1.0

amount of hydrogen doubled ⇨

	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$		
$n_{initial}$ / mol	2.0	3.0	0
Δn / mol	−2.0	−1.0	+2.0
n_{final} / mol	0	2.0	2.0

However, if we double the amount of oxygen (which was already in excess), the amount of hydrogen that reacts and the amount of water produced in the reaction will not change:

	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$		
$n_{initial}$ / mol	1.0	3.0	0
Δn / mol	−1.0	−0.5	+1.0
n_{final} / mol	0	2.5	1.0

amount of oxygen doubled ⇨

	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$		
$n_{initial}$ / mol	1.0	6.0	0
Δn / mol	−1.0	−0.5	+1.0
n_{final} / mol	0	5.5	1.0

Worked example 2

A 16.0 g sample of calcium carbide, $CaC_2(s)$, was placed into a sealed vessel containing 19.3 dm^3 of oxygen. The vessel was heated at 1000 °C until all carbide was converted into calcium oxide, CaO(s), and carbon dioxide, $CO_2(g)$. When the vessel was cooled down, the oxides reacted with each other to produce calcium carbonate, $CaCO_3(s)$. Calculate the mass of calcium carbonate and the volumes of individual gases in the remaining mixture. Assume that all reactions have come to completion. All volumes are measured at **standard temperature and pressure** (STP).

Solution

First, write the balanced equation of the combustion reaction:

$$CaC_2(s) + 2.5O_2(g) \rightarrow CaO(s) + 2CO_2(g)$$

Then, find the amounts of the reactants:

$$n(CaC_2) = \frac{m(CaC_2)}{M(CaC_2)} = \frac{16.0\,g}{64.10\,g\,mol^{-1}} = 0.250\,mol$$

$$n(O_2) = \frac{V(O_2)}{V_m} = \frac{19.3\,dm^3}{22.7\,dm^3\,mol^{-1}} = 0.850\,mol$$

Avogadro's law states that equal volumes of all gases, at the same temperature and pressure, have the same number of molecules (*Structure 1.4*). At standard temperature and pressure (STP), the **molar volume**, $V_m = 22.7\,dm^3\,mol^{-1}$. This can be used to work out the amount of gas when its volume is known, using $n = \frac{V}{V_m}$ (*Structure 1.5*).

According to the equation, calcium carbide and oxygen react in 1 : 2.5 mole ratio. In our case, the ratio of the reactants in the reaction mixture is 0.250 : 0.850 = 1 : 3.4. Therefore, oxygen is in excess, so calcium carbide is the limiting reactant. All calcium carbide will be consumed, while some oxygen will remain unreacted:

	$CaC_2(s)$	+ $2.5O_2(g)$	→ $CaO(s)$	+ $2CO_2(g)$
$n_{initial}$ / mol	0.250	0.850	0	0
Δn / mol	−0.250	−0.625	+0.250	+0.500
n_{final} / mol	0	0.225	0.250	0.500

Next, write the balanced equation for the second reaction, in which the oxides react with each other:

$$CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$$

Calcium oxide and carbon dioxide react in a 1 : 1 mole ratio. However, the ratio of CaO to CO_2 in the reaction mixture is 0.250 : 0.500 = 1 : 2. Therefore, carbon dioxide is in excess, so calcium oxide is the limiting reactant. All calcium oxide will be consumed, while some carbon dioxide will remain unreacted:

	$CaO(s)$	+ $CO_2(g)$	→ $CaCO_3(s)$
$n_{initial}$ / mol	0.250	0.500	0
Δn / mol	−0.250	−0.250	+0.250
n_{final} / mol	0	0.250	0.250

The final mixture will contain 0.250 mol of $CaCO_3(s)$, 0.250 mol of $CO_2(g)$ and 0.225 mol of $O_2(g)$. Therefore:

$$m(CaCO_3) = 0.250\,\text{mol} \times 100.09\,\text{g mol}^{-1} \approx 25.0\,\text{g}$$

$$V(CO_2) = 0.250\,\text{mol} \times 22.7\,\text{dm}^3\,\text{mol}^{-1} \approx 5.68\,\text{dm}^3$$

$$V(O_2) = 0.225\,\text{mol} \times 22.7\,\text{dm}^3\,\text{mol}^{-1} \approx 5.11\,\text{dm}^3$$

Chemical reactions are often carried out in **aqueous solutions**, which are solutions where water is the **solvent**. Aqueous solutions are easier to handle and mix than solids and gases. When solving quantitative problems involving **concentrations** and **volumes** of solutions, you should focus on the amounts, in mol, of the reacting substances and their mole ratios in the balanced chemical equation. Typical problems involving reactions in aqueous solutions are discussed in the following worked examples.

Practice questions

3. Hydrogen and chlorine react with each other to produce hydrogen chloride, HCl(g). A mixture of 4.54 dm^3 of hydrogen, $H_2(g)$, and 2.27 dm^3 of chlorine, $Cl_2(g)$, was heated until the reaction was complete. All volumes are measured at STP.
   a. Deduce the balanced equation for this reaction.
   b. Determine the limiting reactant.
   c. Calculate the volumes, in dm^3 at STP, of each substance in the final mixture.

Definitions for solutions, solutes, solvents and concentration are given in *Structure 1.4*.

Worked example 3

Hydrochloric acid is an aqueous solution of hydrogen chloride, HCl(aq). The reaction of hydrochloric acid with a sodium hydroxide solution, NaOH(aq), produces sodium chloride, NaCl(aq). A 1.00 dm^3 sample of 0.500 mol dm^{-3} hydrochloric acid was mixed with an equal volume of a 0.200 mol dm^{-3} sodium hydroxide solution.

a. Formulate the equation for the reaction of hydrochloric acid with sodium hydroxide.

b. Determine the final concentrations, in mol dm^{-3}, of all solutes in the solution once the reaction is complete. Assume that the volumes of solutions are additive.

Solution

a. $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$

You can find the amount of a substance in a solution, n, from the volume of the solution, V, using $n = V \times c$, where is c is the **molar concentration** of that substance in mol dm^{-3}. This was covered in *Structure 1.4*.

b. First, work out the amounts of each solute using $n = V \times c$:

$$n(HCl) = V(HCl) \times c(HCl)$$
$$= 1.00\ dm^3 \times 0.500\ mol\ dm^{-3}$$
$$= 0.500\ mol$$

$$n(NaOH) = V(NaOH) \times c(NaOH)$$
$$= 1.00\ dm^3 \times 0.200\ mol\ dm^{-3}$$
$$= 0.200\ mol$$

According to the equation, HCl and NaOH react in 1 : 1 ratio. In the reaction mixture, the amount of HCl (0.500 mol) is greater than that of NaOH (0.200 mol). Therefore, NaOH is the limiting reactant, so it will be consumed completely while some HCl will be left over:

	$HCl(aq)$	+ $NaOH(aq)$	→ $NaCl(aq)$	+ $H_2O(l)$
$n_{initial}$ / mol	0.500	0.200	0	excess
Δn / mol	−0.200	−0.200	+0.200	+0.200
n_{final} / mol	0.300	0	0.200	excess

The final solution contains 0.300 mol of unreacted HCl(aq) and 0.200 mol of NaCl(aq). The amount of the solvent (water) is not relevant, as it is present in large excess in both the initial and final solutions.

Assuming that the volumes of solutions are additive, $V_{final} = V(HCl) + V(NaOH) = 1.00\ dm^3 + 1.00\ dm^3 = 2.00\ dm^3$. You can use $\frac{n}{V_{final}}$ to calculate the final concentrations:

$$c_{final}(HCl) = \frac{0.300\ mol}{2.00\ dm^3} = 0.150\ mol\ dm^{-3}$$

$$c_{final}(NaCl) = \frac{0.200\ mol}{2.00\ dm^3} = 0.100\ mol\ dm^{-3}$$

Worked example 4

Sulfur dioxide, $SO_2(g)$, is a toxic gas with a strong smell of burnt matches. When released into the atmosphere in large quantities, sulfur dioxide contributes to acid rain by reacting with water vapour to produce unstable sulfurous acid, $H_2SO_3(aq)$.

Sulfur dioxide can be absorbed by alkaline solutions, such as a solution of sodium hydroxide, NaOH(aq), to produce salts of sulfurous acid, such as $Na_2SO_3(aq)$.

a. Formulate equations for the reaction of sulfur dioxide with water vapour and for the reaction of sulfur dioxide with excess aqueous sodium hydroxide.

b. Determine the concentrations, in mol dm^{-3}, of all solutes in a solution produced when 2.84 dm^3 of sulfur dioxide is absorbed by 2.00 dm^3 of a 0.400 mol dm^{-3} solution of sodium hydroxide at STP. Assume that the absorption does not affect the volume of the solution.

Solution

a. $SO_2(g) + H_2O(g) \rightarrow H_2SO_3(aq)$

$SO_2(g) + 2NaOH(aq) \rightarrow Na_2SO_3(aq) + H_2O(l)$

b. Sulfur dioxide is a gas at STP, so use $n = \frac{V}{V_m}$ to determine the amount present:

$$n(SO_2) = \frac{2.84\ dm^3}{22.7\ dm^3\ mol^{-1}} = 0.125\ mol$$

Work out the amount of the sodium hydroxide solute using $n = V \times c$:

$$n(NaOH) = 2.00\ dm^3 \times 0.400\ mol\ dm^{-3} = 0.800\ mol$$

According to the equation, SO_2 and NaOH react in a 1 : 2 ratio. The amount of NaOH required to absorb 0.125 mol of SO_2 is therefore 0.250 mol. The actual amount of NaOH in the reaction mixture (0.800 mol) is greater than 0.250 mol. Therefore, SO_2 is the limiting reactant, so it will be consumed completely, while some NaOH will be left over:

	$SO_2(g)$	+ 2NaOH(aq)	→ $Na_2SO_3(aq)$	+ $H_2O(l)$
$n_{initial}$ / mol	0.125	0.800	0	excess
Δn / mol	−0.125	−0.250	+0.125	+0.125
n_{final} / mol	0	0.550	0.125	excess

The final solution contains 0.550 mol of unreacted NaOH(aq) and 0.125 mol of $Na_2SO_3(aq)$. The volume of the solution at the end of the reaction will be approximately equal to the volume of the initial solution, 2.00 dm^3. Therefore:

$$c_{final}(NaOH) = \frac{0.550\ mol}{2.00\ dm^3} = 0.275\ mol\ dm^{-3}$$

$$c_{final}(Na_2SO_3) = \frac{0.125\ mol}{2.00\ dm^3} = 0.0625\ mol\ dm^{-3}$$

Measurement

All measurements are, to an extent, limited in precision and accuracy. However, sometimes these limitations are negligible. For example, strictly speaking, the volumes of solutions are not additive. If we mix the solutions from worked example 3, the volume of the final solution will be 2.01 dm^3, or approximately 0.5% greater than expected (2.00 dm^3). However, such differences are usually small and comparable with the accuracy of standard volumetric glassware (0.2–0.5%). Therefore, in most practical situations, we can assume that the volume of the solution at the end of the reaction is approximately equal to the sum of volumes of the initial solutions.

Have you come across other negligible limitations to accuracy or precision in chemistry? When do such effects become significant?

Practice questions

4. Equal volumes of 0.100 mol dm^{-3} solutions of potassium hydroxide, KOH(aq), and sulfuric acid, $H_2SO_4(aq)$, were mixed to produce potassium sulfate, $K_2SO_4(aq)$, and water.

   a. Formulate the balanced equation for the neutralization.

   b. Determine the concentrations, in mol dm^{-3}, of all solutes in the final solution.

The volumes of solutions and gases must *never* be added together. The densities of most gases are approximately 1,000 times lower than those of aqueous solutions, so when a gas is absorbed by a solution, the increase in the solution volume is negligible for most practical purposes.

The mole ratio techniques discussed in this topic are commonly used in **titration** calculations. In an acid–base titration, an acid and a base react with each other to form water and a salt. An acid–base indicator is added to the reaction mixture to monitor the reaction progress. The indicator changes colour when the neutralization is complete. The mole ratio is shown by the balanced equation for this neutralization reaction. If the volumes of both acid and base solutions are known, and the concentration of one of the reactants is also known, the unknown concentration of the second reactant can be determined.

The method for performing a titration is discussed in the *Skills* chapter.



▲ Figure 1 An acid–base titration with a coloured indicator

Worked example 5

20.0 cm^3 of sulfuric acid solution, $H_2SO_4(aq)$, of unknown concentration was titrated with a 0.200 mol dm^{-3} solution of sodium hydroxide, NaOH(aq), in the presence of an acid–base indicator. The colour change was observed when 14.3 cm^3 of the sodium hydroxide solution was added.

a. Formulate the balanced equation for the neutralization reaction.

b. Determine the concentration, in mol dm^{-3}, of sulfuric acid in the analysed solution.

Solution

a. $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

b. Start by converting the volume of sodium hydroxide solution to dm^3:

$V(NaOH) = 14.3\,cm^3 = 0.0143\,dm^3$

Then, find the amount of sodium hydroxide solute using $n = V \times c$:

$n(NaOH) = 0.0143\,dm^3 \times 0.200\,mol\,dm^{-3} = 0.00286\,mol$

According to the equation, H_2SO_4 and NaOH react in 1 : 2 ratio, so $n(H_2SO_4) = 0.00143\,mol$

Convert the volume of sulfuric acid solution to dm^3:

$V(H_2SO_4) = 20.0\,cm^3 = 0.0200\,dm^3$

Then, calculate the concentration of sulfuric acid using $c = \frac{n}{V}$:

$$c(H_2SO_4) = \frac{0.00143\,mol}{0.0200\,dm^3} = 0.0715\,mol\,dm^{-3}$$

You can also use the formula $\frac{c_1V_1}{x_1} = \frac{c_2V_2}{x_2}$, where x_1 and x_2 are the stoichiometric coefficients before the reactants in the balanced equation.

This way, you can skip the step where you find the amounts of the reactants used in titration.

Practice question

5. Complete neutralization of 10.0 cm^3 of a sodium hydroxide solution required 9.10 cm^3 of a 0.500 mol dm^{-3} hydrochloric acid. Determine the concentration, in mol dm^{-3}, of sodium hydroxide in the initial solution.

Back titration

When an analyte cannot be easily quantified with a titration, a back titration is employed. In a back titration, the substance being analysed is first neutralized by a reagent of known volume and concentration. The excess of this reagent is then determined by titration.

Eggshells contain calcium carbonate. Calcium carbonate is insoluble in water, so its quantity can be conveniently determined using a back titration method. The eggshells are first crushed and reacted with a known amount of hydrochloric acid taken in excess. The leftover acid is then titrated against a standard sodium hydroxide solution to find out how much of the acid was consumed in the reaction with the eggshells. The amount of calcium carbonate in the eggshells can then be found using stoichiometric calculations.

Relevant skills

- Tool 1: Applying techniques: titration
- Tool 3: Propagate uncertainties in processed data
- Inquiry 2: Collect and record sufficient relevant data
- Inquiry 2: Carry out correct and accurate data processing

Instructions

1. Using the list of equipment shown below, design a method to analyse the mass percentage of calcium carbonate in eggshells. The equipment list is not definitive: you do not have to use all the items in the list, and you can add further items or substances if you wish.
    - Calcium carbonate chips, $CaCO_3(s)$
    - A variety of different types of eggshells
    - 1.00 mol dm^{-3} hydrochloric acid, HCl(aq)
    - 1.00 mol dm^{-3} sodium hydroxide, NaOH(aq)
    - Phenolphthalein indicator solution
    - Burette
    - Burette clamp and stand
    - Conical flasks
    - Drying oven
    - Pestle and mortar
    - Top-pan balance
    - Volumetric pipette and filler
    - Wash bottle containing distilled water
    - Waste container(s)
    - Standard lab safety equipment
2. Conduct a risk assessment of your method and show it to your teacher for approval.
3. Collect sufficient relevant data, including qualitative observations. You will have to decide how much data will be sufficient.
4. Process the data to determine the mass percentage of calcium carbonate in each of the eggshell samples.
5. Propagate measurement uncertainties to obtain the uncertainties of the calculated mass percentages found above.
6. Graph your data using an appropriate software, including error bars.
7. If possible, compare your data to literature values. Comment on the accuracy and precision of your results.
8. Discuss at least two sources of error in your investigation and propose improvements that could minimize these errors.

Linking questions

How does the molar volume of a gas vary with changes in temperature and pressure? (Structure 1.5)

In what ways does Avogadro's law help us to describe, but not explain, the behaviour of gases? (Structure 1.4)

The reaction yield (*Reactivity 2.1.4*)

In any chemical reaction, the total mass of the reaction products is equal to the total mass of the consumed reactants. This principle, known as the **law of conservation of mass**, follows from atomic theory. Since atoms cannot be created or destroyed, their total number and therefore mass cannot be affected by chemical changes.

Science as a shared endeavour

"Nothing comes from nothing" was one of the basic principles in ancient Greek philosophy. The first recorded experimental evidence for the conservation of mass was obtained in 1748 by the Russian scientist Mikhail Lomonosov, who carried out chemical reactions in sealed vessels and found that chemical changes did not affect the total mass of the mixtures. However, the works of Lomonosov were almost unknown in western European countries, so it was over 25 years before the French chemist Antoine Lavoisier formulated this principle in 1774 and was commonly credited for its discovery.




▲ Figure 2 Left: Mikhail Lomonosov; right: Antoine Lavoisier

This example shows the importance of sharing knowledge and ideas: a valuable discovery that remains hidden from the global scientific community may go unnoticed for some time and have little impact on the advancement of science. What conditions facilitate an individual's access to the international scientific community? How can the number of valuable contributions of scientists from around the world be maximized?

Activity

Calculate the masses, in g, of the initial and final mixtures from worked example 2 and verify the validity of the law of conservation of mass.

When chemical reactions are carried out experimentally, the amounts of the reaction products are usually lower than the theoretical amounts predicted by the equation. This can happen for many reasons, including incomplete conversion of the reactants or simply because some product was lost during its isolation and purification. In such cases, we can calculate the **percentage yield**, which is the ratio of the experimental and theoretical amounts of the product. Since the amount of an individual substance is proportional to its mass ($n = \frac{m}{M}$), the yield can also be found as the ratio of the experimental and theoretical masses of the product:

$$\text{percentage yield} = \frac{n_{\text{experimental}}}{n_{\text{theoretical}}} \times 100\% = \frac{m_{\text{experimental}}}{m_{\text{theoretical}}} \times 100\%$$

The term "yield" can also refer to the amount or mass of the reaction product. In this sense, the **theoretical yield** is the amount or mass of the product that could be obtained from a particular reaction if the reaction went to completion and no product was lost. In contrast, the **experimental yield** is the actual amount or mass of the product isolated from a particular laboratory experiment. The experimental yield can be lower than or equal to the theoretical yield, but it can never exceed it.

The percentage yield, and theoretical and experimental yields of a reaction, are related as follows:

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

Worked example 6

A mixture of 10.0 g of calcium metal and 9.62 g of elemental sulfur was heated for some time to produce 17.0 g of calcium sulfide, CaS(s).

a. Calculate the theoretical yield, in g, of calcium sulfide.

b. Using your answer to part **a**, calculate the percentage yield of calcium sulfide.

Solution

a. First, write the balanced equation for the reaction:

$Ca(s) + S(s) \rightarrow CaS(s)$

Then calculate the amounts of the reactants and products using $n = \frac{m}{M}$:

$$n(\text{Ca}) = \frac{10.0\text{ g}}{40.08\text{ g mol}^{-1}} \approx 0.250\text{ mol}$$

$$n(\text{S}) = \frac{9.62\text{ g}}{32.07\text{ g mol}^{-1}} \approx 0.300\text{ mol}$$

$$n(\text{CaS}) = \frac{17.0\text{ g}}{72.15\text{ g mol}^{-1}} \approx 0.236\text{ mol}$$

The limiting reactant is calcium, as calcium and sulfur are consumed in a 1 : 1 ratio. The theoretical amount of calcium sulfide is 0.250 mol (the same as the initial amount of the limiting reactant, calcium). Therefore, the theoretical yield of calcium sulfide is:

$$m_{\text{theoretical}}(\text{CaS}) = 0.250\text{ mol} \times 72.15\text{ g mol}^{-1} \approx 18.0\text{ g}$$

b. The reaction produced 0.236 mol of calcium sulfide, while the theoretical yield of this compound was 0.250 mol. Therefore:

$$\text{percentage yield} = \frac{n_{\text{experimental}}}{n_{\text{theoretical}}} \times 100\% = \frac{0.236\text{ mol}}{0.250\text{mol}} \times 100\% = 94.4\%$$

The same result could be obtained through the experimental and theoretical masses of calcium sulfide:

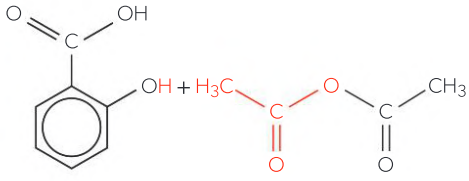
$$\text{percentage yield} = \frac{m_{\text{experimental}}}{m_{\text{theoretical}}} \times 100\% = \frac{17.0\text{ g}}{18.0\text{ g}} \times 100\% = 94.4\%$$

Practice question

6. Calculate the theoretical and percentage yields of the reaction between 9.443 g of aluminium and 7.945 dm^3 (STP) of oxygen that produced 17.13 g of aluminium oxide, $Al_2O_3(s)$.

Data-based questions

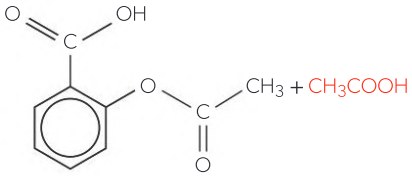
Salicylic (2-hydroxybenzoic) acid was isolated from the bark of willow trees in the first half of the 19th century and used as a pharmaceutical drug for pain and fever relief. However, pure salicylic acid caused severe stomach irritation and bleeding. These side effects could be significantly reduced by the use of a chemically modified salicylic acid, which is now known as acetylsalicylic acid or aspirin.



salicylic acid (2-hydroxybenzoic acid) M_r 138.13

ethanoic anhydride M_r 102.10





aspirin (acetylsalicylic acid) M_r 180.17

ethanoic acid M_r 60.06

The reaction of salicylic acid with ethanoic anhydride can be used in a school laboratory for the preparation of aspirin. In a typical experiment, salicylic acid is mixed with excess ethanoic anhydride and several drops of the catalyst (concentrated phosphoric acid, H_3PO_4). The mixture is heated for a short time, then diluted with water and allowed to cool down slowly, producing crystals of aspirin. The crystals are isolated by filtration and dried in the open air.

The aspirin obtained in this way is usually impure, so it needs to be recrystallized. To do so, impure aspirin is dissolved in hot ethanol, and the solution is allowed to cool down. Pure aspirin precipitates while most of the impurities remain in the solution. The crystals of pure aspirin are filtered and dried until all the ethanol evaporates.

Each step of the procedure outlined causes some loss of the product and therefore reduces the yield of aspirin. In particular, the reaction of salicylic acid with ethanoic anhydride is never 100% complete, some aspirin stays in the solution, small crystals of aspirin can be trapped in the paper filter, and so on. The results of three experiments performed by students are shown in table 2.

Experimental data	Experiment 1	Experiment 2	Experiment 3
mass of salicylic acid / g	2.06	2.57	2.35
mass of ethanoic anhydride / g	2.49	2.85	3.11
mass of aspirin / g	3.42	2.31	1.87
recrystallization from ethanol	none	once	twice
melting point of product / °C	122–128	133–135	135–136

▲ Table 2 Synthesis of aspirin data. The literature value for the melting point of pure aspirin is 136 °C

Impurities usually lower the melting point and widen the melting range of substances. The most common example is sea water, which freezes between −2.5 and −1.8 °C instead of at 0 °C like pure water.

Questions

1. Calculate the amounts of the reactants for each experiment and determine the limiting reactant in each case.
2. Determine the percentage yield of aspirin in each experiment.
3. Identify which experiment produced an impossible percentage yield of aspirin. Using the data in table 2, suggest a reason for this and provide supporting data for your answer.
4. Suggest which experiment produced the purest sample of aspirin. Explain your answer.

The identity of aspirin can be confirmed by IR spectroscopy, NMR spectroscopy, mass spectrometry (*Structure 3.2, AHL*) or by determining its melting point (*Structure 1.1*) and comparing the result with the literature data.

Gravimetric analysis of a precipitation reaction

The reaction between calcium chloride and sodium carbonate solutions produces a precipitate of calcium carbonate, which can be removed by filtration:

$$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaCl(aq)$$

In this practical, you will carry out this reaction, isolate the resulting calcium carbonate and calculate the yield.

Relevant skills

- Tool 1: Measuring mass and volume
- Tool 1: Drying to a constant mass
- Tool 3: Applying general mathematics
- Inquiry 2: Identify and record relevant qualitative observations
- Inquiry 2: Collect and record sufficient relevant quantitative data
- Inquiry 2: Carry out relevant and accurate data processing

Safety

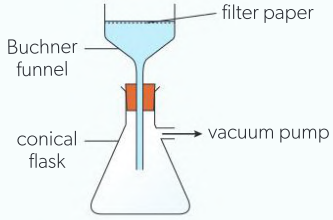
- Wear eye protection
- Sodium carbonate solution is an irritant, wear gloves when handling it

Materials

- two 25 cm^3 graduated pipettes or burettes
- 100 cm^3 beaker
- stirring rod
- vacuum filtration apparatus
- top-pan balance
- filter paper
- wash bottle containing distilled water
- drying oven (if available)
- 0.50 mol dm^{-3} calcium chloride solution, $CaCl_2(aq)$
- 1.0 mol dm^{-3} sodium carbonate solution, $Na_2CO_3(aq)$

Instructions

1. Write your initials on the filter paper in pencil and pre-weigh it.
2. In a 100 cm^3 beaker, add 25 cm^3 of 0.50 mol dm^{-3} calcium chloride solution to 15 cm^3 of 1.0 mol dm^{-3} sodium carbonate solution.
3. Stir the reaction mixture. Remember to record relevant qualitative observations throughout.
4. Filter the mixture (figure 3). Rinse the beaker and stirring rod with distilled water to completely transfer all traces of the precipitate onto the filter paper.
5. Allow the filter paper to dry undisturbed, inside a drying oven if possible. Once it looks dry, weigh it. Repeat until the mass remains constant.
6. Clean up as instructed by your teacher.



▲ Figure 3 Vacuum filtration

Questions

1. Determine the limiting and excess reactants.
2. Calculate the theoretical yield of calcium carbonate.
3. Calculate the experimental yield of calcium carbonate and hence, the percentage yield.
4. Compare and contrast your results to your classmates' results.
5. Outline two ways to improve the efficiency of this synthesis.

ATL Self-management skills

The following steps show a general approach to solving problems involving stoichiometric calculations. To help you remember all the steps, describe each step in three words, and then again in one word.

1. Formulate and balance the chemical equation(s) for any chemical changes mentioned in the problem.
2. Calculate the amounts of as many substances as you can using the formulas $n = \frac{m}{M}$ and $n = \frac{V}{V_m}$. If you know a mass percentage, convert it first to mass (if possible) and then to amount.
3. Write all known amounts of substances below their formulas in chemical equations. Any missing values may suggest the next step of the solution.
4. Determine the limiting reactant and use the mole ratio shown by stoichiometric coefficients in the balanced equation to calculate the changes in amount for all substances. Remember that the mole ratio does not reflect the initial or final amounts of reactants and products.
5. Check the mass balance. The total mass of products must be equal to the total mass of reactants. If it is not the case, revise your solution.
6. Check that the answer makes sense. All percentages should add up to 100%, and no individual percentage can be greater than 100%. The yield of the final product must not exceed 100%. Finally, treat very low or very high percentages with caution.

In practice, chemists often have to find a balance between yield and purity. For example, most medical drugs must have a purity of 99% or even higher, as any impurities could have unpredicted effects on the patient's health. As a result, the strict safety requirements reduce the yields and increase the prices of medical-grade products. At the same time, the acceptable purity for many industrial and agricultural materials, such as detergents and fertilizers, could be well below 50%, as long as the impurities do not compromise the safety and intended use of the product.

The yields of chemical reactions are particularly important in the chemical industry, where the loss of even a small percentage of the final product could mean a significant drop in profit. Low reaction yields increase the amount of waste, which needs to be disposed of safely or reused. The development of highly efficient synthetic procedures with low environmental impact is reflected in the concept of **green chemistry**, which is now adopted by the majority of commercial and research organizations around the world.

Linking question

What errors may cause the experimental yield to be i) higher and ii) lower than the theoretical yield? (Tool 1, Inquiry 1, 2, 3)

In figure 69, the measured length is 3.67 cm. The first two figures, 3 and 6, are certain, as the length is greater than 3.6 but less than 3.7 cm. The last figure, 7, is uncertain, as the actual length could be 3.66 or 3.68 cm. There is absolutely no way of getting the fourth figure using this ruler, as we are not even sure about the third figure.

As measurements are never exact, neither are the results of the calculations involving these measurements. There are two rules of thumb, which depend on the type of calculation being carried out:

1. Adding and subtracting measured values: the answer should be recorded to the least number of decimal places present in the values used.
2. Multiplying and dividing measured values: the answer should be recorded to the least number of significant figures present in the values used.

Worked example 2

At the beginning of a titration, the initial burette reading was 1.03 cm^3. At the end point, the final burette reading was 24.13 cm^3. Calculate the volume of the titrant used.

Solution

Volume used = final volume − initial volume

$= 24.13\,cm^3 - 1.03\,cm^3$

$= 23.10\,cm^3$ (2 dp)

Both values used have two decimal places. They are subtracted and therefore the answer should also be stated to two decimal places.

Worked example 3

Calculate the amount of magnesium oxide, MgO, in mol, in a 0.500 g sample of pure magnesium oxide.

Solution

Molar mass of MgO = $M(MgO) = 24.31 + 16.00 = 40.31\,g\,mol^{-1}$

$$n = \frac{m}{M}$$

$$n(MgO) = \frac{0.500\,g}{40.31\,g\,mol^{-1}}$$

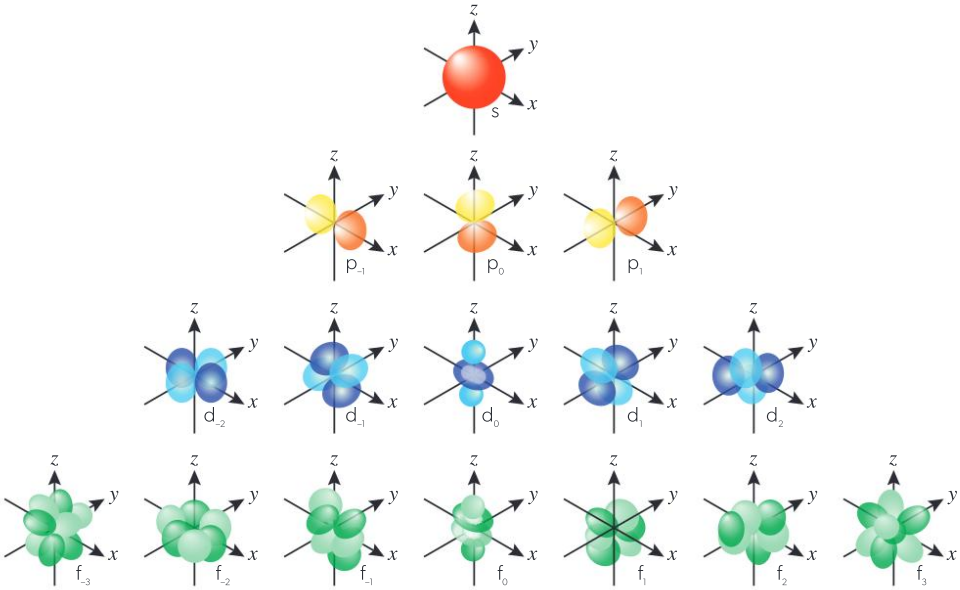
$= 0.012404\ldots$ mol ≈ 0.0124 mol (3 sf)

This operation involves a multiplication, so the answer should be rounded to the least number of significant figures in the values used. The answer is therefore rounded to three significant figures.

Many calculations require several steps. Rounding too early could lead to incorrect answers due to rounding errors. Round the final answer to the required number of decimal places or significant figures, but not the answers to intermediate steps. Always carry two or three extra significant figures through intermediate calculations.

Electron configurations (*Structure 1.3.5*)

Each atomic orbital type has a characteristic shape and energy. The s orbital is spherical and it has the lowest possible energy. There are three p orbitals, each oriented differently. There are five d orbitals and seven f orbitals, and these are higher in energy than s or p.



▲ Figure 16 The shapes of the s, p, d and f orbitals. Only the shapes of s and p orbitals need to be known

Each energy level defined by the principal quantum number, n, can hold n types of orbitals (table 1). For $n = 1$, only the s orbital exists. For $n = 2$, there are two types of orbital: s and p. For $n = 3$, there are three types: s, p, and d. For $n = 4$, there are four types: s, p, d, and f.

Principal quantum number (n)	Type of sublevel	Number of orbitals per type	Total number of orbitals per energy level (n^2)	Maximum number of electrons within energy level ($2n^2$)
1	s	1	1	2
2	s	1	4	8
	p	3		
3	s	1	9	18
	p	3		
	d	5		
4	s	1	16	32
	p	3		
	d	5		
	f	7		

◀ Table 1 Each energy level, defined by n, can hold $2n^2$ electrons. The number of sublevels, or atomic orbital types, is equal to n. For $n = 4$ there are four types of orbitals (s, p, d, and f) with 16 atomic orbitals in total occupied by a maximum of $2(4)^2 = 32$ total electrons

Activity

State the following for the energy level with $n = 5$:

a. the sublevel types

b. the number of atomic orbitals in each sublevel

c. the total number of atomic orbitals

d. the maximum number of electrons at that energy level.

Orbital diagrams

For convention, an "arrow in box" notation called an **orbital diagram** is used to represent how electrons are arranged in atomic orbitals (figure 17). The arrangement of electrons in orbitals is called **electron configuration**.

s sublevel (one box representing an s orbital)



p sublevel (three boxes representing the three p orbitals p_x, p_y, and p_z)



d sublevel (five boxes representing the five d orbitals)



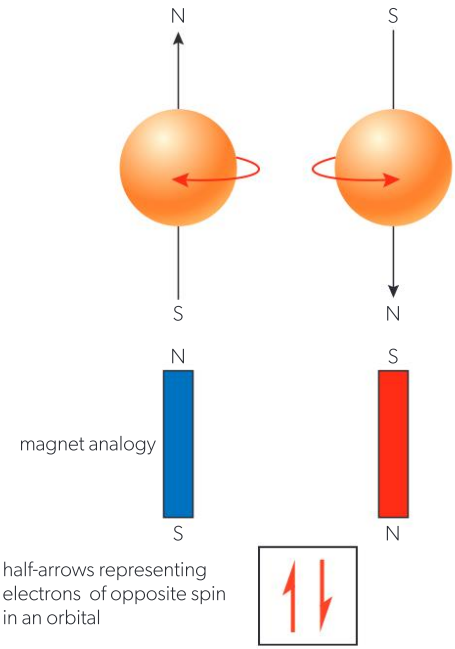
f sublevel (seven boxes representing the seven f orbitals)



▶ Figure 17 In orbital diagrams, each box represents an orbital. This diagram shows the number of orbitals for each sublevel. Arrows are drawn in the boxes to represent electrons. A maximum of two electrons can occupy each orbital, so each box has a maximum of two "arrows"

Atomic orbitals are regions of space where there is a high probability of finding electrons. Electrons are charged negatively, and like charges repel each other, so two electrons should not be able to occupy the same region of space. Quantum mechanics solves this problem by using a ± spin notation for each electron. A pair of electrons with opposite spins behave like magnets facing in opposite directions. Hence each orbital box is shown with an upwards half-arrow, ↿, and one downwards half-arrow, ⇂ (figure 18). This is known as the **Pauli exclusion principle**:

Only two electrons can occupy the same atomic orbital and those electrons must have opposite spins.



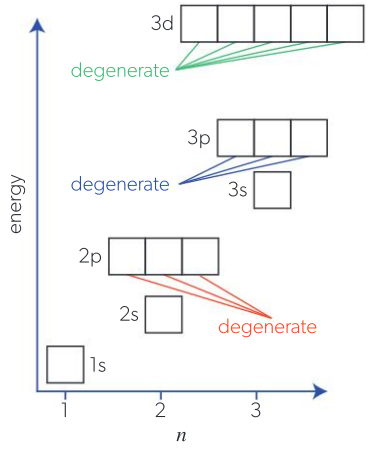
◀ Figure 18 Electron spin is represented by an arrow pointing up (positive spin) or down (negative spin)

TOK

Electron spin is often interpreted as the rotation of the electron around its own axis. However, this interpretation has no physical basis: electrons in atoms behave like waves, and a wave cannot rotate. Unfortunately, neither the spin nor the wave-like behaviour of electrons can be visualized in any way, as they have no analogues in our everyday life and can be expressed only in mathematical form. This lack of visualization does not undermine the quantum theory but rather shows the limits of human perception and, at the same time, the power of mathematics as the language of science.

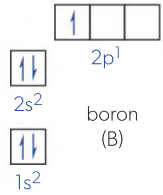
To what extent does mathematics support knowledge development in the natural sciences?

Each of the atomic orbitals of the same type in one sublevel are of equal energy. Orbitals with the same energy are referred to as **degenerate** orbitals (figure 19).



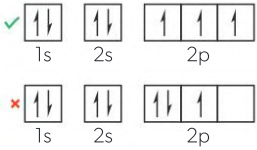
▲ Figure 19 The three 2p orbitals are degenerate as they have the same energy. These three degenerate atomic orbitals have lower energy than the three 3p orbitals

An atom of boron (B) has five electrons, and its orbital diagram is drawn as follows:



The single 2p electron in boron can occupy any of the three orbitals, as they have equal energies. The degenerate 2p orbitals are represented by boxes joined together to show their energy equivalence. Traditionally, the half-arrow is drawn in the leftmost box, although it is a matter of personal preference.

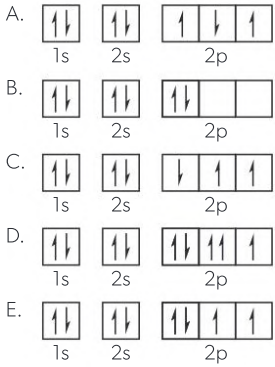
Hund's rule states that every **degenerate orbital in a sublevel is singly occupied before any orbital is doubly occupied** and that **all electrons in singly occupied orbitals have the same spin**. This means that the three p orbitals must have one electron with the same spin in each of them before any orbital can become doubly occupied with an electron of opposite spin (figure 20).



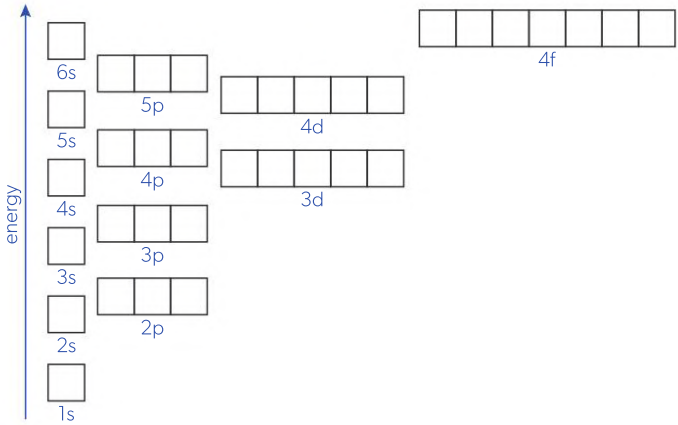
▲ Figure 20 The electrons are evenly distributed across the three degenerate 2p orbitals in nitrogen before an orbital is doubly occupied

Practice questions

1. Look at figure 20. The 1s and 2s orbitals are fully occupied by electrons. Why do you think the 1s and 2s orbitals are filled before the 2p orbitals?
2. State which of the diagrams below represents a correct electron configuration based on Hund's rule and the Pauli exclusion principle. State the reason for the four incorrect diagrams being wrong.

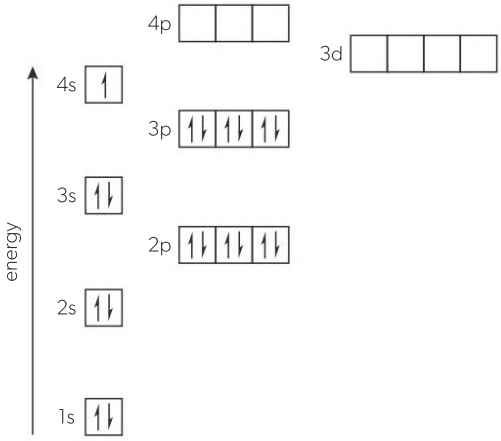


The **Aufbau principle** states that as electrons are added to atoms, **the lowest available energy orbitals fill before higher energy orbitals do**. The third and fourth energy levels contain d and f orbitals (figure 21). These orbitals are typically filled after the s orbitals of the following levels because they are higher in energy. As shown in figure 21, the 3d sublevel has a higher energy than 4s but lower than 4p, so 4s is filled with electrons first, followed by 3d and finally 4p. For the same reason, 4d orbitals are filled after 5s, and 4f orbitals are filled only after 6s.



◀ Figure 21 The 4s sublevel has a lower energy and will fill before the 3d sublevel

This is consistent with experimental data that show that potassium, K, and calcium, Ca have electrons in the 4s sublevel, not in 3d.



◀ Figure 22 Potassium orbital filling diagram showing the outermost electron in the 4s orbital because 3d orbitals are higher in energy

Generally, the following order is observed:

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p \ldots$

In the IB Diploma Programme, only the electron configurations of atoms and ions up to $Z = 36$ will be assessed. Electrons in these species can fill sublevels up to 4p.

Electron sharing and transfer are fundamental to understanding chemical reactions, so it is important to know the electron configuration of an atom or an ion. There are three ways to show the electron configuration:

1. Full electron configuration
2. Condensed electron configuration
3. Orbital filling diagram ("arrows in boxes" notation)

The orbital filling diagram for potassium is given in figure 22.

Activity

Copy the orbital diagram from figure 21 and complete it for the following elements in their ground states:

a. aluminium, Al
b. chlorine, Cl
c. iron, Fe

Refer to the periodic table at the back of this book to deduce the number of electrons in each atom.

Full electron configurations

To write a full electron configuration, we use the periodic table, and "build up" the electrons in successive orbitals according to the Aufbau principle, Hund's Rule and the Pauli exclusion principle.

Worked example 1

Determine the full electron configuration for the calcium atom.

Solution

The Aufbau principle states that as electrons are added to atoms, the lowest available energy orbitals fill before higher energy orbitals do. From the Pauli exclusion principle, we know that each orbital will have a maximum of two electrons.

The atomic number of calcium is 20. Let's split the 20 electrons evenly across each orbital, starting with the lowest energy first. When writing electron configurations, the number of electrons within each sublevel is given in superscript, next to the sublevel:

- The 1s orbital has two electrons: $1s^2$
- The 2s orbital also has two electrons: $2s^2$
- The three 2p orbitals have two electrons each, six in total: $2p^6$
- The 3s orbital has two electrons: $3s^2$
- The three 3p orbitals have three electrons each, six in total: $3p^6$

This brings us up to 18 electrons, with two left over to go into the orbital with the next lowest energy, 4s: $4s^2$
So, for calcium, the full electron configuration is $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2$.

Practice question

3. Determine the full electron configuration for the phosphorus atom.

Condensed electron configurations

As the atomic number of an element increases, the full electron configuration gets longer and it can be time-consuming to write. The chemistry of atoms and ions is mostly determined by their **valence electrons**, that is, the outermost electrons, rather than the **inner core electrons.** A more convenient way of writing electron configurations is to highlight the valence electrons and represent the inner core electrons as having the same electron configuration as the previous group 18 (known as the **noble gases**) element in the periodic table:

Condensed electron configuration = [previous noble gas] + valence electrons

Table 2 shows some more examples of full and condensed electron configurations.

▶ Table 2 Examples of full and condensed electron configurations for selected elements

Element	Atomic number	Full electron configuration	Condensed electron configuration
O	8	$1s^2\,2s^2\,2p^4$	[He] $2s^2\,2p^4$
Ne	10	$1s^2\,2s^2\,2p^6$	[Ne]
Mn	25	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^5$	[Ar] $4s^2\,3d^5$
Br	35	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^{10}\,4p^5$	[Ar] $4s^2\,3d^{10}\,4p^5$

Worked example 2

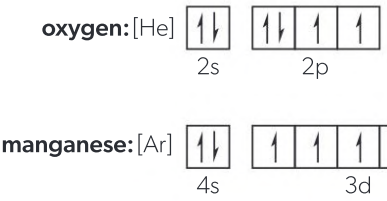
Determine the condensed electron configuration for the calcium atom.

Solution

In worked example 1, we determined the full electron configuration of calcium to be $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2$.

The previous noble gas in the periodic table is argon, which has an atomic number of 18. Argon has an electron configuration of $1s^2\,2s^2\,2p^6\,3s^2\,3p^6$, and we can therefore write the condensed electron configuration of calcium as $[Ar]\,4s^2$.

Orbital diagrams can also sometimes be shortened by using a condensed electron configuration. The condensed orbital diagrams for oxygen and manganese are shown below.



Practice question

4. Determine the condensed electron configuration for the phosphorus atom.

The periodic table is structured according to the type of sublevel that valence electrons of elements appear in. This is discussed further in *Structure 3.1.*

ATL Self-management skills

The ideas in this topic span a range of concepts and skills: from the quantum mechanical model of the atom, to how to write electron configurations.

Write a chapter summary, no longer than a sheet of A4 paper.

Write three key takeaways from the chapter.

List the key vocabulary you should know from this chapter.

Write five brief questions that test your understanding of the ideas in this chapter. Make an answer key, then try them out on one of your peers.

Exceptions to the Aufbau principle

The Aufbau principle correctly predicts the order of filling of atomic orbitals for most elements. However, when atoms lose electrons to form ions, the electrons in the sublevel with the highest principal quantum number (n) are lost first.

So, for Mn, with electron configuration $[Ar]\,4s^2\,3d^5$, the 4s electrons will be lost first. This gives the manganese ion, Mn^{2+}, with electron configuration $[Ar]\,3d^5$, **not** $[Ar]\,4s^2\,3d^3$.

All elements with 3d valence electrons tend to lose two 4s electrons to form 2+ ions. These are known as the 3d **transition elements**, or **transition metals**. If you look at the periodic table at the back of this book, these elements are from scandium (Sc) to copper (Cu). These transition metals can also have **variable oxidation states** in compounds.

There are some exceptions. With only one electron in its 3d orbital, scandium readily forms only Sc^{3+} ions, by losing this 3d electron and the two 4s electrons.

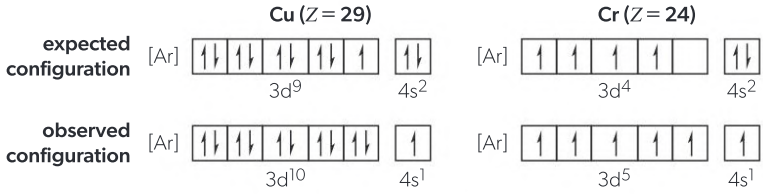
Ionization and oxidation are covered in *Structure 2.1* and *Structure 3.1*.

Activity

Deduce the electron configuration of the Cu^{2+} cation.

The ground state configurations of copper and chromium are also different from those predicted by Aufbau principle.

The predicted electron configuration of copper is [Ar] $4s^2\,3d^9$, as the Aufbau principle suggests that the lower-energy 4s orbital should be filled first. However, the observed ground-state electron configuration for copper is [Ar] $4s^1\,3d^{10}$ (figure 23). For chromium, the predicted configuration is [Ar] $4s^2\,3d^4$ and the observed is [Ar] $4s^1\,3d^5$ (figure 23). In each case, promoting a 4s electron to a 3d level leads to a more stable electron configuration. In the case of copper, this gives a full d sublevel, and in the case of chromium, there are no paired electrons but rather six half-occupied orbitals, each containing an electron with the same spin.



▶ Figure 23 The expected and observed electron configurations of copper and chromium

The electron configurations of chromium (Cr) and copper (Cu) are the only two exceptions that you need to know. In all other elements up to $Z = 36$, the sublevels are filled with electrons according to the general order.

AHL

Ionization energy (*Structure 1.3.6 and 1.3.7*)

The quantum mechanical model of the atom helps to explain the trends and discontinuities in the first **ionization energies (*IE*)** of elements. Ionization energy is the minimum energy required to eject an electron out of a neutral atom or molecule in its ground state.

$$X(g) + \text{energy} \rightarrow X^+(g) + e^-$$

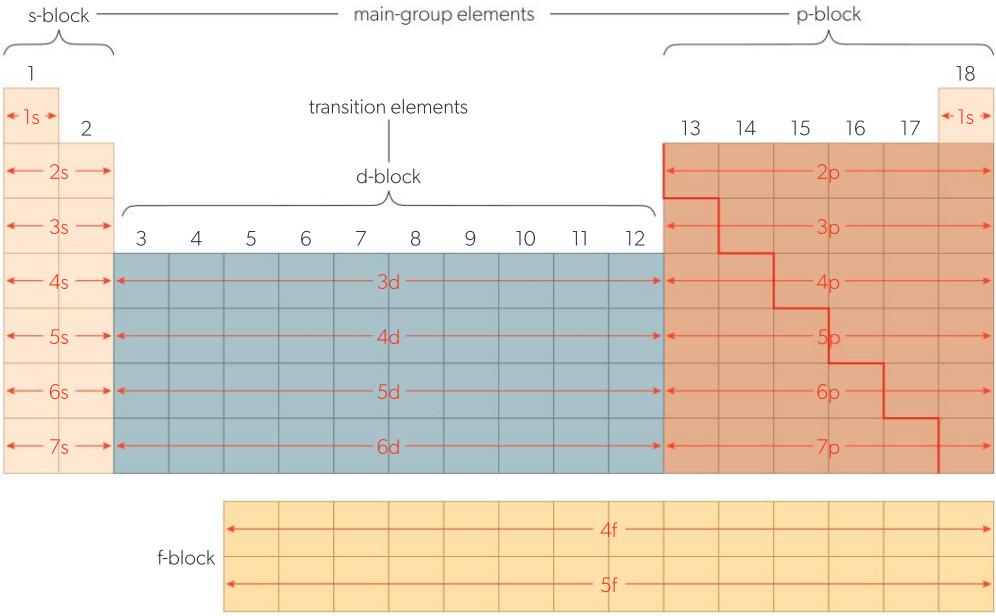
Ionization energy and periodic table trends are discussed further in *Structure 3.1*.

The columns in the periodic table are known as groups, and the rows are known as periods. Going across the periodic table, the groups are numbered from 1 to 18. The periodic table can be shown as four blocks corresponding to the four sublevels s, p, d, and f (figure 24). The sublevels holding the outermost valence electrons for each element are also shown.

First ionization energy (IE_1) generally decreases down the groups of the periodic table and increases across the periods.

Going down a group, the number of sublevels increases. The outermost electrons are shielded from the pull of the nucleus by the electrons in the lower energy sublevels (so-called "inner electrons"). The more sublevels, the greater the shielding and therefore less energy is required to remove electrons from the outermost sublevel.

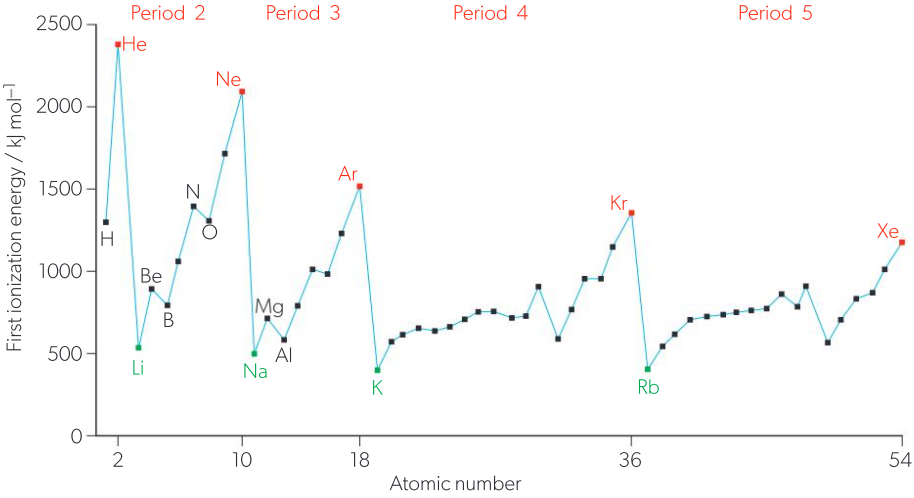
Going across a period, the number of protons in the nucleus increases, so the outermost electrons are held closer to the nucleus by the **increased nuclear charge**. At the same time, the shielding effect remains nearly constant because the number of inner electrons does not change.



▲ Figure 24 The blocks of the periodic table correspond to the sublevels s, p, d and f

Therefore, more energy is required to remove outermost electrons, so ionization energy increases across the period.

The general trend of decreasing ionization energy down a group and increasing across a period is shown in figure 25:



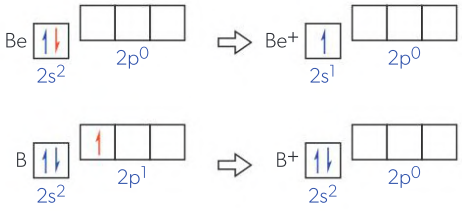
▲ Figure 25 Plot of first ionization energy against atomic number for the elements from hydrogen to xenon

AHL

There are two clear discontinuities across the period:

1. Between the group 2 and group 3 elements

The valence electron configuration of beryllium is $2s^2$ while for boron it is $2s^2\,2p^1$. The paired $2s^2$ electrons shield the single 2p electron in boron from the nucleus, making the electron slightly easier to remove.



Patterns and trends

Scientists look out for patterns and trends in the data they collect. The presence of discrepancies — results that do not fit the overall pattern — allows further conclusions to be drawn.

What can be inferred from the patterns in successive ionization energies?

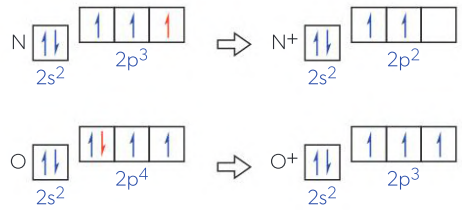
The same trend can be observed in comparing group 2 to group 3 elements in any period. For example, the $3s^2$ electrons shield the lone $3p^1$ electron in aluminium, so the first ionization energy of aluminium is lower than that of magnesium.

Suppose you have a two-story building and you need to remove one floor to meet new height regulations. Which floor would you remove? Obviously, it will be the top floor, as the building would collapse otherwise! The same reasoning can be applied to the ionization of atoms — electrons are removed first from the highest occupied energy level, and from the highest energy sublevel within that level.

2. Between the group 15 and 16 elements

From group 15 to 16 there is also a drop in ionization energy. The electron configuration of nitrogen is $1s^2\,2s^2\,2p^3$ while for oxygen it is $1s^2\,2s^2\,2p^4$.

Nitrogen has a more stable electron configuration than oxygen as it has a half-filled p sublevel, and therefore more energy is required to remove an electron from nitrogen (figure 26). This is because the paired electrons in oxygen occupy the same region of space and have increased repulsion. However, in nitrogen, the three electrons in the 2p orbitals do not come into close proximity.

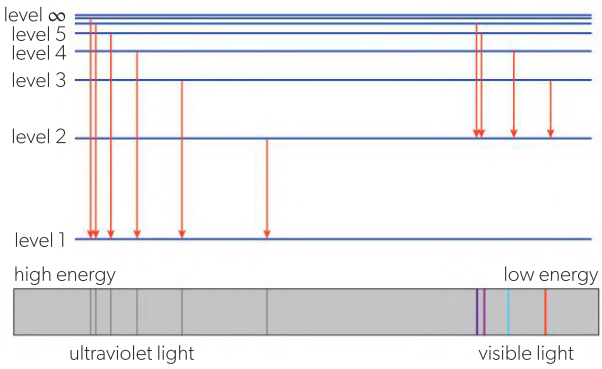


▶ Figure 26 A half-filled p sublevel is more stable than p sublevels with 2 or 4 electrons

The most stable p orbital configuration is p^6, a completely filled p sublevel, followed by p^3, a half-filled sublevel. This is generally true for other sublevels. For example, d^{10} and d^5 electron configurations are also stable, which partly explains why chromium and copper do not obey the Aufbau principle (figure 23).

Calculating ionization energy from spectral data

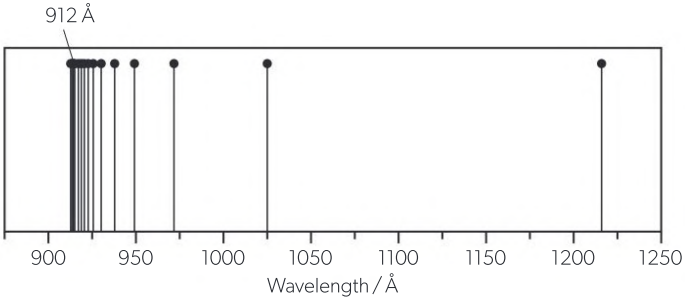
As the principal quantum number of energy levels increases, the distance between the levels converges to a continuum. This can be observed by spectral lines converging in the hydrogen emission spectrum, shown in figure 27.



◀ Figure 27 Ultraviolet and visible light transitions in hydrogen and the resulting emission spectrum

The spectral lines in the hydrogen emission spectrum converge at 9.12×10^{-8} m, or 912 Å (figure 28). This represents the wavelength of light at which the hydrogen atom is ionized.

This wavelength can be used to calculate the first ionization energy of hydrogen.



◀ Figure 28 Hydrogen is ionized at the wavelength where the spectral lines converge in the emission spectrum

AHL

Worked example 3

Spectral lines converge at 9.12×10^{-8} m in the emission spectrum of the hydrogen atom. Calculate the first ionization energy of hydrogen in kJ mol^{-1}.

Solution

First, calculate the frequency of radiation using $c = f \times \lambda$, where c is the speed of light, approximately equal to 3.00×10^8 m s^{-1}.

$3.00 \times 10^8 \text{ m s}^{-1} = f \times 9.12 \times 10^{-8} \text{ m}$

$f = 3.29 \times 10^{15} \text{ Hz (s}^{-1})$

Then, calculate the energy using Planck's constant and the equation $E = h \times f$

$E = 6.63 \times 10^{-34} \text{ J s} \times 3.29 \times 10^{15} \text{ s}^{-1}$

$= 2.18 \times 10^{-18} \text{ J}$

Alternatively, these two steps can be merged into one by using the equation $E = \frac{h \times c}{\lambda}$

This represents the energy of a single photon of light which would be absorbed in exciting the electron in a hydrogen atom to the convergence level, or removing one electron from the atom.

Ionization energies are usually given in kJ mol^{-1}. You can convert the ionization energy value to kJ mol^{-1} using Avogadro's constant (N_A, the number of atoms in 1 mol) and the following equation:

The ionization energy in kJ mol^{-1}

$$= \frac{(\text{energy needed to remove one electron from an atom}) \times N_A}{1000}$$

$$= \frac{2.18 \times 10^{-18} \text{ J} \times 6.02 \times 10^{23} \text{ mol}^{-1}}{1000}$$

$$= 1.31 \times 10^3 \text{ kJ mol}^{-1}$$

The values of the speed of light, Planck's constant and Avogadro's constant are given in the data booklet. The mole and Avogadro's constant are discussed further in *Structure 1.4*.

Worked example 4

The first ionization energy of Na is 496 kJ mol^{-1} as given by the IB data booklet. Calculate the wavelength of convergence for the sodium atom spectrum in Å.

Solution

First, find the energy of ionization for one atom by converting the given value from kJ to J and dividing it by Avogadro's constant.

$496\,000 \text{ J mol}^{-1} / 6.02 \times 10^{23} \text{ mol}^{-1} = 8.24 \times 10^{-19} \text{ J}$

Then calculate the wavelength of light using $E = \frac{h \times c}{\lambda}$

$$8.24 \times 10^{-19} \text{ J} = \frac{6.63 \times 10^{-34} \text{ J s} \times 3.00 \times 10^8 \text{ m s}^{-1}}{\lambda}$$

$$\lambda = 2.41 \times 10^{-7} \text{ m}$$

$$= 2410 \text{ Å}.$$

This corresponds to the UV region in the electromagnetic spectrum.

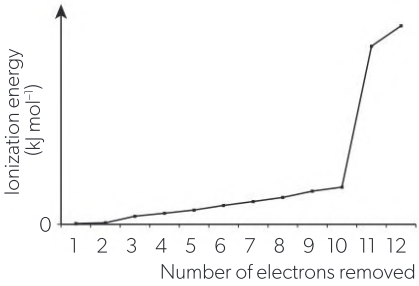
Practice questions

5. In the emission spectrum of the helium atom, the spectral lines converge at 5.05×10^{-8} m. Calculate the first ionization energy, in kJ mol^{-1}, of helium.
6. The first ionization energy of calcium is 590 kJ mol^{-1}. Calculate the wavelength of convergence, in Å, for the calcium atomic spectrum.

Successive ionization energies

It requires more energy to remove the second and **successive** electrons from an atom because the number of protons exceeds the number of remaining electrons while the electron–electron repulsion decreases.

As a result, electron clouds are pulled closer to the nucleus and held tighter by the increased electrostatic attraction. Once all the valence electrons are removed so that only the stable noble gas configuration remains, the energy required to remove the next electron increases sharply, as shown in figure 29.



◀ Figure 29 Removing 10 electrons from magnesium gives the noble-gas configuration $1s^2$ or [He]. There is a considerable increase in energy required to remove the 11th electron

Worked example 5

The first five successive ionization energies for an unknown element X have the following values: 403, 2633, 3860, 5080 and 6850 kJ mol^{-1}. Deduce the group of the periodic table in which element X is likely to be found.

Solution

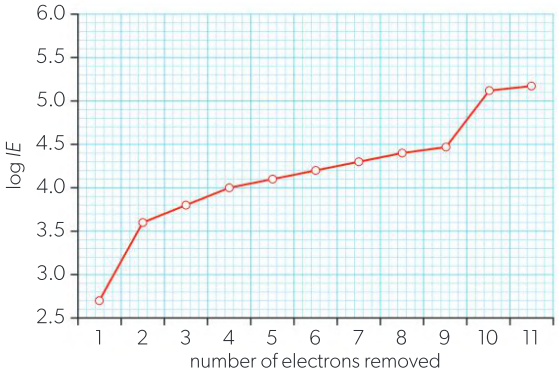
The largest increase in energy occurs from the first ionization (403 kJ mol^{-1}) to the second (2633 kJ mol^{-1}). This means that the second electron is likely to be removed from a stable noble gas configuration of the atom. Therefore, the outermost energy level of the element contains one electron, so the element belongs to group 1 of the periodic table.

Practice question

7. The first five successive ionization energies of an unknown element have the following values: 801, 2427, 3660, 25 026 and 32 827 kJ mol^{-1}. Deduce the group of the periodic table in which this element is likely to be found.

Data-based question

Using figure 30 and the periodic table, explain the two large jumps in the successive ionization energies for sodium.



▲ Figure 30 Successive ionization energies for sodium

AHL

Ionization energy data

Relevant skills

- Tool 2: Extract data from databases
- Tool 2: Use spreadsheets to manipulate data and represent data in graphical form
- Tool 3: Construct and interpret graphs

Instructions

Part 1: Data collection

1. Identify a database that contains successive ionization energy data for the elements (for example, WebElements).
2. Choose one of the following elements: sulfur, chlorine, argon, potassium or calcium.
3. Collect successive ionization energy data in a spreadsheet, labelling the columns as follows:

	A	B
1	**element name:**	
2		
3	**ionization**	**ionization energy/ kJ mol^{-1}**
4	**1**	
5	**2**	
	6 etc	

Part 2: Graphing successive ionization energies

4. Plot a line graph of ionization energy vs ionization. Make sure that you present your graph suitably, with axis labels, suitable scales and a descriptive title.
5. Answer the following questions:
   a. Identify the IE values that correspond to the innermost and outermost electrons.
   b. Explain why ionization energy increases with each successive electron.
   c. Explain how the graph provides evidence for the existence of main energy levels in the atom.

Part 3: Graphing the logarithm of the ionization energies

6. Title the third column in your spreadsheet "log (ionization energy)" as shown below:

	A	B	C
1	**element name:**		
2			
3	**ionization**	**ionization energy/ kJ mol^{-1}**	**log (ionization energy)**
4	**1**		
5	**2**		
	6 etc		

7. Compute the logarithm of each ionization energy using the spreadsheet LOG (or LOG10) function.
8. Construct a graph showing the logs of successive ionization energies by plotting log (ionization energy) vs ionization number.
9. Answer the following questions:
   a. Identify the large increases in ionization energy that indicate a change in main energy level.
   b. Why is it useful to plot the logs of the ionization energies?

Part 4: Evidence for the existence of sublevels

10. Construct a graph that will allow you to closely examine the electrons in energy level $n = 2$. Enlarge the graph and "zoom in" to inspect the increases closely.
11. Can you see any unusually large increases in ionization energy? Explain how they relate to the existence of sublevels.
12. Experimental data is often organized into tables and later transformed into graphical forms. What is the role of graphical representations in the advancement of scientific knowledge? Are graphical representations employed in other subject areas?

TOK

Linking questions

How does the trend in IE values across a period and down a group explain the trends in properties of metals and non-metals? (Structure 3.1)

Why are log scales useful when discussing $[H^+]$ and ionization energies? (Tool 3, Reactivity 3.1)

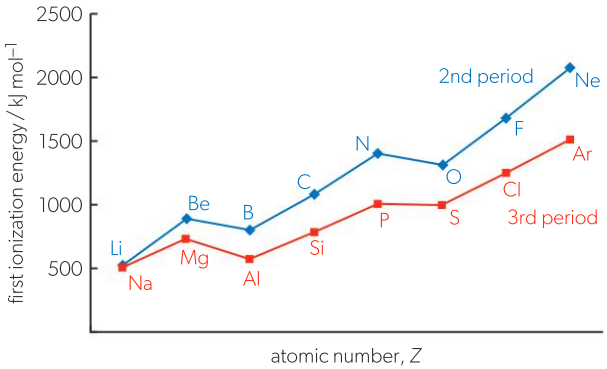
AHL

How do patterns of successive ionization energies of transition elements help to explain the variable oxidation states of these elements? (Structure 3.1)

AHL

Discontinuities in ionization energy trends (*Structure 3.1.7*)

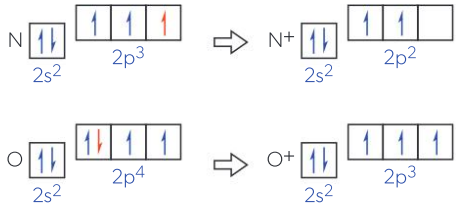
Earlier in this topic, you saw that there was a trend of increasing ionization energy going across a period. You may have noticed two discontinuities from group 2 to group 13, and from group 15 to group 16 (figure 15).



▲ Figure 15 The ionization energies of the second and third period elements

The first discontinuity, between beryllium and boron, can be explained by looking at their electron configurations. The electron configuration of boron is $1s^2\,2s^2$ and the electron configuration of beryllium is $1s^2\,2s^2\,2p^1$. Despite the greater nuclear charge in boron, it is easier to remove an electron from this element as the valence electron is in a higher energy sublevel, which is shielded from the nucleus by the 2s sublevel. This drop in ionization energy therefore provides evidence for the existence of s and p sublevels.

The second discontinuity, between nitrogen and oxygen, can be explained by electron repulsion in orbitals. The electron configuration of nitrogen is $1s^2\,2s^2\,2p^3$ while for oxygen it is $1s^2\,2s^2\,2p^4$. In nitrogen, the three electrons in the 2p orbitals do not come into close proximity. The paired electrons in oxygen occupy the same region of space and have increased repulsion. The higher-energy paired electron is therefore easier to remove. This explains the drop in ionization energy between the two groups.



▲ Figure 16 A half-filled p sublevel is stable due to decreased electron repulsion, as none of the electrons are paired

Structure 2.2 The covalent model

What determines the covalent nature and properties of a substance?

Covalent bonds lead to a vast range of different substances. From water to diamond to nitrogen gas, from oils to plastics to polyatomic ions, these species contain atoms held together by strong covalent bonds. Covalent bonds lead to the formation of two different types of structure: **covalent network structures** (also known as **giant covalent structures**) and **molecular covalent structures**.

In general, covalent substances are poor electrical conductors.

Substances with covalent network structures are also characterized by high melting points and boiling points, low volatility and poor solubility in water. Substances with molecular structures, on the other hand, generally have low melting points and boiling points. Their solubility and volatility vary greatly depending on their intermolecular forces.

In *Structure 2.2*, you will learn what a covalent bond is, how we represent molecules as well as how we describe and explain their shapes and intermolecular forces. You will also learn about covalent network structures.

Understandings

Structure 2.2.1 — A covalent bond is formed by the electrostatic attraction between a shared pair of electrons and the positively charged nuclei.

The octet rule refers to the tendency of atoms to gain a valence shell with a total of 8 electrons.

Structure 2.2.2 — Single, double and triple bonds involve one, two and three shared pairs of electrons respectively.

Structure 2.2.3 — A coordination bond is a covalent bond in which both the electrons of the shared pair originate from the same atom.

Structure 2.2.4 — The Valence Shell Electron Pair Repulsion (VSEPR) model enables the shapes of molecules to be predicted from the repulsion of electron domains around a central atom.

Structure 2.2.5 — Bond polarity results from the difference in electronegativities of the bonded atoms.

Structure 2.2.6 — Molecular polarity depends on both bond polarity and molecular geometry.

Structure 2.2.7 — Carbon and silicon form covalent network structures.

Structure 2.2.8 — The nature of the force that exists between molecules is determined by the size and polarity of the molecules. Intermolecular forces include London (dispersion), dipole–induced dipole, dipole–dipole and hydrogen bonding.

Structure 2.2.9 — Given comparable molar mass, the relative strengths of intermolecular forces are generally: London (dispersion) forces < dipole–dipole forces < hydrogen bonding.

Structure 2.2.10 — Chromatography is a technique used to separate the components of a mixture based on their relative attractions involving intermolecular forces to mobile and stationary phases.

AHL

Structure 2.2.11 — Resonance structures occur when there is more than one possible position for a double bond in a molecule.

Structure 2.2.12 — Benzene, C_6H_6, is an important example of a molecule that has resonance.

Structure 2.2.13 — Some atoms can form molecules in which they have an expanded octet of electrons.

Structure 2.2.14 — Formal charge values can be calculated for each atom in a species and used to determine which of several possible Lewis formulas is preferred.

Structure 2.2.15 — Sigma bonds (σ) form by the head-on combination of atomic orbitals where the electron density is concentrated along the bond axis.

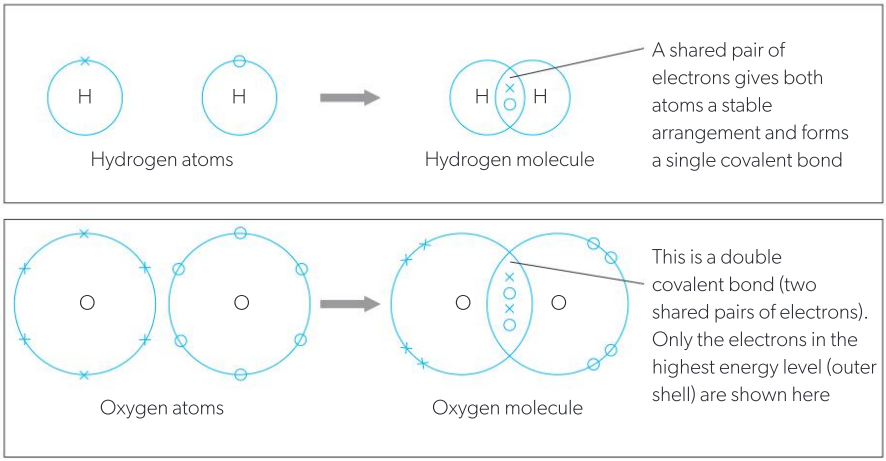
Pi bonds (π) form by the lateral combination of p orbitals where the electron density is concentrated on opposite sides of the bond axis.

Structure 2.2.16 — Hybridization is the concept of mixing atomic orbitals to form new hybrid orbitals for bonding.

Covalent bonds and molecules (*Structure 2.2.1*)

Covalent bonds are formed when atoms share pairs of valence electrons. A covalent bond results from the electrostatic attraction between a shared pair of electrons and the positively charged nuclei of the atoms involved in the bond.

Diatomic hydrogen, H_2, is the simplest covalent molecule. It consists of two hydrogen atoms held together by a covalent bond made from one electron from each hydrogen atom (figure 1). Atoms can share one, two or three pairs of electrons, forming single, double or triple bonds, respectively. Figure 1 also shows the formation of the double covalent bond in diatomic oxygen, O_2.



▲ Figure 1 Covalent bond formation in diatomic hydrogen, H_2, and diatomic oxygen, O_2



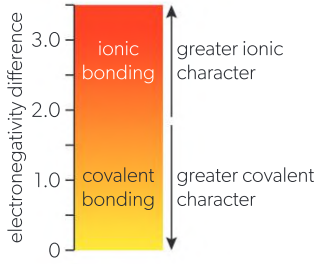
▲ Figure 2 Examples of substances that contain covalent bonds: plastics, graphite (in pencil leads), fizzy water, and the carbon dioxide bubbles in it

Activity

Draw a diagram to show the triple covalent bond in diatomic nitrogen, N_2.

Electronegativity

Covalent bonds generally form between atoms of relatively high electronegativity, typically non-metals. In *Structure 2.1*, we saw that ionic bonds form when the electronegativity difference between two atoms is greater than 1.8. When the difference in electronegativity between the atoms is less than 1.8 (figure 3), the bond between them is predominantly covalent.



▲ Figure 3 Electronegativity differences that are lower than 1.8 suggest that covalent bonding is present

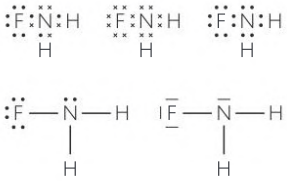
Unlike ionic substances, which are always compounds, covalent substances can be either elements or compounds. When two non-metal atoms of the same element bond together, the electronegativity difference is zero, so they form a covalent bond between them.

Electronegativity increases across periods and up groups. If two atoms of relatively high electronegativity are found close to each other in the periodic table, the bonds they form are likely to be covalent. For example, oxygen and fluorine form covalent bonds with each other.

As discussed in *Structure 2.1*, there is no sharp borderline between ionic and covalent bonding. Bonding occurs along a continuum, with greater ionic character at one end and greater covalent character at the other. *Structure 2.4* addresses this continuum in more detail with the **bonding triangle**.

Practice questions

1. Determine whether the following pairs of elements bond covalently or ionically, by referring to their electronegativity values:
   a. carbon and oxygen
   b. sodium and oxygen
   c. carbon and hydrogen
   d. chlorine and oxygen
   e. iodine and iodine
   f. aluminium and fluorine



▲ Figure 4 These are all acceptable Lewis formulas of fluoroamine, NH_2F

Lewis formulas

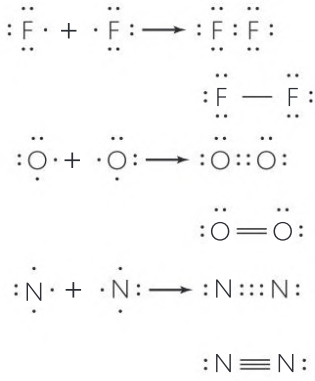
Groups of atoms that are covalently bonded together are called **molecules**. The way valence electrons are arranged in a molecule can be shown using **Lewis formulas**. In Lewis formulas, pairs of valence electrons are represented as dashes, pairs of dots or pairs of crosses, or a combination of all three. Consider the various Lewis formulas of fluoroamine, NH_2F, in figure 4.

Here are the rules for drawing Lewis formulas regardless of whether dots, crosses or dashes are used:

1. Only valence electrons are shown.
2. Electrons are arranged in pairs.
3. Each pair of electrons shared between two atoms represents a covalent bond.
4. Electrons in a bond (termed **bonding electrons**) are positioned in the region between the two atoms involved in the bond.
5. **Non-bonding electrons** (also referred to as **lone pairs**) are positioned away from the region between the two atoms involved in the bond.

Atoms in Lewis formulas generally have noble gas electron configurations. This is known as the **octet rule** because noble gases often have eight valence electrons. Noble gases already have full octets and do not readily gain, lose or share electrons. Hence, they are usually found in an unbonded, monatomic form.

The octet rule is a useful rule of thumb, but it has limitations. It does not explain why some noble gases form bonds in certain situations. Sometimes atoms can form stable molecules even with fewer than an octet of electrons. Conversely, larger atoms can form **expanded octets** (this is seen at AHL). Species with odd numbers of valence electrons are also exceptions to the octet rule.



▲ Figure 5 By sharing a pair of electrons, two fluorine atoms can obtain a full octet of valence electrons. Oxygen atoms can do this by sharing two pairs of electrons, and nitrogen atoms share three pairs

Following these steps will help you draw Lewis formulas in most cases:

1. Work out the total number of the valence electrons for each atom in the molecule.
2. Divide the total number of valence electrons by two to work out how many pairs of electrons there are.
3. Arrange the atoms by drawing their symbols on the page. The element with the least number of atoms is usually found in the centre. Hydrogen atoms always surround the central atom(s).
4. Bond the central and peripheral atoms together by drawing single bonds between them. Each single bond represents an electron pair.
5. Assign non-bonding pairs of electrons to the peripheral atoms. Keep going until they achieve noble-gas configurations.
6. Assign any remaining electron pairs to the central atom(s).
7. Check that the central atom has a full octet. If it does not, try the following two methods:
    - Reassign non-bonding pairs on the peripheral atoms to become additional bonds to the central atom
    - Check that the molecule you are looking at is not an exception to the octet rule (see page 123 for examples)

Worked example 1

Draw the Lewis formulas of each of the following molecules:

a. water, H_2O b. nitrogen trichloride, NCl_3 c. carbon dioxide, CO_2

Solution

Follow the steps above for each molecule:

	a. water, H_2O	b. nitrogen trichloride, NCl_3	c. carbon dioxide, CO_2
Step 1 Count valence electrons	hydrogen: $1 \times 2 = 2$ oxygen: 6 Total: $2 + 6 = 8$	nitrogen: 5 chlorine: $7 \times 3 = 21$ Total: $5 + 21 = 26$	carbon: 4 oxygen: $6 \times 2 = 12$ Total: $4 + 12 = 16$
Step 2 Calculate the number of electron pairs	$\frac{8}{2} = 4$ pairs	$\frac{26}{2} = 13$ pairs	$\frac{16}{2} = 8$ pairs
Step 3 Arrange the atoms	H O H Hydrogens on the periphery. There is only one oxygen, so it is likely to be in the centre.	Cl Cl N Cl There is only one nitrogen, so it is likely to be in the centre.	O C O There is only one carbon, so it is likely to be in the centre.
Step 4 Draw the single bonds	H—O—H 2 pairs used so far...	Cl Cl—N—Cl 3 pairs used so far...	O—C—O 2 pairs used so far...
Step 5 Put non-bonding pairs on peripheral atoms	The peripheral hydrogen atoms already have noble gas configurations because the first energy level only holds up to two electrons. They do not need any more electrons.	:Cl: :Cl—N—Cl: The chlorine atoms now have full octets. 12 pairs used so far...	:O—C—O: The oxygen atoms now have full octets. 8 pairs used so far...
Step 6 Put any remaining electron pairs on the central atom	H—O—H The 2 remaining electron pairs are assigned to the oxygen.	:Cl: :Cl—N—Cl: The remaining electron pair is assigned to the nitrogen.	We have used all 8 available electron pairs. None are available for the central carbon.
Step 7 Check that the central atom has a full octet	Oxygen has 4 electron pairs in the Lewis formula, therefore a full octet.	Nitrogen has 4 electron pairs in the Lewis formula, therefore a full octet.	:O—C—O: The carbon atom has only 2 electron pairs in this Lewis formula. We need to reassign two of the non-bonding pairs on the oxygen atoms to form double bonds: :O=C=O: Now the carbon atom has 4 electron pairs and therefore a full octet.

Polyatomic ions are charged groups of covalently bonded atoms. The magnitude of the charge indicates how many electrons have been lost or gained. Lewis formulas of polyatomic ions are enclosed in square brackets, with the charge indicated with a superscript outside the brackets. When counting the valence electrons, you also need to factor in the charge: for every additional negative charge, you add an electron, and for every additional positive charge, you subtract an electron. This is illustrated in the next set of worked examples.

Worked example 2

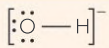
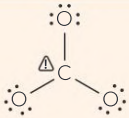
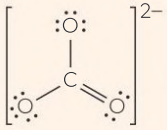
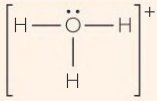
Draw the Lewis formulas of the following ions:

a. hydroxide ion, OH^- b. carbonate ion, CO_3^{2-} c. hydronium ion, H_3O^+

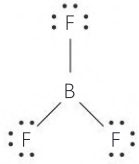
Solution

Follow the steps above for each ion:

	a hydroxide ion, OH^-	b carbonate ion, CO_3^{2-}	c hydronium ion, H_3O^+
Step 1 Count valence electrons	oxygen: 6 hydrogen: 1 This polyatomic ion has a 1– charge, meaning that it has one additional electron. Total: 6 + 1 + 1 = 8	carbon: 4 oxygen: 6 × 3 = 18 This polyatomic ion has a 2– charge, meaning that it has two additional electrons. Total: 4 + 18 + 2 = 24	hydrogen: 1 × 3 = 3 oxygen: 6 This polyatomic ion has a 1+ charge, meaning that it has one fewer electron. Total: 3 + 6 – 1 = 8
Step 2 Calculate the number of electron pairs	$\frac{8}{2} = 4$ pairs	$\frac{24}{2} = 12$ pairs	$\frac{8}{2} = 4$ pairs
Step 3 Arrange the atoms	O H	O C O O	H O H H
Step 4 Draw the single bonds	O—H 1 pair used so far…	O | C / \\ O O 3 pairs used so far…	H—O—H | H 3 pairs used so far…
Step 5 Put non-bonding pairs on peripheral atoms	The hydroxide ion has only two atoms in it, so you do not need to distinguish between central and peripheral atoms here. Skip this step.	:Ö: | C / \\ :Ö: :Ö: The oxygen atoms now have full octets. 12 pairs used so far…	The peripheral hydrogen atoms have noble gas configurations and therefore do not need any more electrons.
Step 6 Put any remaining electron pairs on the central atom	:Ö:—H The remaining 3 electron pairs are assigned to the oxygen atom. The hydrogen already has a noble gas configuration.	We have used all 12 available electron pairs. None are available for the central carbon.	H—Ö—H | H The fourth electron pair is assigned to the oxygen atom.

	a hydroxide ion, OH^-	b carbonate ion, CO_3^{2-}	c hydronium ion, H_3O^+
Step 7 Check that the central atom has a full octet	 All atoms have noble-gas configurations. Note the square brackets and charge, which are used in the Lewis formulas of polyatomic ions.	 The carbon atom has only 3 pairs in this Lewis formula. We need to reassign one of the non-bonding pairs on the oxygen atoms to form a double bond:  Now the carbon atom has 4 electron pairs and therefore a full octet. Brackets and charge are added to complete the Lewis formula.	 All atoms have noble gas configurations. Brackets and charge are added to complete the Lewis formula.

Sometimes molecules contain atoms that do not follow the octet rule. At SL, you should be aware of Lewis formulas that contain atoms with fewer than eight valence electrons. Consider the arrangement of the 12 electron pairs of boron trifluoride, BF_3 in the Lewis formula shown in figure 6. Here, the boron atom has only three pairs of electrons around it. The boron atom is **electron deficient**.



▲ Figure 6 Lewis formula of boron trifluoride

Similar to boron, many other elements of groups 2 and 13 form stable electron-deficient molecules. These include beryllium, magnesium and aluminium. In Lewis formulas of such molecules, group 2 elements (Be and Mg) have only two bonding electron pairs while group 13 elements (B and Al) have three bonding electron pairs.

Practice questions

2. Draw the Lewis formula of each of the following:
   a. BH_3
   b. BCl_3
   c. $BeCl_2$
   d. $AlCl_3$
   e. CH_3^+
3. Draw the Lewis formula of each of the following:
   a. HBr
   b. OF_2
   c. O_2
   d. O_3
   e. HCN
   f. CH_3Cl
   g. CH_2O
   h. N_2H_4
4. Draw the Lewis formula of each of the following:
   a. NH_4^+
   b. NO_3^-
   c. NO_2^+
   d. NO_2^-
   e. OCl^-

Linking questions

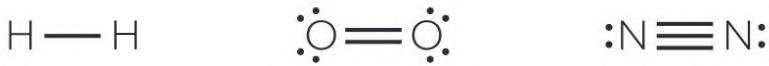
What are some of the limitations of the octet rule?

Why do noble gases form covalent bonds less readily than other elements? (Structure 1.3)

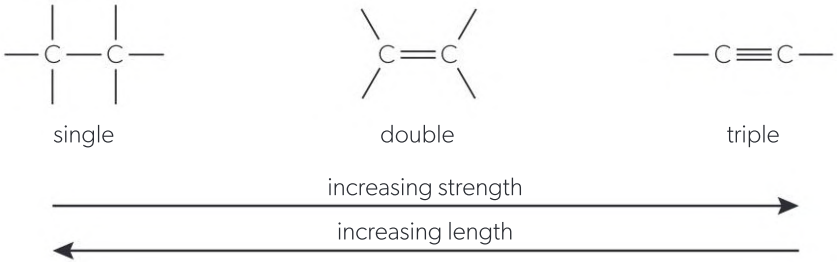
Why do ionic bonds only form between different elements while covalent bonds can form between atoms of the same element? (Structure 2.1)

Bond order (*Structure 2.2.2*)

The hydrogen molecule, H_2, shown in figure 7, contains a **single bond**. The oxygen molecule, O_2, contains a **double bond**. Diatomic nitrogen, N_2, contains a **triple bond**.



▲ Figure 7 Hydrogen, oxygen and nitrogen molecules with differing bond order



▲ Figure 8 The relationship between bond order, bond length and bond strength

The number of bonded electron pairs between two atoms is referred to as **bond order**. Single bonds (bond order 1), double bonds (bond order 2) and triple bonds (bond order 3) differ in their strength and length. Double bonds are stronger than single bonds, and triple bonds are stronger still. Triple bonds hold atoms closer together than double bonds, and hence triple bonds are shorter than double bonds. Double bonds are, in turn, shorter than single bonds. Figure 8 shows this pattern for carbon–carbon bonds.

Bond enthalpies are discussed in *Reactivity 1.2*.

Bond strength and length can be quantified. The stronger a bond is, the larger its bond enthalpy.

Bond length is defined as the average distance between two bonded nuclei. Table 1 shows bond enthalpy and bond length data for carbon–carbon single, double and triple bonds. The carbon–carbon triple bond is the shortest but also the strongest.

Bond	Bond enthalpy (at 298 K) / kJ mol^{-1}	Bond length / 10^{-12} m
C–C	346	154
C=C	614	134
C≡C	839	120

◀ Table 1 Bond enthalpies and bond lengths of carbon–carbon bonds

Data-based question

The bond enthalpy of a carbon–nitrogen triple bond is 890 kJ mol^{-1} and its bond length is 116×10^{-12} m. Use these data and the information in table 1 to predict the bond enthalpies and bond lengths of single and double carbon–nitrogen bonds. Then compare your predictions with the values in the data booklet.

ATL Communication skills

Graphs often complement written explanations. The paragraph below explains why a covalent bond forms between two hydrogen atoms by describing the relationship between potential energy and the distance between two hydrogen atoms. Read it and try to represent what it is describing by sketching a graph of potential energy (y-axis) vs distance between hydrogen nuclei (x-axis).

> *Two hydrogen atoms have no effect on each other if they are separated by a sufficiently large distance. As the two atoms approach each other, the electrostatic attraction between the hydrogen nuclei and each other's electrons increases. This process leads to a decrease in potential energy. As the atoms get closer, they reach a point where the two nuclei attract the two electrons in the pair, effectively sharing them. This arrangement occurs at a potential energy minimum, meaning that the molecule is energetically stable. If the atoms were to get any closer together, the resulting repulsion between the two positively charged nuclei would outweigh the attraction for the shared pair of electrons, leading to a rise in potential energy.*

When you have finished, compare your sketch graph to a potential energy curve. You can find one by typing "hydrogen molecule potential energy curve" into a search engine.

What can graphical representations add to explanations such as the one above? To what extent is a graphical representation better than a textual description?

Linking question

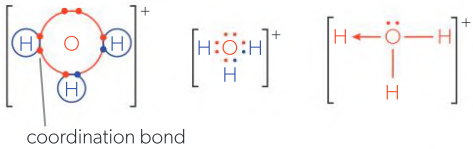
How does the presence of double and triple bonds in molecules influence their reactivity? (Reactivity 2.2)

The formation of coordination bonds is a feature of electron pair sharing reactions, as discussed in *Reactivity 3.4.*

Coordination bonds (*Structure 2.2.3*)

We have seen that a covalent bond is formed when each of the two atoms in the bond contributes an electron to the bond. Sometimes both the electrons in the covalent bond come from the same atom. The resulting bonds are called **coordination bonds.** For example, when a hydrogen cation encounters a water molecule, a coordination bond is formed, leading to the formation of a hydronium ion (figure 9). Coordination bonds are often indicated with an arrow along the bond. The direction of the arrow shows which atom donated the electron pair to the bond, and which atom accepted the electron pair.

▶ Figure 9 Different representations of the hydronium ion, showing the coordination bond



Coordination bonds help to explain how bonding occurs in certain molecules. Once formed, coordination bonds are indistinguishable from any other covalent bond.

Practice questions

5. Draw the Lewis formulas of the following molecules, indicating the coordination bonds clearly.
   a. ammonium ion, NH_4^+
   b. ozone, O_3
   c. carbon monoxide, CO
   d. ammonia boron trifluoride, NH_3BF_3

AHL

The properties of complex ions and their reactions are discussed in *Reactivity 3.4.*

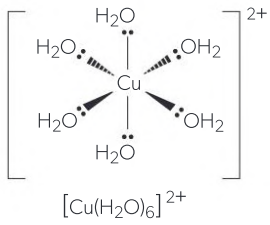
Linking question

Why do Lewis acid–base reactions lead to the formation of coordination bonds? (Reactivity 3.4)

Coordination bonds in transition metal complexes

Transition metals can form complex ions such as the one shown in Figure 10. These complex ions contain coordination bonds, which hold together the central metal cation and the surrounding atoms or groups of atoms, called **ligands**.

A common feature of all ligands is a lone pair of electrons, which can be used to form the coordination bond to the metal ion.





▲ Figure 10 Water molecules form coordination bonds with copper(II) cations, leading to the formation of a blue hexaaquacopper(II) complex ion

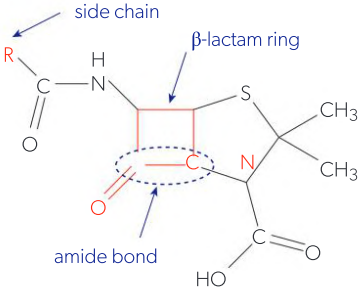
The valence shell electron pair repulsion model (VSEPR) (*Structure 2.2.4*)

You might have come across a Lewis formula for water that shows the two hydrogen atoms at an angle. Such formulas better illustrate the geometry of the molecule. This can also be shown by ball-and-stick and space-filling models (figure 11).



▲ Figure 11 Space-filling model of a water molecule

Two-dimensional representations of molecules such as Lewis formulas do not reflect the three-dimensional arrangement of atoms in a molecule. The three-dimensional shape of a molecule is termed **molecular geometry**, and it is an essential feature of molecules which contributes to their properties. For example, the molecular geometry within a structure known as a beta-lactam ring is key to its antibiotic properties. This is shown in figure 12.



▲ Figure 12 The beta-lactam ring (in red above) is a common feature of antibiotic molecules such as penicillins. The geometry of this structure is vital to its reactivity. The amide bond hydrolyses readily due to the strain caused by the 90° bond angles. This means that it can attach to enzymes responsible for building bacterial cell walls.

Molecular geometry can be explored using the **VSEPR** model:

Valence Shell Electron Pair Repulsion

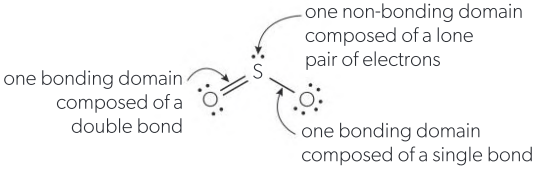
The VSEPR model is based on the following premises:

1. Electron pairs repel each other and therefore arrange themselves as far apart from each other as possible.
2. Non-bonding (or lone) electron pairs occupy more space than bonding pairs (single bonds).
3. Double and triple bonds occupy more space than single bonds.

The term **electron domain** will be useful in these discussions. An electron domain is a region of high electron density due to the presence of electron pairs. An electron domain can be:

- a non-bonding pair of electrons (known as a **lone pair**)
- a bonding pair of electrons (single bond)
- a double or triple bond (which involve multiple pairs of electrons).

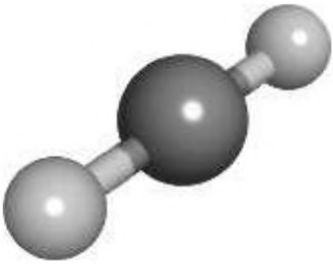
The Lewis formula in figure 13 illustrates this point: the central sulfur atom has three electron domains, of which one is a non-bonding domain, and two are bonding domains. One of the bonding domains is a double bond, the other is a single bond.



◀ Figure 13 The sulfur atom in sulfur dioxide has three electron domains

We will now explore geometries involving two, three and four electron domains using VSEPR. Predicting the VSEPR shape of a molecule involves two steps:

1. Count the number of electron domains around a central atom to deduce the electron domain geometry.
2. Determine how many of these are **bonding domains** and how many are **non-bonding domains**.



▲ Figure 14 Linear geometry has two electron domains at a 180° angle to each other

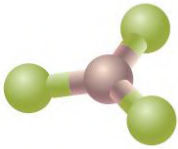
Two domains: Linear geometry

If there are two electron domains, the electron pairs in those domains repel each other. They therefore adopt positions at 180° (figure 14). The angle between bonding pairs in a molecule is known as the **bond angle**. This **electron domain geometry** is called **linear** to illustrate that the central atom and both domains are on a straight line.

Molecules with this electron domain geometry also have a linear **molecular geometry**. Examples include beryllium chloride, $BeCl_2$, carbon dioxide, CO_2, and ethyne, C_2H_2. These are shown in figure 15.



▲ Figure 15 Examples of linear molecules, in which the central atom has two electron domains. Note that double and triple bonds count as one domain



▲ Figure 16 Trigonal planar geometry has three electron domains at 120° angles to each other

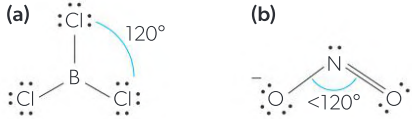
Three domains: Trigonal planar geometry

If there are three bonding domains, the electron pairs adopt positions at 120° from each other. This electron domain geometry is called **trigonal planar**. Trigonal, because the domains form a triangle, and planar, because the atoms lie flat on a plane.

A trigonal planar electron domain geometry can have two possible molecular geometries, depending on the presence of non-bonding domains:

- When all three domains are bonding domains, the molecule has **trigonal planar** geometry.
- When only two of the three domains are bonding domains, and one is a non-bonding domain, the molecule has **bent** (or **V-shaped**) geometry.

Examples of each type are shown in figure 17.



▲ Figure 17 A trigonal planar electron domain geometry can lead to (a) trigonal planar molecular geometry if all three domains are bonding domains, or (b) bent (V-shaped) molecular geometry if one of the domains is a lone pair of electrons

The bond angle in NO_2^- is smaller than the predicted 120°. This is because the non-bonding pair (or lone pair) exerts a stronger repulsion than the bonding domains, and therefore takes up more space.

Four domains: Tetrahedral geometry

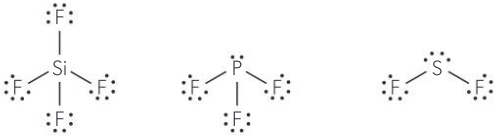
If there are four bonding domains, the electron pairs adopt positions at 109.5° from each other. This electron domain geometry is called **tetrahedral** because the ends of the domains form the corners of a tetrahedron.

You might expect four electron domains to arrange themselves in a square configuration. However, this would result in angles of 90° between the domains. The domains arrange themselves tetrahedrally to maximize the bond angles and distances between them.

A tetrahedral electron domain geometry gives rise to three possible molecular geometries, depending on the presence of non-bonding domains:

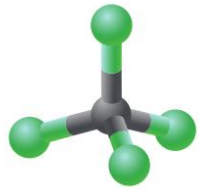
- When all four domains are bonding domains, the molecule has **tetrahedral** geometry.
- When three of the four domains are bonding domains and one is a non-bonding domain, the molecule has **trigonal pyramidal** geometry.
- When only two of the four domains are bonding domains and two are non-bonding domains, the molecule has **bent** (or **V-shaped**) geometry.

Examples of each type are shown in figure 20.



tetrahedral trigonal pyramidal bent or V-shaped

◀ Figure 20 A tetrahedral electron domain geometry leads to tetrahedral, trigonal pyramidal or bent (V-shaped) molecular geometries



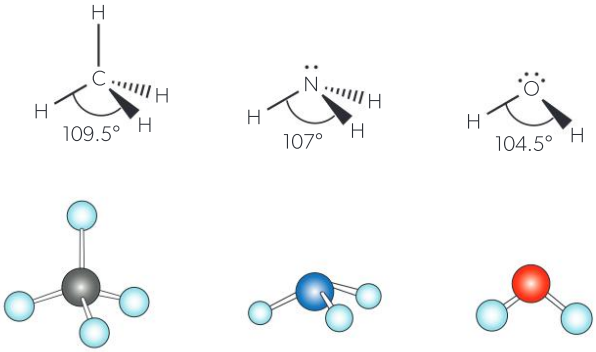
▲ Figure 18 Tetrahedral geometry has four electron domains at 109.5° angles to each other



▲ Figure 19 Tetrahedral food packages are convenient because they can be easily filled, stacked and packed. Their invention revolutionized the food packaging industry in the mid-20th century

Non-bonding pairs occupy more space than bonding domains, which leads to decreased bond angles. Methane, CH_4, has no non-bonding pairs, and therefore the bond angle corresponds to the predicted 109.5°. Ammonia, NH_3, has one non-bonding pair, and therefore has a smaller bond angle (107°). Water, H_2O, has two non-bonding pairs and therefore has an even smaller bond angle of 104.5°.

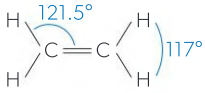
The angles and geometries are shown in figure 21. Note that some bonds are represented using **wedges** (▬) and **dashes** (⁞⁞⁞) to better convey their 3D shape. Wedges represent bonds that are coming out of the plane of the page at an angle and dashes represent bonds that are going into the plane of the page.



◀ Figure 21 In methane, CH_4, ammonia, NH_3, and water, H_2O, the bond angles are all slightly different due to the stronger repulsion exerted by the lone pairs

Multiple bonds

Multiple bonds (double and triple bonds) count as one domain. A double bond represents one domain, but it contains two electron pairs. Similarly, a triple bond is one domain composed of three electron pairs. Since multiple bonds contain more than one pair of electrons, they exert a greater repulsion than single bonds. The increased repulsion causes the bond angles in the molecule to deviate from predicted values.



▲ Figure 22 Multiple bonds generally exert a greater repulsion than single bonds

For example, each carbon atom in ethene has three bonding electron domains and hence its molecular geometry is trigonal planar. This would suggest 120° bond angles. However, the H–C–H bond angle is 117° and the H–C=C bond angle is 121.5°, as the greater repulsion exerted by the double C=C bond pushes away both C–H bonding domains. This is shown in figure 22.

Table 2 shows a summary of the geometry of molecules with two, three and four electron domains.

Electron domain geometry	Number of domains	Number of bonding domains	Number of non-bonding domains	Molecular geometry	Bond angle	Examples
linear	2	2	0	linear	180°	$BeCl_2$ CO_2 HCN
trigonal planar	3	3	0	trigonal planar	120°	BF_3, NO_3^-, SO_3
		2	1	bent (or V-shaped)	<120°	O_3, SO_2, NO_2^-
tetrahedral	4	4	0	tetrahedral	109.5°	CH_4, NH_4^+, ClO_4^-
		3	1	trigonal pyramidal	<109.5°	NH_3, H_3O^+, PBr_3
		2	2	bent (or V-shaped)	<109.5°	H_2O, NH_2^-, SF_2

▲ Table 2 Summary of the geometries for two, three and four electron domains

ATL Thinking skills

You can model electron domains using balloons. If you hold several balloons at a central point by their knots, they will arrange themselves as far away from each other as possible. The central point where the balloons meet represents the central atom in the molecule. How else might you model electron domains?



VSEPR

Unlike Lewis formulas, molecular models represent the 3D geometry of molecules. You will explore the models in a VSEPR database to develop your understanding of molecular geometry.

Relevant skills

- Tool 1: Digital molecular models
- Tool 2: Identify and extract data from a database

Instructions

1. Write a list of the molecular geometries that you need to know.
2. Access a VSEPR database (such as those suggested in the *Tools for chemistry* chapter) and extract two examples of species belonging to each molecular geometry. If possible, rotate the images to get a sense of their three-dimensional shape.
3. For each, write down the following:
   - electron domain geometry
   - molecular geometry
   - molecular formula
   - bond angle data (if available)
   - 3D sketch of the molecule, including bonding and non-bonding domains.
4. Draw the Lewis formula of each. Then deduce the number of bonding domains and non-bonding domains, and whether the molecule is polar or non-polar.
5. Organize your data into a suitable table.
6. If bond angle data is available, search for examples that illustrate:
   - the effect of non-bonding domains on the bond angle
   - the effect of multiple bonds (e.g. double or triple bonds) on bond angle.

Practice questions

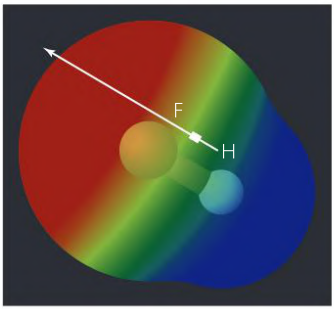
6. Identify the electron domain geometry and molecular geometry given the following numbers of bonding and non-bonding domains.
   a. four bonding domains
   b. two bonding domains and one non-bonding domain
   c. two bonding domains only
   d. two bonding domains and two non-bonding domains
   e. three bonding domains and one non-bonding domain.
7. Identify the number of bonding and non-bonding domains around atoms with the following molecular geometries.
   a. linear
   b. trigonal pyramidal
   c. tetrahedral
   d. trigonal planar
   e. bent (there are two possible answers here).
8. Deduce the electron domain geometry and molecular geometry of each of the following:
   a. NF_3 b. CH_2Cl_2
   c. BF_3 d. BeH_2
   e. HCN f. SO_3
   g. SF_2 h. NO_2^-
9. Predict the bond angles in each of the species in question 8.
10. Methanal is an organic compound with the molecular formula CH_2O.
    a. Draw the Lewis formula for methanal.
    b. Deduce the molecular geometry of methanal.
    c. Suggest values for the bond angles in methanal. Explain your reasoning.

Linking question

How useful is the VSEPR model at predicting molecular geometry?

Bond polarity (*Structure 2.2.5*)

Polarity is related to the way electrons are distributed within bonds and molecules. The shared pair of electrons in a covalent bond is not necessarily shared equally between the two atoms in the bond. **Bond polarity** results from the difference in the electronegativities of the bonded atoms.





▲ Figure 23 Representation of the molecular electrostatic potential of the polar covalent bond in HF. Note that the bond dipole in HF can be represented as a vector (arrow)

In the case of identical atoms, such as the two fluorine atoms in F_2, there is an equal sharing of the electrons in the bonding pair. This is not the case, however, in HF. The fluorine atom has a much greater electronegativity than the hydrogen atom, so it pulls the shared electron pair more strongly than hydrogen does. This leads to what we describe as a **polar covalent bond**, with one atom (fluorine) adopting a partial negative charge, δ–, and the other atom (hydrogen) adopting a partial positive charge, δ+.

The separation of charge between two non-identical bonded atoms is called a **dipole moment**. The dipole moment can be represented by a vector (figure 23). This vector is shown as an arrow along the bond. The arrow starts as a plus sign at the less electronegative atom in the bond, and points towards the more electronegative atom in the bond.

The shared pair of electrons is shifted towards the more electronegative atom in the covalent bond. The greater the difference in electronegativity between the two atoms, the greater this shift.

If the two atoms involved in the formation of the covalent bond are identical, the bond is said to be a **pure covalent bond.** The charge is distributed evenly about the covalent bond, meaning that it is non-polar and has no dipole moment.

As seen in *Structure 2.1*, if the difference of electronegativity between the two atoms is very large (greater than 1.8), then the bonding pair of electrons is shifted almost entirely towards the more electronegative atom. Such bonds are considered ionic because electrons are transferred, not shared.

Practice questions

11. Rank the following bonds in order of increasing polarity:
    C–H, C–C, C–F, C–O
12. Explain why Br_2 is non-polar, whereas HBr is polar.

Linking question

What properties of ionic compounds might be expected in compounds with polar covalent bonding? (Structure 2.1)

AHL

Practice questions

42. Describe two pieces of evidence—one physical and one chemical—for the structure of benzene.
43. How many delocalized electrons does each of the carbon atoms in a benzene ring contribute to the delocalized electron cloud?
44. State the bond angle in benzene and the molecular geometry around each of the carbon atoms.

Linking questions

How does the resonance energy in benzene explain its relative unreactivity? (Reactivity 2.1, 2.2)

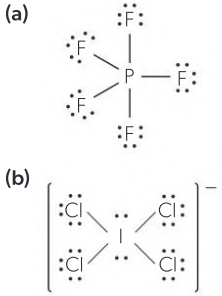
What are the structural features of benzene that favour it undergoing electrophilic substitution reactions? (Reactivity 3.4)

Expanded octets (*Structure 2.2.13*)

Lewis formulas and VSEPR were discussed in the SL section of this chapter. At HL you should also be familiar with molecules containing atoms with more than eight valence electrons. This is known as an **expanded octet**. The central atoms in these molecules can have five or six pairs of electrons, amounting to 10 or 12 electrons, respectively. Examples of such species include phosphorus pentafluoride, PF_5, and the tetrachloroiodide ion, ICl_4^-, shown in figure 73.

Drawing Lewis formulas for expanded octets

You can follow the same steps used for drawing Lewis formulas at SL for molecules containing expanded octets. However, in this case, we omit the final step—we do not check that the central atom has a full octet.



▲ Figure 73 Two species with expanded octets: PF_5 and ICl_4^-, with five and six pairs of electrons around the central atom, respectively

Worked example 5

Draw the Lewis formulas of each of the following species with expanded octets:

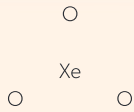
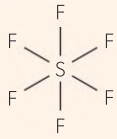
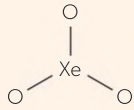
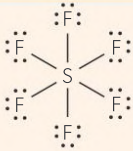
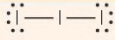
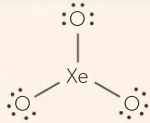
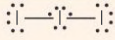
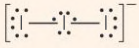
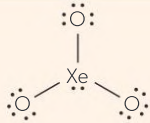
a. sulfur hexafluoride, SF_6 b. triiodide ion, I_3^- c. xenon trioxide, XeO_3

Solution

Follow the steps above for each molecule:

	a sulfur hexafluoride, SF_6	b triiodide ion, I_3^-	c xenon trioxide, XeO_3
Step 1 Count valence electrons	Sulfur: 6 Fluorine: $7 \times 6 = 42$ Total: $6 + 42 = 48$	Iodine: $7 \times 3 = 21$ This polyatomic ion has a −1 charge meaning that it has an additional electron. Total: $21 + 1 = 22$	Xenon: 8 Oxygen: $6 \times 3 = 18$ Total: $8 + 18 = 26$
Step 2 Calculate the number of electron pairs	$\frac{48}{2} = 24$ pairs	$\frac{22}{2} = 11$ pairs	$\frac{26}{2} = 13$ pairs

AHL

	a sulfur hexafluoride, SF_6	b triiodide ion, I_3^-	c xenon trioxide, XeO_3
Step 3 Arrange the atoms			
Step 4 Draw the single bonds	 6 pairs used so far...	 2 pairs used so far...	 3 pairs used so far...
Step 5 Put non-bonding pairs on peripheral atoms	 The fluorine atoms now have full octets. 24 pairs used.	 The peripheral iodine atoms now have noble gas configurations and therefore do not need any more electrons 8 pairs used so far...	 The oxygen atoms now have full octets. 12 pairs used so far...
Step 6 Put any remaining electron pairs on the central atom	We have used all 24 available electron pairs, so the structure above is final. There are 6 pairs of electrons around the central sulfur atom. It has an expanded octet.	 The remaining three pairs of electrons are assigned to the central iodine atom.  This species is an ion, so square brackets and the charge are needed. We can see that the central iodine atom has an expanded octet, with 5 pairs of electrons around it.	 We assign the final electron pair to the xenon atom. All 13 available electron pairs have been used.

In the structure deduced for XeO_3, each atom has eight electrons and none have an expanded octet. However, this structure is incorrect, and an exception to the method we use for deducing Lewis formulas.

Later in this chapter, we will discuss the concept of **formal charge**: this will help us deduce the true structure of XeO_3 and explain why it is considered to have an expanded octet.

Practice questions

45. Draw the Lewis formula of each of the following:
    a. PCl_5
    b. BrF_3
    c. IF_5
    d. ICl_2^-
    e. ICl_2^+
    f. SOF_4

Molecular geometry of expanded octets

In the SL section of this chapter, we discussed the geometries that arise when atoms have two, three or four electron domains. Several additional geometries occur for expanded octets containing five or six electron pairs.

Five domains: trigonal bipyramidal geometry

If there are five electron domains, the electrons in those domains repel each other and therefore adopt positions at 120° and 90° from each other. This electron domain geometry is called **trigonal bipyramidal**. The five domains form a shape consisting of two pyramids that share the same triangular base.

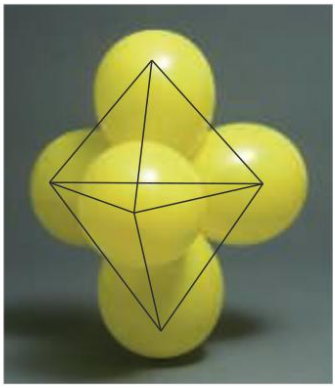
The electron domains in a trigonal bipyramidal molecule fall into two categories:

1. **Equatorial**: the bonds forming the triangular base of the pyramids, at 120° from each other.
2. **Axial**: the bonds that form the axis of the molecule, at 90° from the plane of the triangular base.

A trigonal bipyramidal electron domain geometry gives rise to four possible molecular geometries, depending on the presence of non-bonding domains:

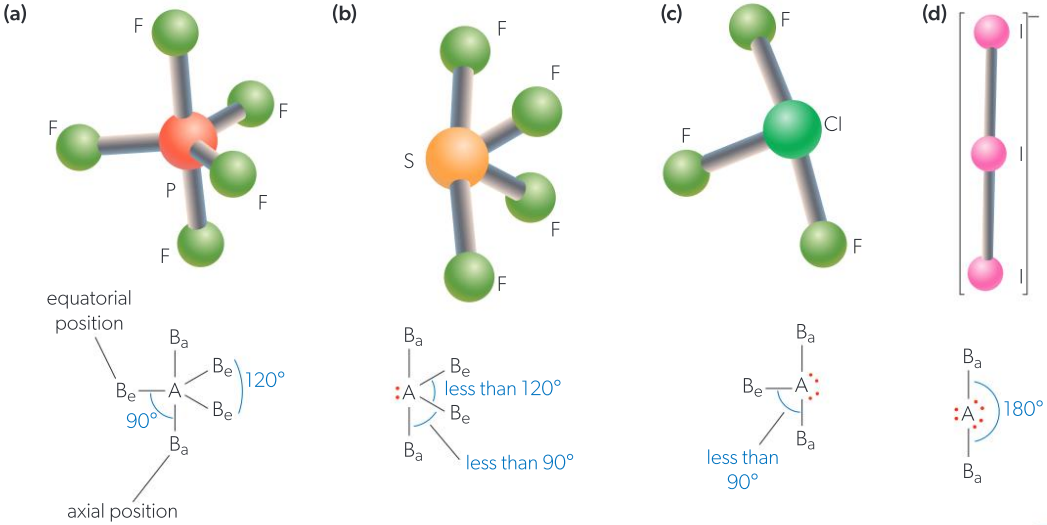
- When all five domains are bonding domains, the molecule has **trigonal bipyramidal** geometry.
- When four of the five domains are bonding domains, and one is a non-bonding domain, the molecule has **seesaw** geometry.
- When three of the five domains are bonding domains, and two are non-bonding domains, the molecule has **T-shaped** geometry.
- When only two of the five domains are bonding domains, and three are non-bonding domains, the molecular has **linear** geometry.

Examples of each type are shown in figure 75.



▲ Figure 74 A balloon model of the trigonal bipyramidal electron domain geometry

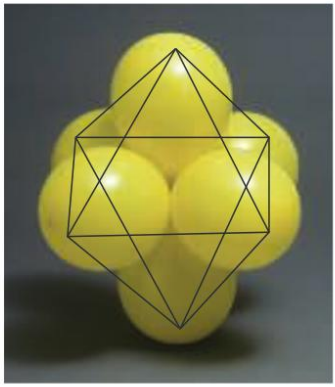
▼ Figure 75 The four types of trigonal bipyramidal electron domain geometry: (a) trigonal bipyramidal (b) seesaw (c) T-shaped (d) linear



AHL

Practice question

46. Consider the seesaw and T-shaped molecular geometries. Why do you think the non-bonding domains are located in equatorial positions, not in axial positions?



▲ Figure 76 A balloon model of the octahedral electron domain geometry

Like other geometries, the 120° and 90° bond angles in a trigonal bipyramidal shape are distorted by the presence of non-bonding pairs, due to their greater repulsion.

Six domains: octahedral geometry

If there are six domains, the electron pairs in those domains adopt positions at 90° from each other. This electron domain geometry is called **octahedral**. This is because the domains form an octahedron, which is an eight-sided polyhedron consisting of eight identical equilateral triangles.

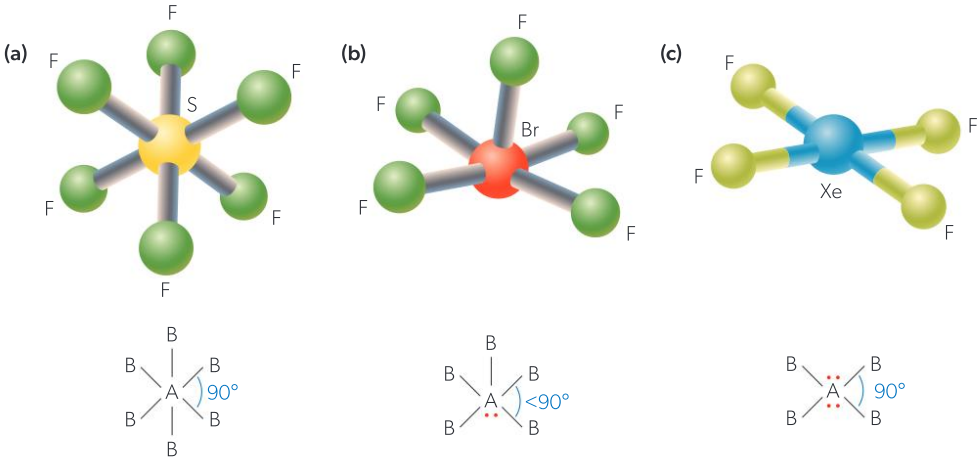
The electron domains in an octahedral molecule are all at 90° to each other, so there is no need to distinguish between axial and equatorial domains.

An octahedral electron domain geometry gives rise to three common molecular geometries, depending on the presence of non-bonding domains:

- When all six domains are bonding domains, the molecule has **octahedral** geometry.
- When five of the six domains are bonding domains, and one is a non-bonding domain, the molecule has **square pyramidal** geometry.
- When four of the six domains are bonding domains, and two are non-bonding domains, the molecule has **square planar** geometry.

Examples of each type are shown in figure 77.

The 90° bond angles in octahedral molecular geometries are distorted by the presence of non-bonding pairs, due to their greater repulsion.



▲ Figure 77 The three types of octahedral electron domain geometry: (a) octahedral (b) square pyramidal (c) square planar

Electron domain geometry	Number of domains	Number of bonding domains	Number of non-bonding domains	Molecular geometry	Bond angle	Example
trigonal bipyramidal	5	5	0	trigonal bipyramidal	90°, 120°	PF_5
		4	1	seesaw	<90°, <120°	SF_4
		3	2	T-shaped	<90°	ClF_3
		2	3	linear	180°	I_3^-
octahedral	6	6	0	octahedral	90°	SF_6
		5	1	square pyramidal	<90°	BrF_5
		4	2	square planar	90°	XeF_4

▲ Table 15 Summary of molecular geometries with five and six electron domains

Activity

1. Identify the electron domain geometry and molecular geometry of a species with the following bonding and non-bonding domains:
   a. six bonding domains
   b. four bonding domains and one non-bonding domain
   c. five bonding domains
   d. two bonding domains and three non-bonding domains
   e. four bonding domains and two non-bonding domains
2. Identify the number of bonding and non-bonding domains around atoms with the following molecular geometries:
   a. trigonal bipyramidal
   b. T-shaped
   c. octahedral
   d. square planar
   e. seesaw
   f. square pyramidal
   g. linear (there are two possible answers here)
3. Deduce the electron domain geometry and molecular geometry of each of the following:
   a. PCl_5
   b. BrF_3
   c. IF_5
   d. ICl_2^-
   e. ICl_2^+
   f. SOF_4
   g. SiF_6^{2-}
   h. IF_4^+
   j. SCl_4
4. Predict the bond angles in molecules **a** to **j** above.

Linking question

How does the ability of some atoms to expand their octet relate to their position in the periodic table? (Structure 3.1)

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Formal charge (*Structure 2.2.14*)

Formal charge is the charge an atom would have if all the bonding electrons in the molecule were shared equally, and if its non-bonding electrons were not shared at all. We can assign formal charge to atoms in a molecule to select the best Lewis formula when several are possible. Formal charge is calculated by looking at the electrons an atom has before and after bonding:

Formal charge = number of electrons before bonding – number of electrons after bonding

$$FC = VE - (NBE + \tfrac{1}{2}BE)$$

Where:

FC = formal charge

VE = number of valence electrons of the atom before bonding

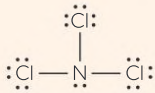
NBE = number of non-bonding electrons assigned to the atom in the Lewis formula

BE = number of bonding electrons assigned to the atom in the Lewis formula

The sum of the formal charges of all atoms in a molecule or ion should be equal to the overall charge of that molecule or ion.

Worked example 6

Show that the overall charge of nitrogen trichloride, NCl_3, is zero by calculating the formal charges of the atoms in the molecule. The Lewis formula of NCl_3 is shown below.



Solution

The formal charge of the nitrogen atom can be calculated as follows:

$VE = 5$ (because nitrogen is in group 15)

$NBE = 2$ (because in the Lewis formula the nitrogen atom has two non-bonding electrons)

$BE = 6$ (because in the Lewis formula the nitrogen atom has three bonds, which are equivalent to six bonding electrons)

$FC(\text{N}) = 5 - (2 + 0.5 \times 6) = 0$

Therefore, the formal charge of the nitrogen atom in this Lewis formula is zero.

The same process can be repeated to determine the formal charge of each chlorine atom:

$VE = 7$ (because chlorine is in group 17)

$NBE = 6$ (because in the Lewis formula each chlorine atom has six non-bonding electrons)

$BE = 2$ (because in the Lewis formula each chlorine atom has one bond, therefore two bonding electrons)

$FC(\text{Cl}) = 7 - (6 + 0.5 \times 2) = 0$

Therefore, the formal charge of each chlorine atom in this Lewis formula is zero.

The sum of the formal charges is zero, which is also the overall charge of the molecule.

The following conditions are generally favourable:

1. formal charges close to zero
2. a difference in formal charges in the molecule close to zero
3. negative formal charge(s) assigned to the more electronegative atom(s).

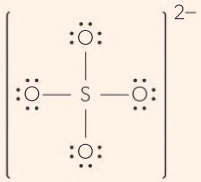
When presented with alternative Lewis formulas, we look for the one that satisfies these requirements most.

Worked example 7

Assign formal charges to the atoms in the sulfate ion, SO_4^{2-}, to deduce its Lewis formula.

Solution

We start by drawing the Lewis formula of SO_4^{2-} in a way that follows the octet rule:



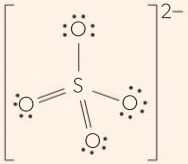
Now we can calculate the formal charges of the sulfur and oxygen atoms based on this Lewis formula:

$FC(\text{S}) = 6 - (0 + 0.5 \times 8) = +2$

$FC(\text{O}) = 6 - (6 + 0.5 \times 2) = -1$ (all the oxygen atoms are equivalent in this Lewis formula and therefore each have the same formal charge of –1)

The formal charges in this Lewis formula are not close to zero, and the difference between them is 3: these conditions are not favourable.

We know that sulfur is prone to forming expanded octets, so the following Lewis formula is proposed:



Calculating the formal charges of the sulfur and oxygen atoms in this Lewis formula gives:

$FC(\text{S}) = 6 - (0 + 0.5 \times 12) = 0$

$FC(\text{O singly bonded}) = 6 - (6 + 0.5 \times 2) = -1$

$FC(\text{O doubly bonded}) = 6 - (4 + 0.5 \times 4) = 0$

The more electronegative oxygen has the negative charge, same as that in the previously suggested structure. However, the formal charge values in this second Lewis formula are closer to zero and have a range of 1: these conditions are more favourable. Therefore, the second Lewis formula is preferred.

In both cases, the sums of the formal charges are –2, which is equal to the charge of the polyatomic ion.

Practice questions

47. Determine the formal charge of every atom in each of the molecules below. You will need to draw the Lewis formulas first.
    a. H_2O
    b. CO_2
    c. OH^-
    d. NO_3^-
    e. SF_6
    f. BH_3

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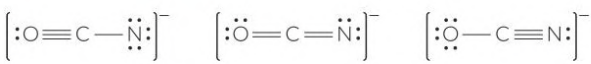
Linking question

What are the different assumptions made in the calculation of formal charge and of oxidation states for atoms in a species? (Structure 3.1, Reactivity 3.2)

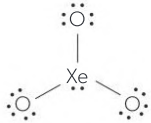
Practice questions

48. Based on formal charge considerations, determine which Lewis formula is preferred:

a. 

b. 

49. Determine the formal charge of each atom in the XeO_3 molecule shown below and suggest an alternative Lewis formula in which the formal charges are all zero.



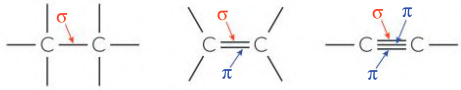
50. Draw two alternative Lewis formulas for ClO_3^-. Determine, using formal charge, which of the two is favourable.

Sigma bonds (σ) and pi bonds (π) (*Structure 2.2.15*)

Earlier in this topic, we defined covalent bonding as the electrostatic attraction between two nuclei and a shared pair of electrons. In this section, we will delve deeper into how pairs of electrons occupying atomic orbitals on neighbouring atoms are shared using **valence bond theory**. This is the idea that valence electrons form bonds when orbitals in neighbouring atoms overlap with each other, allowing the electrons to pair their **spins**.

We know that bond strength increases with greater bond order. Consider the average bond enthalpy for a single C–C bond (346 kJ mol^{-1}) and contrast it with that of a double C=C bond (614 kJ mol^{-1}). The average bond enthalpy for the double bond is significantly higher but ***not double*** that of the single bond. This is because the two bonds in the double bond are not the same: one is a **sigma bond (σ)** and the other is a **pi bond (π)**. Pi bonds are generally weaker than sigma bonds. This is why the strength of the double bond is not exactly double the strength of the single bond.

In a triple carbon–carbon bond, one of the three bonds is a sigma bond and the other two are pi bonds (figure 78).



▲ Figure 78 Single bonds are sigma bonds. Double bonds contain one sigma and one pi bond. Triple bonds contain one sigma and two pi bonds

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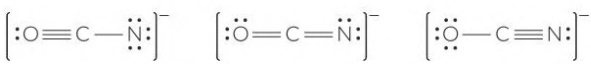
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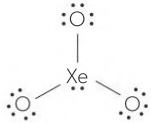
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b. 

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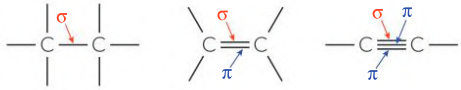
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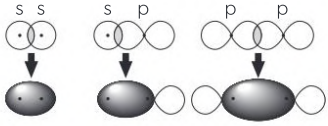


▲ Figure 78 Single bonds are sigma bonds. Double bonds contain one sigma and one pi bond. Triple bonds contain one sigma and two pi bonds

Sigma bonds (σ)

Imagine two hydrogen atoms approaching each other. They each have one unpaired electron, in the 1s orbital. Eventually their 1s orbitals will overlap. When they do so, the electrons will pair up forming a covalent bond.

Imagine an invisible line between the nuclei of the two bonding atoms. This line is known as the **bond axis** or **internuclear axis.** The two 1s orbitals will overlap along this axis to form a sigma bond. This means that there is a region of high electron density along the bond axis. Figure 79 shows that sigma bonds can form between pairs of orbitals of different types.



▲ Figure 79 Sigma bonds are formed when orbitals overlap along the bond axis

Single bonds are always sigma bonds.

The shapes of s and p orbitals are discussed in *Structure 1.3*.

ATL Thinking skills

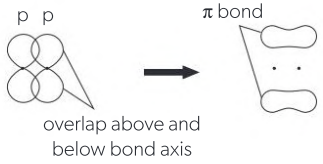
Sigma bonds get their name from the Greek letter σ, which is where the letter ***s*** came from in the Latin alphabet. When you look directly along the bond axis, the sigma bond looks spherical, like an s orbital.

Pi bonds get their name from the Greek letter π, which corresponds to the letter ***p***. When you look along the bond axis, the pi bond looks like a p orbital, with lobes above and below the axis.

What other things in science are named after things they resemble?

Pi bonds (π)

If p orbitals are present in two neighbouring atoms, they can overlap sideways, above and below the bond axis. This type of overlap forms a pi bond (figure 80). In pi bonds, the electron density is concentrated at opposite sides of the bond axis.

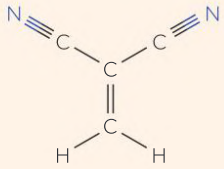


▲ Figure 80 Pi bonds are formed when p orbitals overlap above and below the bond axis

One pi bond consists of ***two*** lobes at opposite sides of the bond axis, as shown in figure 80. A pi bond contains two electrons, which occupy both lobes.

Worked example 8

Determine the number of sigma and pi bonds in the molecule $(NC)_2C{=}CH_2$. The structural formula is given below.



Solution

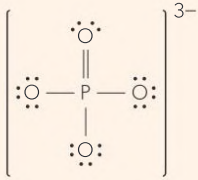
First, count the sigma bonds in the structural formula. Remember that single bonds are sigma bonds and each of the multiple bonds contains one sigma bond. Therefore, there are 7 sigma bonds.

Then, count the pi bonds. Each triple bond contains two pi bonds. Each double bond contains one pi bond. There are 5 pi bonds in total.

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Worked example 9

Determine the number of sigma and pi bonds in the phosphate anion, PO_4^{3-}. The Lewis formula is given below.



Solution

The three single bonds and one of the bonds in the double bond are all sigma bonds, giving a total of 4 sigma bonds. There is one pi bond: the second bond in the double bond.

Practice questions

51. Compare and contrast sigma bonds (σ) and pi bonds (π).
52. Explain why two overlapping s orbitals cannot form a pi bond.

Activity

Look up the Lewis formula of each of the following molecules. Determine their number of sigma and pi bonds.

a. ethanoic acid, CH_3COOH
b. propyne, CH_3CCH
c. ethanenitrile, CH_3CN
d. carbonate ion, CO_3^{2-}
e. ammonium ion, NH_4^+
f. phosphorus trichloride, PCl_3
g. azide ion, N_3^-
h. methanoate ion, $HCOO^-$

Theories

While valence bond theory helps to explain molecular geometry, it fails to give information on electron energies or properties such as magnetism. Many discussions of bonding amongst chemists are based on a more sophisticated theory, **molecular orbital theory**. This theory assumes that electrons are spread across the entire molecule rather than being localized in specific bonds.

Sometimes a phenomenon is best described using both theories. For example, the aromatic stability of benzene can be explained by two models:

- Valence bond theory describes the strong, hexagonal sigma bond framework of benzene.
- Molecular orbital theory shows that delocalization gives benzene a distinct energetic advantage.

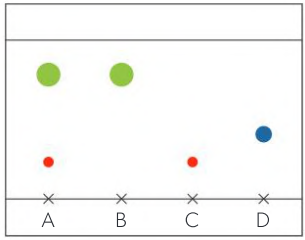
Molecular orbital theory draws heavily from quantum mechanics. Can you think of any other examples where we use the other sciences to enhance our understanding of chemistry?

Hybridization (*Structure 2.2.16*)

By now you should know that carbon is **tetravalent**, meaning it can form four covalent bonds. If these bonds are single bonds, they are usually arranged in a tetrahedral structure around the carbon atom at approximately 109.5° angles from one another. For example, methane, CH_4, has a tetrahedral structure.

Practice questions

36. Calculate the retardation factor, R_F, for the green spot at 6.0 cm in figure 58.
37. Suggest why the baseline in paper chromatograms is drawn in pencil.
38. Chromatographic analysis of a mixture of dyes (A) and three individual pigments (B, C and D) produced the chromatogram on the right.
    Identify and explain which pigments are present in A.



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Resonance (*Structure 2.2.11*)

Sometimes, a molecule cannot be described by a single Lewis formula. Instead, there are two or more possible **resonance structures** that collectively represent the molecule. Resonance often happens when there is more than one position for a double or triple bond in a molecule. For example, two Lewis formulas can be drawn for ozone, O_3 (figure 59). Note that a double-headed arrow is used to show alternative resonance structures.



▲ Figure 59 Ozone has two possible resonance structures

These structures might suggest that the two oxygen–oxygen bonds in ozone are different: one single and one double. In reality, the ozone molecule is a hybrid of these two resonance structures. Bond length and strength data (table 13) show that the two bonds in ozone are:

- identical to each other
- intermediate between a single O–O bond and a double O=O bond in terms of bond strength and length
- have a bond order of 1.5, because there are 3 bonding electron pairs distributed across two domains.

Bond	Length / 10^{-12} m	Average bond enthalpy / kJ mol^{-1}	Bond order
O–O single bond	148	144	1
O⋯O bond in ozone	128	362	1.5
O=O double bond	121	498	2

▲ Table 13 Bond data for oxygen–oxygen bonds

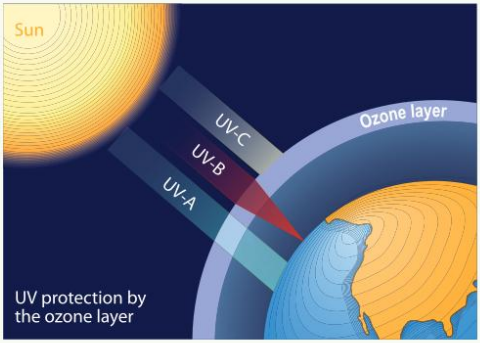
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Global impact of science

Diatomic oxygen, O_2, and ozone, O_3, are oxygen allotropes that absorb different wavelengths of electromagnetic radiation corresponding to the ultraviolet (UV) section of the spectrum. The double bond in O_2, with bond order 2, is stronger and therefore has a higher bond enthalpy than the oxygen–oxygen bond in O_3, with bond order 1.5. Breaking the oxygen–oxygen double bond requires higher energy.

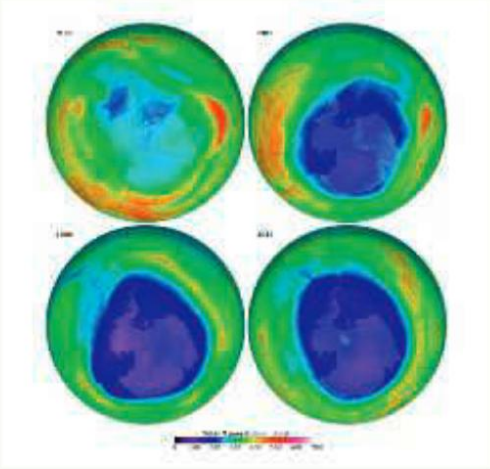
The dissociation of the double oxygen–oxygen bond in O_2 can be brought about by electromagnetic radiation with wavelengths of 240 nm and below. The bonds in ozone can be broken by wavelengths of 330 nm and below. These wavelengths correspond to the UV-C and UV-B regions of the electromagnetic spectrum, respectively.

The presence of O_3 in the stratosphere prevents nearly all solar UV-C and UV-B radiation from reaching the Earth's surface (figure 60).



▲ **Figure 60** The ozone layer in the stratosphere absorbs UV-C radiation and most UV-B radiation

UV-C and UV-B radiation can cause DNA mutations, resulting in skin cancer (melanomas) and cataracts. Ozone layer depletion caused by chlorofluorocarbons (CFCs) in the atmosphere means that more UV radiation can reach the Earth's surface. International collaboration involving the 1987 Montreal Protocol and the banning of ozone-depleting substances has slowed down the rate of depletion. Evidence in recent years suggests the concentration of ozone in the stratosphere has increased, resulting in the recovery of the ozone layer (figure 61).



▲ **Figure 61** Satellite images of a hole in the ozone layer over Antarctica from 1979 to 2011. The largest ozone hole to date was recorded in 2006. Since then, data suggests that the hole has been getting smaller again



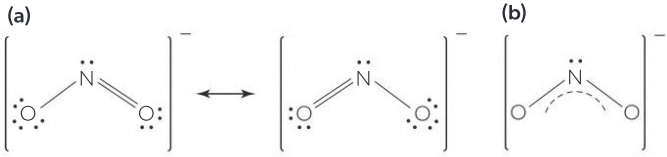
◀ **Figure 62** Former US president Barack Obama presents the Presidential Medal of Freedom to Mexican chemist and environmental scientist Mario Molina. Molina helped to uncover the link between CFCs and ozone layer depletion

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Delocalization

Resonance structures can be described as a single structure using the concept of **delocalization**. In covalent bonds, two atoms share a pair of electrons, which are localized between the two atoms. Delocalization occurs when electrons are shared by more than two atoms in a molecule or ion, as opposed to being localized between a pair of atoms.

Consider the ion NO_2^-. It is represented by two resonance structures (figure 63a). The nitrogen–oxygen bond has bond order 1.5, and its length and strength are between those of a single N–O bond and a double N=O bond. One pair of electrons is delocalized across the two N–O bonding domains. For this reason, the ion can be represented with a structure containing a dashed line that shows the delocalization (figure 63b).



▲ Figure 63 (a) The ion NO_2^- can be represented by two Lewis formulas (b) An alternative representation of the ion shows the delocalized electron pair

Note that the structure in Figure 63b is not a Lewis formula because it does not specify locations of bonding and non-bonding electron pairs.

Resonance does not involve the interconversion between the various resonance structures through rotation or bond movement. Instead, resonance structures collectively describe a single molecule that cannot be represented by a single Lewis formula due to the presence of electron delocalization. Therefore, the actual molecule is somewhere "in-between" the various resonance structures, which is best represented by the single delocalized structure.

Practice questions

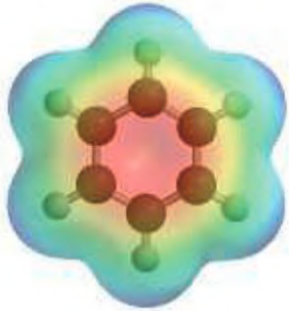
39. Draw the structure of ozone that shows the delocalized electron pair.
40. Draw the delocalized structure of the carbonate ion, CO_3^{2-}, and explain why the C–O bond order in this ion is 1.33.
41. Compare the C–O bond lengths and strengths in the carbonate ion to those in carbon dioxide and methanol, CH_3OH.

Linking question

Why are oxygen and ozone dissociated by different wavelengths of light? (Structure 1.3)

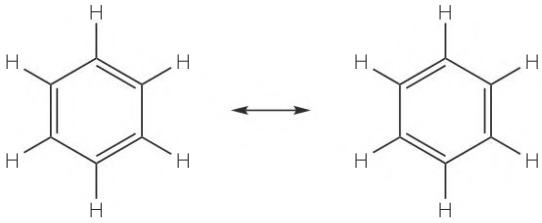
Benzene and resonance (*Structure 2.2.12*)

Benzene, C_6H_6, is an important example of a molecule that demonstrates resonance.



▲ Figure 64 The ball-and-stick structure and electron cloud of benzene, C_6H_6

The Lewis formula for benzene contains a six-membered ring of carbon atoms connected through alternating single and double bonds. Two resonance structures are possible (figure 65).



◀ Figure 65 Resonance structures of benzene

Some of the electrons in the carbon–carbon bonds in benzene are delocalized around the ring. For this reason, these delocalized electrons are often represented with a circle:

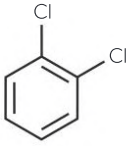


In skeletal formulas, straight lines represent bonds. The ends and vertices (corners) of the bonds represent carbon atoms, and hydrogen atoms are not drawn. Detailed instructions for drawing skeletal formulas of molecules are given in *Structure 3.2*.

Benzene is an aromatic hydrocarbon. The term "**aromatic**" is used to describe cyclic molecules that are planar and stabilized due to the presence of delocalized electrons. Benzene is the simplest aromatic hydrocarbon, and structures similar to it can be found in a large range of substances (figure 66).

Activity

1. One of the resonance structures of 1,2-dichlorobenzene is shown below. Draw the other resonance structure.



2. Draw a structure of 1,2-dichlorobenzene that contains a circle to represent the delocalized electrons.
3. Research the structures of the following molecules and identify the benzene rings in each:



(a) curcumin



(b) Kevlar



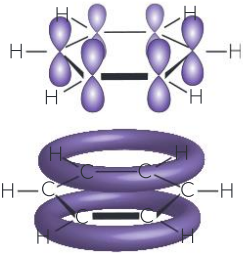
(c) estradiol

▲ Figure 66 (a) Curcumin, found in the yellow food spice turmeric (b) Kevlar is used for making personal armour (c) Synthetic estradiol is used as medication in menopausal hormone therapy

TOK

Skeletal formulas are an example of "chemistry shorthand" that are used extensively by chemists. To what extent are chemical symbols, formulas and equations like a language? What is gained or lost using chemical notation?

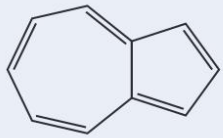
AHL



▲ Figure 67 The delocalized electrons in the p orbitals of benzene form a ring of electron density above and below the plane of the molecule

ATL Communication skills

In ordinary conversation, the term "aromatic" can be used to describe something as having a pleasant smell. In chemistry, however, this term is used to indicate a benzene derivative or the presence of a benzene ring in a compound. However, some aromatic compounds are non-benzenoids. For example, azulene—a component of some blue pigments found in nature—contains alternating single and double bonds in two planar rings that are joined together:

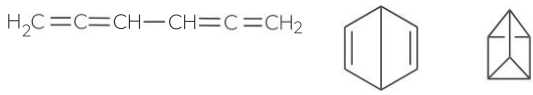


Can you think of other examples of chemical terms that have different meanings in everyday life?

In benzene, one electron in each of the six carbon atoms occupies a p orbital. The six p orbitals overlap, forming a ring of electron density above and below the plane of the molecule. The electrons in this ring are delocalized (figure 67). The carbon atoms are sp^2 hybridized (see ***Structure 2.2.16***).

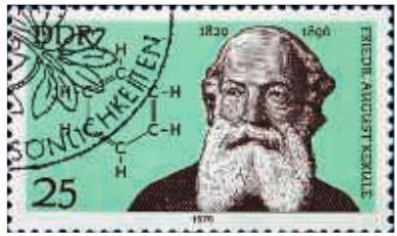
Physical evidence for the structure of benzene

Once it was established that C_6H_6 was the molecular formula for benzene, the next step was to deduce its structure (figure 68). The 1:1 ratio of carbon to hydrogen suggested the presence of multiple bonds. Eventually a cyclic arrangement of carbon atoms with alternating single and double bonds was proposed (figure 69). August Kekulé was credited with the structure, which is said to have come to him in a dream. The structure proposed was a lop-sided hexagon because of the differing carbon-carbon bond lengths.

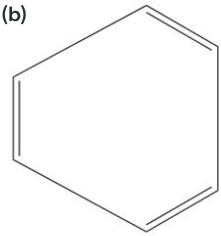


▲ Figure 68 Some of the 19th century suggestions for the structure of benzene

(a)



(b)



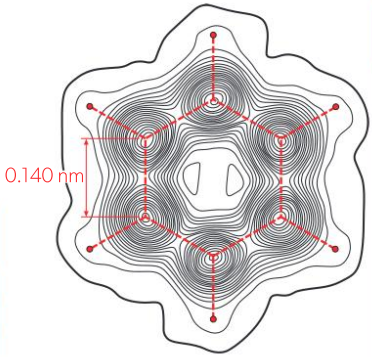
▶ Figure 69 (a) August Kekulé on an East German postage stamp (b) Kekulé's suggestion for the structure of benzene

X-ray diffraction patterns later showed a regular hexagonal arrangement of carbon atoms in benzene where all carbon–carbon bonds had the same length, suggesting they were all equivalent (figure 70).

The carbon–carbon bond length in benzene is intermediate between that of a single and a double carbon–carbon bond (table 14). Bond enthalpy data shows that they are also of intermediate strength.

Bond	Bond length / 10^{-12} m	Average bond enthalpy / kJ mol^{-1}
C–C single bond	154	346
C═C in benzene	140	507
C=C double bond	134	614

▲ Table 14 Bond length and strength data for the carbon–carbon bond in benzene compared to single and double carbon–carbon bonds



▲ Figure 70 A technique called X-ray diffraction shows a contour map of the electron density in an individual benzene molecule. Each carbon–carbon bond length is 0.140 nm

Chemical evidence for the structure of benzene

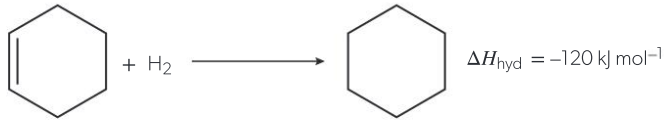
The chemical behaviour of benzene differs from what would normally be expected from alkenes. Like alkenes, benzene is unsaturated, but unlike alkenes, it does not readily undergo addition reactions. Alkenes undergo addition reactions, where the double bond breaks and the carbon atoms on either end form bonds with new species. In benzene, addition would disrupt the stabilizing effect of the delocalized electron ring, which would be energetically unfavourable. Instead, benzene tends to undergo substitution reactions.

Addition and substitution reactions are discussed in *Reactivity 3.4*.

Resonance energy of benzene

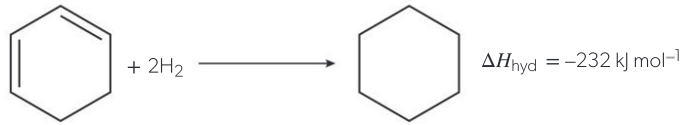
Resonance increases the stability of the benzene molecule. This is quantified by a value called **resonance energy**. For benzene it is 152 kJ mol^{-1}. This value is established by comparing the thermochemistry data for benzene and those for cyclic six-membered alkenes.

For example, cyclohexene contains one double bond. When we add hydrogen to this double bond, the change in energy is –120 kJ mol^{-1}.



cyclohexene cyclohexane

This reaction is called a hydrogenation reaction, and the change in energy is referred to as the **enthalpy of hydrogenation**, ΔH_{hyd}. 1,3-cyclohexadiene is a similar molecule with two double bonds, so its enthalpy of hydrogenation is nearly double: –232 kJ mol^{-1}.

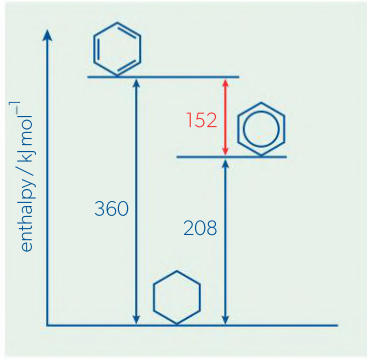


1,3-cyclohexadiene cyclohexane

AHL

Changes in energy across a reaction are known as enthalpy changes. Enthalpy changes are discussed in *Reactivity 1.1*. Naming cyclic compounds is covered in *Structure 3.2*.

The next molecule in the series matches Kekulé's suggested structure and is the theoretical molecule, 1,3,5-cyclohexatriene. 1,3,5-cyclohexatriene contains three double bonds and its enthalpy of hydrogenation is predicted to be $-120 \times 3 = -360\ kJ\ mol^{-1}$ (figure 71). However, the actual value for benzene is $-208\ kJ\ mol^{-1}$. This is due to the added stability provided by resonance in the benzene ring. The difference between these enthalpy values is $152\ kJ\ mol^{-1}$, which is the resonance energy.

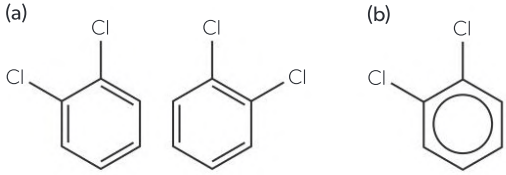


▶ Figure 71 The resonance energy of benzene ($152\ kJ\ mol^{-1}$) compared to the theoretical compound 1,3,5-cyclohexatriene

Hydrogenation of benzene is less **exothermic** than expected. This is because the reaction has an additional energetic cost to break the delocalized electron rings.

1,2-disubstituted benzene compounds

Consider the Kekulé structure of benzene, containing alternating double and single bonds. When chlorine, Cl_2, reacts with benzene to form 1,2-dichlorobenzene, you would expect that two possible **isomers** can be formed: one with the two chlorine atoms at either end of a carbon–carbon single bond, whereas in the other isomer, the chlorines would be at either end of a carbon–carbon double bond (figure 72a). However, because all electrons are evenly distributed, these two isomers represent the same molecule (figure 72b).



▲ Figure 72 (a) The two theoretical isomers of 1,2-dichlorobenzene for the Kekulé structure of benzene (b) The actual structure of 1,2-dichlorobenzene has only one isomer because the carbon atoms are all chemically equivalent

AHL

Practice questions

42. Describe two pieces of evidence—one physical and one chemical—for the structure of benzene.
43. How many delocalized electrons does each of the carbon atoms in a benzene ring contribute to the delocalized electron cloud?
44. State the bond angle in benzene and the molecular geometry around each of the carbon atoms.

Linking questions

How does the resonance energy in benzene explain its relative unreactivity? (Reactivity 2.1, 2.2)

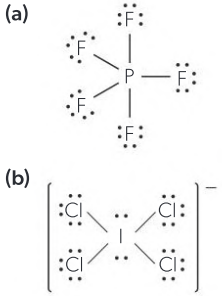
What are the structural features of benzene that favour it undergoing electrophilic substitution reactions? (Reactivity 3.4)

Expanded octets (*Structure 2.2.13*)

Lewis formulas and VSEPR were discussed in the SL section of this chapter. At HL you should also be familiar with molecules containing atoms with more than eight valence electrons. This is known as an **expanded octet**. The central atoms in these molecules can have five or six pairs of electrons, amounting to 10 or 12 electrons, respectively. Examples of such species include phosphorus pentafluoride, PF_5, and the tetrachloroiodide ion, ICl_4^-, shown in figure 73.

Drawing Lewis formulas for expanded octets

You can follow the same steps used for drawing Lewis formulas at SL for molecules containing expanded octets. However, in this case, we omit the final step—we do not check that the central atom has a full octet.



▲ Figure 73 Two species with expanded octets: PF_5 and ICl_4^-, with five and six pairs of electrons around the central atom, respectively

Worked example 5

Draw the Lewis formulas of each of the following species with expanded octets:

a. sulfur hexafluoride, SF_6 b. triiodide ion, I_3^- c. xenon trioxide, XeO_3

Solution

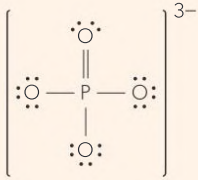
Follow the steps above for each molecule:

	a sulfur hexafluoride, SF_6	b triiodide ion, I_3^-	c xenon trioxide, XeO_3
Step 1 Count valence electrons	Sulfur: 6 Fluorine: $7 \times 6 = 42$ Total: $6 + 42 = 48$	Iodine: $7 \times 3 = 21$ This polyatomic ion has a −1 charge meaning that it has an additional electron. Total: $21 + 1 = 22$	Xenon: 8 Oxygen: $6 \times 3 = 18$ Total: $8 + 18 = 26$
Step 2 Calculate the number of electron pairs	$\frac{48}{2} = 24$ pairs	$\frac{22}{2} = 11$ pairs	$\frac{26}{2} = 13$ pairs

AHL

Worked example 9

Determine the number of sigma and pi bonds in the phosphate anion, PO_4^{3-}. The Lewis formula is given below.



Solution

The three single bonds and one of the bonds in the double bond are all sigma bonds, giving a total of 4 sigma bonds. There is one pi bond: the second bond in the double bond.

Practice questions

51. Compare and contrast sigma bonds (σ) and pi bonds (π).
52. Explain why two overlapping s orbitals cannot form a pi bond.

Activity

Look up the Lewis formula of each of the following molecules. Determine their number of sigma and pi bonds.

a. ethanoic acid, CH_3COOH
b. propyne, CH_3CCH
c. ethanenitrile, CH_3CN
d. carbonate ion, CO_3^{2-}
e. ammonium ion, NH_4^+
f. phosphorus trichloride, PCl_3
g. azide ion, N_3^-
h. methanoate ion, $HCOO^-$

Theories

While valence bond theory helps to explain molecular geometry, it fails to give information on electron energies or properties such as magnetism. Many discussions of bonding amongst chemists are based on a more sophisticated theory, **molecular orbital theory**. This theory assumes that electrons are spread across the entire molecule rather than being localized in specific bonds.

Sometimes a phenomenon is best described using both theories. For example, the aromatic stability of benzene can be explained by two models:

- Valence bond theory describes the strong, hexagonal sigma bond framework of benzene.
- Molecular orbital theory shows that delocalization gives benzene a distinct energetic advantage.

Molecular orbital theory draws heavily from quantum mechanics. Can you think of any other examples where we use the other sciences to enhance our understanding of chemistry?

Hybridization (*Structure 2.2.16*)

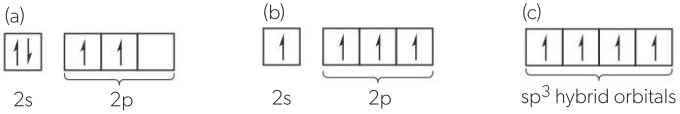
By now you should know that carbon is **tetravalent**, meaning it can form four covalent bonds. If these bonds are single bonds, they are usually arranged in a tetrahedral structure around the carbon atom at approximately 109.5° angles from one another. For example, methane, CH_4, has a tetrahedral structure.

However, the ground state electron configuration of carbon, $1s^2\,2s^2\,2p^2$, contradicts these observations. Carbon contains two unpaired 2p electrons (figure 81a) and therefore should form two bonds (one involving each unpaired electron), not four. Furthermore, the two occupied 2p orbitals are at 90° from one another, not 109.5°. This means that the atomic orbitals must undergo certain changes when they form bonds.

You can deduce the electron configuration of an element from its atomic number. This is covered in *Structure 1.3*.

Hybridization is the concept of mixing atomic orbitals to form new **hybrid orbitals** for bonding. An example of this process is shown in figure 81 for the carbon atom. There are two steps:

1. **Promotion**: A 2s electron is promoted to one of the 2p orbitals.
2. **Hybridization**: The singly occupied 2s and 2p atomic orbitals are hybridized, meaning they combine and give rise to orbitals of new shapes. The resulting orbitals are called hybrid orbitals and they all have the same energy.

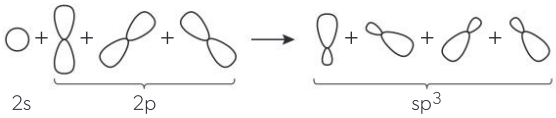


▲ Figure 81 (a) The ground state electron configuration of carbon
(b) One of the 2s electrons is promoted to a 2p orbital
(c) The four atomic orbitals are hybridized to form four sp^3 hybrid orbitals of identical energy

Overall, hybridization is energetically favourable. The promotion step absorbs energy, but the energy released by the subsequent bond formation outweighs this. Promotion does not require much energy because it relieves the 2s electron of the repulsion it experiences when paired.

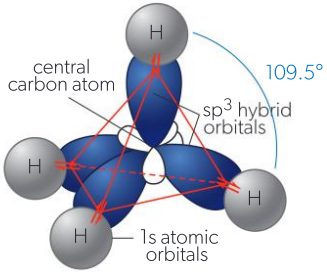
sp^3 hybrid orbitals

The number of resulting hybrid orbitals is equal to the number of atomic orbitals combined to make them. If four atomic orbitals (one 2s and three 2p) are combined in carbon, four equivalent **sp^3** hybrid orbitals are produced. Each of these hybrid orbitals is a mixture composed of one part 2s and three parts 2p, and therefore has 25% s character and 75% p character (figure 82).



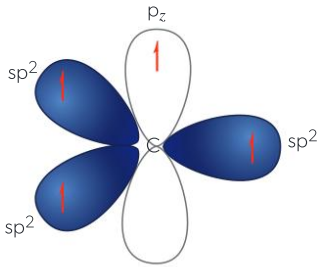
▲ Figure 82 One 2s and three 2p atomic orbitals combine to form four equivalent sp^3 hybrid orbitals

The four sp^3 hybrid orbitals arrange themselves tetrahedrally, leading to the 109.5° bond angles. In carbon, these orbitals are each occupied by one electron and therefore can form four sigma bonds. For example, in methane, CH_4, each of the sp^3 hybrid orbitals overlaps with a 1s atomic orbital on a hydrogen atom, forming four covalent bonds (figure 83).



▲ Figure 83 In methane, each carbon sp^3 orbital overlaps with a hydrogen 1s orbital. The geometry around the carbon atom is tetrahedral

AHL

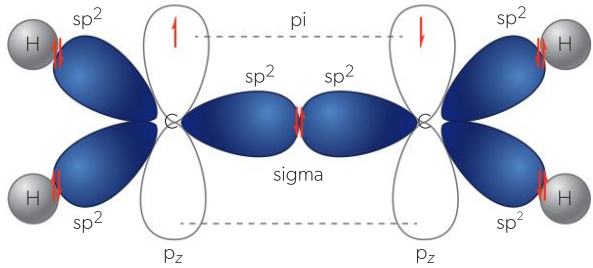


▲ Figure 84 Each of the three sp^2 hybrid orbitals (blue) contains one electron. The unhybridized p_z atomic orbital (white) also contains one electron

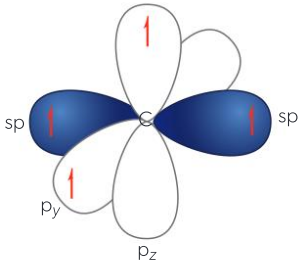
sp^2 hybrid orbitals

Carbon can also become **sp^2** hybridized. The combination of one 2s and two 2p atomic orbitals produces three sp^2 hybrid orbitals. These sp^2 hybrid orbitals arrange themselves in a trigonal planar fashion, at 120° from one another (figure 84).

Each hybridized orbital contains one electron, so they can form sigma bonds. The remaining unhybridized p_z orbital can then go on to form a pi bond with a parallel p_z orbital on a different atom (figure 85).



▲ Figure 85 In ethene, C_2H_4, the two carbon atoms each have three sp^2 hybrid orbitals (blue) which form sigma bonds with each other and two hydrogen atoms. The remaining unhybridized p_z orbitals (white) overlap side-by-side and form a pi bond, resulting in a carbon–carbon double bond

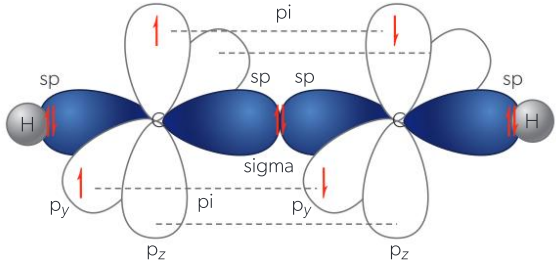


▲ Figure 86 Each of the two sp hybrid orbitals (blue) contains one electron. The two unhybridized p_y and p_z atomic orbitals (white) also contain one electron each

sp hybrid orbitals

The combination of one 2s and one 2p atomic orbital leads to the formation of two **sp** hybrid orbitals. They adopt a linear arrangement, with 180° between them (figure 86).

The hybrid orbitals can form sigma bonds. The remaining unhybridized p_y and p_z orbitals can form two pi bonds with parallel p orbitals on a neighbouring atom (figure 87).



▲ Figure 87 In ethyne, C_2H_2, the one sp–sp overlap and the two 1s–sp overlaps form three sigma bonds. The four unhybridized p orbitals form two pi bonds, resulting in a carbon–carbon triple bond

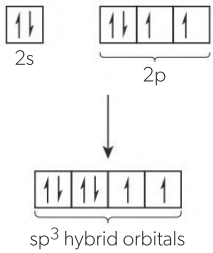
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Hybridization in other atoms

Hybridization also occurs in atoms other than carbon. The number of electrons in the p orbitals will differ, but the general principles are the same. For example, consider the oxygen atom in water, H_2O, which is sp^3 hybridized:

- The oxygen atom's ground state electron configuration is $1s^2\,2s^2\,2p^4$.
- We can distinguish between the three 2p orbitals to show that one of them contains a pair of electrons and the other two are singly occupied: $1s^2\,2s^2\,2p_x^2\,2p_y^1\,2p_z^1$.
- The 2s and 2p orbitals hybridize to form four equivalent sp^3 hybrid orbitals containing six electrons in total (figure 88). Two of the sp^3 hybrid orbitals contain a pair of electrons each and therefore do not form bonds: they constitute the two lone pairs on oxygen that you are familiar with. The remaining two hybrid orbitals are singly occupied and go on to form sigma bonds with the s orbitals on hydrogen atoms.

Since the four hybrid orbitals are tetrahedrally arranged, the bond angles are close to 109.5°. However, two of the hybrid orbitals are already full, so the oxygen atom has two lone pairs. This corresponds to bent molecular geometry. The fact that the bond angle is not exactly 109.5° but rather 104.5° suggests that the hybridization is close to but not exactly sp^3.



▲ Figure 88 The 2s and 2p atomic orbitals in oxygen are hybridized to form four sp^3 hybrid orbitals

Practice questions

53. Draw orbital box diagrams to show the promotion and hybridization process for the oxygen atom in water, described above.
54. Describe the changes undergone by the atomic orbitals of nitrogen when it bonds to hydrogen to form ammonia, NH_3.

Hybridization and geometry

Hybridization and sigma bonding are closely linked to electron domain geometry. The number of hybrid orbitals formed by an atom is equal to the number of its electron domains.

You will remember from the discussion of VSEPR that double and triple bonds are treated as single additional electron domains when determining molecular shape. This is because pi bonds do not involve hybridized orbitals, so they do not have a large effect on the geometry of a molecule.

Hybridization	Number of hybrid orbitals	Number of electron domains	Electron domain geometry	Number of non-bonding domains	Molecular geometry
sp^3	4	4	tetrahedral	0	tetrahedral
				1	trigonal pyramidal
				2	bent
sp^2	3	3	trigonal planar	0	trigonal planar
				1	bent
sp	2	2	linear	0	linear

▲ Table 16 Hybridization, electron domain geometry and molecular geometry

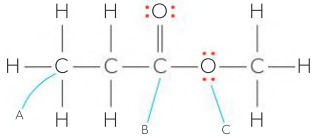
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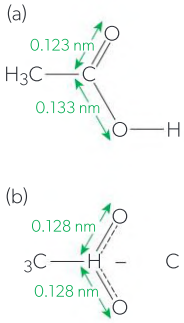
Practice questions

55. State the hybridization of:
    a. an atom with tetrahedral electron domain geometry
    b. the carbon atom in ethene, C_2H_4
    c. the oxygen atom in oxygen difluoride, OF_2
    d. the nitrogen atom in molecular nitrogen, N_2
    e. an atom with trigonal pyramidal molecular geometry
    f. an atom with bent molecular geometry (two answers are possible here).

56. Consider the following species: CF_4, HCN, N_2H_2. For each of the carbon and nitrogen atoms in these species, deduce the following:
    a. number of electron domains
    b. electron domain geometry
    c. hybridization.

57. The Lewis formula of methyl propanoate is shown below. Deduce the hybridization, number of electron domains, electron domain geometry and molecular geometry for atoms A, B and C.





▲ Figure 89 When ethanoic acid loses a hydrogen ion, the C–O bonds become equivalent

Hybridization and delocalization

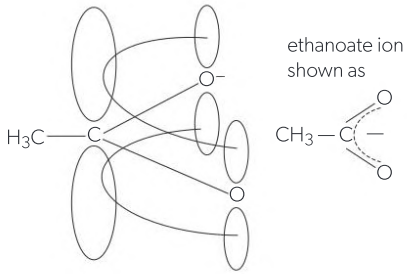
Consider the ethanoate ion, CH_3COO^-. It forms when ethanoic acid loses a hydrogen ion. By doing so, the carbon–oxygen bond lengths change: the bond orders change from 2 and 1 to 1.5 each. The electrons in the double bond become delocalized across the two carbon–oxygen domains. This is shown in figure 89.

In the C=O bond of ethanoic acid, CH_3COOH, the carbon and oxygen are both sp^2 hybridized. The unhybridized 2p orbital electron in each atom forms the pi bond between them. The other oxygen atom, in –OH, has sp^3 hybridization.

When the –OH hydrogen is lost to form ethanoate, the remaining oxygen atom adopts an sp^2 hybridization. As a result, the oxygen atoms and the carbon atom each have three electron domains around them. Three electron domains correspond to sp^2 hybridization and one unhybridized 2p orbital each. The orbitals overlap allowing the p electrons in them to be delocalized as shown in figure 90.

Activity

Draw the two resonance structures of the ethanoate ion.



▲ Figure 90 The delocalization of electrons due to resonance in the ethanoate anion

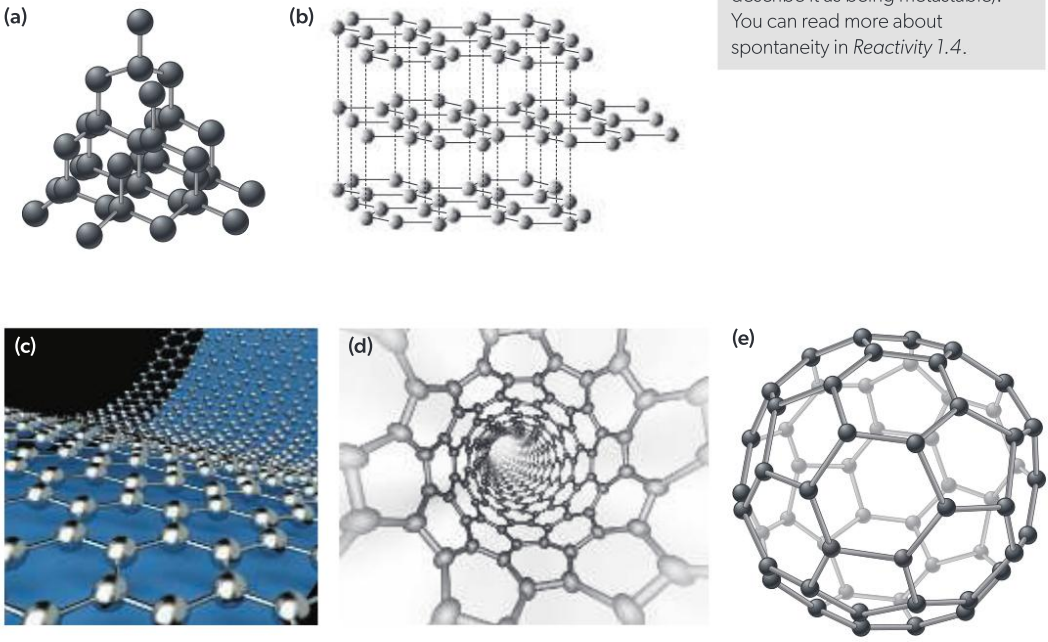
Covalent network structures (*Structure 2.2.7*)

Covalent bonds give rise to two different types of structure: **molecular** and **covalent network** (also known as **giant covalent**). The atoms in both cases are joined by covalent bonds, but the two structures are very different in terms of their properties. Molecular substances consist of discrete groups of covalently bonded atoms called molecules. By contrast, covalent network structures contain atoms that are held together by covalent bonds in a continuous three-dimensional lattice. Examples of covalent network structures include silicon, silicon dioxide, and most of the allotropes of carbon, such as diamond and graphite.

Allotropes of carbon

Some elements have different structural forms known as **allotropes**. Carbon is one such element, and its allotropes include diamond, graphite, graphene, and a group of substances known as fullerenes. They are all composed of carbon atoms but have different chemical and physical properties due to their different structural arrangements.

Chemists talk about two types of stability: kinetic (*Reactivity 2.2*) and thermodynamic (*Reactivity 1*), and diamond illustrates the difference well. At high pressures, such as those occurring deep inside the Earth's crust, diamond is the most stable allotrope. At standard temperature and pressure, graphite is the most thermodynamically stable allotrope of carbon. Under these conditions, the transformation of diamond into graphite is thermodynamically spontaneous, but is so slow that diamonds effectively last forever. Diamond is kinetically stable but thermodynamically unstable (we describe it as being metastable). You can read more about spontaneity in *Reactivity 1.4*.

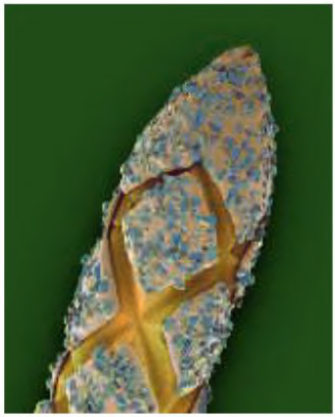


▲ Figure 25 The allotropes of carbon: (a) diamond, (b) graphite, (c) graphene, (d) carbon nanotubes and (e) buckminsterfullerene, C_{60}

In **diamond**, each carbon atom is bonded to four other atoms in a tetrahedral arrangement. It is one of the hardest substances known. For this reason, it is often used in heavy-duty cutting tools such as saws, polishing tools and dental drills. Diamond's high refractive index and durability mean that it is also used to make jewelry. Diamond is a poor electrical conductor because it has no mobile charged particles—the electrons are all localized in the bonds. It is however an excellent thermal conductor: vibrational energy carriers called phonons travel well through the highly regular lattice and strong covalent bonds.



▶ Figure 26 The Big Hole: a former diamond mine in Kimberley, South Africa



▲ Figure 27 A microscope image of a dentist's drill showing the diamond chips

Graphite is composed of layers of sheets made of carbon atoms. Each carbon atom is bonded to three other carbon atoms in a hexagonal arrangement where the geometry around each carbon atom is trigonal planar. The carbon atoms in graphite are bonded such that one electron per carbon atom is delocalized. These delocalized electrons are free to move in the planes above and below each sheet and therefore graphite is a good electrical conductor. While the covalent bonds between the carbon atoms within the sheets are strong, the forces of attraction (called **London (dispersion) forces**) between the sheets are weak. This means that the sheets can be separated easily, making graphite a good lubricant, as well as an ideal material for pencil leads. When you use a pencil to write, the force applied causes parts of the graphite sheets to come off, leaving a mark on the paper.

Graphene is essentially a single sheet of graphite. Graphene is thus one-atom thick and is therefore said to be two-dimensional. Like graphite, it is an excellent electrical conductor. It is also flexible, lightweight, transparent and, at the same time, extraordinarily strong. First isolated in 2004, graphene is a promising new material, with a vast range of potential applications, from desalination technology to bendable electronic displays.

Fullerenes are a group of carbon allotropes with atoms arranged in interlinking hexagonal and pentagonal rings. Some fullerenes form long hollow cylinders: these are known as **carbon nanotubes**. Due to their size, carbon nanotubes are utilized in **nanotechnology**. Nanotechnology involves the use of atoms, molecules and objects with dimensions of less than 100 nm (about 1000 atoms or less across).

Carbon nanotubes are used to reinforce composite materials. Their exceptional strength is due to the strong covalent bonds holding the carbon atoms together. Carbon nanotubes have potential applications in electronics because, like graphite and graphene, the presence of delocalized electrons means that they are good electrical conductors.

Buckminsterfullerenes, or **buckyballs**, have a covalent molecular structure. With the formula C_{60}, the atoms in buckminsterfullerene are arranged in hexagons and pentagons to suggest a very familiar shape: a football. Along with other spherical fullerenes, buckyballs could have exciting new roles in medicine as drug carriers. Because of its molecular structure, C_{60} has a low boiling point: overcoming the weak intermolecular forces of attraction does not require much thermal energy.

Silicon and carbon are found in group 14 hence they have some similar properties. Other patterns in the periodic table are explored in *Structure 3.1*.

Silicon and silicon dioxide

Silicon forms a three-dimensional lattice where each silicon atom is bonded to four other silicon atoms in a tetrahedral arrangement. This arrangement is similar to that of the carbon atoms in diamond.

The extensive covalent bonds in the lattice result in high strength, as well as high melting point and boiling point. Table 4 contrasts some of the features of carbon and silicon. The Si–Si bond is weaker than its C–C counterpart as silicon has a larger atomic radius. Therefore, the Si–Si bond is more reactive than the C–C bond. This difference in strength also helps to explain the different melting and boiling points.

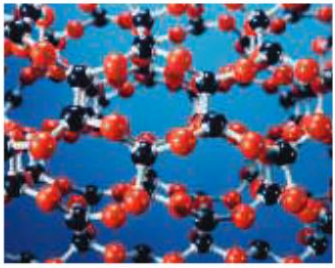
	C (graphite)	Si
Atomic radius / 10^{-12} m	75	114
X–X bond enthalpy / kJ mol^{-1}	346	226
Melting point / °C	3500	1414
Boiling point / °C	4827	3265
First ionization energy / kJ mol^{-1}	1086	787
X–O bond enthalpy	358	466

▲ Table 4 A selection of features of carbon and silicon, where X = Si or C

Interestingly, despite belonging to the same periodic table group, the properties of carbon and silicon are not as similar as one might anticipate. For example, carbon is a non-metal, while silicon is a metalloid. Diamond is a poor electrical conductor, while silicon is a semiconductor. Double and triple bonds are common between carbon atoms but unusual between silicon atoms.

Comparing the Si–O and C–O bond strengths (table 4) shows that single bonds between Si and O are very strong. Silicon is one of the most abundant elements in the Earth's crust and much of it is found in species containing Si–O bonds. **Silica**, or **silicon dioxide**, is a mineral that forms a covalent network containing Si and O in a 1:2 ratio, hence the formula SiO_2. Each silicon atom is bonded covalently to four oxygen atoms and each oxygen atom is bonded covalently to two silicon atoms. The crystalline form of silicon dioxide is quartz (figure 28). Other forms of silica include sand and glass.

The properties of silicon dioxide are those we associate with sand: it is hard, insoluble in water and has a high melting point. In its solid crystalline form, quartz, it is a poor conductor of heat and electricity.





▲ Figure 28 Structure (left) and crystals of silicon dioxide, SiO_2, known as quartz. The photograph also contains crystals of iron disulfide, FeS_2, commonly known as "fool's gold"

Practice questions

16. Explain why graphite and graphene conduct electricity, but diamond does not.
17. Explain why diamond and silicon dioxide have high melting and boiling points.

Linking question

Why are silicon–silicon bonds generally weaker than carbon–carbon bonds? (Structure 3.1)

Activity

Summarize the properties of some of the covalent substances using a table like the one below.

	Diamond	Graphite	Graphene	C_{60} fullerene	Silicon dioxide
Element(s)					
Arrangement of atoms					
Electrical conductivity					



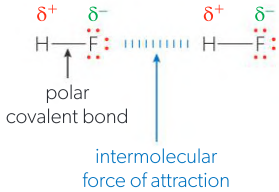
▲ Figure 29 Intermolecular forces hold the molecules together in water and inside the ice cubes

Intermolecular forces (*Structure 2.2.8*)

So far, we have been discussing forces of attraction that hold atoms and ions together, which are collectively referred to as **bonding**. We will now turn our attention to the forces of attraction between molecules, known as **intermolecular forces.** As their name suggests, intermolecular forces apply to molecular substances. The type and strength of intermolecular forces depend on the size and polarity of the molecules. Types of intermolecular forces include **London (dispersion) forces**, **dipole–induced dipole forces**, **dipole–dipole forces** and **hydrogen bonding**. Collectively the first three intermolecular forces are termed **van der Waals forces**.

Consider a glass of water. You know that the water molecules contain two hydrogen atoms each, bonded to a central oxygen atom through strong covalent bonds. But what forces cause the water molecules to stay close to one another? How about the water molecules inside ice cubes? What exactly is being overcome when some of the water molecules in ice break away from the cube and join the liquid water surrounding it? The answer in each case is intermolecular forces. Intermolecular forces are weaker than chemical bonds, but they affect physical properties such as volatility, solubility and boiling point of molecular substances.

Intermolecular forces are weak electrostatic forces of attraction that occur between molecules. Figure 30 highlights the difference between intermolecular and **intramolecular** forces.



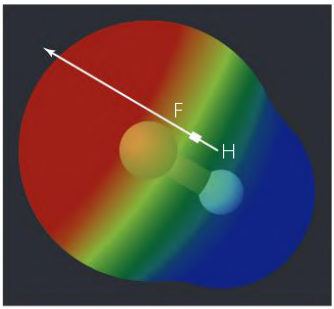
▶ Figure 30 Intramolecular forces, such as covalent bonds, occur within molecules. Intermolecular forces occur between molecules

Linking question

How useful is the VSEPR model at predicting molecular geometry?

Bond polarity (*Structure 2.2.5*)

Polarity is related to the way electrons are distributed within bonds and molecules. The shared pair of electrons in a covalent bond is not necessarily shared equally between the two atoms in the bond. **Bond polarity** results from the difference in the electronegativities of the bonded atoms.





▲ Figure 23 Representation of the molecular electrostatic potential of the polar covalent bond in HF. Note that the bond dipole in HF can be represented as a vector (arrow)

In the case of identical atoms, such as the two fluorine atoms in F_2, there is an equal sharing of the electrons in the bonding pair. This is not the case, however, in HF. The fluorine atom has a much greater electronegativity than the hydrogen atom, so it pulls the shared electron pair more strongly than hydrogen does. This leads to what we describe as a **polar covalent bond**, with one atom (fluorine) adopting a partial negative charge, δ–, and the other atom (hydrogen) adopting a partial positive charge, δ+.

The separation of charge between two non-identical bonded atoms is called a **dipole moment**. The dipole moment can be represented by a vector (figure 23). This vector is shown as an arrow along the bond. The arrow starts as a plus sign at the less electronegative atom in the bond, and points towards the more electronegative atom in the bond.

The shared pair of electrons is shifted towards the more electronegative atom in the covalent bond. The greater the difference in electronegativity between the two atoms, the greater this shift.

If the two atoms involved in the formation of the covalent bond are identical, the bond is said to be a **pure covalent bond.** The charge is distributed evenly about the covalent bond, meaning that it is non-polar and has no dipole moment.

As seen in *Structure 2.1*, if the difference of electronegativity between the two atoms is very large (greater than 1.8), then the bonding pair of electrons is shifted almost entirely towards the more electronegative atom. Such bonds are considered ionic because electrons are transferred, not shared.

Practice questions

11. Rank the following bonds in order of increasing polarity:
    C–H, C–C, C–F, C–O
12. Explain why Br_2 is non-polar, whereas HBr is polar.

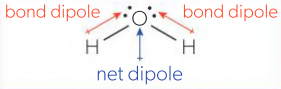
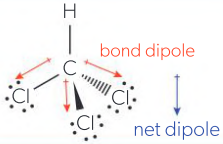
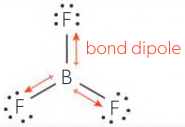
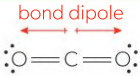
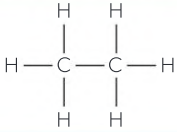
Linking question

What properties of ionic compounds might be expected in compounds with polar covalent bonding? (Structure 2.1)

Molecular polarity (*Structure 2.2.6*)

Molecular polarity is similar to bond polarity, but it describes the electron distribution throughout the whole molecule. A molecule is polar when the electron distribution leads to a partial negative charge on one end of the molecule, and a partial positive charge on the other. This results in a **dipole moment**, ***μ***. The value of the dipole moment is often reported in the unit debye, D. Polarity is an important trait of molecules because it gives rise to many characteristic properties such as volatility, solubility and boiling point.

Molecular polarity depends on both bond polarity and molecular geometry. Some examples are given in table 3.

Molecule	Lewis formula	Polarity	Dipole moment / D
water, H_2O		polar	1.85
trichloromethane, $CHCl_3$		polar	1.01
boron trifluoride, BF_3		non-polar	0
carbon dioxide, CO_2		non-polar	0
ethane, C_2H_6		non-polar	0

▲ Table 3 Examples of polar and non-polar molecules

Molecules are polar when their bond dipoles do not cancel each other out. This may occur due to the geometry of the molecule, or because the bonds have different polarities. Both water and trichloromethane (table 3) are polar because they contain polar bonds that do not cancel out. This results in a net dipole across the molecule.

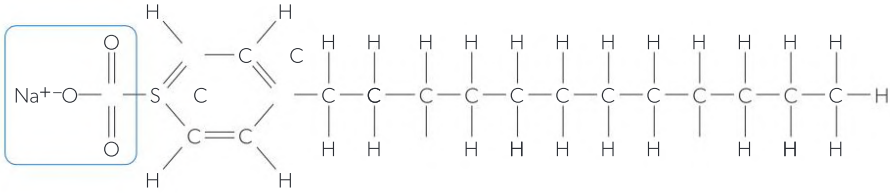
Molecules are non-polar when their bond dipoles cancel each other out. This happens when the bond dipoles are equal and their arrangement results in no net dipole. Examples include boron trifluoride and carbon dioxide. Both molecules contain polar bonds because of the difference in electronegativity of the atoms involved. These polar bonds are positioned in such a way that they cancel each other out. This means that boron trifluoride and carbon dioxide are both non-polar, despite the presence of polar bonds.

Molecules are non-polar when all their bonds are non-polar. Hydrocarbons (compounds of hydrogen and carbon) such as ethane contain two types of bonds: carbon–carbon bonds and carbon–hydrogen bonds. The carbon–carbon bonds are non-polar because both atoms have the same electronegativity. The carbon–hydrogen bonds are virtually non-polar because of the small electronegativity difference between the two atoms. In addition, the tetrahedral geometry around each carbon atom results in a net dipole of zero, or very close to zero. Therefore, hydrocarbons are generally non-polar.

So far, we have only discussed small molecules. Long molecules can have a polar region and a non-polar region, leading to many interesting applications including emulsifiers, soaps and detergents, as well as the phospholipid bilayer in cell membranes. Figure 24 shows a typical detergent molecule. Can you identify the polar and non-polar regions in the molecule?

Practice questions

13. Determine whether each of the following molecules is polar or non-polar:
    a methane, CH_4
    b ammonia, NH_3
    c hydrogen cyanide, HCN
    d hexane, C_6H_{14}
14. Explain why CO_2 is non-polar.
15. State and explain whether PH_3 is polar or non-polar.



▲ Figure 24 A diagram of a synthetic detergent molecule, showing the hydrophobic and hydrophilic regions

Linking question

AHL What features of a molecule make it "infrared (IR) active"? (Structure 3.2)

Activity

Summarize the properties of some of the covalent substances using a table like the one below.

	Diamond	Graphite	Graphene	C_{60} fullerene	Silicon dioxide
Element(s)					
Arrangement of atoms					
Electrical conductivity					



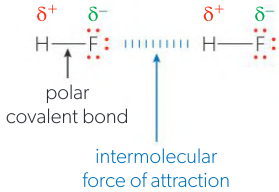
▲ Figure 29 Intermolecular forces hold the molecules together in water and inside the ice cubes

Intermolecular forces (*Structure 2.2.8*)

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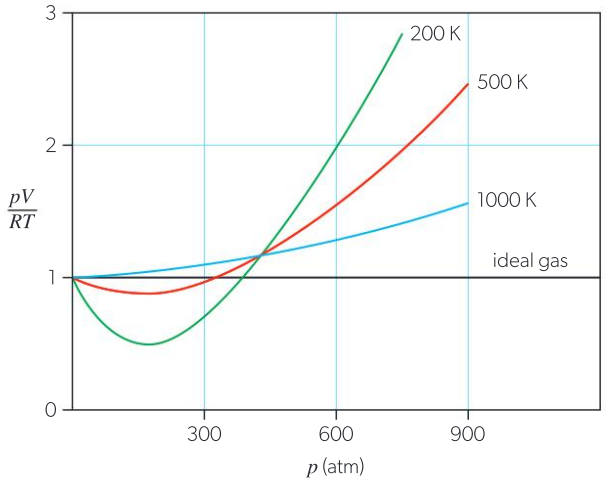
Intermolecular forces are weak electrostatic forces of attraction that occur between molecules. Figure 30 highlights the difference between intermolecular and **intramolecular** forces.



▶ Figure 30 Intramolecular forces, such as covalent bonds, occur within molecules. Intermolecular forces occur between molecules

When molecular substances melt, boil or sublime, the intermolecular forces between the molecules are overcome. Therefore, melting and boiling points are often used as indicators of intermolecular force strength. The covalent bonds in the molecule do not break during phase changes (figure 31).

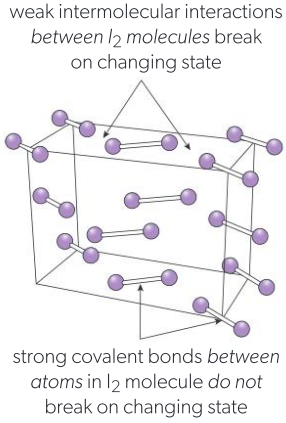
In ***Structure 1.5,*** we discussed the ideal gas law, which carries the assumption that no intermolecular forces are present in the gas. This is generally true at low pressures and high temperatures, for which the ideal gas equation can frequently be used. When a gas deviates from ideal behaviour, we must correct for the presence of intermolecular forces using a **real gas** model. Real gas models also take into account the actual volume of gas particles, which is considered negligible in the ideal gas model.



▲ Figure 32 Gases deviate more from ideal gas behaviour at lower temperatures. The horizontal black line represents an ideal gas. The curves show the deviations from ideal gas behaviour at different temperatures for a real gas

London (dispersion) forces (LDFs)

All molecules experience London (dispersion) forces (or LDFs), which are intermolecular forces resulting from temporary **instantaneous dipoles**. A molecule has a dipole when one side carries a partial positive charge, and the other a partial negative charge. London (dispersion) forces involve induced dipoles, which means that one molecule causes (or **induces**) a temporary dipole in another molecule. These induced dipoles occur due to the random movement of electrons around the molecule.

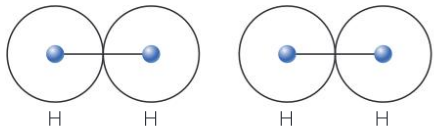


▲ Figure 31 Intermolecular forces of attraction are overcome when a molecular substance changes state, but the covalent bonds remain intact



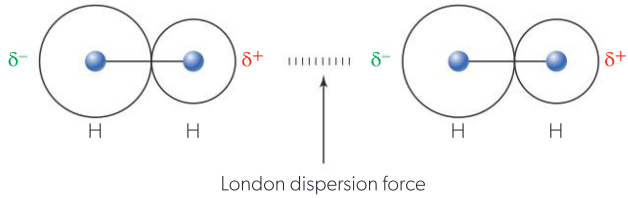
▲ Figure 33 Intermolecular forces of attraction are overcome when iodine sublimes, and are formed when iodine is deposited on a surface

In a simple non-polar molecule such as hydrogen, H_2, the electron distribution is on average symmetrical (figure 34). However, electrons are constantly moving around within the molecule. If we could somehow freeze time and take a photo of electrons in a molecule, the electron distribution would very unlikely be perfectly symmetrical. Instead, we would see a somewhat unequal electron density, with one region of the molecule having more electrons, rendering it slightly negative (δ–). There would also be a region of lower electron density elsewhere in the molecule with a positive partial charge (δ+).

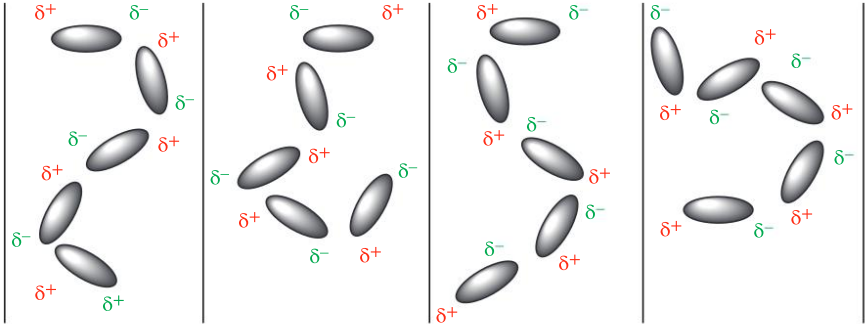


▲ Figure 34 On average, the electron distribution in H_2 is symmetrical

This temporary dipole then induces further temporary dipoles in the molecules around it. Due to electrostatic repulsion, a region of partial negative charge (δ–) on one molecule will repel the electrons in neighbouring molecules. This temporarily creates regions of partial positive charge (δ+). The resulting electrostatic attraction between the δ– region on one molecule and the δ+ region on the next is termed a London dispersion force. However, this arrangement is temporary. In the next moment of time a different pattern of induced dipoles will emerge.



▲ Figure 35 At any given moment in time, a temporary dipole is present in a molecule, inducing a dipole in another. A weak London dispersion force forms between the opposite partial charges



▲ Figure 36 Different arrangements of LDFs between molecules, which result from interactions between an instantaneous dipole on one molecule and an induced dipole on an adjacent molecule

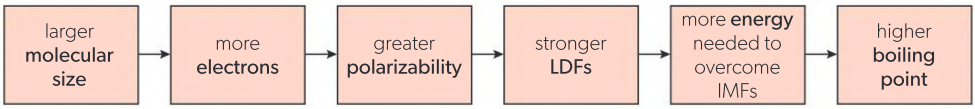
There are two main factors that increase the strength of LDFs: the number of electrons and the molecular shape. They affect the **polarizability** of the molecule: how easily the electron distribution is distorted by an electric field. The greater the polarizability, the stronger the dispersion forces.

Consider the group 17 elements, all of which form non-polar diatomic molecules. The number of electrons increases going down the group. As a result, the electron clouds become larger and less attracted to the nuclei. This means that the electrons become more easily polarized and the strength of LDFs increases, leading to higher boiling points as you descend the group (table 5).

Substance	Boiling point / °C
fluorine, F_2	–188.1
chlorine, Cl_2	–34.04
bromine, Br_2	58.78
iodine, I_2	184.4

Increasing LDF strength

▲ Table 5 Boiling point increases down group 17 due to an increasing number of electrons, resulting in stronger LDFs



▲ Figure 37 The effect of molecular size on LDF strength and boiling point

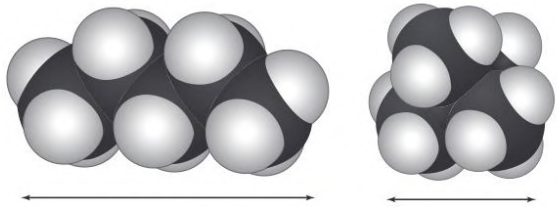
The boiling point also increases for successive members of a **homologous series** of organic compounds: each molecule has one more CH_2 than the one before it, hence greater molecular size and a larger number of electrons. Molecular size can be quantified in terms of mass, so we often refer to molecular mass when comparing molecules in terms of their LDFs.

Homologous series and isomers of organic compounds are discussed in *Structure 3.2*.

Patterns and trends

Scientists look for patterns and try to understand relationships between variables. For example, we often say LDF strength increases with increasing molecular mass, due to an increasing number of electrons. This may seem contradictory because electrons have negligible mass. But here the relationship between molecular mass and the number of electrons is not **causal**. Rather, the greater mass of larger molecules is *accompanied by* a greater number of electrons. The greater number of protons and neutrons is responsible for the increase in molecular mass. However, each proton requires an electron to maintain the overall neutrality of the molecule. Therefore, any increase in the number of protons always leads to a proportional increase in the number of electrons.

Compare the boiling points of the two isomers of C_5H_{12}: pentane, $CH_3CH_2CH_2CH_2CH_3$ and 2,2-dimethylpropane $(CH_3)_4C$ (table 6 and figure 38).



▶ Figure 38 Space-filling models show that pentane (top) has a greater area of contact for LDFs than 2,2-dimethylpropane (bottom)

Isomer	Boiling point / °C
pentane	36.1
2,2-dimethylpropane	9.5

▲ Table 6 Boiling points of the two isomers of pentane

Both isomers contain the same number of electrons and are non-polar. The difference is in their shapes. Pentane molecules are long and are therefore able to interact with each other across the full length of the molecule; that is, there is a large area of interaction because of the better contact between molecules of pentane. As a result, pentane has a relatively high boiling point and is a liquid at room temperature. Molecules of 2,2-dimethylpropane are rounder and more compact, so they pack together less efficiently. As the interaction between the molecules is limited, the LDFs in 2,2-dimethylpropane are weaker, so it is a gas at room temperature.

Data-based questions

1. Boiling point data for the first four alkanes are shown below. Describe and explain the boiling point trend you observe.

Alkane	Boiling point / °C
methane, CH_4	–161
ethane, C_2H_6	–88
propane, C_3H_8	–42.1
butane, C_4H_{10}	–0.5

2. The boiling points of 2,2-dimethylpropane and pentane are 9.5 °C and 36.1 °C, respectively. Predict a value for the boiling point of 2-methylbutane, $CH_3CH_2CH(CH_3)_2$. Explain your answer.

LDFs occur between all types of molecules because all molecules contain electrons. If we have a substance made up of only non-polar molecules, only LDFs will occur.

Dipole–induced dipole forces

LDFs are forces of attraction between temporary, or instantaneous, dipoles. **Dipole–induced dipole forces** are a type of related intermolecular force occurring between a polar molecule and a nearby non-polar molecule. The presence of a permanent dipole in the polar molecule induces the formation of a temporary dipole in the neighbouring non-polar molecule. For example, this type of intermolecular force attracts non-polar oxygen molecules, O_2, to polar water molecules. Dipole–induced dipole forces are weak, which explains why the aqueous solubility of oxygen is relatively low.

Activity

Compare and contrast London (dispersion) forces and dipole-induced dipole forces.

Dipole–dipole forces

Dipole–dipole forces involve permanent dipoles, whereas LDFs result from temporary dipoles. When a molecule is polar, it has a permanent dipole and therefore experiences dipole–dipole forces of attraction with neighbouring polar molecules (figure 39).



▲ Figure 39 Hydrogen chloride, HCl, molecules are polar and attract each other due to dipole–dipole forces

Hydrogen chloride, HCl, and diatomic fluorine, F_2, have similar sizes and comparable molecular masses (table 7). They both therefore experience LDFs of a similar strength. However, HCl molecules are polar and therefore experience dipole–dipole forces in addition to LDFs. The intermolecular forces between HCl molecules are stronger than those between F_2 molecules, so HCl has a higher boiling point.

Molecule	Molecular mass	Boiling point / °C	Intermolecular forces
F_2	38.00	–188.1	London (dispersion)
HCl	36.46	–85.05	London (dispersion) and dipole–dipole

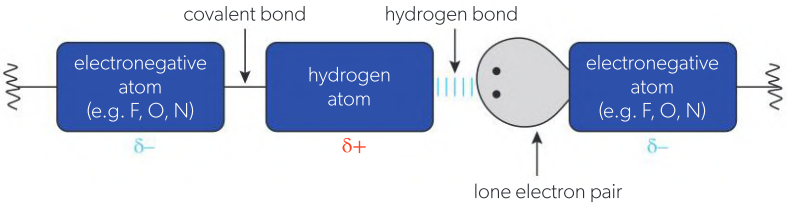
▲ Table 7 Diatomic fluorine and hydrogen chloride have similar molecular masses but different boiling points

Practice questions

18. Which of the following species experience dipole–dipole forces?
    A. oxygen, O_2
    B. carbon dioxide, CO_2
    C. carbon monoxide, CO
    D. carbonate ion, CO_3^{2-}
19. The molecular masses of ICl and Br_2 are very similar, while their boiling points are 97.4 °C and 58.8 °C, respectively. Explain this difference.
20. Draw a diagram to show how propanone molecules, CH_3COCH_3, might arrange themselves when near one another. Label any dipole–dipole forces that occur.

Hydrogen bonding

Hydrogen bonds are strong intermolecular forces that form when a molecule contains a strong dipole involving hydrogen. When a hydrogen atom is covalently bonded to a highly electronegative atom such as oxygen, nitrogen or fluorine, the bond between them is very polar. The electrons in the covalent bond are drawn towards the more electronegative atom, resulting in a considerable partial positive charge (δ+) on the hydrogen. This hydrogen atom can then form a strong electrostatic interaction—a hydrogen bond—with the electrons of another electronegative atom. This is usually found on a different molecule, but intramolecular hydrogen bonds can also exist in certain situations.

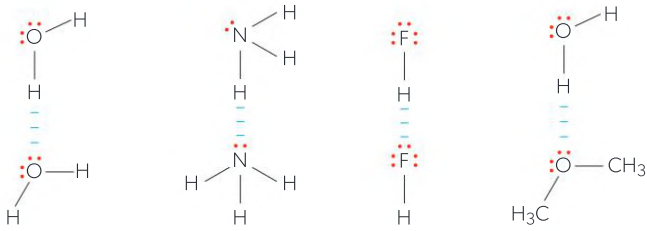


◀ Figure 40 Hydrogen bonds form between the hydrogen in a very polar bond and electrons on a highly electronegative atom

Hydrogen bonds occur, for example, between:

1. water molecules
2. ammonia (NH_3) molecules
3. hydrogen fluoride (HF) molecules
4. water molecules and dimethyl ether molecules, $(CH_3)_2O$

These hydrogen bonds are depicted with blue dashes in figure 41.



▶ Figure 41 Hydrogen bonds can form between a variety of different molecules

Despite the name suggesting otherwise, hydrogen bonds are not chemical bonds: they are intermolecular forces. They are generally stronger than other types of intermolecular force, but much weaker than covalent bonds.

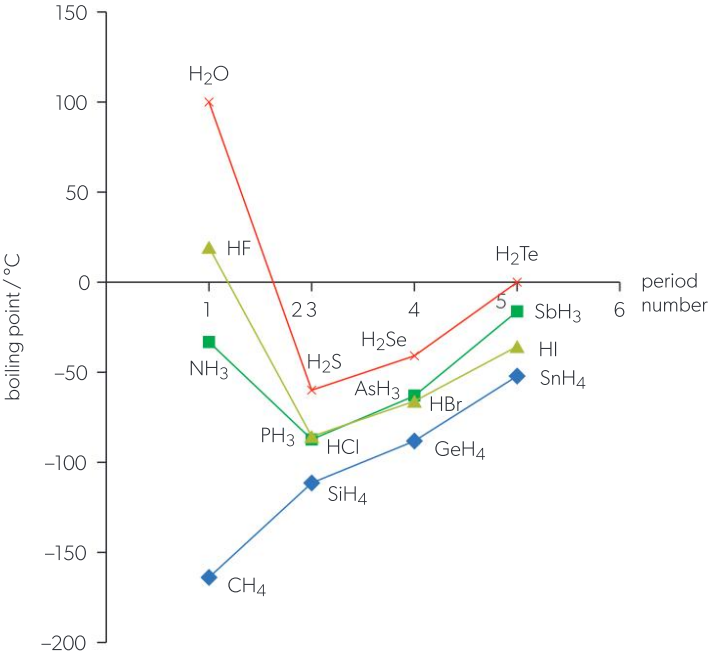
Bond or intermolecular force type	Typical bond or intermolecular force enthalpy / kJ mol^{-1}
Hydrogen bond	20–40
Single covalent bonds	150–600

▶ Table 8 Enthalpies of hydrogen bonds and single covalent bonds compared

A good way to remember that hydrogen bonds are weaker than chemical bonds is to look at what happens when water boils. At 100 °C, the energy available is sufficient to overcome the hydrogen bonds between water molecules (~20 kJ mol^{-1}), but not the covalent bonds between oxygen and hydrogen atoms in the water molecules themselves (463 kJ mol^{-1}).

Practice questions

21. Which **two** of the following substances can form hydrogen bonds between their molecules?
    A. hydrogen fluoride, HF
    B. fluorine, F_2
    C. methanol, CH_3OH
    D. fluoromethane, CH_3F
22. Draw a diagram to show a hydrogen bond between two methanol molecules, CH_3OH.
23. Draw a diagram to show a hydrogen bond between a water molecule and a methanamine molecule, CH_3NH_2.
24. Draw a diagram to show a hydrogen bond between a water molecule and a methanal molecule, CH_2O.



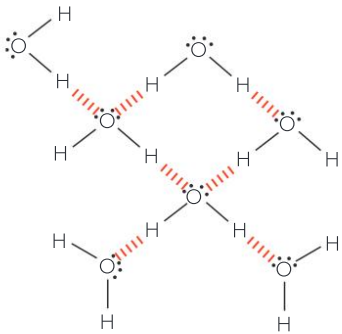
◀ Figure 42 The boiling points of the group 14 to 17 hydrides show the effect of hydrogen bonding

How do the boiling points of the group 14 hydrides (CH_4, SiH_4, GeH_4 and SnH_4) change as you go down the group? The molecular mass increases down the group, increasing the number of electrons and polarizability of the electron cloud. Descending the group, there are stronger LDFs, and therefore an increasing boiling point is expected. This is precisely what the data show in the bottom curve of figure 42.

Now look at the trend of the group 15 hydrides, NH_3, PH_3, AsH_3 and SbH_3, in the next curve up. We see the expected rise in boiling point from phosphine, PH_3, to stibane, SbH_3, due to increasing molecular mass. Extrapolating the trend to ammonia predicts a boiling point of around –130 °C, when in fact it is 100 °C higher, approximately –30 °C. The higher-than-expected boiling point is due to the presence of hydrogen bonds between ammonia molecules.

The trends in the group 16 and group 17 hydrides are similar to that of group 15: the period 2 hydrides, water, H_2O, and hydrogen fluoride, HF, have very high boiling points. Again, this is due to the presence of hydrogen bonds between their molecules.

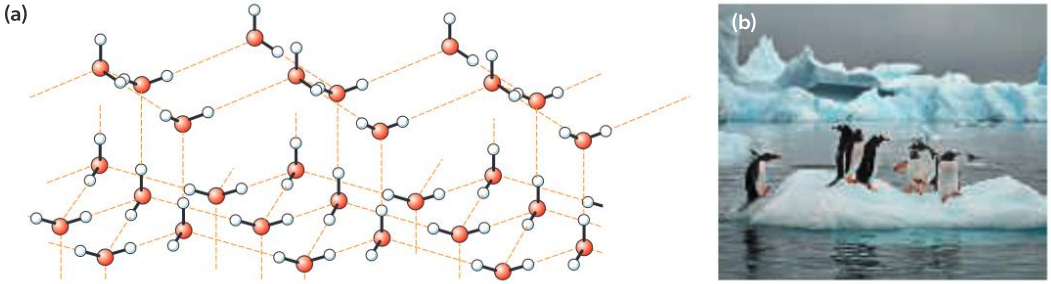
A single water molecule can form up to four hydrogen bonds (figure 43) and therefore the boiling point of water is very high: 100 °C.



▲ Figure 43 Water molecules can form up to four hydrogen bonds each

Hydrogen bonds give water its notable properties. Most substances have higher densities when solid than when liquid. Solid water (ice) floats on liquid water because it is less dense than liquid water. This unusual property leads to the formation of layers of ice on lake surfaces during the winter. Insulation from this ice layer prevents lakes freezing into solid blocks of ice, allowing aquatic ecosystems to survive.

The hydrogen bonding in ice results in the formation of a regular, very ordered open-cavity network of molecules (figure 44). In the liquid phase, the hydrogen bonding is more random, which results in a higher density compared to ice. Note that O–H covalent bonds in water are shorter than H - - - O hydrogen bonds between water molecules.



▲ Figure 44 **(a)** The open-cavity structure of ice causes it to be less dense than water **(b)** Ice floats on water because of this



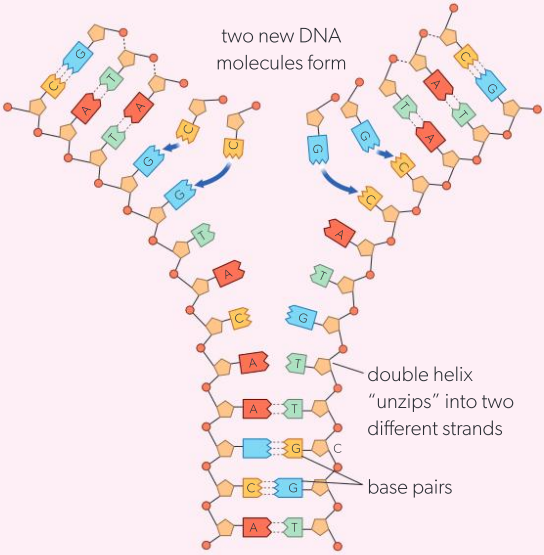
▲ Figure 45 The hexagonal arrangement of water molecules in the ice lattice leads to the characteristic shape of snowflakes



▲ Figure 46 Extensive hydrogen bonding is responsible for the high surface tension of water, which allows small insects to walk on the surface without sinking or becoming even partly submerged

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DNA (deoxyribonucleic acid) molecules store genetic information. A highly accurate DNA-copying mechanism is essential for this information to be passed along between generations. DNA has a double-helix structure composed of two strands of organic molecules held together by hydrogen bonds. DNA replication involves breaking these hydrogen bonds to "unzip" the two strands, allowing copies to be made.



▲ Figure 47 The DNA double helix is held together by hydrogen bonds, which are broken during replication. Hydrogen bonds are shown by black dashes

What other types of information storage systems exist in the natural world, in science and in other areas of knowledge?

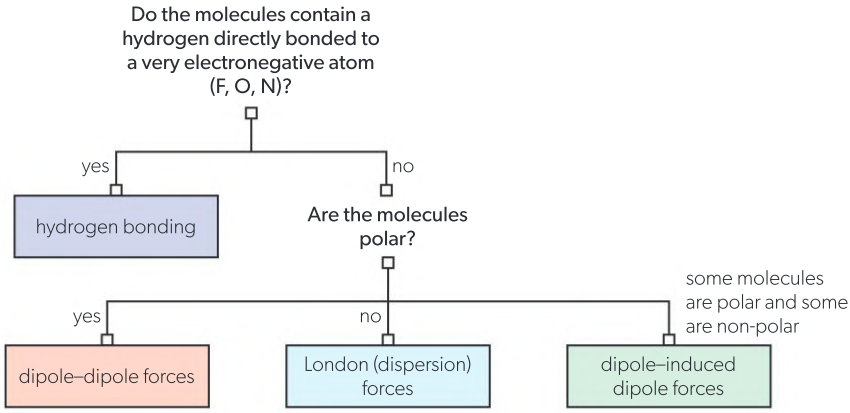
In summary, intermolecular forces are electrostatic attractions that keep molecules together. We have discussed the four types of intermolecular forces, summarised in table 9.

	Type of intermolecular force	Where does it occur?
van der Waals forces	London (dispersion) forces (LDFs)	between all molecules
	dipole–induced dipole forces	between a polar molecule and a non-polar molecule
	dipole–dipole forces	between polar molecules
	hydrogen bonding	between a highly electronegative atom (F, O or N) and a hydrogen covalently bonded to another highly electronegative atom

increasing strength

▲ Table 9 Summary of types of intermolecular forces

Figure 48 shows the overall method for determining the strongest intermolecular force present in a substance.



▲ Figure 48 Figuring out the strongest intermolecular force present between a pair of interacting molecules

Worked example 3

For each of the following substances, determine the intermolecular forces that attract the molecules to each other.

a. butane, $CH_3CH_2CH_2CH_3$
b. ethoxyethane, $CH_3CH_2OCH_2CH_3$
c. carbon tetrafluoride, CF_4
d. ethanamine, $CH_3CH_2NH_2$
e. fluoroethane, CH_3CH_2F
f. chloromethane, CH_3Cl, and methane, CH_4
g. water, H_2O, and hydrogen fluoride, HF

Solution

All the substances in this worked example form London (dispersion) forces.

a. Butane is non-polar, so it only forms London (dispersion) forces.
b. Ethoxyethane is polar, so it forms dipole–dipole forces.
c. Carbon tetrafluoride is non-polar, so it only forms LDFs.
d. Ethanamine can form hydrogen bonds because it contains a hydrogen atom directly bonded to nitrogen, which is a very electronegative atom. It is also polar, so it forms dipole–dipole forces.
e. Fluoroethane is polar, so it forms dipole–dipole forces. Note that it does not form hydrogen bonds. The structural formula might suggest that fluorine is directly bonded to hydrogen, but this is not the case. If you draw the Lewis formula, you will see that fluorine is bonded to a carbon atom.
f. Chloromethane is polar while methane is non-polar. They would therefore form dipole–induced dipole forces.
g. Hydrogen bonds are formed between water and hydrogen fluoride molecules because they both contain a highly electronegative atom that is directly bonded to a hydrogen. They are both polar, so they also experience dipole–dipole forces.

Practice questions

25. For each of the following substances, determine the intermolecular forces that attract molecules to each other:
    a. propane, $CH_3CH_2CH_3$
    b. hydrogen bromide, HBr
    c. hydrogen fluoride, HF
    d. buckminsterfullerene, C_{60}
    e. ammonia, NH_3, and diatomic oxygen, O_2
    f. ethanal, CH_3CHO
26. State and explain which of the following species can form hydrogen bonds with water molecules:
    a. ammonia, NH_3
    b. propane, $CH_3CH_2CH_3$
    c. ethanoic acid, CH_3COOH

Linking questions

To what extent can intermolecular forces explain the deviation of real gases from ideal behaviour? (Structure 1.5)

How do the terms "bonds" and "forces" compare? (Structure 1.1, Structure 2.1, Structure 2.3)

How can advances in technology lead to changes in scientific definitions, e.g. the updated International Union of Pure and Applied Chemistry (IUPAC) definition of the hydrogen bond?

The properties of covalent substances (*Structure 2.2.9*)

Relative strength of intermolecular forces

Given comparable molar mass, the relative strengths of intermolecular forces are generally:

London (dispersion) forces < dipole–dipole forces < hydrogen bonding.

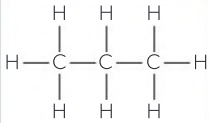
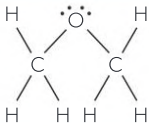
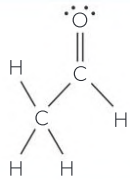
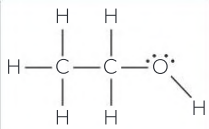
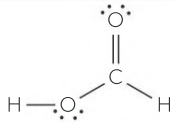
Table 10 shows typical enthalpy ranges associated with overcoming each of these forces. The trend in enthalpy reflects the trend in strength. The strength of dipole–induced dipole forces lies somewhere in-between London (dispersion) and dipole–dipole forces.

Intermolecular force type	Typical enthalpy / kJ mol^{-1}
London (dispersion) forces	1 to 5
sipole–dipole forces	3 to 25
hydrogen bonds	20 to 40

▲ Table 10 Comparison of the strength of various intermolecular forces

Data-based questions

The relative strengths of intermolecular forces are best illustrated by comparing molecules that have similar molecular masses. Table 11 shows molecules with molecular masses in the range of 44.0 to 46.1 and wildly different boiling points.

Name and molecular formula	Structural formula	Molecular mass	Molecular dipole moment / D	Boiling point / °C	Intermolecular forces		
					London (dispersion)	Dipole–dipole	Hydrogen bond
propane C_3H_8		44.11	0.084	–42.1	yes	no	no
methoxymethane C_2H_6O		46.08	1.30	–24.8	yes	yes	no
ethanal C_2H_4O		44.06	2.69	20.2	yes	yes	no
ethanol C_2H_6O		46.08	1.69	78.2	yes	yes	yes
methanoic acid CH_2O_2		46.03	1.41	100.7	yes	yes	yes

▲ Table 11 Polarity, boiling point and intermolecular force data for organic compounds of similar molecular masses

Interpreting and explaining data are important skills in science. What do the data in table 11 suggest? For example, they show that all molecules experience London (dispersion) forces because they all contain electrons that move around, causing temporary dipoles that then induce further temporary dipoles in the surrounding molecules.

Write down four more statements describing the relationship between molecular structure and intermolecular forces that are shown by the data in table 11. Attempt to explain each statement using your knowledge from this topic.

Physical properties of covalent substances

There are two types of covalent structures: covalent network and molecular covalent. These two structures have very different properties. Most of the properties of molecular substances are governed by intermolecular forces. Substances with a covalent network structure are not comprised of molecules, so their properties are dictated by their lattice features.

We will look at physical properties of covalent substances including volatility, electrical conductivity and solubility.

Volatility

Substances with a covalent network structure are solids at room temperature and pressure. Vaporizing them requires a lot of energy because of the strong covalent bonds holding the structure together. They are therefore non-volatile and have very high melting and boiling points.

In order to vaporize molecular substances, the intermolecular forces between the molecules need to be overcome. Since intermolecular forces are relatively weak, the energy required to overcome them is low and therefore molecular substances are generally volatile.

Due to the variety of sizes and intermolecular forces, there is, however, a large variation in the volatility of molecular substances. Many of the molecular substances we have considered in this chapter consist of small molecules, for example, N_2, CO_2, H_2O and some hydrocarbons. Many of these are gases or liquids at room temperature.

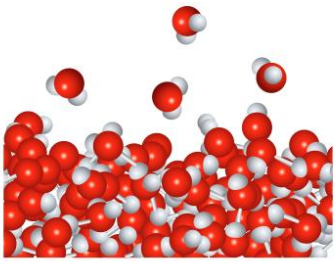
However, substances that consist of much larger molecules have lower volatility and higher melting and boiling points. This is because larger molecules have stronger LDFs between each other. Examples include buckminsterfullerene (C_{60}), which sublimes at 500–600 °C, and a component of candle wax, tetracosane ($C_{24}H_{50}$), which boils at 391 °C.

Electrical conductivity

Covalent substances, both network and molecular, are usually not electrical conductors. To conduct electricity, a substance needs to contain charged particles that are free to move. Covalent substances generally do not contain such particles, as their electrons are "locked up" in localized covalent bonds, and they do not contain ions.

There are some notable exceptions, such as graphite, which is a good electrical conductor due to the presence of delocalized electrons. Silicon is a semiconductor, meaning that its intermediate conductivity places it between conductors and insulators.

Solar panels consist of photovoltaic cells, which convert solar energy into electricity. The cells contain semiconductors, such as silicon.



▲ Figure 49 Molecular covalent substances vaporize when the intermolecular forces are overcome

Practice questions

27. Identify which species in each pair has the highest boiling point.
    a. He or Kr
    b. CH_4 or NH_3
28. The boiling point of carbon tetrachloride, CCl_4, is 76.7 °C. The boiling point of fluoromethane, CH_3F, is –78.2 °C. Comment on this difference using your knowledge of intermolecular forces.



▲ Figure 50 Solar panels containing silicon provide this satellite with electrical power

Solubility

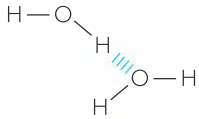
When a substance dissolves, forces of attraction are formed between the substance (known as the solute) and the solvent. Substances with covalent network structures are insoluble in most solvents because of the strong covalent bonds holding their atoms together.

A molecular substance is likely to dissolve in a solvent if the intermolecular forces between the solute and solvent are stronger than the attraction between the solute molecules.

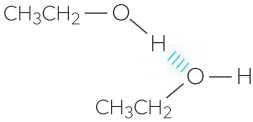
It is often said in chemistry that "like dissolves like". In other words, non-polar solutes are likely to dissolve in non-polar solvents. Similarly, polar solutes are likely to dissolve in polar solvents. Dissolving is unlikely to occur if the solute is polar and the solvent is non-polar, or vice versa.

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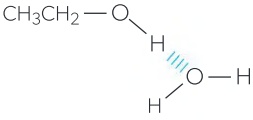
"Like dissolves like" is a useful rule of thumb, which is helpful in many cases but not all, and it has no explanatory power. In a small group, discuss the advantages and disadvantages of having rules like this. You may want to consider: what are the alternatives? Do such rules have an "expiry date"? What rules of thumb do you use in other areas of knowledge?



(a) between water molecules



(b) between ethanol molecules



(c) between ethanol and water

▲ Figure 51 Ethanol and water can form hydrogen bonds. Therefore, ethanol is water soluble

Iodine, I_2, is non-polar. It dissolves readily in non-polar solvents such as hexane. Iodine is almost insoluble in water because it cannot associate strongly with water molecules. Substances that can form hydrogen bonds, such as ethanol, are more likely to dissolve in water (figure 51). Ethanol and water are said to be **miscible**, as they can form a solution when mixed in any proportion.

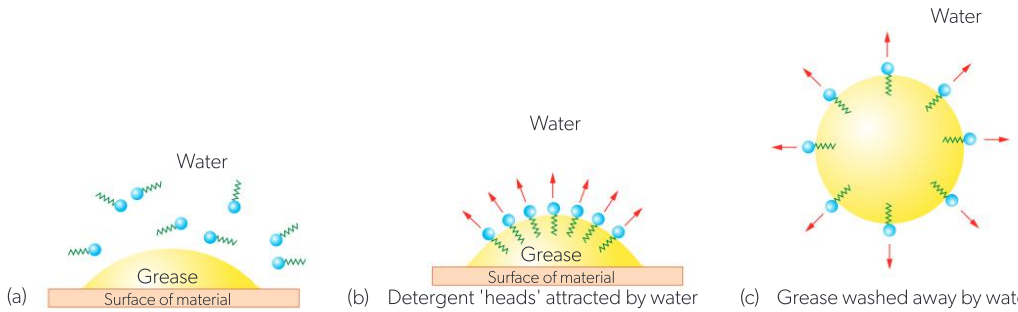
Methanol, ethanol and propan-1-ol readily dissolve in water. The solubility of the primary alcohols, however, decreases with increasing carbon chain length (table 12).

Primary alcohol	Structural formula	Solubility g / 100 g of H_2O
ethanol	CH_3CH_2OH	miscible
butan-1-ol	$CH_3(CH_2)_3OH$	6.8
nonan-1-ol	$CH_3(CH_2)_8OH$	0.014

▲ Table 12 The aqueous solubility of primary alcohols decreases with increasing hydrocarbon chain length

Hydrocarbon chains are non-polar and do not form hydrogen bonds with water. Fats and oils are largely non-polar and therefore do not dissolve in water. This is easily observable when vinegar (an aqueous solution of ethanoic acid) is mixed with oil to make salad dressing. The two liquids form two layers and are said to be **immiscible**, which means they do not mix, due to their differing polarities.

Long hydrocarbon chains, such as those in fats and oils, are said to be **hydrophobic** (water-hating). Soap and detergent molecules often have a hydrophobic tail and a **hydrophilic** (water-loving) head. They work because they associate with both water and the non-polar fats and oils frequently found in greasy stains (figure 52). Molecules with a hydrophobic tail and hydrophilic head are called **surface-active agents**, or **surfactants**.



▲ Figure 52 In a surfactant molecule, the hydrophilic (water-loving) head and a hydrophobic (water-hating) tail lift greasy stains off materials

Global impact of science

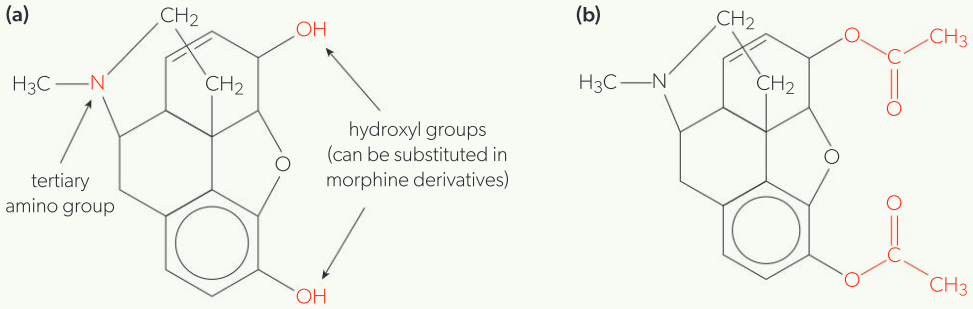
Morphine and its derivatives, known as opiates, are strong painkillers with a high potential for misuse and addiction. The effect they have on the body depends on two factors:

1. their ability to cross the blood-brain barrier
2. their ability to bind to opioid receptors in the brain.

The blood-brain barrier consists of a series of cell membranes that coat the blood vessels in the brain and prevent polar molecules from entering the central nervous system. These membranes are lipophilic, which means they are permeable to lipid molecules. The presence of one amino and two hydroxyl groups in the morphine molecule (figure 53) makes it sufficiently polar to be soluble in water but at the same time reduces its solubility in lipids and therefore limits its ability to reach the opioid receptors in the brain.

In diamorphine, also known as heroin, both hydroxyl groups are substituted with ester groups, which greatly reduces the polarity of the molecule. Diamorphine is soluble in lipids and can easily cross the blood-brain barrier. In the brain, diamorphine is quickly metabolized into morphine, which binds to the opioid receptor.

This mechanism of action makes diamorphine about five times more potent an analgesic than morphine. At the same time, diamorphine has more severe side effects, including tolerance, addiction and central nervous system depression. In most countries the use of diamorphine is either banned or restricted to terminally ill patients with certain forms of cancer or central nervous system disorders.



▲ Figure 53 The structures of (a) morphine and (b) diamorphine (heroin)

Physical properties of ionic and covalent substances

You can figure out the structure of a substance by testing its physical properties such as electrical conductivity, aqueous solubility and melting point.

Relevant skills

- Tool 1: Recognize and address the relevant safety, ethical or environmental issues in an investigation.
- Inquiry 1: Demonstrate independent thinking, initiative or insight.
- Inquiry 2: Identify and record relevant qualitative observations.

Instructions

You are presented with three bottles labelled A, B and C. Each bottle contains a crystalline solid. One is sodium chloride, another is sugar, and the third is white sand, but you do not know which is which. Design a safe method that could be used to identify A, B and C.

Possible equipment and substances you could use might include:

spatulas, beakers, distilled water, pipettes, power pack, leads, ammeter, lamp, conductivity probe, electrodes, crucible, Bunsen burner, pipeclay triangle, heat-proof mat, mortar and pestle, melting point apparatus, fume hood, combustion spoon.

Complete a risk assessment and show this and your proposed method to your teacher for approval. If you have time, try it out.

Extension

What if the ionic substance were not water soluble? How would you modify your method?

Linking questions

What experimental data demonstrate the physical properties of covalent substances? (Tool 1, Inquiry 2)

To what extent does a functional group determine the nature of the intermolecular forces? (Structure 3.2)

To summarize, substances with covalent network structures are typically:

- non-volatile because they are held together by strong covalent bonds between atoms
- poor electrical conductors (exceptions include graphite and graphene) because they do not contain mobile charged particles
- insoluble in all solvents because their dissolution would require breaking the strong covalent bonds.

Substances with molecular covalent structures are typically:

- volatile because their molecules are held together by intermolecular forces, which are weak
- poor electrical conductors because they do not contain mobile charged particles
- of varying solubility, depending on the strength of the intermolecular forces they form with solvent molecules.

Practice questions

29. Explain why covalent compounds are not electrical conductors, except for graphite and graphene.
30. Describe and explain the relationship between volatility and boiling point.
31. Explain why covalent network substances are non-volatile, and molecular substances are volatile, although they both contain covalent bonds.
32. Explain why ethanol, CH_3CH_2OH, is miscible with water.
33. Which **two** of the following substances are readily soluble in water?
    A. ethanoic acid, CH_3COOH
    B. octane, $CH_3(CH_2)_6CH_3$
    C. octan-1-ol, $CH_3(CH_2)_6CH_2OH$
    D. propanone, CH_3COCH_3
34. Explain why ionic substances, such as sodium chloride, do not dissolve in oil.
35. Explain why diatomic bromine, Br_2, is more soluble in hexane than in water.

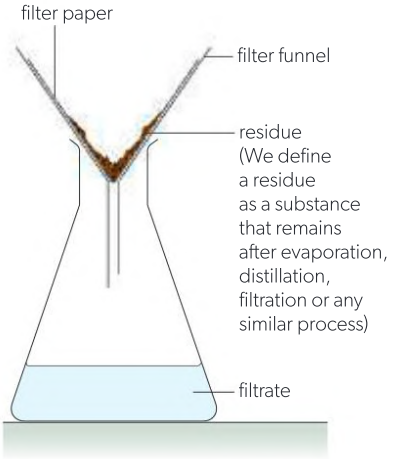
Separating mixtures

Mixtures can be separated by physical means because each component of the mixture has unique properties. A mixture of iron and sulfur powders can be separated using a magnet. Iron is magnetic while sulfur is not. This difference in property is used to separate them. The compound iron(II) sulfide, FeS, is not magnetic and does not have a sulfurous smell. It maintains none of the properties of the components as it is a new, individual pure substance.

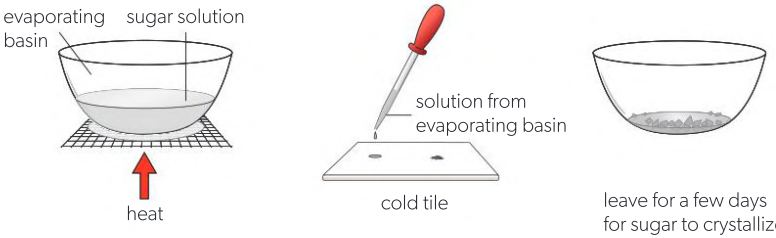
Two solids can usually be separated if we understand their intermolecular forces. Sand can be separated from sugar because sugar will dissolve in water, due to the intermolecular attractions between sugar and water.

The solid mixture of sand and sugar is placed in water and the sugar dissolves. The solution can then be poured through filter paper placed inside a funnel, a process called **filtration** (figure 6). The large sand particles will not pass through and remain on the filter paper, whereas the sugar dissolved in the water will pass through the filter paper. The wet sand is dried, and the water evaporates leaving behind the pure sand. The sugar can be obtained by evaporating the water from the **filtrate** — the solution which passed through the filter paper. Sugar crystals will form in this **crystallization** process (figure 7).

Intermolecular forces are discussed in *Structure 2.2*.

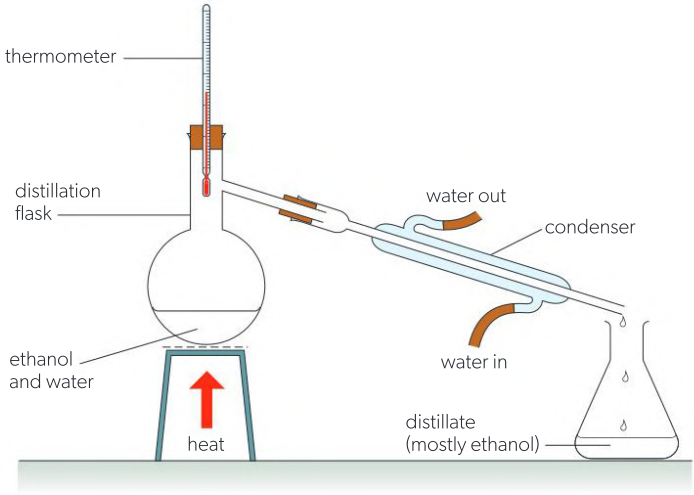


▲ Figure 6 Filtration apparatus



◀ Figure 7 The crystallization process

Distillation can be used to separate **miscible** liquids with different boiling points, such as ethanol and water. Ethanol has a lower boiling point and will evaporate first. Once the vapours rise up a cooling column, they can be condensed to a liquid. As shown in figure 8, cold water surrounds the condenser and allows the vapours to condense to liquid ethanol. The water remains mostly in the distillation flask.

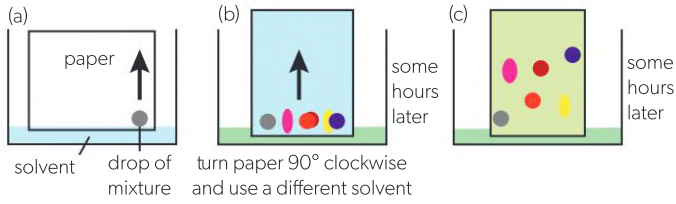


▶ Figure 8 Distillation apparatus

Paper chromatography will be discussed in more detail in *Structure 2.2*.

Paper chromatography can be used to separate substances such as components in inks. A piece of chromatography paper is spotted with the mixture. The bottom of the paper, below the spot, is placed in a suitable solvent as in figure 9(a).

The substances in the mixture have different affinities for the solvent (the **mobile phase**) and the paper (the **stationary phase**). The affinity depends on the **intermolecular forces of attraction** between the pure substances in the mixture and the solvent or the paper. Figure 9(c) shows a mixture that was composed of five pure substances.



▶ Figure 9 The stages in 2D paper chromatography

Data-based questions

Look at figure 9.

1. Which colour dot had the strongest affinity for both solvent 1 and solvent 2?
2. Which colour dots had a stronger affinity for solvent 1 than solvent 2?
3. Which had a stronger affinity for solvent 2 than solvent 1?

Table 3 shows a summary of the separation techniques discussed.

Technique	Description	Components	
		removed	left
filtration	mixture is poured through a paper filter or other porous material	liquid(s)	solid(s)
dissolution (solvation)	mixture is added to water or an organic solvent	soluble substance(s)	insoluble substance(s)
crystallization	mixture is dissolved in hot water or an organic solvent, the solution cools down, and the crystals formed are isolated by filtration	more soluble substance(s)	less soluble substance(s)
evaporation or distillation	mixture is heated up until one or more of its components vaporize(s)	volatile liquid(s)	solid(s) and/or non-volatile liquid(s)
paper chromatography	mixture is placed on a piece of paper; one side of the paper is submerged in water or a solvent; components move along the paper	more soluble component(s) move(s) faster	less soluble component(s) move(s) slower or stay(s) in place

▲ Table 3 Summary of separation techniques

Activity

Suggest a suitable method for separating each of the following mixtures:

a. salt and pepper
b. several water-soluble dyes
c. sugar and water
d. iron and copper filings

For each mixture, describe the separation technique and outline how each component is isolated.



▲ Figure 10 An advanced filtration technique called reverse osmosis extracts salt from seawater, providing fresh water for millions of people. However, this process requires vast amounts of energy, most of which is currently provided by fossil fuels. Why might it be important to consider alternative energy sources?

Planning experiments and risk assessments

Relevant skills

- Tool 1: Separation of mixtures
- Tool 1: Addressing safety of self, others and the environment

Instructions

1. Using the ideas in this chapter, devise a method that would allow you to separate a mixture containing sand, salt, iron filings and powdered calcium carbonate. In doing so, you must consider the physical and chemical properties of each of these four substances.
2. Once you have decided on a method, identify the hazards and complete a risk assessment protocol in which you:
   - Identify the **hazards**
   - Assess the level of **risk**
   - Determine relevant **control measures**
   - Identify suitable disposal methods aligned with your school's health and safety policies.
3. If you have time, try it out! Remember that your teacher should validate your methodology and risk assessment beforehand.

Extension

You could evaluate the effectiveness of your method by comparing the mass of each component (sand, salt, iron filings, and calcium carbonate) before and after the separation. Measure the mass of each component prior to mixing them together. Then mix them together, carry out your separation, make sure the components are all dry, and measure the mass of each again. Compare the masses before and after to calculate the percentage recovery of each component.

Linking questions

What factors are considered in choosing a method to separate the components of a mixture? (Tool 1)

How can the products of a reaction be purified? (Tool 1)

How do intermolecular forces influence the type of mixture that forms between two substances? (Structure 2.2)

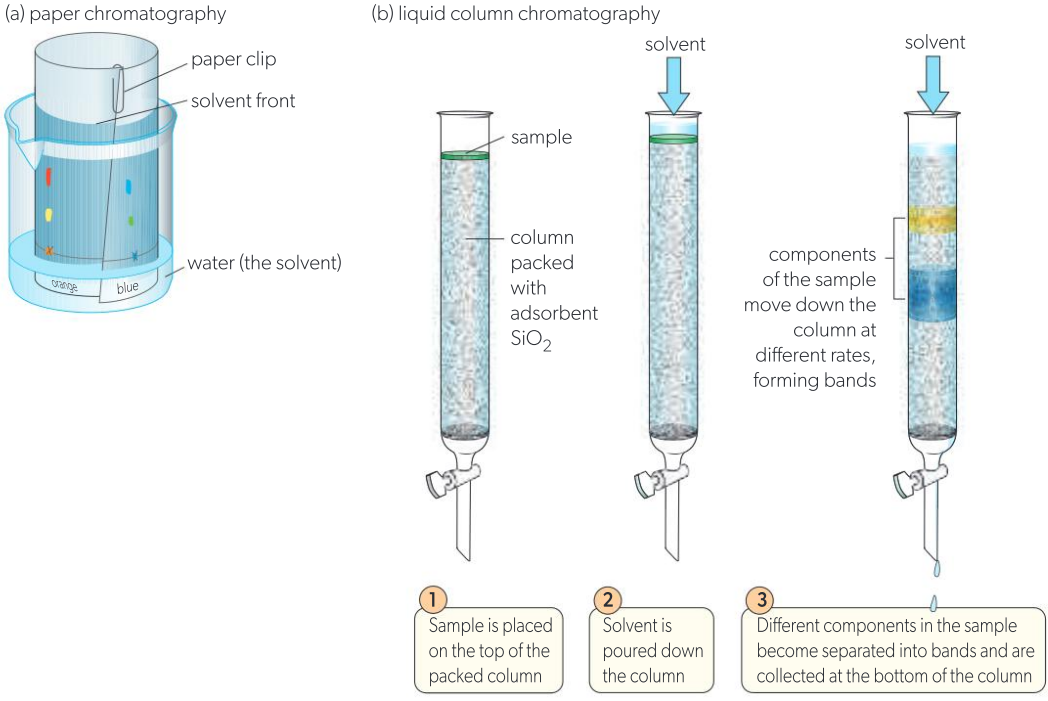
Why are alloys generally considered to be mixtures, even though they often contain metallic bonding? (Structure 2.3 and Structure 2.4)

Chromatography (*Structure 2.2.10*)

The components of a mixture can often be separated and identified using **chromatography**. All forms of chromatography (figure 54) have the same underlying principle: the existence of a **mobile phase** and a **stationary phase**. In most forms of chromatography, the components of the mixture are separated based on their different affinities for each of the two phases. These affinities involve intermolecular forces.

The mobile phase moves through the stationary phase, carrying the components of the mixture. The components all start at the same point but are transported through the stationary phase at different rates due to their differing affinities for each of the two phases. The components are eventually separated.

Methods for separating mixtures are discussed in *Structure 1.1.*



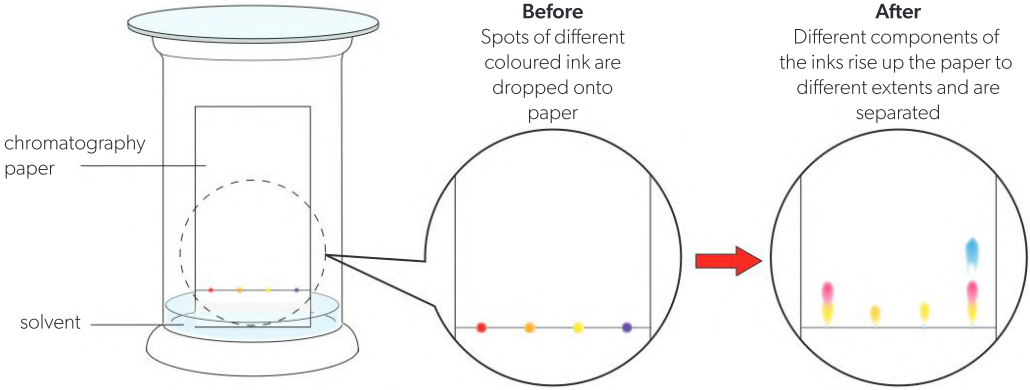
◀ **Figure 54** All types of chromatography involve a mobile phase and a stationary phase

ATL Research skills

Chromatographic methods can be classified into categories. They can be grouped according on their format (for example, **planar** vs **column chromatography**), or the mechanism of separation (for example, partition, adsorption, size exclusion or ion exchange chromatography). In this chapter, we will focus on two types of planar chromatography: paper and thin-layer. The former separates mixtures based on the principle of partition, whereas the latter is based on adsorption. Research at least one other method of chromatography (such as gas chromatography) and compare and contrast the different methods. Organize your notes in a Venn diagram.

Paper chromatography

Separating mixtures of pigments is a simple way to demonstrate **paper chromatography** (figure 55). The stationary phase is a rectangular piece of chromatography paper (made of hydrated cellulose) and the mobile phase is a suitable solvent. The mixture to be separated is dotted onto a start line near the bottom of the paper. The paper is then placed in a chamber such as a beaker, containing a small amount of solvent at the bottom. Placing a lid on top of the beaker allows the atmosphere in the chamber to become saturated with the solvent vapour and prevents solvent loss through evaporation.



▼ Figure 55 Simple paper chromatography experiment using different inks. You can see that the purple ink at the right-hand side is a mixture containing red, yellow and blue dyes

The solvent is allowed to move up the paper until it reaches a point near the top. The components of the mixture move different distances up the paper according to their relative affinities for the stationary and mobile phases. Components with a greater affinities for the solvent will dissolve more readily in it and are therefore transported further up the paper. The results of a chromatography experiment are known as **chromatogram.**

Chromatography paper is composed of hydrated cellulose, which contains many –OH groups. These groups are very polar and attract water molecules tightly, saturating the surface of cellulose with water. When coupled with a less polar organic solvent, this technique can be used to investigate mixtures such as leaf pigments or amino acid mixtures. The mixture components are **partitioned** between the water layer and the solvent: less polar components dissolve in the solvent and get carried further up the paper, whereas more polar components stay predominantly in the water layer. The separation can be optimized by using different solvents, or mixtures of solvents.

Thin layer chromatography (TLC)

Thin layer chromatography (TLC) is more expensive than its paper counterpart, but it offers greater sensitivity. In TLC, the stationary phase is a rectangular plate made of glass or metal coated with silica (silicon dioxide, SiO_2) or alumina (aluminium oxide, Al_2O_3). The silica or alumina surfaces are very polar and contain many hydroxyl groups. The mixture is spotted onto the plate and placed into an eluting chamber containing a non-polar organic solvent. The polar substances in the mixture travel up the plate slowly because their tendency to adsorb onto the silica or alumina surface is greater than their tendency to dissolve in the solvent. Conversely, non-polar components dissolve in the solvent and travel further up the plate.

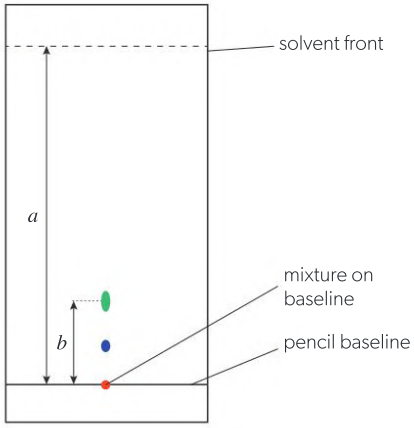
Experiments

Experimental techniques are adapted to suit different purposes. Coloured substances are easily located on a chromatogram. When a substance is colourless, an additional step is needed. For example, when a mixture contains amino acids, the chromatogram can be sprayed with a locating agent. A locating agent is a substance that reacts with the spots to form coloured products. Other colourless compounds can be located in different ways. For example, some might show up under UV light. Can you think of any other examples of how existing scientific methods were adapted to solve a problem?

Retardation factor (R_F)

The results of a chromatography experiment can be quantified by calculating the **retardation factor (R_F)**. The R_F value for a spot on a chromatogram is the ratio of the distance travelled by the spot (b) to the distance travelled by the solvent (a):

$$R_F = \frac{b}{a}$$



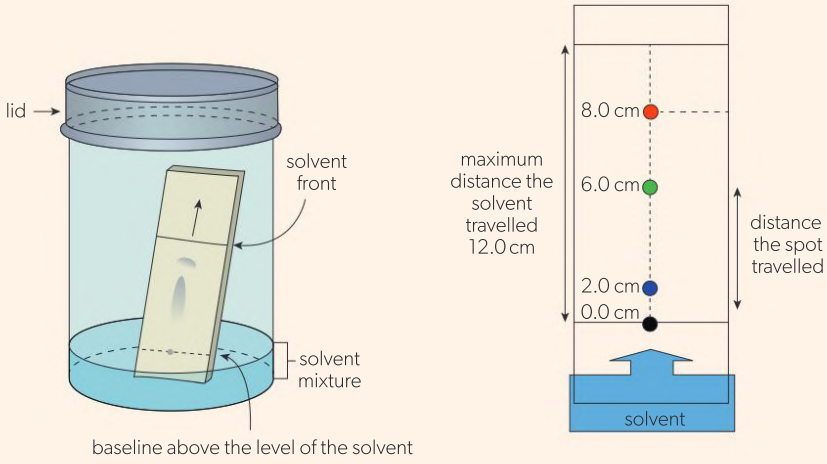
◀ Figure 56 Retardation factors, R_F, are calculated from the distances measured on a chromatogram

Since the spots cannot travel further than the solvent that transports them, R_F values are always in the range of 0 to 1. The distance travelled by a spot can vary depending on factors such as the composition of the solvent, temperature, pH and type of paper used (in paper chromatography). R_F values are reproducible and therefore can be used to identify substances by comparing experimental values to accepted values, provided that the conditions are the same.

Worked example 4

Consider the thin layer chromatogram shown in figure 57. Calculate the retardation factor for the red and blue spots.

Given that the stationary phase is silica and the solvent is a hydrocarbon, determine which component is more polar.



▲ Figure 57 The apparatus for thin layer chromatography (left) and the resulting chromatogram (right)

Solution

Figure 58 shows that the mixture contains three components, which appear as distinct spots at different distances from the baseline: at 8.0 cm (red), 6.0 cm (green) and 2.0 cm (blue). The distance between the baseline and the solvent front is 12.0 cm.

You can calculate the retardation factors (R_F) for the red and blue spots (top and bottom) as follows:

$$R_F(\text{red}) = \frac{\text{distance travelled by the red spot}}{\text{distance travelled by the solvent}}$$

$$= \frac{8.0\,\text{cm}}{12.0\,\text{cm}}$$

$$= 0.67 \text{ (2 significant figures)}$$

$$R_F(\text{blue}) = \frac{\text{distance travelled by the blue spot}}{\text{distance travelled by the solvent}}$$

$$= \frac{2.0\,\text{cm}}{12.0\,\text{cm}}$$

$$= 0.17 \text{ (2 significant figures)}$$

R_F(red) is larger than R_F(blue). This means that the component(s) of the red spot are more soluble in the mobile phase (non-polar hydrocarbon solvent) and the blue spot is composed of substances that have a greater affinity for the stationary phase. The stationary phase (silica) is polar, so the blue spot contains substances that are more polar than the substances in the red spot.

Chromatograms composed of clear dots are frequently shown in books. If you do a chromatography experiment, you will often obtain a chromatogram in which each colour blends into the next. If you want a clearer separation between substances in the mobile phase, you can try using different solvents.



▶ Figure 58 This experimental chromatogram has less clear separation between substances than the previous examples we looked at

Thin-layer chromatography of plant pigments

The pigments found in plants can be separated by chromatography, depending on their relative affinities to the mobile and stationary phases. The separation can be optimized by changing the phases. In this investigation, you will compare the chromatograms obtained when the mobile and stationary phases are changed.

Relevant skills

- Tool 1: Measuring length
- Tool 1: Paper or thin layer chromatography
- Tool 3: Record uncertainties in measurements
- Tool 3: Propagate uncertainties in processed data
- Inquiry 3: Discuss the impact of uncertainties on the conclusions

Safety

- Wear eye protection.
- Flammable solvents. Keep away from flames and other sources of ignition.
- Organic solvents are toxic to the environment.
- Work in a well-ventilated space and dispense solvents in a fume cupboard.

Materials

- Pestle and mortar
- Chromatography spotter
- Chromatography paper
- TLC plate
- Pencil
- Ruler
- Chromatography tank or beaker
- Lid or aluminium foil for the chromatography tank
- Propanone
- Fresh green leaves
- Chromatography solvents (9:1 petroleum ether:propanone mixture)

Method

Part 1: Preparation of the chromatography plate

1. Grind the leaves with a few cm^3 of propanone until you have obtained a concentrated pigment extract.
2. Handle the TLC plate carefully by the edges. Draw a line in pencil, across the plate, 1 to 2 cm from the bottom.
3. Spot the plant pigment extract onto the starting line, taking care to create a small, concentrated spot of pigment. Allow it to dry.

Part 2: Chromatography

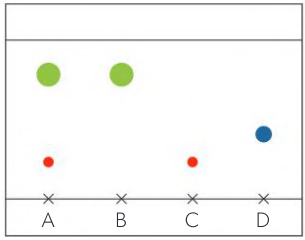
4. Place a few cm^3 of the chromatography solvent into the tank. Place the lid on top, to ensure that the atmosphere in the chamber becomes saturated with the solvent vapours. You can place a paper towel soaked in the solvent vertically inside the tank to speed up the process.
5. Then, lower the TLC plate into the solvent, taking care to keep the solvent below the pigment spot on the starting line. Ensure the plate does not touch the sides of the beaker. If you are using paper, you must make sure that it is upright in the tank by attaching it to the lid or to a splint placed across the top of the tank.
6. Remove the plate from the tank once the solvent is near the top. Quickly draw a line in pencil over the solvent front before it evaporates. Draw circles around the pigment spots that are visible in the chromatogram. Allow the plate to dry.
7. Measure the distance travelled by the solvent. Measure the distance travelled by each pigment from the base line to the centre of the spot.
8. Determine a reasonable uncertainty for the distances you have measured.
9. Calculate the R_F value of each pigment, and the uncertainty of each.
10. Compare your results to those of other members of your class. Compare your results to literature values for the R_F values, if available. Search online for information that will help you identify each of the pigments.
11. Comment on the precision and, if possible, accuracy, of your class's results. Discuss the impact of measurement uncertainty on the validity of your conclusions.

Part 3: Changing the phases

12. Consider how the results would differ if you used a different solvent system, or a different stationary phase. Then, try it out.
13. Clear up as instructed by your teacher, taking care not to dispose of the solvents down the sink.

Practice questions

36. Calculate the retardation factor, R_F, for the green spot at 6.0 cm in figure 58.
37. Suggest why the baseline in paper chromatograms is drawn in pencil.
38. Chromatographic analysis of a mixture of dyes (A) and three individual pigments (B, C and D) produced the chromatogram on the right.
    Identify and explain which pigments are present in A.



AHL

Resonance (*Structure 2.2.11*)

Sometimes, a molecule cannot be described by a single Lewis formula. Instead, there are two or more possible **resonance structures** that collectively represent the molecule. Resonance often happens when there is more than one position for a double or triple bond in a molecule. For example, two Lewis formulas can be drawn for ozone, O_3 (figure 59). Note that a double-headed arrow is used to show alternative resonance structures.



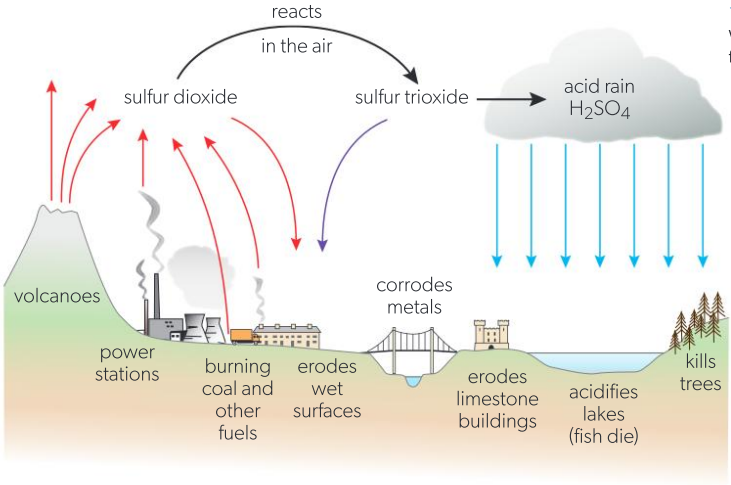
▲ Figure 59 Ozone has two possible resonance structures

These structures might suggest that the two oxygen–oxygen bonds in ozone are different: one single and one double. In reality, the ozone molecule is a hybrid of these two resonance structures. Bond length and strength data (table 13) show that the two bonds in ozone are:

- identical to each other
- intermediate between a single O–O bond and a double O=O bond in terms of bond strength and length
- have a bond order of 1.5, because there are 3 bonding electron pairs distributed across two domains.

Bond	Length / 10^{-12} m	Average bond enthalpy / kJ mol^{-1}	Bond order
O–O single bond	148	144	1
O⋯O bond in ozone	128	362	1.5
O=O double bond	121	498	2

▲ Table 13 Bond data for oxygen–oxygen bonds

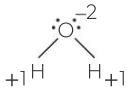


◀ Figure 12 Sulfur dioxide reacts with water and oxygen in the atmosphere to form sulfuric acid, causing acid rain

Oxidation states (*Structure 3.1.6*)

The bonding triangle introduced in *Structure 2.4* shows that bonding is often not purely ionic or covalent, but rather occurs on a continuum with unequal sharing of electrons. The concept of **oxidation state**, or **oxidation number**, is used to represent the charge on an atom in a compound if it were composed of ions. It also describes the number of electrons shared or transferred when forming a bond.

Water, for example, is a predominantly covalent molecule: H_2O. If it were an ionic compound, then oxygen would gain two electrons and each hydrogen atom would lose one, as oxygen is more electronegative than hydrogen. Therefore, oxygen is assigned an oxidation state of –2 and each hydrogen has an oxidation state of +1.



▲ Figure 13 The oxidation states of elements in water

Deducing oxidation states in compounds

In compounds, the more electronegative atoms are assigned negative oxidations states and the less electronegative atoms are assigned positive oxidation states. There are a number of useful rules for deducing the oxidation states of atoms:

Rule 1 The oxidation state of any free element is zero.

Rule 2 The sum of the oxidation states of all atoms in a neutral compound is zero.

Rule 3 The sum of the oxidation states of all atoms in a polyatomic ion is equal to the charge of the ion.

Rule 4 The oxidation state of fluorine is –1 in all compounds.

Rule 5 The oxidation state of the group 1 metals is always +1 and the oxidation state of the group 2 metals is always +2.

Rule 6 The oxidation state of oxygen is –2 except in OF_2, where fluorine is more electronegative, so oxygen has an oxidation state of +2, and in peroxides such as H_2O_2, where it is –1.

Rule 7 The oxidation state of hydrogen is +1 when it is combined with more electronegative elements (most non-metals), and –1 when it is combined with less electronegative elements (metals). Hydrogen in a –1 oxidation state is called a hydride ion.

The first rule states that the oxidation state of an element is always zero. This is because atoms of the same element have equal electronegativity, so the electrons are shared evenly across bonds in elemental compounds. For example, consider elemental oxygen, O_2: the molecule is symmetrical, and the electronegativity of the atoms is identical. Therefore, the oxidation state of each atom is zero.

Worked example 1

Deduce the oxidation state of

a. Mn in $KMnO_4$
b. the atoms in H_2, Cu, Zn, Xe, and Au
c. P in the phosphate ion, PO_4^{3-}
d. all the elements in $LiAlF_4$

Solution

a. According to rule 2, as $KMnO_4$ is a neutral compound, the sum of the oxidation states of its atoms is equal to zero:

   (oxidation state of K) + (oxidation state of Mn) + (oxidation state of O × 4) = 0

   We multiply the oxidation state value for oxygen by four, as there are four oxygen atoms in the compound.

   You can assign oxidation states to O and K, and then find the value for Mn that would make the sum of the oxidation states zero.

   According to rule 5, the oxidation state of potassium is +1. According to rule 6, the oxidation state of oxygen is –2. Substituting these values into the above equation above gives:

   1 + (oxidation state of Mn) + (–2 × 4) = 0

   oxidation state of Mn = +7

b. These are all elemental species, so each atom has an oxidation state of 0 (rule 1).

c. According to rule 3, as PO_4^{3-} is a polyatomic ion, the sum of the oxidation states of its atoms is equal to the charge on the ion:

   (oxidation state of P) + (oxidation state of O × 4) = –3

   According to rule 6, the oxidation state of oxygen is –2. Substituting this value into the above equation above gives:

   (oxidation state of P) + (–2 × 4) = –3

   oxidation state of P = +5

d. Using rule 2: (oxidation state of Li) + (oxidation state of Al) + (oxidation state of F × 4) = 0

   According to rule 5, the oxidation state of lithium is +1. According to rule 4, the oxidation state of fluorine is –1. Substituting these values into the above equation above gives:

   1 + (oxidation state of Al) + (–1 × 4) = 0

   oxidation state of Al = +3

Practice questions

4. Assign oxidation states to each atom in the following compounds:
   a. $K_2Cr_2O_7$
   b. CH_4
   c. CO_2
   d. $HClO_4$
   e. OF_2
   f. SO_3^{2-}
   g. NaH
   h. Fe_2O_3
   j. CH_3OH
   k. CH_2O
   l. Na_2O_2

Naming of oxyanions

Oxyanions are polyatomic anions that include oxygen atoms. When naming oxyanions, you should include the oxidation state of the non-oxygen atom as a Roman numeral at the end of the name. The Roman numerals are enclosed in brackets with no space between the ion name and the bracket.

For example, consider the anion MnO_4^-. The anion is named manganate(VII), as Mn has an oxidation state of +7. The neutral compound $KMnO_4$ is called potassium manganate(VII). This compound is also known under its traditional name, potassium permanganate.

While the Roman numerals are often used to refer to the oxidation state of a transition metal, the non-oxygen atom in an oxyanion can also be a non-metal. $KClO_4$, for example, is called potassium chlorate(VII).

In practice, the Roman numeral is omitted for some common oxyanions. These are shown in table 3.

Ionic formula	Common name	Systematic name
NO_2^-	nitrite	nitrate(III)
NO_3^-	nitrate	nitrate(V)
SO_3^{2-}	sulfite	sulfate(IV)
SO_4^{2-}	sulfate	sulfate(VI)

▲ Table 3 The common names and systematic names of some oxyanions. Both are acceptable in examinations

Practice questions

5. Write the chemical formulas for the following compounds:
   a. hydrogen nitrate
   b. sodium phosphate(V)
   c. magnesium bromate(III)
   d. potassium sulfite

Linking question

How can oxidation states be used to analyse redox reactions? (Reactivity 3.2)

ATL Communication skills

Scientists have used symbols to identify different elements for a long time (figure 14). Nowadays, we use the symbols in the periodic table, and the nomenclature conventions encouraged by the International Union of Pure and Applied Chemistry (IUPAC). Your study of chemistry will involve using symbols as they are stated in the data booklet, writing chemical formulas correctly (including subscripts and brackets where necessary) and naming substances according to IUPAC guidance.

Test your chemical communication skills by finding the error in each of the following and writing the correct answer:

1. HCL
2. NE
3. O3
4. MgOH2
5. cuprum(II) oxide



▲ Figure 14 The symbols used by alchemists for copper and iron, respectively

Reactivity 3.2 Electron transfer reactions

What happens when electrons are transferred?

In a reaction where electrons are transferred, one species will lose electrons—the species is **oxidized**—and a different species will gain electrons—the species is **reduced**. It is impossible to have **oxidation** of one species without **reduction** of another. These reactions are referred to as **redox** reactions.

Some redox reactions are spontaneous and exothermic. If the electron transfer from one species to another is made to occur through a wire, such as in electrochemical cells or batteries, the energy released in the process can be used to power appliances. Other redox reactions are not spontaneous and require energy to occur, such as in electroplating, or the reduction of aluminium oxide, Al_2O_3, to produce aluminium metal.

Understandings

Reactivity 3.2.1—Oxidation and reduction can be described in terms of electron transfer, change in oxidation state, oxygen gain/loss or hydrogen loss/gain.

Reactivity 3.2.2—Half-equations separate the processes of oxidation and reduction, showing the loss or gain of electrons.

Reactivity 3.2.3—The relative ease of oxidation and reduction of an element in a group can be predicted from its position in the periodic table. The reactions between metals and aqueous metal ions demonstrate the relative ease of oxidation of different metals.

Reactivity 3.2.4—Acids react with reactive metals to release hydrogen.

Reactivity 3.2.5—Oxidation occurs at the anode and reduction occurs at the cathode in electrochemical cells.

Reactivity 3.2.6—A primary (voltaic) cell is an electrochemical cell that converts energy from spontaneous redox reactions to electrical energy.

Reactivity 3.2.7—Secondary (rechargeable) cells involve redox reactions that can be reversed using electrical energy.

Reactivity 3.2.8—An electrolytic cell is an electrochemical cell that converts electrical energy to chemical energy by bringing about non-spontaneous reactions.

Reactivity 3.2.9—Functional groups in organic compounds may undergo oxidation.

Reactivity 3.2.10—Functional groups in organic compounds may undergo reduction.

Reactivity 3.2.11—Reduction of unsaturated compounds by the addition of hydrogen lowers the degree of unsaturation.

AHL

Reactivity 3.2.12—The hydrogen half-cell $H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$ is assigned a standard electrode potential of zero by convention. It is used in the measurement of standard electrode potential, $E^\ominus$.

Reactivity 3.2.13—Standard cell potential, $E^\ominus_{cell}$, can be calculated from standard electrode potentials. $E^\ominus_{cell}$ has a positive value for a spontaneous reaction.

Reactivity 3.2.14—The equation $\Delta G^\ominus = -nFE^\ominus_{cell}$ shows the relationship between standard change in Gibbs energy and standard cell potential for a reaction.

Reactivity 3.2.15—During electrolysis of aqueous solutions, competing reactions can occur at the anode and cathode, including the oxidation and reduction of water.

Reactivity 3.2.16—Electroplating involves the electrolytic coating of an object with a metallic thin layer.

Oxidation and reduction (*Reactivity 3.2.1*)

Reduction and oxidation can be defined in several ways:

1. in terms of the loss and gain of oxygen
2. in terms of the gain and loss of hydrogen
3. in terms of electron transfer
4. in terms of oxidation state.

1. Oxidation and reduction in terms of oxygen gain/loss

According to the first definition, oxidation is a reaction where a substance combines with oxygen. Examples of this type of oxidation reaction include the combustion of metals to form metal oxides:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

During aerobic respiration, oxygen reacts with glucose to form carbon dioxide and water. This can also be described as an oxidation reaction:

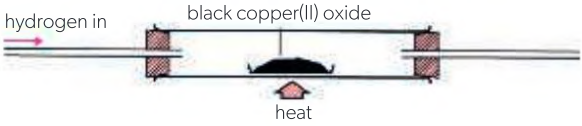
$$C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$$

According to the first definition, reduction is a reaction where oxygen is removed from a substance. Examples of this type of reduction reaction include the reduction of nickel(II) oxide by carbon to produce pure nickel and carbon monoxide:

$$NiO(s) + C(s) \rightarrow Ni(s) + CO(g)$$

In all redox reactions, one species is oxidized, and another is reduced. If the substance being oxidized gains oxygen, then the substance being reduced loses oxygen. In the experiment in figure 2, copper(II) oxide is being reduced, losing oxygen, and hydrogen gas is being oxidized, gaining oxygen:

$$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$$



▲ Figure 2 Experimental set-up for the reduction of copper(II) oxide by hydrogen

In some cases, one substance can be simultaneously reduced and oxidized in the same reaction. This is known as **disproportionation**. For example, potassium chlorate, $KClO_3$, decomposes on heating as follows:

$$4KClO_3(s) \rightarrow 3KClO_4(s) + KCl(s)$$

In this reaction, some formula units of $KClO_3$ are reduced to KCl by losing oxygen while other formula units are oxidized to $KClO_4$ by gaining oxygen.

TOK

In chemistry, the same concept can be defined in several ways by drawing on different knowledge. For example, in *Reactivity 3.1*, you met three separate definitions of acids and bases: the Arrhenius, Brønsted–Lowry, and Lewis theories. Each of these theories are informed by a different aspect of a species' chemistry. What other concepts in chemistry can be defined in different ways?



▲ Figure 1 In the oxidation of iron, iron(III) oxide, or rust, is produced

2. Oxidation and reduction in terms of hydrogen loss/gain

Oxidation can also be considered as a loss of hydrogen. For example, in the reaction between hydrogen chloride and oxygen, hydrogen chloride loses hydrogen to form chlorine gas:

$$4HCl(g) + O_2(g) \rightarrow 2Cl_2(g) + 2H_2O(g)$$

Reduction can be considered as the addition of hydrogen to a species. An example of such a reaction is the hydrogenation of ethene:

$$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$$

This reaction typically requires Ni(s) as a heterogeneous catalyst.

The process of heterogeneous catalysis is detailed in *Structure 3.1*.

3. Oxidation and reduction in terms of electron transfer

During a reaction, if a species loses electrons, it is oxidized, and if a species gains electrons, it is reduced. A useful mnemonic for remembering this is **OIL RIG**:

OIL: **O**xidation **I**s **L**oss of electrons

RIG: **R**eduction **I**s **G**ain of electrons

Consider the reaction between sodium metal and chlorine gas:

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

This reaction cannot be described in terms of the gain or loss of hydrogen and oxygen, as these elements are not involved in the reaction. However, we can instead describe the oxidation and reduction occurring in terms of the transfer of electrons between sodium and chlorine.

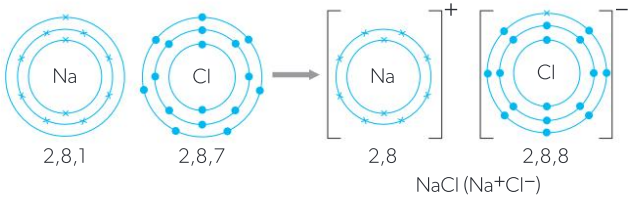
Sodium metal loses electrons to form sodium cations, so it is oxidized.

$$2Na(s) \rightarrow 2Na^+ + 2e^-$$

These electrons are transferred to chlorine gas, reducing it to chloride anions:

$$Cl_2(g) + 2e^- \rightarrow 2Cl^-$$

The sodium cations and chloride anions are held together by ionic bonds in a three-dimensional lattice structure, NaCl(s) (figure 3).



▲ Figure 3 Sodium atoms are oxidized (lose electrons) and chlorine atoms are reduced (gain electrons) in the formation of sodium chloride

The formation of ionic lattice structures is detailed in *Structure 2.1*.

Case study: Redox reactions in optometry

Optometrists often prescribe glasses with photochromic lenses. These lenses darken in the presence of ultraviolet light (from sunlight); this change is based on a redox reaction.

Ordinary glass is composed of silicates while photochromic lenses contain copper(I) chloride, CuCl, and silver chloride, AgCl. The chloride ions are oxidized to chlorine atoms on exposure to ultraviolet light (hf).

$$Cl^- \xrightarrow{hf} Cl + e^-$$

Electron transfer then takes place, reducing the silver cations to silver atoms.

$$Ag^+ + e^- \rightarrow Ag$$

The silver atoms turn the lenses dark, inhibiting the transmission of light. The darkening process is reversed by copper(I) chloride allowing the lenses to become transparent again. When the lenses are no longer exposed to ultraviolet light, the following reaction takes place:

$$Cu^+ + Cl \rightarrow Cu^{2+} + Cl^-$$

The chlorine atoms formed by the exposure to light are reduced by the Cu^+ ions. In turn, Cu^+ ions are oxidized to Cu^{2+} ions. These Cu^{2+} ions then oxidize silver atoms to Ag^+ ions:

$$Cu^{2+} + Ag \rightarrow Cu^+ + Ag^+$$

As a result, the lenses become transparent again and the silver and chlorine atoms return to the initial species, Ag^+ and Cl^-.



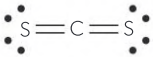
▲ Figure 4 Photochromic lenses

4. Oxidation and reduction in terms of oxidation state change

Some redox reactions do not involve a transfer of electrons between species. For example, in the formation of carbon disulfide, electrons are shared between the carbon and sulfur atoms, forming a covalently bonded molecule:

$$C(s) + 2S(s) \rightarrow CS_2(l)$$

Carbon disulfide has the following Lewis structure:



We can describe oxidation and reduction in terms of the change in oxidation state of the atoms in the reacting species. The **oxidation state** represents the charge that an atom would have in a compound if the compound were composed of ions. In other words, all polar covalent bonds are treated as ionic, so all shared electrons in each bond are formally transferred to the more electronegative atom.

A compound is oxidized if the oxidation state of an atom in that compound increases, and reduced if the oxidation state of an atom in that compound decreases.

You learned the rules for assigning oxidation states to atoms in covalent compounds in *Structure 3.1*.

In the carbon disulfide example, the oxidation state of carbon increases from 0 in the reactant, C(s), to +4 in the product, $CS_2(l)$, where carbon shares four electrons with the more electronegative sulfur. Carbon is therefore oxidized.

Elemental sulfur has an oxidation state of 0. In carbon disulfide, the oxidation state of each sulfur atom decreases to −2. Sulfur is therefore reduced.

TOK

The underlying assumption when assigning oxidation states is that electrons are transferred between atoms, not shared. However, oxidation states are not real because they do not represent the actual charges on the atoms in a molecule.

If you study the HL course, you learned about formal charges in *Structure 2.2*. Formal charges are also artificial and do not represent actual charges. When assigning formal charges, we assume that the electrons in covalent bonds are shared equally between the atoms involved.

In spite of these simplifying assumptions, oxidation states and formal charges are both useful tools that help us to explain redox reactions and Lewis formulas, respectively.

Worked example 1

Sodium chloride, sulfuric acid, and manganese(IV) oxide react according to the following chemical equation.

$4NaCl(s) + 2H_2SO_4(aq) + MnO_2(s) \rightarrow 2Na_2SO_4(aq) + MnCl_2(aq) + 2H_2O(l) + Cl_2(g)$

a. Deduce the change in oxidation states for each atom.

b. State which atom is oxidized, and which atom is reduced.

c. Identify the oxidizing agent and the reducing agent.

Solution

a. First, review the species on each side of the equation and assign oxidation states to each atom:

On the left-hand side	On the right-hand side
In NaCl, Na: +1, Cl: −1 In H_2SO_4, H: +1, S: +6, O: −2 In MnO_2, Mn: +4, O: −2	In Na_2SO_4, Na: +1, S: +6, O: −2 In $MnCl_2$, Mn: +2, Cl: −1 In H_2O, H: +1, O: −2 In Cl_2, Cl: 0

All the oxidation states stay the same, except for manganese, where the oxidation state changes from +4 to +2, and chlorine, where the oxidation state changes from −1 to 0.

b. The oxidation state of manganese decreases, so manganese is reduced. The oxidation state of chlorine increases, so chlorine is oxidized.

c. As manganese has been reduced and caused chlorine to be oxidized, manganese(IV) oxide, $MnO_2(s)$, is the oxidizing agent. Chlorine has been oxidized and caused manganese to be reduced. That makes sodium chloride, NaCl(s), the reducing agent.

Worked example 2

Consider the following balanced equation:

$$Fe(s) + 2HBr(aq) \rightarrow FeBr_2(aq) + H_2(g)$$

a. Deduce the oxidation states of iron and hydrogen in the reactants and products.

b. State which species is oxidized, and which species is reduced.

c. Identify the oxidizing agent and the reducing agent.

Solution

a.

On the left-hand side	On the right-hand side
Fe: 0 H in HBr: +1 Br in HBr: −1	Fe in $FeBr_2$: +2 H in H_2: 0 Br in $FeBr_2$: −1

b. The oxidation state of Fe increases, so Fe(s) is oxidized. The oxidation state of H decreases, so HBr(aq) is reduced.

c. The oxidizing agent is HBr(aq) and the reducing agent is Fe(s)

An **oxidizing agent** causes another species to be oxidized, with the oxidizing agent itself being reduced in the process. A **reducing agent** causes another species to be reduced, with the reducing agent itself being oxidized in the process.

The use of Roman numerals to represent oxidation states of oxyacids and transition element compounds was covered in *Structure 3.1*.

Practice questions

1. Consider the following balanced equation:

   $$Cl_2(aq) + 2KI(aq) \rightarrow 2KCl(aq) + I_2(aq)$$

   a. Deduce the oxidation states of chlorine and iodine in the reactants and products.

   b. State which element is oxidized, and which element is reduced.

   c. Identify the oxidizing agent and the reducing agent.

2. Identify the oxidizing agents and the reducing agents in the following reactions:

   a. $PbO(s) + H_2(g) \rightarrow Pb(s) + H_2O(l)$

   b. $2CuO(s) + C(s) \rightarrow CO_2(g) + 2Cu(s)$

   c. $Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$

   d. $2KI(aq) + Br_2(aq) \rightarrow I_2(aq) + 2KBr(aq)$

When writing oxidation states, the sign is always placed before the number and not after it. In worked example 2, the oxidation state of hydrogen in HBr is +1 and not 1+.

Remember to describe the *compounds* as being oxidized and reduced, not the individual atoms in the compounds. This also applies to describing the oxidizing and reducing agents. In worked example 2, HBr(aq) is reduced and is the oxidizing agent, not H.

Half-equations (*Reactivity 3.2.2*)

Consider the reaction between sodium metal and chlorine gas, discussed in the previous section:

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

You saw that the processes of oxidation and reduction could be separated into two equations:

$$Na(s) \rightarrow Na^+ + e^-$$

$$Cl_2(g) + 2e^- \rightarrow 2Cl^-$$

These equations are called **half-equations**. Separating a redox process into two half-equations helps to show the transfer of electrons. Half-equations can also make it easier to balance the full equation for the redox reaction. The steps for writing redox reactions in aqueous solutions are:

1. Identify the species being oxidized and reduced.
2. Separate the equation into an oxidation half-equation and reduction half-equation.
3. Balance any atoms being oxidized or reduced.
4. For the oxidation half-equation, write the electrons lost on the right-hand side of the equation. The number of electrons should be equal to the magnitude of the change in oxidation state of the oxidized species.
5. For the reduction half-equation, write the electrons gained on the left-hand side of the equation. The number of electrons should be equal to the magnitude of the change in oxidation state of the reduced species.
6. Balance these half-equations so that the number of electrons lost in oxidation equals the number of electrons gained in reduction.
7. Add the two half-equations together and cancel the electrons.
8. If the reaction is occurring in acidic solution, add $H_2O(l)$ to balance any oxygen atoms and $H^+(aq)$ to balance any hydrogen atoms.
9. For neutral or basic solutions, add $OH^-(aq)$ to balance oxygen atoms and $H_2O(l)$ to balance hydrogen atoms.
10. Finally, add up the charges and check that the sum is equal to zero.

Worked example 3

Iron metal, Fe(s), will react with a solution of silver(I) ions, $Ag^+(aq)$, to form aqueous iron(III) ions, $Fe^{3+}(aq)$, and silver metal, Ag(s).

Write the balanced equation for this redox reaction.

Solution

First, write the unbalanced equation for the reaction:

$$Fe(s) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$$

The oxidation state of iron increases from 0 to +3, so Fe(s) loses electrons and is therefore oxidized.

The oxidation state of silver decreases from +1 to 0, so $Ag^+(aq)$ gains electrons and is therefore reduced.

Write the oxidation and reduction half-equations, ensuring that the atoms are balanced:

oxidation: $Fe \rightarrow Fe^{3+}$

reduction: $Ag^+ \rightarrow Ag$

To save time, you can omit states of reacting species in all steps except the final answer.

Add the electrons to make sure that the number of electrons in each half-equation is equal to the magnitude of the change in oxidation state:

$$Fe \rightarrow Fe^{3+} + 3e^-$$

$$Ag^+ + e^- \rightarrow Ag$$

Then, multiply the reduction half-equation by three, so that there are three electrons in each half-equation:

$$Fe \rightarrow Fe^{3+} + 3e^-$$

$$3Ag^+ + 3e^- \rightarrow 3Ag$$

Add the two half-equations together and cancel the electrons:

$$Fe + 3Ag^+ + \cancel{3e^-} \rightarrow Fe^{3+} + 3Ag + \cancel{3e^-}$$

$$Fe(s) + 3Ag^+(aq) \rightarrow Fe^{3+}(aq) + 3Ag(s)$$

Finally, check that the charges are balanced: on the left-hand side, there are three silver(I) ions with a 1+ charge each, and on the right, there is one iron ion with a 3+ charge. So, the total charge on each side of the equation is 3+ and therefore the equation is balanced. Do not forget to add states to all species in the final equation.

Worked example 4

Iron(II) ions, $Fe^{2+}(aq)$, and dichromate(VI) ions, $Cr_2O_7^{2-}(aq)$, react in acidic solution to form iron(III) ions, $Fe^{3+}(aq)$, and chromium(III) ions, $Cr^{3+}(aq)$.

Deduce the balanced redox equation for this reaction.

Solution

First, write the unbalanced equation:

$$Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) \rightarrow Fe^{3+}(aq) + Cr^{3+}(aq)$$

The oxidation state of iron increases from +2 to +3, therefore $Fe^{2+}(aq)$ loses electrons and is oxidized.

In $Cr_2O_7^{2-}(aq)$, chromium has an oxidation state of +6. The oxidation state of chromium decreases from +6 to +3, therefore $Cr_2O_7^{2-}(aq)$ gains electrons and is reduced.

Write the oxidation and reduction half-equations, ensuring that all the atoms that change their oxidation states are balanced:

oxidation: $Fe^{2+} \rightarrow Fe^{3+}$

reduction: $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

Add the electrons such that number of electrons in each half-equation is equal to the magnitude of the change in oxidation state (remembering that there are two Cr^{3+} ions in the second half-equation):

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

$$Cr_2O_7^{2-} + 6e^- \rightarrow 2Cr^{3+}$$

Then, multiply the oxidation half-equation by six, so that there are six electrons in each half-equation:

$$6Fe^{2+} \rightarrow 6Fe^{3+} + 6e^-$$

$$Cr_2O_7^{2-} + 6e^- \rightarrow 2Cr^{3+}$$

Add the two half-equations together and cancel the electrons:

$$6Fe^{2+} + Cr_2O_7^{2-} + \cancel{6e^-} \rightarrow 6Fe^{3+} + 2Cr^{3+} + \cancel{6e^-}$$

$$6Fe^{2+} + Cr_2O_7^{2-} \rightarrow 6Fe^{3+} + 2Cr^{3+}$$

As this reaction is taking place in acidic solution, we need to balance the oxygen and hydrogen atoms. There are seven oxygen atoms on the left-hand side of the equation, so add seven moles of water to the right-hand side of the equation to balance the oxygen atoms:

$$6Fe^{2+} + Cr_2O_7^{2-} \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

There are now 14 hydrogen atoms on the right-hand side of the equation, so add 14 moles of H^+ to the left-hand side of the equation to balance the hydrogen atoms:

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

Finally, check that the charges are balanced. On the left-hand side, there are six Fe^{2+}, $Cr_2O_7^{2-}$ and 14 H^+ ions:

$$(6 \times 2) + (-2) + (14 \times 1) = 24$$

On the right-hand side, there are six Fe^{3+} and two Cr^{3+} ions:

$$(6 \times 3) + (2 \times 3) = 24$$

So, the total charge on each side of the equation is 24+ and therefore the equation is balanced. As usual, do not forget to add states to all reacting species in the final equation:

$$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

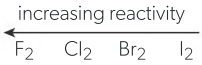
Practice questions

3. Write balanced equations for the following reactions that occur in acidic solutions:
   a. $Zn(s) + SO_4^{2-}(aq) \rightarrow Zn^{2+}(aq) + SO_2(g)$
   b. $MnO_4^-(aq) + Br^-(aq) \rightarrow Mn^{2+}(aq) + BrO_3^-(aq)$
   c. $I_2(s) + OCl^-(aq) \rightarrow IO_3^-(aq) + Cl^-(aq)$
   d. $Cr_2O_7^{2-}(aq) + C_2O_4^{2-}(aq) \rightarrow Cr^{3+}(aq) + CO_2(g)$

Oxidation and reduction of metals and halogens (*Reactivity 3.2.3*)

Relative ease of reduction of halogens

Halogens can act as oxidizing agents in redox reactions. In these reactions, halogens in their elemental state gain electrons, being reduced to singly charged halogen anions. In *Structure 3.1*, you learned that the reactivity of halogens increases going up the group.



This means that fluorine is the strongest oxidizing agent among the halogens, and the most easily reduced, followed by chlorine, and then bromine. For example, chlorine, Cl_2, can oxidize bromide ions, Br^-:

$$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$$

However, chlorine cannot oxidize fluoride ions because fluorine is a stronger oxidizing agent:

$$Cl_2 + 2F^- \not\rightarrow \text{no reaction}$$

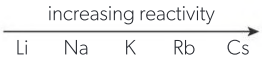
Instead, the reverse reaction between fluorine and chloride ions is possible:

$$F_2 + 2Cl^- \rightarrow 2F^- + Cl_2$$

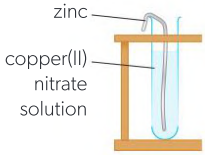
Iodine is the weakest oxidizing agent among the halogens, so it cannot be reduced by the other halogens. However, iodine will oxidize many metals and other strong reducing agents.

Relative ease of oxidation of metals

Group 1 metals can act as reducing agents because they lose their valence electron easily. The reactivity of group 1 metals increases going down the group, and therefore the relative ease of oxidation increases going down the group.



For other metals, you can deduce their relative ease of oxidation by placing a pure metal into a solution of ions of a different metal. If a reaction occurs, then the pure metal is more easily oxidized and it is a stronger reducing agent. If no reaction occurs, then the metal comprising the ionic solution is more easily oxidized.



▲ Figure 5 Zinc will displace copper in solution, changing the colour of the solution from blue to colourless and forming a red copper precipitate

Consider the reaction between zinc metal and copper(II) nitrate solution. In the copper nitrate solution, there are copper(II) ions, $Cu^{2+}(aq)$. In this reaction, zinc metal will dissolve to form a zinc nitrate solution, and copper(II) ions will be reduced to copper metal, which will precipitate out as a solid:

$$Zn(s) + Cu(NO_3)_2(aq) \rightarrow Zn(NO_3)_2(aq) + Cu(s)$$

Therefore, zinc is a stronger reducing agent than copper, and is more easily oxidized. Conversely, this means that copper(II) ions are a stronger oxidizing agent and are more easily reduced than zinc ions. You can track the transfer of electrons in this reaction using half-equations:

$$\text{oxidation: } Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

$$\text{reduction: } Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

You can repeat this experiment with several different metals to obtain a **reactivity series**. In a reactivity series, the most easily oxidized metal is listed first and the least easily oxidized metal is listed last.

Worked example 5

Strips of five different metals, zinc, iron, magnesium, copper and silver, were each added to solutions of their metal nitrate counterparts. The mixtures were observed for a period of time to see whether a reaction has occurred or not. These observations were recorded in table 1.

	$Zn(NO_3)_2(aq)$	$Fe(NO_3)_2(aq)$	$Mg(NO_3)_2(aq)$	$Cu(NO_3)_2(aq)$	$AgNO_3(aq)$
Zn(s)	–	Yes	No	Yes	Yes
Fe(s)	No	–	No	Yes	Yes
Mg(s)	Yes	Yes	–	Yes	Yes
Cu(s)	No	No	No	–	Yes
Ag(s)	No	No	No	No	–

▲ Table 1 Summary of reactions between metals and metal ion solutions

Use table 1 to deduce the reactivity series of the five metals.

Solution

Magnesium reacts with all four solutions and is therefore the most easily oxidized and the most reactive. Silver metal does not react with any solution and is the least easily oxidized. However, $Ag^+(aq)$ ions are the most easily reduced, making $AgNO_3(aq)$ the best oxidizing agent on the list.

Completing the list by inspection gives the following activity series, from the most easily oxidized to the least easily oxidized: Mg(s), Zn(s), Fe(s), Cu(s), Ag(s).

Redox reactions of acids and metals (*Reactivity 3.2.4*)

Reactive metals, such as magnesium, zinc, and iron, are readily oxidized by strong acids, such as hydrochloric acid, HCl(aq), and sulfuric acid, H_2SO_4(aq). In dilute solutions, these reactions produce hydrogen gas and a metal salt:

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

$$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

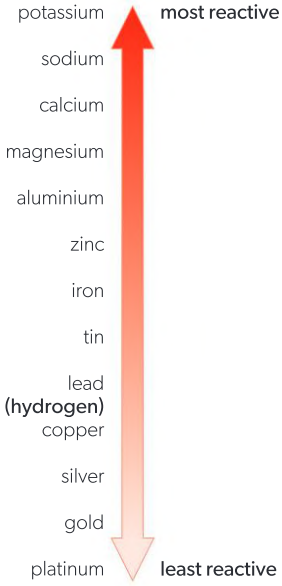
In these reactions, the oxidation state of zinc changes from 0 to +2, and the oxidation state of hydrogen changes from +1 to 0. The electron transfer is shown by the following half-equations:

oxidation: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

reduction: $2H^+(aq) + 2e^- \rightarrow H_2(g)$

Therefore, in the reaction between a metal and an acid, the metal is the reducing agent, and the acid is the oxidizing agent.

Copper and silver are less easily oxidized than magnesium, zinc and iron, so they do not react with dilute solutions of common acids (figure 7).



▲ Figure 7 A reactivity series showing the most easily oxidized metals to the least easily oxidized. Metals above hydrogen on the list can react with common acids, those below can not



▲ Figure 6 The reaction of metals with acids can be detected by the "pop" test: the gas released from the reaction mixture is collected in an inverted test tube, and a lit splint is held close to the test tube opening. A small explosion ("pop") suggests the presence of hydrogen gas that reacts with oxygen in the air

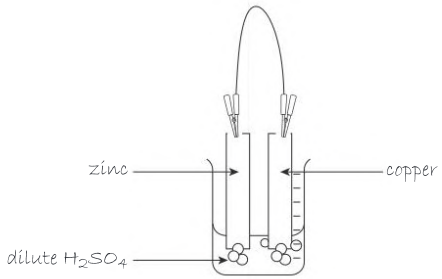


▲ Figure 8 Gold is at the bottom of the reactivity series of metals, so it is not oxidized easily. Therefore, it is the most likely to be found in its reduced form, with zero oxidation state: elemental gold. It is impossible to "pan for lithium" because it is at the top of the activity series

ATL Thinking skills

A student investigating the reactivity of zinc and copper noted the following qualitative observations:

1. *Copper wire was placed in dilute sulfuric acid. No change was observed. A zinc strip was placed in dilute sulfuric acid. Bubbles appeared.*
2. *Some copper wire was wrapped around one end of the zinc strip. This end of the strip and the surrounding copper were placed in dilute sulfuric acid. Bubbles evolved quickly on the surface of the copper.*
3. *Strips of copper and zinc were placed in dilute sulfuric acid and connected to each other. Bubbles evolved on the surface of the copper.*



Explain the student's observations, using your knowledge of metal reactivity, electron transfer reactions, reactions of acids and metallic bonding.

Write two or three linking questions relating the concepts illustrated by this experiment.

Redox titration

Iron supplements are used to treat iron deficiency. You will determine the iron content in iron tablets by titrating them with a solution of potassium manganate(VII) of known concentration. Manganate(VII) ions, $MnO_4^-(aq)$, are powerful oxidizing agents. In their reaction with iron(II) in acidic solution, purple manganate(VII) ions are reduced to pale pink manganese(II) ions, $Mn^{2+}(aq)$. In this process, iron(II) ions are oxidized to iron(III) ions.

Relevant skills

- Tool 1: Titration
- Tool 3: General mathematics
- Tool 3: Record and propagate uncertainties
- Inquiry 2: Assess reliability and validity of results

Safety

- Wear eye protection
- Sulfuric acid, $H_2SO_4(aq)$, is an irritant
- Dilute potassium manganate(VII), $KMnO_4(aq)$, is an irritant and will stain skin and fabrics

Materials

- iron tablets (or other source of iron(II), $Fe^{2+}(aq)$, ions)
- 1.0 mol dm^{-3} sulfuric acid, $H_2SO_4(aq)$
- distilled water
- 0.020 mol dm^{-3} potassium manganate(VII) solution, $KMnO_4(aq)$
- pestle and mortar
- top pan balance
- two 250 cm^3 conical flasks
- 100 cm^3 measuring cylinder
- 250 cm^3 volumetric flask
- funnel
- filter paper
- reagent jar
- burette
- 25 cm^3 volumetric pipette
- white tile

Instructions

Part A Preparation of acidified iron tablet extract

1. Grind four iron tablets into a fine powder using a pestle and mortar.
2. Weigh the iron tablet powder and transfer it into a conical flask.
3. Add 100 cm^3 of 1.0 mol dm^{-3} sulfuric acid, $H_2SO_4(aq)$, to the tablet powder and leave for 24 to 48 hours.
4. In the meantime, review the titration and uncertainty propagation sections in the *Tools for chemistry* chapter before starting part B.

Part B Titration against potassium manganate(VII)

5. Filter the iron tablet extract. Transfer the filtrate into a volumetric flask and make up to the 250 cm^3 mark with distilled water. Store in a labelled reagent jar.
6. Fill the burette with 0.020 mol dm^{-3} potassium manganate(VII), $KMnO_4(aq)$.
7. Transfer 25.0 cm^3 of iron tablet solution to a clean conical flask. Place this flask on a white tile under the burette.
8. Perform a rough titration of the iron tablet solution, stopping when the solution in the flask permanently turns pale pink.
9. Repeat the titration several times until you obtain two concordant values.
10. Clear up according to the directions given by your teacher.

Questions

1. Write the oxidation and reduction half-equations for this reaction.
2. Deduce the redox equation for this reaction.
3. Using your results, determine the mass of iron in one tablet.
4. Propagate the measurement uncertainties to obtain the uncertainty of the mass of iron per tablet.
5. Compare your result to the iron content reported on the tablet packaging. Calculate the percentage error.
6. Comment on the reliability and validity of your result.
7. Describe at least two sources of systematic error and suggest improvements that would minimize these sources of error.
8. Explain why you used four iron tablets in this analysis, not just one.
9. Explain why you left the iron tablet powder in dilute acid solution for 24 to 48 hours.
10. Suggest why redox titrations such as this are said to be "self-indicating".

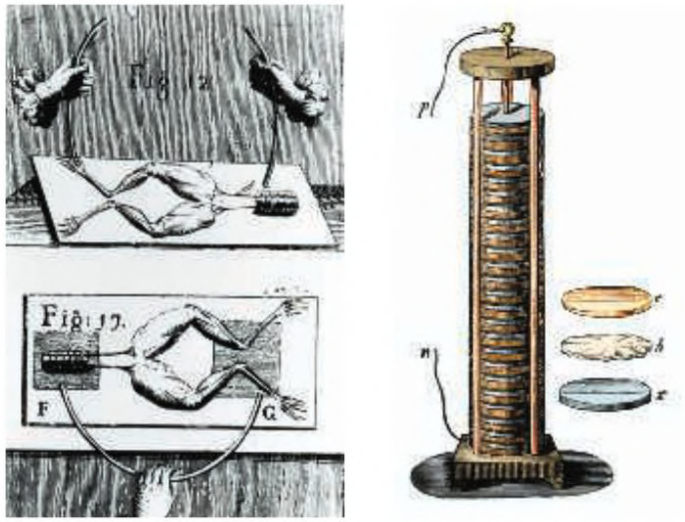
Electrochemical cells (*Reactivity 3.2.5* and *Reactivity 3.2.6*)

An **electrochemical cell** interconverts electrical and chemical energy. There are two types of electrochemical cell:

1. In **primary (voltaic) cells**, **secondary (rechargeable) cells**, and **fuel cells** the energy produced by spontaneous chemical reactions is used to generate electrical energy.
2. In **electrolytic cells**, electrical energy is used to drive forward non-spontaneous chemical reactions.

In redox reactions, electrons move from the substance being oxidized to the substance being reduced. Nearly all spontaneous redox reactions are exothermic, and the energy released in these chemical changes can be used to produce electrical energy. Redox reactions used in primary (voltaic) cells are irreversible while secondary (rechargeable) cells utilize reversible redox reactions.

In the 18th century, the Italian scientist Luigi Galvani discovered accidentally that an electric current could be produced by two dissimilar metals connected by a moist substance. He noticed that he could cause an amputated frog leg to twitch by touching it with two dissimilar metals. Alessandro Volta, another Italian scientist, doubted that animal legs were integral to produce electricity. He showed that chemical reactions can produce electricity and made the first **battery**.



▲ Figure 9 (a) Galvani's frog legs experiment (b) Volta's voltaic pile, the first modern type of electric battery

Any two dissimilar metals and their ions can participate in redox reactions. The one higher in the activity series will oxidize to ions and the one lower in the series will have its ions reduced to the pure metal. For example, zinc can be oxidized by copper(II) ions, to form zinc ions. The Cu^{2+} ions get reduced, and act as an oxidizing agent.

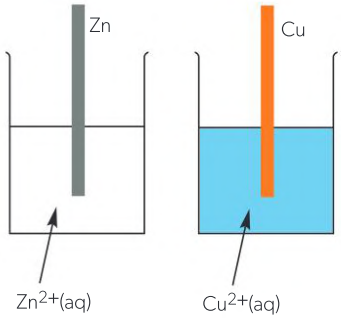
$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

Separating this into two half-equations gives:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

These two processes can occur in separate beakers, called **half-cells**.



▲ Figure 10 The zinc and copper half-cells

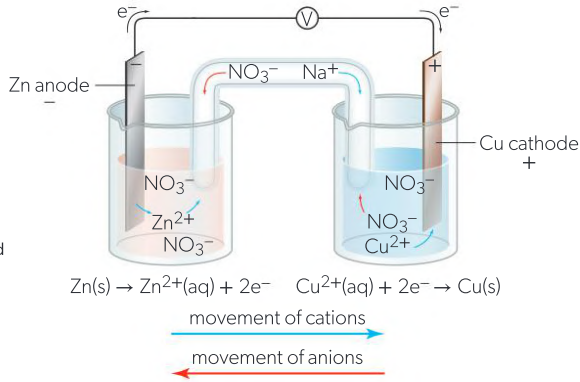
The two half-cells can be connected with a wire to form an electrochemical cell. In electrochemical cells, oxidation always occurs at the **anode**, and reduction at the **cathode**. **RED CAT** is a useful mnemonic: **RED**uction at **CAT**hode. Therefore, in this case, Zn(s) is the anode and Cu(s) the cathode. This electrochemical cell can be represented by the following equations:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- \,||\, Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

When the half-cells are connected, the electrons produced by the oxidation of zinc at the anode flow through the wire to the copper half-cell to reduce copper(II) ions. As a result, the zinc half-cell loses two electrons and now takes on a slight positive charge while the copper half-cell gains two electrons, taking on a slightly negative charge. Any further oxidation of zinc is prevented by the slight negative charge on the copper half-cell, which repels the electrons. The electrons therefore remain in the slightly positively charged zinc half-cell. The cell becomes **polarized**, and the redox reaction stops.

To get around this, a salt bridge is used to connect the solutions in the two half-cells and complete the electrical circuit. A salt bridge consists of an ionic salt solution, such as sodium sulfate, Na_2SO_4, or potassium nitrate, KNO_3. It allows the positive ions (cations) to flow toward the slightly negatively charged half-cell with the cathode and negative ions (anions) to flow toward the slightly positively charged half-cell with the anode.

Consider the addition of a sodium sulfate salt bridge to our zinc–copper cell. The slightly negatively charged copper half-cell attracts $Na^+(aq)$ and $Zn^{2+}(aq)$ cations through the salt bridge and solution. The slightly positively charged zinc half-cell attracts $SO_4^{2-}(aq)$ anions through the salt bridge and solution. This neutralizes the charge in each half-cell, so the redox reaction can continue. This is now a complete primary cell (figure 11). This kind of primary cell is known as the Daniell cell, named after its inventor, the British chemist John Frederic Daniell.



▶ Figure 11 A primary cell consisting of a zinc metal anode (labelled as negative because it is the source of electrons) dipped in $ZnSO_4(aq)$, a copper metal cathode (labelled positive as it attracts electrons) dipped in $CuSO_4(aq)$, an electrical connecting wire, a voltmeter and a salt bridge

As the reaction continues, the blue colour of the copper(II) sulfate solution fades, the copper bar increases in size as it becomes coated in more copper, and the zinc bar gets thinner. Once there is a significant build-up of $Zn^{2+}(aq)$ ions on the cathode side, the cell ceases to function.

Cell diagrams are used as a short-hand way to represent primary cells. In this convention, the cathode is always written on the right-hand side and the anode is always written on the left-hand side. The salt bridge is represented by two parallel vertical lines. You can use the following general template to write cell diagrams for metal–ion primary cells:

anode being oxidized | product of oxidation || species being reduced | product of reduction/cathode

Therefore, the cell diagram for the Daniell cell would be written as follows:

$$Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$$

The species on the left-hand side, $Zn(s) \mid Zn^{2+}(aq)$, represent the anode, with zinc metal being oxidized to $Zn^{2+}(aq)$. The species on the right-hand side, $Cu^{2+}(aq) \mid Cu(s)$, represent the cathode, with $Cu^{2+}(aq)$ being reduced to $Cu(s)$.

Worked example 6

Manganese metal reacts with nickel(II) ions to form manganese(II) ions and nickel metal.

a. Write the redox reaction that occurs between nickel(II) ions and manganese metal.

b. Assuming that this redox reaction occurs in a manganese–nickel primary cell, write the half-equations that occur in each half-cell.

c. Write the cell diagram to represent the primary cell for this redox reaction.

d. Sketch a primary cell for this reaction and identify the anode, cathode, direction of electron flow and direction of ion flow.

Solution

a. $Mn(s) + Ni^{2+}(aq) \rightarrow Mn^{2+}(aq) + Ni(s)$

b. At the cathode: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$
At the anode: $Mn(s) \rightarrow Mn^{2+}(aq) + 2e^-$

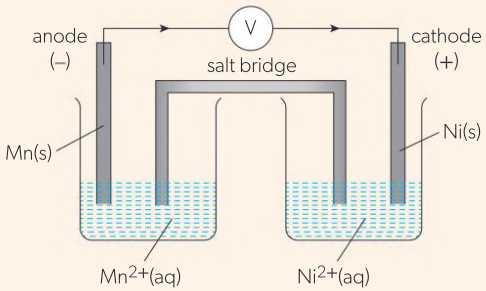
c. You can use the general template for cell diagrams:

anode being oxidized | product of oxidation || species being reduced | product of reduction/cathode

This gives:

$Mn(s) \mid Mn^{2+}(aq) \parallel Ni^{2+}(aq) \mid Ni(s)$

d.



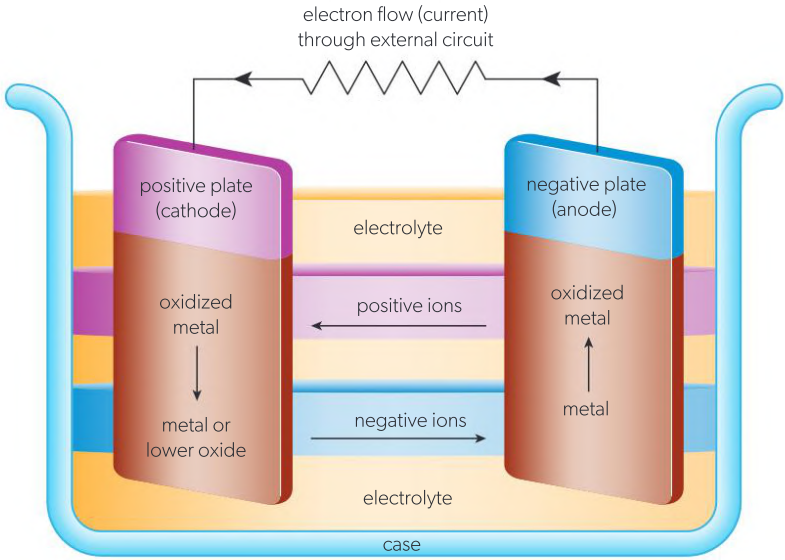
Electrons flow from left to right (from the anode to the cathode) through the electrical wire. $Mn^{2+}(aq)$ and the cations of the salt in the salt bridge also flow from left to right through the salt bridge toward the cathode. The anions of the salt flow from right to left through the salt bridge toward the anode.

Secondary cells (*Reactivity 3.2.7*)

A **battery** is a series of two or more electrochemical cells, typically enclosed in a single container. In a primary electrochemical cell, ultimately the reaction materials will be consumed, and the reaction is not reversible. Either the anode, the electrolyte, or both, need to be replaced, or the battery will be thrown away. Typically, the anode (negative electrode) is oxidized and can no longer be used.

As previously discussed, the ions travelling through the solution and salt bridge can polarize the cell, which causes the chemical reaction to stop. Polarization can also cause a build-up of hydrogen bubbles on the surface of the anode. These can increase the internal resistance of the cell and reduce its electrical output. Primary cells do not operate well under high-current demands, such as in flash photography or electric cars, but are suitable for low-current household devices.

In a **secondary cell**, or a **rechargeable cell**, the chemical reactions that generate electricity can be reversed by applying an electric current to the cell. Secondary cells can satisfy higher current demands than primary cells but have a higher rate of self-discharge. For example, batteries in cell phones are made of secondary cells that can be recharged using electrical energy. When you purchase a replacement battery for a phone, you need to charge it before use because it will have self-discharged during the storage and transportation.



▲ Figure 12 Structure of an electrochemical cell. The negative anode is oxidized and the flow of ions causes polarization. This process cannot be reversed in a primary cell, but it can be reversed in a secondary (rechargeable) battery

Case study: Lead–acid battery

Car batteries are made of secondary cells. The electrical energy from the battery is used to power the motor that starts the engine, and to power any electrical systems in the car. This is known as **discharge**. Some of the chemical energy from the combustion in the engine is used to reverse the chemical reactions that occur during discharge, therefore recharging the battery. An idling engine runs slowly and does not provide enough energy to recharge the battery, and so the battery will gradually discharge.

A typical car uses a lead–acid battery. This battery is composed of a lead anode, Pb(s), a lead(IV) oxide cathode, $PbO_2(s)$, and sulfuric acid, $H_2SO_4(aq)$. Sulfuric acid is a strong acid, so it will ionize into $H^+(aq)$ and $HSO_4^-(aq)$. When the battery is powering the motor and the car's electrical systems, $HSO_4^-(aq)$ will oxidize Pb(s) at the anode, and $H^+(aq)$ will reduce $PbO_2(s)$ at the cathode. This gives rise to the following discharge reactions:

anode (oxidation): $Pb(s) + HSO_4^-(aq) \rightarrow PbSO_4(s) + H^+(aq) + 2e^-$

cathode (reduction): $PbO_2(s) + 3H^+(aq) + HSO_4^-(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

overall cell reaction: $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$

Note that the $H^+(aq)$ and $HSO_4^-(aq)$ ions are ultimately produced by the dissociation of sulfuric acid, so in the overall cell reaction they are combined together and shown as $H_2SO_4(aq)$.

Worked example 7

Determine the reactions that occur at the anode and cathode in a car battery cell, and the overall cell reaction, when the engine is charging the battery.

Solution

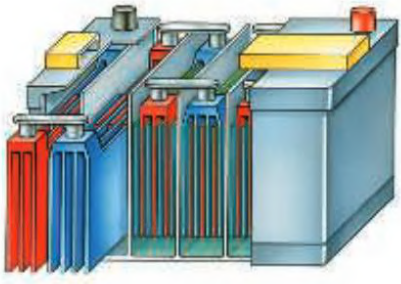
The charging reactions are the reverse of the discharge reactions:

anode: $PbSO_4(s) + H^+(aq) + 2e^- \rightarrow Pb(s) + HSO_4^-(aq)$

cathode: $PbSO_4(s) + 2H_2O(l) \rightarrow PbO_2(s) + 3H^+(aq) + HSO_4^-(aq) + 2e^-$

overall cell reaction: $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$

The continual charging of a lead–acid battery tends to produce hydrogen and oxygen from water. Therefore, non-sealed car batteries occasionally need to be topped up with distilled water.



◀ Figure 13 A lead–acid battery consists of a series of secondary cells with lead(IV) oxide plates, lead plates, and sulfuric acid electrolyte

Case study: Lithium-ion batteries

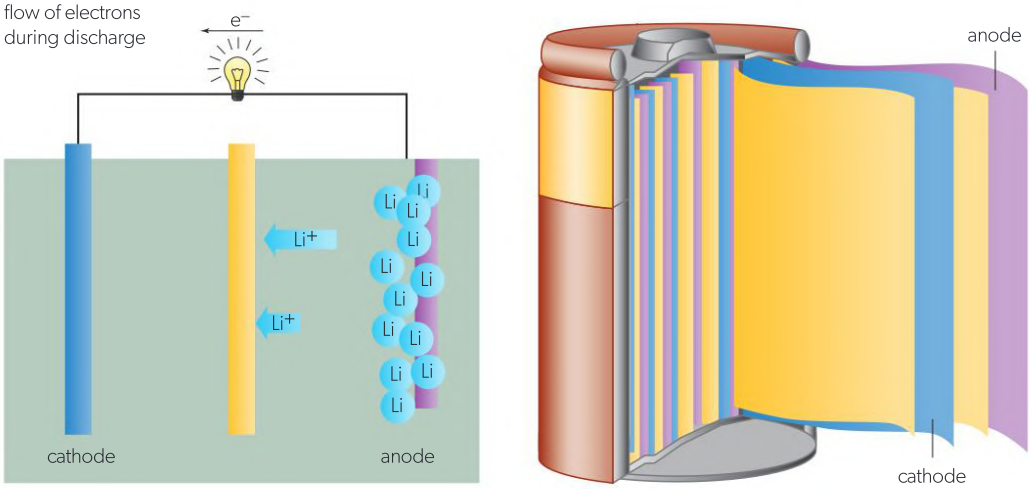
Lithium-ion rechargeable batteries use lithium atoms embedded in a lattice of graphite electrodes, rather than pure lithium metal for the anode. The cathode is a lithium–cobalt oxide complex, $LiCoO_2$. The lithium atoms are oxidized to lithium ions during discharge.

During charging, the opposite process occurs: the lithium ions in the complex migrate through the battery medium to the anode, where they accept electrons and are reduced to lithium atoms. These atoms become embedded in the graphite lattice. The battery medium must be completely non-aqueous, usually a polymer gel, as lithium reacts vigorously with water. Table 2 summarizes these reactions:

Electrode	Discharging reaction	Charging reaction
anode	$Li(s) \rightarrow Li^+ + e^-$	$Li^+ + e^- \rightarrow Li(s)$
cathode	$Li^+ + e^- + CoO_2(s) \rightarrow LiCoO_2(s)$	$LiCoO_2(s) \rightarrow Li^+ + e^- + CoO_2(s)$

▲ Table 2 Anode and cathode reactions in the lithium-ion battery

The overall cell reaction during discharge is $Li(s) + CoO_2(s) \rightarrow LiCoO_2(s)$.



▲ Figure 14 Structure of a typical lithium-ion rechargeable battery. The battery consists of a series of secondary cells composed of cathodes and anodes with a layer of polymer (yellow) separating them

When the lithium-ion battery is in use, electrons flow from the anode to the cathode through the external circuit while lithium ions flow from the anode to the cathode through the polymer gel inside the cell. When no more lithium ions are left on the anode, the battery is flat. To recharge it, the process is reversed, transferring lithium ions back to the anode.

Practice questions

4. During the discharge of a nickel–cadmium battery, the following reactions occur in the cells:

   anode: $Cd(s) + 2OH^-(aq) \rightarrow Cd(OH)_2(s) + 2e^-$
   cathode: $NiO(OH)(s) + H_2O(l) + e^- \rightarrow Ni(OH)_2(s) + OH^-(aq)$

   a. Write the overall equation for the cell.

   b. Write the cell diagram.

   c. Determine the charging reactions that occur in the cell.

Fuel cells

Fuel cells are a type of electrochemical cell that convert hydrogen, methanol, or ethanol and oxygen into water, carbon dioxide and heat. They cause little pollution and are very efficient. Like primary cells, they are not rechargeable, but unlike primary cells, they require a steady supply of fuel and oxygen, so the reactions in the cell can continue indefinitely.

For example, in a **hydrogen fuel cell**, hydrogen gas is supplied to the anode while oxygen gas is supplied to the cathode. The following reactions occur in a hydrogen fuel cell:

anode: $H_2(g) \rightarrow 2H^+(aq) + 2e^-$

cathode: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

overall cell equation: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

In a **direct methanol fuel cell** (**DMFC**), methanol is supplied to the anode. The following reactions occur in a DMFC:

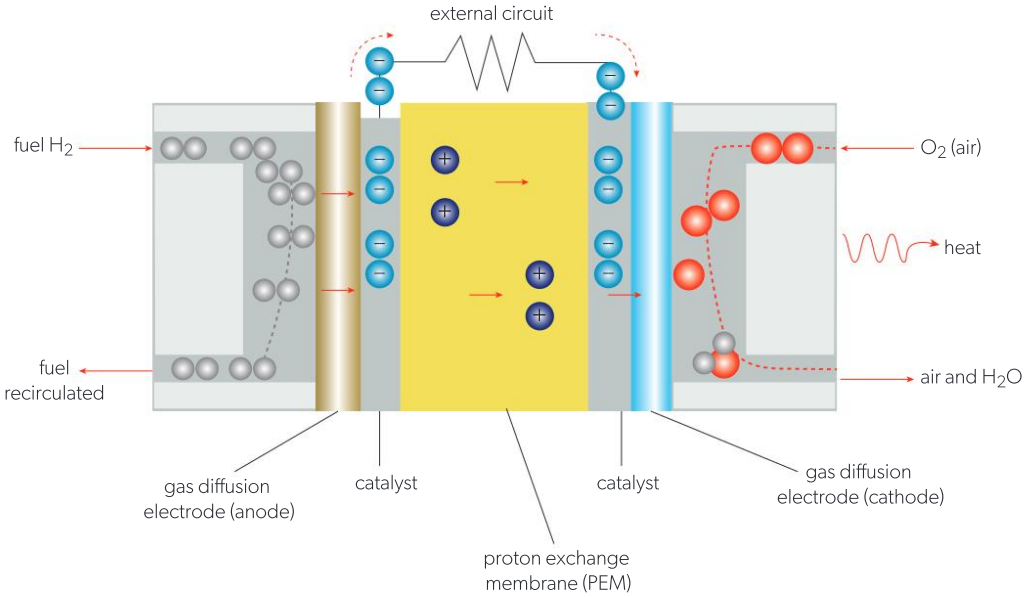
anode: $CH_3OH(l) + H_2O(l) \rightarrow CO_2(g) + 6H^+(aq) + 6e^-$

cathode: $\frac{3}{2}O_2(g) + 6H^+(aq) + 6e^- \rightarrow 3H_2O(l)$

overall cell equation: $CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

A typical fuel cell has the following key components:

- **Electrolyte** or **separator**: this keeps components from mixing. For example, a **proton exchange membrane** (**PEM**) is a polymer that allows H^+ ions to diffuse through but prevents the diffusion of other ions, electrons or molecules.
- **Electrodes**: The electrodes are made of a catalyst that allows for the chemical reactions to occur. There is an oxidizing electrode (anode) and reducing electrode (cathode).
- **Bipolar plate**: This conducts the electrical current from cell to cell and ensures uniform distribution of the fuel gas.



▲ Figure 15 In a hydrogen fuel cell, $H_2(g)$ is oxidized at the anode and $O_2(g)$ is reduced at the cathode. The proton exchange membrane (PEM) allows H^+ ions to diffuse but prevents the diffusion of other ions, molecules and electrons

Hydrogen fuel cells use hydrogen gas as the fuel. These cells do not produce greenhouse gases. The heat formed in the exothermic reaction $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ can be utilized, increasing the efficiency of the cell. The oxygen gas can be obtained from the air.

The catalysts used for the electrodes are often made of platinum or other expensive metals, which makes fuel cells expensive to run on a commercial scale. The hydrogen gas used must be very pure to prevent the poisoning of the catalyst and therefore the reduction of the cell electrical output. There are two main sources of hydrogen gas:

1. Clean hydrogen can be produced by the electrolysis of water. Solar cells or wind generators provide the cleanest form of energy for powering the electrolysis.
2. Hydrogen gas can also be obtained by the reaction of hydrocarbons, especially methane, with steam: $CH_4(g) + H_2O(g) \longrightarrow 3H_2(g) + CO(g)$

DMFCs have the advantage of not needing to extract hydrogen gas because they use methanol as the fuel. However, they produce carbon dioxide, $CO_2(g)$, which is a greenhouse gas.

Practice question

5. Outline the function of the proton exchange membrane (PEM) in fuel cells.

The effect of greenhouse gases on atmospheric temperatures is discussed in *Structure 3.2*.

AHL

Standard electrode potentials (*Reactivity 3.2.12*)

The ease of oxidation and reduction of a species in an electrochemical cell can be described numerically using a **standard electrode potential**, or **standard reduction potential**, $E^{\ominus}$. Standard electrode potentials are defined relative to a hydrogen-based half-cell where the following reaction occurs:

$$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g) \qquad E^{\ominus} = 0\,V$$

This half-cell is known as the **standard hydrogen electrode** (**SHE**) and is assigned a standard electrode potential of zero. An inert platinum electrode is used in the SHE because the reduced species is hydrogen gas and not a metal.

Species with a more negative standard electrode potential will have greater ease of oxidation, and greater tendency to reduce other species. They will be higher in the reactivity series. Species with a more positive standard electrode potential will have greater ease of reduction and greater tendency to oxidize other species. They will be lower in the reactivity series. Of all the elements, lithium has the greatest ease of oxidation and fluorine has the greatest ease of reduction.

The reactivity series and the corresponding standard electrode potentials are given in section 19 of the data booklet. SATP conditions are provided in section 4.

Standard electrode potentials are measured in volts (V). They are correct for standard temperature and pressure (SATP) conditions (temperature = 298 K, pressure = 100 kPa). All aqueous species present in the half-cell equation must also have a concentration of 1.0 mol dm^{-3}.

Practice question

8. The standard electrode potentials of four metals are given in table 3.

Metal	$E^{\ominus}$ / V
tin, Sn(s)	−0.14
calcium, Ca(s)	−2.87
lithium, Li(s)	−3.04
aluminium, Al(s)	−1.66

▲ Table 3 $E^{\ominus}$ values for selected metals

Order these metals according to their ease of oxidation, with the most readily oxidized first.

Standard cell potentials (*Reactivity 3.2.12*)

The voltage of an electrochemical cell depends on the identity of the electrodes in each half-cell. The further apart the species are on the reactivity series, the greater the voltage. This voltage, known as the **standard cell potential**, $E^{\ominus}_{cell}$, can be calculated by finding the difference between the standard electrode potentials of the half-cells:

$$E^{\ominus}_{cell} = E^{\ominus}(\text{reduced species}) - E^{\ominus}(\text{oxidized species})$$

For a reaction in an electrochemical cell to be spontaneous, $E^{\ominus}_{cell}$ must have a positive value. In that case, the reduction will occur at the electrode with the more positive value of $E^{\ominus}$ (the cathode) and the oxidation will occur at the electrode with the more negative value of $E^{\ominus}$ (the anode). In other words:

$$E^{\ominus}_{cell} = E^{\ominus}(\text{cathode}) - E^{\ominus}(\text{anode})$$

Consider an electrochemical cell composed of $Fe^{2+}|Fe$ and $Cu^{2+}|Cu$ half-cells. The data booklet states the half-cell reduction reactions and their standard electrode potentials as follows:

$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$ $\quad E^{\ominus} = -0.45\,V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^{\ominus} = +0.34\,V$

The copper electrode has the more positive value of $E^{\ominus}$, so $Cu^{2+}(aq)$ will be reduced to Cu(s), and the half-equation at that electrode will proceed in the forward direction.

The iron electrode has the more negative value of $E^{\ominus}$, so Fe(s) will be oxidized to $Fe^{2+}(aq)$, and the half-equation at that electrode will proceed in the reverse direction to that stated above:

$$Fe(s) \rightleftharpoons Fe^{2+}(aq) + 2e^-$$

Now you can determine the overall standard potential of the cell:

$$E^{\ominus}_{cell} = E^{\ominus}(\text{reduced species}) - E^{\ominus}(\text{oxidized species})$$
$$= (+0.34\,V) - (-0.45\,V)$$
$$= 0.79\,V$$

To find the overall equation for the electrochemical cell, add the two half-equations together, cancelling the electrons and ensuring that the equation is balanced:

$$Fe(s) + Cu^{2+}(aq) \rightleftharpoons Fe^{2+}(aq) + Cu(s)$$

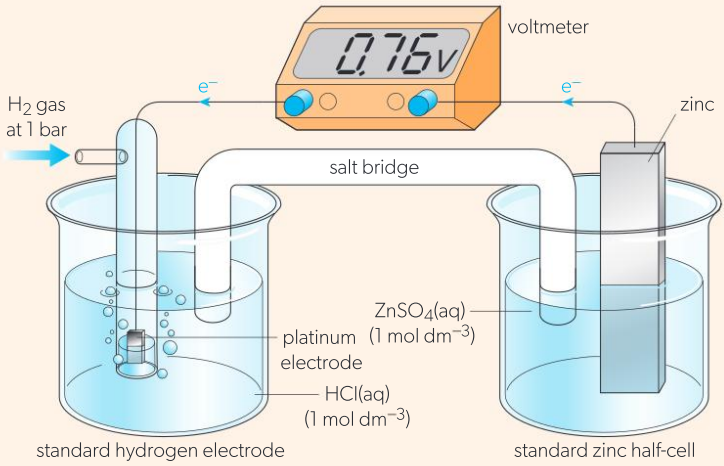
This reaction is described as being spontaneous in the forward direction. You can therefore use $E^{\ominus}$ data to predict which direction will be spontaneous for a reversible redox reaction in an electrochemical cell.

AHL

Worked example 9

An electrochemical cell composed of a $Zn^{2+}|Zn$ half-cell and a standard hydrogen electrode (SHE) half-cell is shown in figure 22. The redox reaction in the cell can be written as follows:

$$2H^+(aq) + Zn(s) \rightleftharpoons Zn^{2+}(aq) + H_2(g)$$



▲ Figure 22 The electrochemical cell with a SHE half-cell and a $Zn|Zn^{2+}$ half-cell. The voltmeter is set up to measure the overall cell potential

a. State which half-cell contains the anode and which contains the cathode.

b. Predict whether the forward or backwards reaction is spontaneous.

c. Show that the standard cell potential, $E^{\ominus}_{cell}$, is +0.76 V using section 19 of the data booklet.

Solution

a. First, write the half-cell reduction reactions and standard electrode potentials from the data booklet:

$$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s) \qquad E^{\ominus} = -0.76\,V$$

$$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g) \qquad E^{\ominus} = 0.00\,V$$

The SHE half-cell has the more positive value of $E^{\ominus}$, so it will involve a reduction reaction and therefore contains the cathode.

The zinc half-cell has the more negative value of $E^{\ominus}$, so it will involve an oxidation reaction and therefore contains the anode.

b. Reduction occurs at the SHE half-cell, so its half-equation will proceed in the forward direction as given in the data booklet.

Oxidation occurs at the zinc half-cell, so its half-equation will proceed in the reverse direction:

$$Zn(s) \rightleftharpoons Zn^{2+}(aq) + 2e^-$$

Before you add the half-equations together, you first need to double the stoichiometric coefficients in the hydrogen half-equation to ensure that there is an equal number of electrons in each half-equation:

$$2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$$

Now add the half-equations together, cancelling the electrons:

$$2H^+(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + H_2(g).$$

Therefore, the reaction will be spontaneous in the forward direction for the equation given in the question.

c. $E^{\ominus}_{cell} = E^{\ominus}(\text{reduced species}) - E^{\ominus}(\text{oxidized species})$:

$$(0.00\,V) - (-0.76\,V)$$

$$= 0.76\,V$$

Worked example 10

An electrochemical cell comprises a $Cu^{2+}|Cu$ half-cell and an $Ag^{+}|Ag$ half-cell.

a. State which half-cell contains the anode, and which contains the cathode.

b. Write the spontaneous reaction that occurs in the electrochemical cell.

c. Calculate the standard cell potential, $E^{\ominus}_{cell}$.

Solution

a. Copy the half-equations and standard reduction potentials for each half-cell from the data booklet:

$$Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s) \qquad E^{\ominus} = +0.80\,V$$

$$Cu^{2+}(aq) + 2e^{-} \rightleftharpoons Cu(s) \qquad E^{\ominus} = +0.34\,V$$

The silver half-cell has the more positive value, so it contains the cathode, and the copper half-cell contains the anode.

b. The silver half-cell has the more positive $E^{\ominus}$ value, so reduction occurs in this half-cell and the half-equation is already shown in the correct direction. Oxidation occurs in the copper half-cell, so the half-equation needs to be reversed:

$$Cu(s) \rightleftharpoons Cu^{2+}(aq) + 2e^{-}$$

Before you add the half-equations together, you first need to double the stoichiometric coefficients in the silver half-equation to ensure that there is an equal number of electrons in each half-equation:

$$2Ag^{+}(aq) + 2e^{-} \rightleftharpoons 2Ag(s)$$

Now add the half-equations together, cancelling the electrons:

$$2Ag^{+}(aq) + Cu(s) \rightarrow 2Ag(s) + Cu^{2+}(aq)$$

c. $E^{\ominus}_{cell} = E^{\ominus}(\text{reduced species}) - E^{\ominus}(\text{oxidized species})$:

$$= (+0.80\,V) - (+0.34\,V)$$

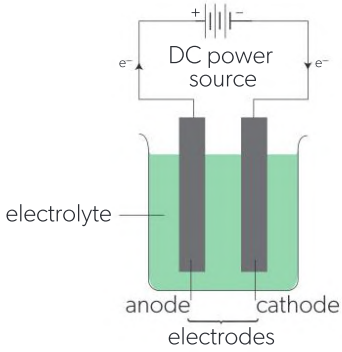
$$= 0.46\,V$$

It is important to note that the half-cell voltage for silver electrode is not multiplied by two. The electrons lost and gained need to be balanced, but the potential of the cell only depends on the chemical nature of the species involved in the half-equation, and not on the way the half-equation is balanced.

Electrolytic cells (*Reactivity 3.2.8*)

An electrolytic cell is an electrochemical cell that converts electrical energy to chemical energy. The oxidation and reduction reactions in an electrolytic cell are non-spontaneous, so they require an external source of electricity to bring about the chemical changes. This process is known as **electrolysis**.

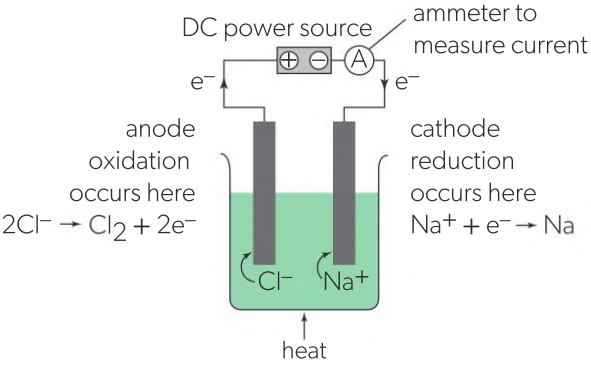
An electrolytic cell consists of a single container filled with an **electrolyte**. The electrolyte can be a solution of an ionic salt, or a molten ionic salt, composed of free-moving cations and anions. Two electrodes (the cathode and the anode) are dipped in the electrolyte, and a direct current (DC) power source is connected to the electrodes.



▲ Figure 16 The structure of an electrolytic cell

In a closed circuit, electrons flow from the negative terminal to the positive terminal of the DC power source. The negative terminal of the DC power source is connected to the cathode. Therefore, electrons flow to the cathode and reduce the cations in the electrolyte. The anions in the electrolyte flow to the anode and undergo oxidation, releasing electrons. The electrons flow to the positive terminal of the DC power source to complete the circuit. The flow of electrons and ions in an electrolytic cell comprises electric **current**.

Consider the electrolysis of molten sodium chloride, NaCl(l), shown in figure 17.



◀ Figure 17 The electrolysis of molten sodium chloride

In the electrolytic cell, molten sodium chloride is the electrolyte, which contains sodium cations, Na^+, and chloride anions, Cl^-. Electrons flow to the cathode, reducing the sodium cations in the electrolyte to form molten sodium metal. The reduction half-equation is as follows:

$$Na^+ + e^- \rightarrow Na(l)$$

At the anode, chloride anions are oxidized, producing chlorine gas and electrons to complete the circuit. The oxidation half-equation is as follows:

$$2Cl^- \rightarrow Cl_2(g) + 2e^-$$

The overall equation for reactions in the electrolytic cell is:

$$2NaCl(l) \rightarrow 2Na(l) + Cl_2(g)$$

This electrolytic cell is therefore useful for the production of sodium metal and chlorine gas.

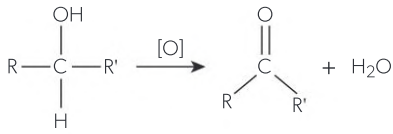
Reactive metals, such as lithium, magnesium, aluminium, and sodium are all obtained by electrolysis of their molten salts. These metals react with oxygen, so the electrolysis must take place in an inert atmosphere.

Practice question

6. Write the half-equations, and full equation, for the electrolysis of molten lead bromide, $PbBr_2$.

Oxidation of organic compounds (*Reactivity 3.2.9*)

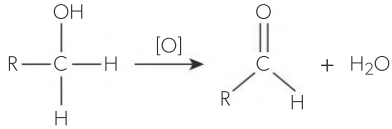
Some **functional groups** in organic compounds can undergo oxidation under certain conditions. For example, in the presence of an oxidizing agent, the **hydroxyl group** in **secondary alcohols** can be oxidized to a **carbonyl group**, forming a **ketone**:



secondary alcohol ketone

The symbol [O] is used to indicate the oxidizing agent, which provides a source of oxygen atoms. In this oxidation reaction, two hydrogen atoms are lost from the compound, forming water with an oxygen atom from the oxidizing agent. An additional bond forms between carbon and oxygen.

Primary alcohols can be oxidized to **carboxylic acids** in a two-step reaction. In the first step, the hydroxyl group is oxidized to a carbonyl group, forming an **aldehyde**:



primary alcohol aldehyde

Like in the oxidation of a secondary alcohol, two hydrogen atoms are lost and an additional bond forms between carbon and oxygen.

The definitions of primary, secondary and tertiary alcohols are given in *Structure 3.2*. Naming compounds with hydroxyl, carboxyl and carbonyl functional groups is also covered in *Structure 3.2*.

Electrolysis of aqueous solutions (*Reactivity 3.2.15*)

We have discussed the electrolysis of molten ionic salts where cations are reduced and anions are oxidized. The electrolysis of aqueous solutions of ionic salts introduces oxidation and reduction reactions involving water, which compete with the redox reactions of the anions and cations of the salt at the anode and cathode.

The reduction of water can proceed as follows:

$$H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq) \qquad E^\ominus = -0.83\,V$$

If the standard electrode potential of the salt cation is more negative than −0.83 V, then water will be preferentially reduced over the salt cation and hydrogen gas will be formed at the cathode.

The other possible competing reaction is the oxidation of water. In the data booklet, the reduction of oxygen is given as follows:

$$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l) \qquad E^\ominus = +1.23\,V$$

Reversing the equation gives the half-equation for the oxidation of water. The reduction potential has to be reversed as well, giving an oxidation potential:

$$H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \qquad E^\ominus = -1.23\,V$$

If the oxidation potential of the salt anion is more negative than −1.23 V, then water will be preferentially oxidized over the salt anion and oxygen gas will be formed at the anode.

For example, consider two electrolytic cells: one composed of molten sodium chloride, NaCl(l), and one composed of an aqueous sodium chloride, NaCl(aq). In the electrolysis of NaCl(l), electrical energy is provided to the cell, resulting in the reduction of sodium ions to form sodium metal at the cathode and the oxidation of chloride ions to form chlorine gas at the anode:

cathode: $Na^+ + e^- \rightarrow Na(l)$

anode: $Cl^- \rightarrow \frac{1}{2}Cl_2(g) + e^-$

overall cell equation: $2NaCl(l) \rightarrow 2Na(l) + Cl_2(g)$

In the electrolysis of aqueous sodium hydroxide, NaCl(aq), the two competing oxidation and reduction reactions involving water are introduced. There are now two species that can potentially be reduced at the cathode: sodium ions from the salt, and water. The two competing reduction reactions are as follows:

$$Na^+(aq) + e^- \rightarrow Na(s) \qquad E^\ominus = -2.71\,V$$

$$H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq) \qquad E^\ominus = -0.83\,V$$

The reduction potential of water is less negative than that of sodium, so water is more easily reduced than the sodium ions. Therefore, the only product formed at the cathode will be hydrogen gas, $H_2(g)$. The sodium ions, $Na^+(aq)$, and hydroxide ions, $OH^-(aq)$, will stay in the solution.

The chloride ions from the salt and water also compete to be oxidized at the anode. The half-equations and standard electrode potentials involving the oxidation of chloride ions and water are shown below.

$$\frac{1}{2}Cl_2(g) + e^- \rightarrow Cl^-(aq) \qquad E^\ominus = +1.36\,V$$

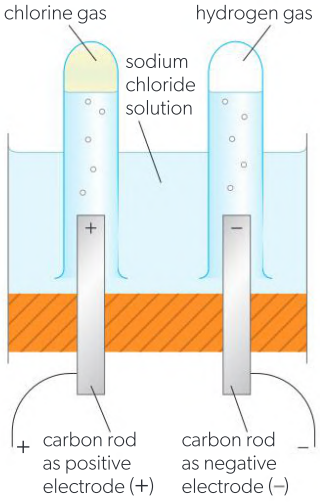
$$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l) \qquad E^\ominus = +1.23\,V$$

The equations need to be reversed to reflect the oxidation of these species:

$$Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^- \qquad E^\ominus = -1.36\,V$$

$$H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \qquad E^\ominus = -1.23\,V$$

The chloride ions have a more negative oxidation potential than water, so we can expect water to be oxidized preferentially compared to chloride ions. However, the difference between the two oxidation potentials is small, so in solutions with high concentration of $Cl^-(aq)$ ions the main product formed at the anode will be chlorine gas, $Cl_2(g)$ (figure 23).



▲ Figure 23 The electrolysis of aqueous sodium chloride

The overall cell equation is as follows:

$$Na^+(aq) + Cl^-(aq) + H_2O(l) \rightarrow \frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) + Na^+(aq) + OH^-(aq)$$

or

$$NaCl(aq) + H_2O(l) \rightarrow \frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) + NaOH(aq)$$

Doubling the stoichiometric coefficients gives the final equation:

$$2NaCl(aq) + 2H_2O(l) \rightarrow H_2(g) + Cl_2(g) + 2NaOH(aq)$$

In dilute solutions of NaCl(aq), the concentration of $Cl^-(aq)$ ions is low, so water is oxidized along with chloride ions. In that case, a mixture of oxygen gas and chlorine gas is produced at the anode. In very dilute solutions, oxygen gas will be the only product at the anode.

Worked example 12

Deduce the products of the electrolysis of aqueous copper(II) sulfate, $CuSO_4(aq)$, with inert electrodes. Write the overall cell equation.

Solution

First, write the half-equations and standard electrode potentials involving the reduction of copper(II) ions and water.

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \qquad E^\ominus = +0.34\,V$$

$$H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq) \qquad E^\ominus = -0.83\,V$$

The reduction potential of copper(II) ions is more positive than that of water, so copper(II) ions will be reduced to copper metal at the cathode.

The two competing species for oxidation at the anode are sulfate ions, $SO_4^{2-}(aq)$, and water. In this case, the sulfate ions cannot be oxidized because sulfur is in its highest oxidation state of +6. Therefore, removing any more electrons from sulfate would result in an impossible electron configuration for sulfur.

AHL

Practice question

10. Deduce the products of the electrolysis of aqueous potassium dichromate, $K_2Cr_2O_7(aq)$, with inert electrodes.

This means that water will be oxidized at the anode and oxygen gas will be released, with hydrogen ions staying in solution:

$$H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^-$$

The overall cell equation is therefore:

$$Cu^{2+}(aq) + SO_4^{2-}(aq) + H_2O(l) \rightarrow Cu(s) + \frac{1}{2}O_2(g) + 2H^+(aq) + SO_4^{2-}(aq)$$

or

$$CuSO_4(aq) + H_2O(l) \rightarrow Cu(s) + \frac{1}{2}O_2(g) + H_2SO_4(aq)$$

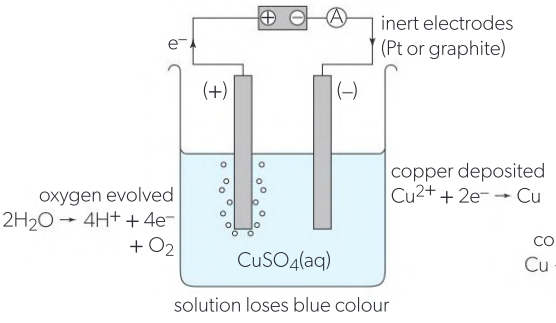
Doubling the stoichiometric coefficients gives the final equation:

$$2CuSO_4(aq) + 2H_2O(l) \rightarrow 2Cu(s) + O_2(g) + 2H_2SO_4(aq)$$

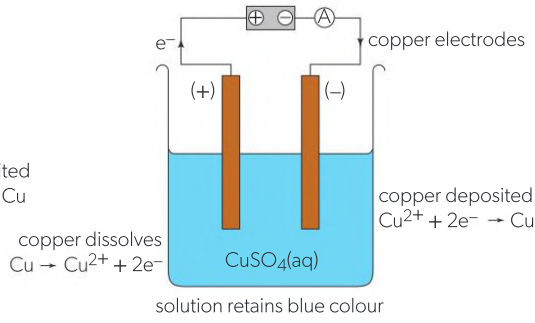
Electroplating (*Reactivity 3.2.16*)

The electrolysis of aqueous copper(II) sulfate described in worked example 12 uses inert carbon electrodes (figure 24, left). However, if the electrodes are made from a reactive metal, the electrolysis of an aqueous solution of an ionic salt will add material to the cathode and take material away from the anode. These processes are known as **plating** and **eroding**, respectively.

For example, consider an electrolytic cell containing a solution of copper sulfate, $CuSO_4(aq)$, with the anode and the cathode each made of copper metal (figure 24, right). When the electric current is applied, the copper anode will erode to form $Cu^{2+}(aq)$ ions while the same ions will be reduced at the cathode to form Cu(s). This process can be used to purify a sample of impure copper metal. The impure copper is used as the anode, which will be eroded to produce $Cu^{2+}(aq)$ ions. These ions will then be reduced at the cathode and plated there as pure copper metal. The impurities will either stay on the anode (if they are less readily oxidized than copper metal) or remain in the solution (if their ions are less readily reduced than copper(II) ions).



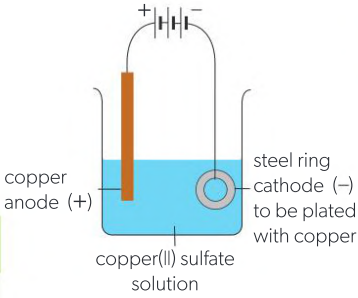
▲ Figure 24 Electrolysis of copper(II) sulfate with inert electrodes reduces copper(II) ions at the cathode and oxidizes water, producing oxygen gas at the anode. The blue colour of the solution fades as copper(II) ions are replaced with hydrogen ions.



Electrolysis using copper electrodes causes the copper anode to dissolve and copper metal to deposit on the cathode. The amount of copper(II) ions in the solution remains constant, so the blue colour of the solution does not fade

Electroplating involves coating an object with a thin layer of pure metal by electrolysis. A metal anode is oxidized to form cations in the solution. The cations travel through the solution to the cathode, where they are reduced to form a thin layer of metal on the cathode.

For example, in an electrochemical cell comprising a copper anode, a steel cathode, and copper(II) sulfate solution, the steel cathode will be plated with a thin layer of copper (figure 25).



▲ Figure 25 In copper electroplating, copper(II) ions are reduced at the cathode to form a thin layer of Cu(s) on the surface of the cathode (in this case, a steel ring)

Observations

Scientists make inferences from their observations. Observation involves use of the senses. Our knowledge of the behaviour of matter then allows us to infer conclusions from observed data. For example, you can observe gas bubbles being generated at an electrode during electrolysis, and you can infer the identity of the gas from your knowledge of the composition of the electrolyte. Similarly, you may observe a brownish-red solid being deposited at the cathode during the electrolysis of aqueous copper(II) sulfate and, from that, infer that copper is reduced at that electrode. What "counts" as an observation in science?

Worked example 13

Deduce the half-equations at each electrode in the electroplating of a steel electrode with copper in a copper(II) sulfate solution, $CuSO_4(aq)$.

Solution

At the anode, copper metal is oxidized to copper(II) ions:

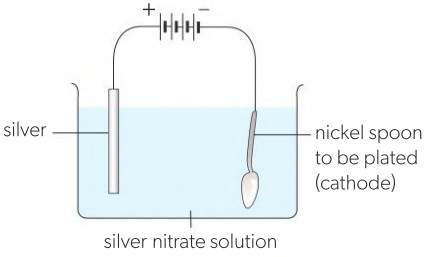
$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

At the cathode, the reverse reaction occurs:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

Practice question

11. A nickel spoon is used in a silver electroplating experiment as shown below.



a. Write the half-equation for the reaction occurring at the:
   i. cathode ii. anode.

b. Describe and explain what would happen to the concentration of the silver(I) nitrate electrolyte.

c. Describe and explain the mass changes at each of the two electrodes.



▲ Figure 26 The trophy for the top prize at the Berlin International Film Festival, the Golden Bear, is made of bronze coated with a thin layer of gold by electroplating

Reactivity 1.3 Energy from fuels

What are the challenges of using chemical energy to address our energy needs?

When we burn fossil fuels, chemical energy is converted into thermal energy. Thermal energy can be used to generate electrical energy to power devices in our homes and in industry. The use of fossil fuels is being increasingly questioned as advances in science and technology reveal the damage that their use is having on our planet. In recent years, the cost of alternative renewable energy sources has decreased and renewable energy technology has advanced.

This, coupled with scientific evidence of the rate of change in global warming, has increased the urgency for governments to commit to net-zero carbon emissions by 2050.

The challenges of using chemical energy from non-renewable, carbon-based fuels, such as crude oil, natural gas and coal, which provide 80% of our energy, are clearly understood by the scientific community. Many industrialized nations depend on the use of fossil fuels to sustain their fast-developing economies. This is a major problem that has to be overcome if the use of renewable, clean sources to meet our global energy needs is to exceed the use of fossil fuels.

Understandings

Reactivity 1.3.1—Reactive metals, non-metals and organic compounds undergo combustion reactions when heated in oxygen.

Reactivity 1.3.2—Incomplete combustion of organic compounds, especially hydrocarbons, leads to the production of carbon monoxide and carbon.

Reactivity 1.3.3—Fossil fuels include coal, crude oil and natural gas, which have different advantages and disadvantages.

Reactivity 1.3.4—Biofuels are produced from the biological fixation of carbon over a short period of time through photosynthesis.

Reactivity 1.3.5—A fuel cell can be used to convert chemical energy from a fuel directly to electrical energy.

Combustion (*Reactivity 1.3.1* and *Reactivity 1.3.2*)

In combustion reactions, substances are burned in oxygen. Metals, non-metals and hydrocarbons all react with oxygen when combusted, producing different products.

Combustion of metals

Oxidation and reduction are defined in terms of electron transfer in *Reactivity 3.2*.

The combustion of reactive metals in the presence of oxygen results in the **oxidation** of the metal, the **reduction** of oxygen, and the formation of an ionic compound. This type of reaction is therefore known as a **redox** reaction. Oxidation can be defined as a gain of oxygen or a loss of electrons. Reduction is defined as a loss of oxygen or a gain of electrons.

The general equation for the reaction of a metal with oxygen is as follows:

metal + oxygen → metal oxide

Lithium burns in oxygen to release heat and produce lithium oxide:

$$4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$$

Magnesium readily reacts with oxygen, producing a brilliant white light and magnesium oxide:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

In both of these reactions, the reactive metals are being oxidized to form metal ions. The half-equations for each of these metals reveal the loss of electrons:

$$Li \rightarrow Li^+ + e^-$$

$$Mg \rightarrow Mg^{2+} + 2e^-$$

The oxygen atoms are reduced to form O^{2-} ions in each reaction:

$$O_2 + 4e^- \rightarrow 2O^{2-}$$

Most metal oxides are ionic compounds. Rules for naming ionic compounds are outlined in *Structure 2.1*.

Combustion of non-metals

Non-metals are also oxidized when combusted in oxygen, forming non-metal oxides:

$$\text{non-metal} + \text{oxygen} \rightarrow \text{non-metal oxide}$$

Sulfur, a non-metal, can be found as impurities in fossil fuels, such as coal and crude oil. Coal may contain up to 3% of sulfur. The combustion of sulfur-containing compounds in oxygen predominantly produces sulfur dioxide, SO_2:

$$S(s) + O_2(g) \rightarrow SO_2(g)$$

Sulfur dioxide can then further react with oxygen in the atmosphere to produce sulfur trioxide:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Sulfur trioxide reacts with water in the atmosphere, to form sulfuric acid:

$$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$$

In industry, sulfur dioxide is produced in vast quantities as feedstock for the synthesis of sulfuric acid. The majority of the sulfuric acid is then used in the production of fertilizers, along with paper, paints, textiles and a wide variety of other products.

Activity

Research the applications of sulfur dioxide, and evaluate the environmental impact of its industrial production. Research other non-metal oxides and their applications and environmental impacts.

Complete combustion of organic compounds

Hydrocarbons are organic compounds composed only of carbon and hydrogen atoms. Alkanes are the simplest hydrocarbons, which are present in fossil fuels, such as crude oil and natural gas. They are relatively inert. This is because they have a low bond polarity, strong covalent carbon–carbon bonds (bond enthalpy = 346 kJ mol^{-1}) and strong carbon–hydrogen bonds (bond enthalpy = 414 kJ mol^{-1}). However, alkanes do participate in some reactions, including combustion.



▲ Figure 1 Liquefied petroleum gas contains liquid propane, and is used as a fuel for gas barbecues

The chemical equation for the combustion of alkanes shows the general formula of the homologous series of alkanes: C_nH_{2n+2}. Homologous series of organic compounds were introduced in *Structure 3.2*.

Alkanes are commonly used as fuels, releasing large amounts of energy in combustion reactions. For combustion to occur, a fuel must be volatile. Volatility is the tendency of a substance to change state from liquid to gas. As the length of the carbon chain increases in the alkane series, the boiling point also increases, and therefore volatility decreases. As a result, short-chain alkanes tend to be used as fuels. Liquefied petroleum gas (LPG) consists predominantly of compressed propane, C_3H_8, while petrol (gasoline) is a mixture of hydrocarbons from butane, C_4H_{10}, to dodecane, $C_{12}H_{26}$.

Alkanes undergo **complete combustion** in the presence of excess oxygen. This reaction is exothermic, and it produces carbon dioxide and water. The general equation for the complete combustion of alkanes is shown below:

$$\text{alkane} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water}$$

$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \rightarrow nCO_2 + (n+1)H_2O$$

In this, and following combustion reactions, the states of all species depend on the reaction conditions. For example, at high temperatures, water can be produced as steam, $H_2O(g)$, while at lower temperatures it will condense into liquid, $H_2O(l)$.

Petrol, also known as gasoline, is a mixture of hydrocarbons obtained from oil, with octane present in the highest proportion. The reaction for the combustion of octane is shown below:

$$C_8H_{18}(l) + \frac{25}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(l) \qquad \Delta H_c^{\ominus} = -5470\,\text{kJ mol}^{-1}$$

Falsification

Up until the late 18th century, combustion was explained in terms of the phlogiston theory. A substance known as "phlogiston" was thought to be present in all combustible materials. Burning was believed to release phlogiston into the air. The combustion of carbon would lead to a mass loss, which was interpreted as the loss of phlogiston. The fact that some substances, such as metals, gain mass when combusted was a puzzling observation—further hypotheses were developed to justify it. Examination of the mass losses and gains when different elements burnt in air, along with the discovery of oxygen, led to the falsification of the phlogiston theory. It was superseded by the theory of combustion that you have learnt about and that explains combustion in terms of oxidation.

The phlogiston theory may seem unreasonable or even peculiar to someone who knows what we know now. If we consider the knowledge base at the time, however, phlogiston was not an implausible concept. Scientific claims, hypotheses or theories can be falsified. This can lead to the development of new theories, which must then be corroborated by experimental evidence in order to eventually become accepted (although theories can never be proven with certainty). Scientists must remain open-minded with respect to new evidence.

Think of a claim, hypothesis or theory you have encountered in your study of science. What type of evidence could challenge it? How much evidence would be needed to falsify it?

Alcohols are another class of organic compounds. They have a wide range of applications, such as fuels, solvents and antiseptics. Like alkanes, alcohols undergo complete combustion reactions releasing carbon dioxide and water.

The general equation for the complete combustion of alcohols is shown below:

$$\text{alcohol} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water}$$

$$C_nH_{2n+1}OH + \frac{3n}{2}O_2 \rightarrow nCO_2 + (n+1)H_2O$$

Ethanol is used in medicine for its antiseptic properties. It is also used as a fuel for vehicle engines. It is known as a **biofuel** because it can be produced by plants, a renewable resource, as opposed to fossil fuels, the supply of which is finite. The reaction for the complete combustion of ethanol is shown below:

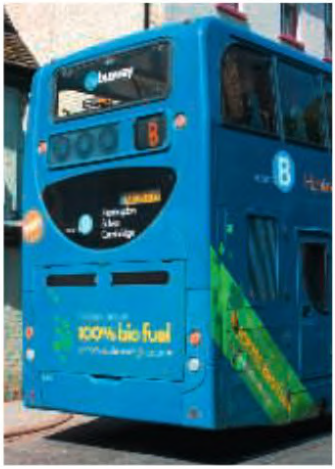
$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H_c^{\ominus} = -1367\,\text{kJ mol}^{-1}$$

This reaction is strongly exothermic. The use of ethanol as a renewable biofuel is increasing. The scientific community is working to resolve the problems associated with the production of biofuels, such as the high cost of production, the impact of using farmland on food supply, and the lower amount of energy produced per unit mass or volume of the fuel.

Global impact of science

Carbon dioxide has a significant environmental impact, as it contributes to global warming. This is the major reason why governments around the world promote clean energy, and legislate for limiting the production of new cars that use fossil fuels. Ethanol is seen as a more environmentally friendly fuel, as it comes from a renewable source.

How can the scientific community, industry and governments work together to reduce the use of fossil fuels? Who has the greatest responsibility to reduce our fossil fuel use?



▲ Figure 2 Ethanol-fuelled public transport is becoming more common as governments look for ways to reduce reliance on fossil fuels

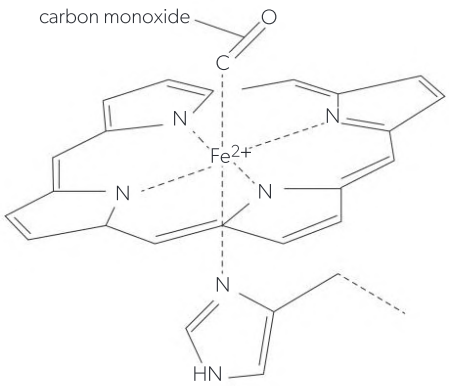
Practice questions

1. Deduce balanced equations for the **complete** combustion of:
   a. methane
   b. propane
   c. pentane
   d. hexane
2. Deduce balanced equations for the **complete** combustion of:
   a. propanol
   b. pentanol
   c. heptanol

Incomplete combustion of organic compounds

If there is a limited supply of a species participating in a chemical reaction, this species is known as the **limiting reactant**. Combustion reactions usually occur in a system where atmospheric air and oxygen are in excess, so the limiting reactant is the organic compound (fuel). However, when this is not the case, such as in a car engine, oxygen becomes the limiting reactant, which causes **incomplete combustion** of the fuel.

In the incomplete combustion of hydrocarbons, carbon monoxide, CO(g), and/or elemental carbon, C(s), can be produced. Carbon monoxide is a poisonous gas that irreversibly binds to hemoglobin in the red blood cells, reducing the oxygen-carrying capacity of the blood (figure 3). Elemental carbon produced in incomplete combustion reactions is often referred to as soot.



▶ Figure 3 The carbon monoxide molecule irreversibly binds to the large hemoglobin molecule

The general equation for the incomplete combustion of an alkane to form carbon monoxide is as follows:

alkane + oxygen → carbon monoxide + water

$$C_nH_{2n+2} + \frac{2n+1}{2}O_2 \rightarrow nCO + (n+1)H_2O$$

The general equation for the incomplete combustion of an alkane to form carbon (soot) is as follows:

alkane + oxygen → carbon + water

$$C_nH_{2n+2} + \frac{n+1}{2}O_2 \rightarrow nC + (n+1)H_2O$$

The incomplete combustion reactions of methane are shown below:

$$CH_4(g) + \frac{3}{2}O_2(g) \rightarrow CO(g) + 2H_2O(l)$$

$$CH_4(g) + O_2(g) \rightarrow C(s) + 2H_2O(l)$$

These incomplete combustion reactions can occur simultaneously with complete combustion reactions in different ratios. Incomplete combustion reactions are less exothermic than the corresponding complete combustion reactions.

Incomplete combustion can be observed in the laboratory, with the appearance of black soot on the bottom of glassware that has been heated over a Bunsen or spirit burner flame.

Practice questions

3. Deduce balanced equations for the **incomplete** combustion of the following alcohols, where carbon monoxide is one of the products:
   a. propanol
   b. pentanol
   c. heptanol

Fossil fuels (*Reactivity 1.3.3*)

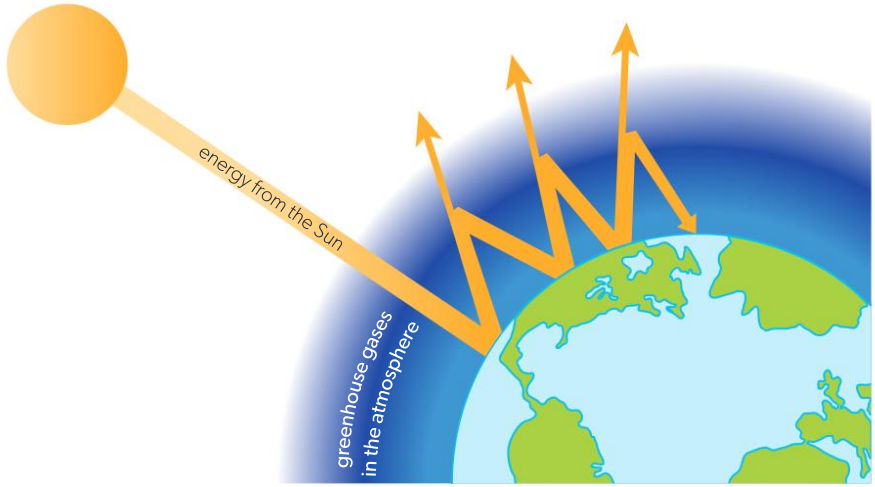
Fossil fuels include crude oil, coal and natural gas. Crude oil, or petroleum, is a non-renewable resource, and its use as a fuel is deeply embedded in the global society. Crude oil is a natural mixture of hydrocarbons, organic compounds containing nitrogen, sulfur and oxygen, and a wide variety of other elements.

Coal, petroleum and natural gas are the main fuels used to generate electricity in power stations and the internal combustion engines of cars. As the global demand for energy increases, so does the consumption of fossil fuels. Globally, governments are making plans and legislating to limit the consumption of fossil fuels and promote the use of renewable energy. However, the transition to clean energy will take many decades.

One consequence of the use of fossil fuels is the release of large quantities of carbon dioxide, a product of the combustion reaction, into the atmosphere. Carbon dioxide is a **greenhouse gas**, which means that it traps heat energy inside the Earth's atmosphere. This is known as the **greenhouse effect** (figure 4).

Linking questions

Why is high activation energy often considered to be a useful property of a fuel? (*Reactivity 2.2*)
Which species are the oxidizing and reducing agents in a combustion reaction? (*Reactivity 3.2*)
What might be observed when a fuel such as methane is burned in a limited supply of oxygen? (*Inquiry 2*)
How does limiting the supply of oxygen in combustion affect the products and increase health risks? (*Reactivity 2.1*)



▲ Figure 4 Greenhouse gases reduce the amount of heat radiated into space

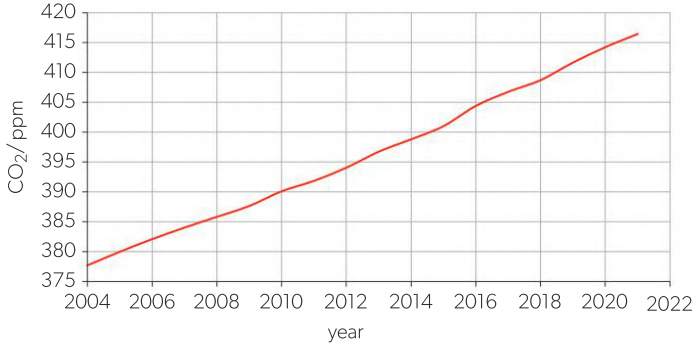
Nitrogen, N_2, and oxygen, O_2, make up over 99% of atmospheric air. Neither nitrogen or oxygen have the ability to absorb infrared radiation that enters the Earth's atmosphere from the Sun. By comparison, carbon dioxide, CO_2, constitutes approximately 0.04% of the atmosphere. Despite the small proportion of carbon dioxide, the increase in the concentration of this greenhouse gas is causing significant damage to our environment. A carbon dioxide molecule can absorb infrared radiation resulting in the vibration of bonds within the molecule. After absorbing the infrared radiation and undergoing vibration, the molecule will emit infrared radiation back into the atmosphere. Some of this radiation will be directed towards the Earth's surface, which will increase the global temperature.

The IR activity of carbon dioxide is discussed in *Structure 3.2*.

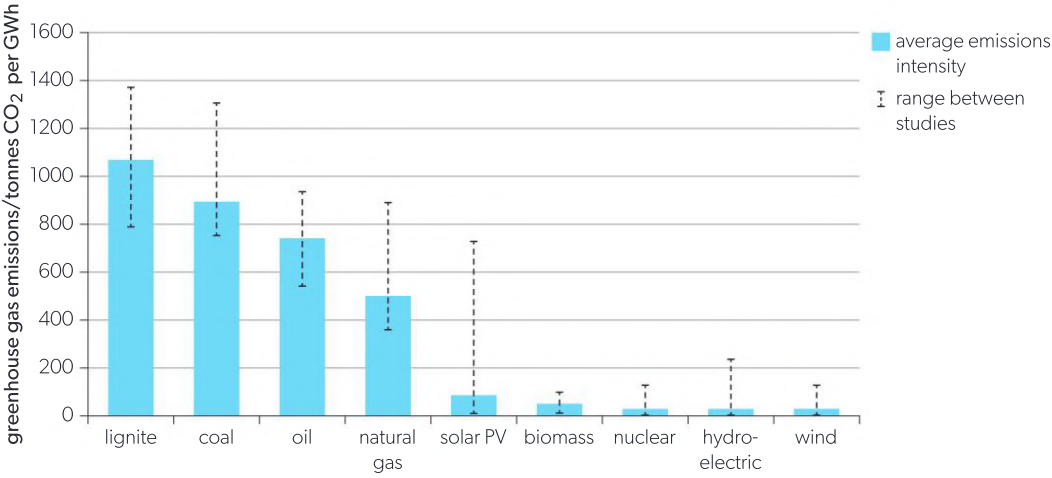
Other greenhouse gases include methane, nitrous oxide, water vapour and fluorinated substances such as hydrofluorocarbons.

As a result of the greenhouse effect, average temperatures around the world are increasing, which is known as global warming. There is widespread agreement in the scientific community that the main cause of global warming is the increase in the levels of greenhouse gases: in particular, carbon dioxide (figure 5), and to a lesser extent, methane.

▶ Figure 5 Annual mean atmospheric CO_2 levels recorded at Mauna Loa Observatory

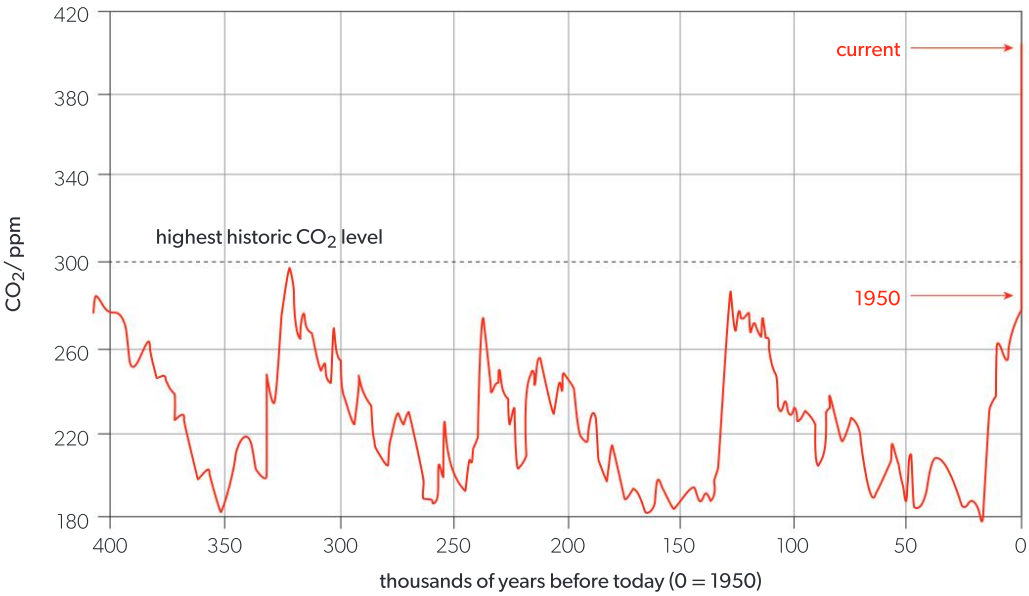


The combustion of different fuels releases different quantities of greenhouse gases (figure 6).



▲ Figure 6 Greenhouse gas emissions by energy source

Figure 7 shows the historic patterns of atmospheric carbon dioxide concentration during the last three glacial cycles, constructed by the analysis of ice-core samples. The current CO_2 levels are much higher than historical peaks, suggesting that human-made emissions are largely responsible for the additional quantities of carbon dioxide in the atmosphere.



▲ Figure 7 Historic CO_2 levels during the last three glacial cycles, as reconstructed from ice cores

TOK

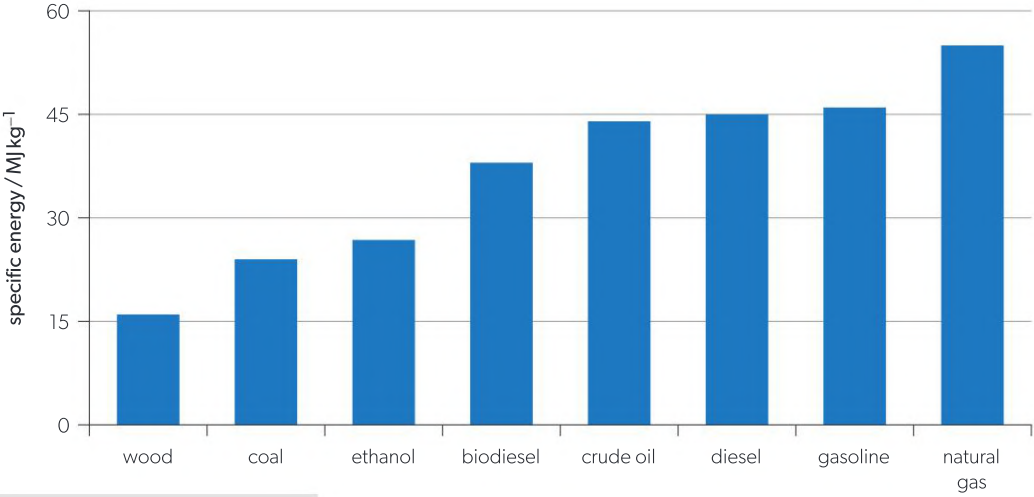
One of the optional themes in TOK is knowledge and indigenous societies. The global population of indigenous people is estimated to be over 450 million, living all over the world. Indigenous communities have a long-standing relationship with, and a highly developed understanding of, the natural environment and biodiversity. Many are particularly vulnerable to the adverse effects of climate change, despite the fact that their contribution to greenhouse gas emissions is exceptionally low. International organizations such as the United Nations (UN) have not only called for awareness and action to protect indigenous peoples, but also recognized that their participation in decision-making processes is of value to all: "indigenous peoples' knowledge should be considered an important element within the international debate regarding adaptation to climate change".



▲ Figure 8 Participants at the United Nations Permanent Forum for Indigenous Issues

What values and assumptions underpin the use of the term "indigenous" knowledge?

Many different sources of fuels are used in everyday life. The choice of fuels depends on the economic development of nations and the natural resources available. Each fuel has a different **specific energy**: the amount of heat energy released per mass of the fuel. Wood, a traditional means of generating energy for cooking and heating, has the lowest specific energy of all common fuels.



▲ Figure 9 The specific energy of different fuels

Practice questions

4. Propane gas C_3H_8 is a hydrocarbon that is classified as a fossil fuel.
   a. Write the balanced chemical equation for the complete combustion of propane and identify the greenhouse gas that is produced.
   b. Enthalpy of combustion of propane is $-2219\,kJ\,mol^{-1}$. Calculate the energy released in burning 1 gram (the specific energy) of propane.
   c. Calculate the mass of carbon dioxide produced per gram of propane burned.
   d. Calculate the carbon footprint, in tonnes of carbon dioxide, from burning 1000 kg (1 tonne) of propane gas. Identify some of the environmental issues caused by the release of this greenhouse gas that was previously locked within the Earth.

Common fuels vary in composition. The chain length of hydrocarbons present in these fuels also varies. In general, the longer the hydrocarbon chain, the greater the tendency of the fuel to undergo incomplete combustion. As discussed, incomplete combustion results in the release of poisonous carbon monoxide and/or elemental carbon (soot). But it also produces a smaller amount of heat energy per unit mass of the fuel when compared to the complete combustion of the same hydrocarbon. Larger hydrocarbons have a reduced volatility due to stronger **London (dispersion) forces (LDFs)**. This affects the way the hydrocarbon molecules interact with the oxygen molecules and the type of combustion that occurs.

Linking questions

Why do larger hydrocarbons have a greater tendency to undergo incomplete combustion? (*Structure 3.2*)

AHL Why is carbon dioxide described as a greenhouse gas? (*Structure 3.2*)

What are some of the environmental, economic, ethical and social implications of burning fossil fuels? (*Reactivity 3.2*)

Air pollution database task

The concepts in this topic are directly and indirectly related to air pollution, including carbon monoxide (released during the incomplete combustion of hydrocarbons), NO_x and SO_x (released in the combustion of substances containing nitrogen and sulfur, which is also discussed in *Structure 3.1*) and greenhouse gases (methane, carbon dioxide and CFCs). In this task you will explore a factor affecting the environmental impact of an atmospheric pollutant of your choice.

Relevant skills

- Tool 2: Identify and extract data from databases
- Inquiry 1: Develop investigations that involve databases
- Inquiry 1: Justify the range and quantity of measurements
- Inquiry 2: Carry out relevant and accurate data processing
- Inquiry 2: Assess reliability and validity
- Inquiry 3: Relate the outcomes of an investigation to the stated research question
- Inquiry 3: Discuss the impact of uncertainties on the conclusions
- Inquiry 3: Evaluate the implications of methodological weaknesses, limitations and assumptions on conclusions

Method

1. Access an air quality database online. Possible sources include:
    - World Health Organization's air pollution data portal
    - World Air Quality Index project
    - Air pollution data shared by the European Environment Agency
    - Your local ministry of the environment
2. Browse the database, carry out some background research, and select one or two pollutants to focus on.
3. Decide how much data to collect and the range of data required to address your aim.
4. Formulate a focused research question that includes the variables, methodology and range of data.
5. Collect data in a well-organized table, including the data sources.
6. Estimate the uncertainty of the data, stating clearly how you have done so and any assumptions you have made.
7. Transform your data by processing and graphing. Include an explanation of how you have processed the data.
8. Discuss and interpret the processed data and their uncertainties.
9. Assess the validity and reliability of the outcome of your investigation.
10. Evaluate the outcome of your investigation: how confident are you in your conclusion?
11. Evaluate the methodology: discuss the strengths and limitations of the data, assumptions and analysis.
12. Outline possible extensions to your investigation. Try to relate one of these proposed extensions to another diploma subject, or TOK.

Biofuels (*Reactivity 1.3.4*)

With the global population continuing to increase at approximately 1% per year, our understanding of how to efficiently use our finite resources is of vital importance. Alternatively, renewable energy resources can be used instead of fossil fuels. Sustainable energy is a UN initiative, with the goal of doubling the share of renewable energy contributed to global energy production by 2030. The cost of renewable energy has decreased over the decades, and now represents a viable option for governments, businesses and individuals.

Renewable energy resources include those that depend on the heat or motion of the Earth (geothermal, wind, and tidal power) or the Sun's radiation (solar power and biomass). Non-renewable energy technology generally uses fossil fuels, such as coal and natural gas, which are finite in their supply.

Biofuels are renewable resources, produced from organic compounds, which in turn are generated from carbon dioxide during biological processes. The production of organic compounds from carbon dioxide is known as **biological carbon fixation**. For example, green plants use photosynthesis to absorb carbon dioxide from the atmosphere and transform it into glucose, which can be converted into ethanol, a biofuel, by fermentation.

In the process of photosynthesis, radiant energy in the form of sunlight is converted by plants into chemical energy. Plants contain chlorophyll molecules, which are capable of absorbing light energy. This light energy is used for photosynthesis, a complex series of reactions that results in the conversion of carbon dioxide and water into glucose, $C_6H_{12}O_6$, and oxygen:

$$6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$$

Glucose stores chemical energy in its bonds. Photosynthesis is an example of biological fixation of carbon.

Global impact of science

Clean, renewable energy has become a global priority, as there is a growing acceptance that the planet is undergoing rapid climate change leading to global warming, more frequent extreme weather events, and often irreversible changes in habitats. The ability of scientists to effectively harness energy and minimize the use of non-renewable energy sources in industry is of paramount importance today.

Scientists are developing new types of catalysts that absorb light energy and transfer this energy to chemical reactions.



▲ **Figure 10** A light-powered catalyst at the Massachusetts Institute of Technology, USA

What would be the benefits of utilizing light energy to promote chemical reactions, in the same way as plants using sunlight in the process of photosynthesis?

The fermentation of glucose produces ethanol, C_2H_5OH, a biofuel:

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

Carbon dioxide, a greenhouse gas, is also produced in the fermentation process. However, this is offset by the absorption of a greater amount of carbon dioxide in the process of photosynthesis.

The industrial production of biofuels in many countries has economic and environmental implications. Brazil has undertaken large-scale production of ethanol from sugarcane for decades. An increased demand for renewable biofuels has both advantages and disadvantages (table 1).

Advantages	Disadvantages
Renewable resource	Use of agricultural land, water resources, fertilizers and pesticides for growing crops
Reduced greenhouse emissions	Diversion of food production to the production of biofuels
Sustainable resource	Monocultures can result in a reduction of biodiversity
Wide range of plant materials and waste can be used for biofuel production	Possible deforestation as the demand for biofuels increases
Economic security with a reduced dependence on imported oil supplies	High cost of production

▲ Table 1 Advantages and disadvantages of the use of biofuels

Practice questions

5. a. Write the balanced chemical equation for the fermentation of glucose.
   b. Explain why ethanol-based fuels are said to have a lower carbon footprint than petroleum-based fuels, even though they both release similar amounts of carbon dioxide upon combustion.
   c. Present the arguments for both the advantages and disadvantages of the production and use of biofuels, if you were asked to address a United Nations Climate Change conference.

Source analysis

In this task, you will research information about biofuel from a variety of sources, and evaluate the quality of the information.

Relevant skills

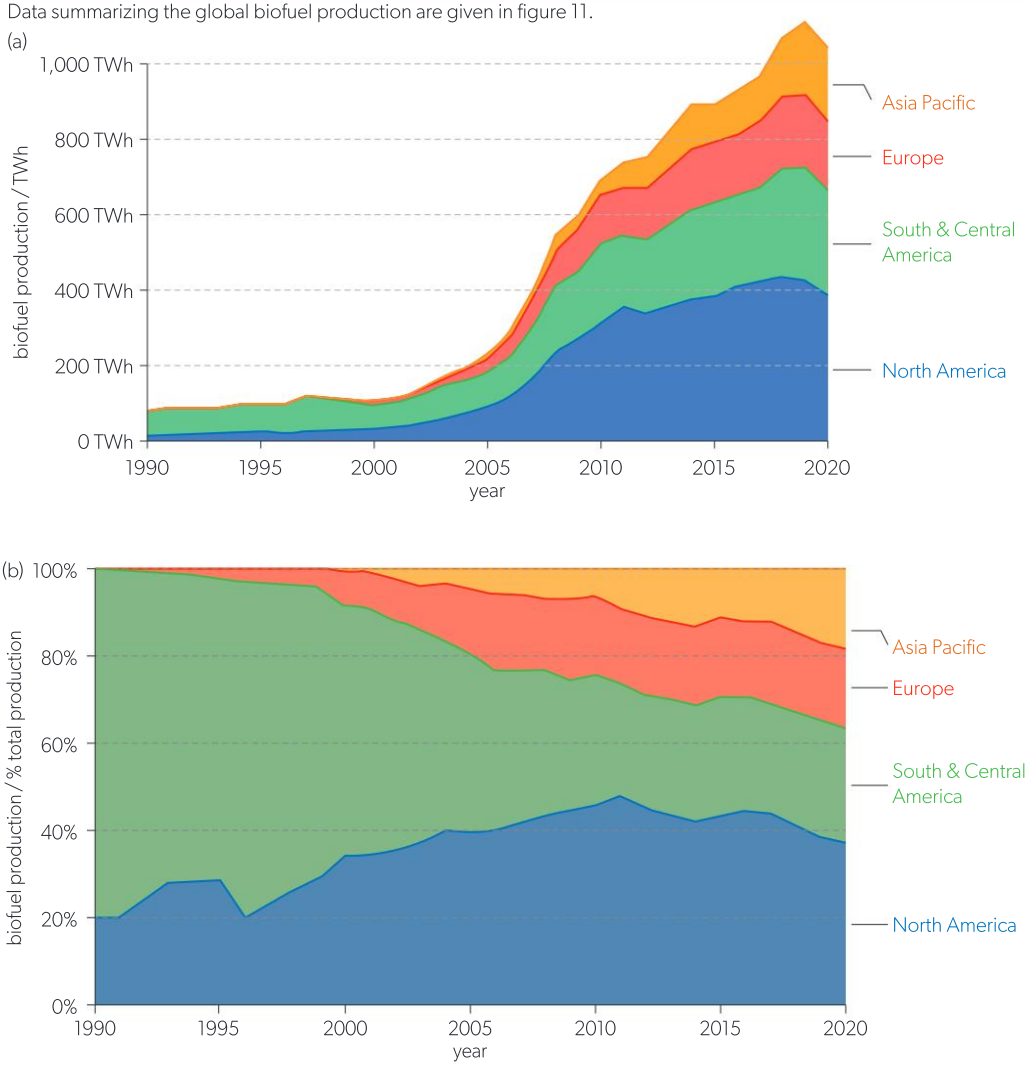
- Inquiry 1: Consult a variety of sources

Method

1. Consult a variety of sources related to biofuels. Try to cover at least three different types of sources, for example:
   - Government or international organization website
   - Academic journal article
   - Secondary data from a database or publicly available data set
   - Newspaper or newspaper article
   - Textbook
   - Video
   - Online image search
2. Each of your sources will provide information about biofuels. Summarize this information.
3. Construct a table assessing the following aspects of each source you have chosen: citation, source type, accuracy, bias, credibility, relevance, advantages, and disadvantages.

Data-based questions

Data summarizing the global biofuel production are given in figure 11.



▲ Figure 11 Biofuel production by world region (a) in terawatt-hours (b) as a percentage of total production. (Source of data: Statistical Review of World Energy, BP, 2021)

1. Describe three conclusions that can be drawn from the graphs in figure 11.
2. Formulate two questions, stemming from the information in the graphs in figure 11.
3. The two graphs were produced from the same data, but they represent these data in slightly different ways. Compare and contrast the purpose of the two graphs.

Fuel cells (*Reactivity 1.3.5*)

In *Reactivity 3.2*, you will learn that a **primary (voltaic) cell** is an electrochemical cell that converts chemical energy from spontaneous redox reactions into electrical energy. These reactions are mostly irreversible, and the cells of this type are mainly used for low-current applications, such as hand torches, long-life fire alarms, calculators and wall clocks. However, primary cells often contain materials that are toxic to humans and the environment if they are not disposed of properly. Primary cells typically contain zinc metal, manganese dioxide and potassium hydroxide.

In a **secondary cell**, the chemical reactions are reversible, so the battery can be recharged. Therefore, secondary cells are more environmentally friendly than primary cells. The lead–acid battery is a type of rechargeable battery, found in the majority of gasoline-powered vehicles. Electric vehicles are powered by lithium-ion batteries, another type of secondary cell.

While secondary cells have advantages over traditional primary cells, there are still problems associated with these types of batteries, such as overheating, limited lifespan and environmental concerns in terms of their disposal.

A **fuel cell** is used to convert chemical energy of a fuel directly into electrical energy. The first fuel cell was invented in 1838 by Welsh scientist, William Groves. The difference between a fuel cell and a voltaic cell is that in a fuel cell, the fuel is continuously supplied from an external source, whereas a voltaic cell contains finite amounts of reactants locked within the cell. Therefore, a fuel cell can produce electricity indefinitely, while a voltaic cell stops working when all reactants within the cell are consumed.

Hydrogen fuel cell

The **hydrogen fuel cell** is an electrochemical cell that uses hydrogen and oxygen gases as fuel. The redox reaction between hydrogen and oxygen produces water, electricity and heat.

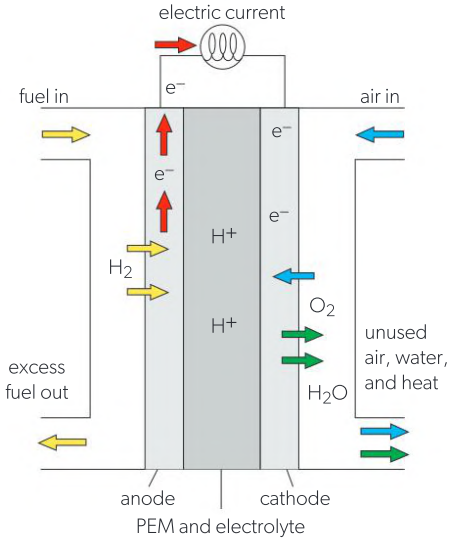
The following steps occur in a hydrogen fuel cell:

1. The **proton exchange membrane (PEM)** selectively allows hydrogen ions to diffuse between the **cathode** and **anode** but prevents the passage of other ions, molecules and electrons between these **electrodes**.
2. Hydrogen gas is oxidized at the anode on the surface of a platinum-based catalyst. The half-equation for the reaction at the anode is as follows:

   $H_2(g) \rightarrow 2H^+(aq) + 2e^-$

3. The electrons cannot move through the PEM, and therefore have to leave the cell through an external circuit, producing the electrical output of the cell.
4. The protons formed at the anode move across the PEM to the cathode, where they combine with the oxygen gas and electrons. Oxygen gas is reduced to form water as a waste product. The half-equation for the reaction at the cathode is as follows:

   $O_2(g) + 4e^- + 4H^+(aq) \rightarrow 2H_2O(l)$



▲ Figure 12 The main components of a commercial hydrogen cell include an electrolyte, a proton exchange membrane (PEM) and the electrodes

The overall redox reaction is:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Hydrogen fuel cells are considered to be a clean energy source because their only product is water. However, hydrogen gas, $H_2(g)$, is not abundant on Earth and needs to be produced by the following methods, one of which has environmental implications.

1. **Electrolysis of water**

The electrolysis of water is the process of using electricity to split water into oxygen and hydrogen gas. This is the reverse process of the reaction that occurs in a hydrogen fuel cell.

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

The electrolysis of water can be powered by renewable energy sources, such as solar energy. This environmentally friendly source of hydrogen gas only makes up a small proportion of the global supply.

2. **Steam reforming of hydrocarbons**

Hydrocarbons, such as natural gas, gasified coal and biomass, diesel and other liquid fuels, undergo steam reforming resulting in the production of toxic carbon monoxide gas and hydrogen gas. The general equation for the steam reforming of alkanes is as follows:

$$C_nH_{2n+2}(g) + nH_2O(g) \rightleftharpoons nCO(g) + (2n + 1)H_2(g)$$

Carbon monoxide then reacts with steam to form the greenhouse gas carbon dioxide:

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

The annual production of hydrogen from the various sources is responsible for carbon dioxide emissions greater than the total annual carbon dioxide emissions of the United Kingdom. The scientific community continues to search for new green sources of hydrogen gas.

Experiments

Alternatives to fossil fuels, ranging from hydrogen fuel cells to biofuels, have arisen from extensive experimentation by numerous scientists over time. Creativity and imagination play a role in experimental design, interpretation and conclusions. Once considered an electrochemical curiosity, fuel cells are now seen as a promising source of renewable energy.

Direct-methanol fuel cell

In a **direct-methanol fuel cell (DMFC)**, the source of the hydrogen ions, H^+, is methanol rather than hydrogen gas. Methanol is a liquid fuel that can be produced from renewable resources through fermentation. Methanol is cleaner than hydrogen because the method of its production has less impact on the environment in terms of formation of greenhouse gases. Methanol also has a far greater energy density (energy per unit volume) than hydrogen gas.

DMFCs use the electrochemical reactions below:

anode half-equation: $CH_3OH(aq) + H_2O(l) \rightarrow CO_2(g) + 6H^+(aq) + 6e^-$

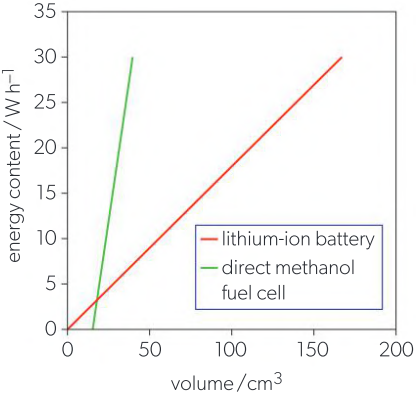
cathode half-equation: $\frac{3}{2}O_2(g) + 6H^+(aq) + 6e^- \rightarrow 3H_2O(l)$

overall reaction: $CH_3OH(aq) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

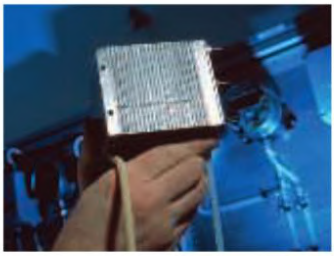
The electrochemical reaction produces carbon dioxide, a greenhouse gas. The reaction at the anode requires a catalyst that contains expensive precious metals, usually ruthenium and palladium. Another disadvantage is the toxicity of methanol.

Methanol can also be used to supply hydrogen gas for hydrogen fuel cells. Steam reforming of methanol at 250 °C produces $CO_2(g)$ and $H_2(g)$, along with a small amount of CO(g).

A distinct advantage of DMFCs is their high energy density (figure 14). The slope of the graph gives the energy per unit volume, showing that the methanol fuel cell has a much higher energy density than the lithium-ion battery.



▲ Figure 14 Comparison of the energy density for the lithium-ion battery and the direct-methanol fuel cell



▲ Figure 13 A portable direct-methanol fuel cell (DMFC), which can be used to power laptops and video cameras

Practice questions

6. a. Outline the differences between a primary and secondary cell.
   b. What are the advantages a hydrogen fuel cell has over the traditional lead-acid battery.
   c. Identify and explain the function of the feature of a hydrogen fuel cell that involves the passage of ions.
   d. A direct methanol fuel cell (DMFC) converts chemical energy to electrical energy.
      i. Deduce the anode and cathode half equations and the overall chemical equation for this electrochemical cell.
      ii. Outline one advantage and one disadvantage of the methanol cell (DMFC) compared to a hydrogen fuel cell.

When comparing fuels, the energy density (energy released per unit volume) and specific energy (energy released per unit mass) can give quite different pictures (table 2).

Fuel source	Energy density / MJ dm^{-3}	Specific energy / MJ kg^{-1}
compressed hydrogen	1.9	120
methanol	16	20
liquefied natural gas	21	50
liquid propane	27	46
gasoline	32	46

▲ Table 2 Comparing fuels in terms of energy density and specific energy

The specific energy of hydrogen is more than double that of any other fuel in table 2. In hydrogen, H_2, 2.02 g contains 1 mol of fuel, compared with 32.05 g for 1 mol of methanol, CH_3OH, or approximately 110 g for 1 mol of gasoline. Therefore, it could be easy to believe that hydrogen is the best fuel choice.

However, fuels need to be stored and transported. The molar volume of a gas at room temperature and 1 atm pressure is approximately 24 dm^3. One mole of gaseous hydrogen under these conditions would require a 24 dm^3 storage tank, which adds to the weight if the device is to be portable, such as in a car. One mole of methanol would occupy 40.4 cm^3, and the same 24 dm^3 storage tank could hold over 545 mol of methanol fuel. Even when compressed, the hydrogen gas occupies a much larger volume, and regulators and compressors further increase the weight of the storage system. Gasoline offers the highest energy density but has associated environmental problems.

ATL Communication skills

The general public needs to have a good understanding of certain scientific issues in order to make informed choices. This presents scientists with the challenge of accurately conveying complex information and specific terminology in a succinct and accessible way. For example, this chapter contains information about fuels that is communicated in different types of tables, diagrams, charts and graphs. What are the advantages and disadvantages of different data communication methods?

Design an infographic to communicate the science behind an issue of your choice to a particular audience (also of your choice). Possible issues related to this chapter include renewable energy, biofuels, fuel cells and incomplete combustion. Possible audiences include younger pupils at your school, your family or your local community. You may even wish to discuss this with your creativity, activity, service (CAS) coordinator. Depending on the nature of the issue and the audience, this task could be connected to a CAS experience.

Linking question

What are the main differences between a fuel cell and a primary (voltaic) cell? (*Reactivity 3.2*)

End of topic questions

Topic review

1. Using your knowledge from the *Reactivity 1.3* topic, answer the guiding question as fully as possible:

   What are the challenges of using chemical energy to address our energy needs?

Exam-style questions

Multiple-choice questions

2. Which of the following statements is false?

   A. The complete combustion of a hydrocarbon forms carbon dioxide and water.

   B. All combustion reactions are endothermic reactions.

   C. The incomplete combustion of hydrocarbons and alcohols form the same products.

   D. Complete combustion reactions require an excess of oxygen gas as a reactant.

3. Which equation(s) represent the incomplete combustion of methane?

   I. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

   II. $CH_4(g) + 1\frac{1}{2}O_2(g) \rightarrow CO(g) + 2H_2O(g)$

   III. $CH_4(g) + O_2(g) \rightarrow C(s) + 2H_2O(g)$

   A. I, II and III C. III only

   B. II and III only D. II only

Extended-response questions

4. a. Write the chemical equation for the incomplete combustion of methane. [1]

   b. Explain how one of the products of incomplete combustion can lead to increased health risks. [2]

   c. Calculate the enthalpy change for the incomplete combustion of methane, in kJ mol^{-1}, using the bond enthalpy data given below. [2]

Bond	Average bond enthalpy / kJ mol^{-1}
C—H	414
O—H	463
O═O	498
C≡O	1077

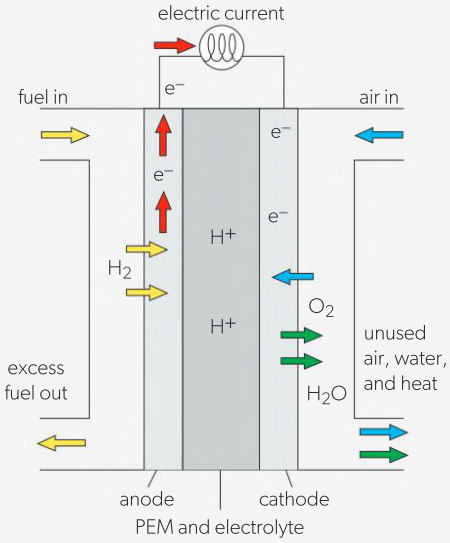
5. Ethanol is a biofuel that can be mixed with gasoline.

   a. Write the equation for the complete combustion of ethanol. [1]

   b. Outline the evidence that relates global warming to increasing concentrations of greenhouse gases in the atmosphere. [3]

   c. Explain, including a suitable equation, why biofuels are considered to be carbon neutral. [2]

6. A hydrogen fuel cell uses pure hydrogen gas as the fuel and a proton exchange membrane as the electrolyte. A diagram of the hydrogen fuel cell is shown below.



   a. Suggest an advantage of the hydrogen fuel cell over the lead–acid battery for use in cars. [1]

   b. Outline **one** advantage and **one** disadvantage of the methanol cell (DMFC) compared with a hydrogen fuel cell. [2]

7. Even though fuel cells, primary cells and rechargeable cells have similar fundamental characteristics, there are important differences between them.

   a. Suggest one common feature that fuel cells, primary cells and rechargeable cells share. [1]

   b. Outline the difference between primary and rechargeable cells. [1]

   c. Identify **one** factor that affects the voltage of a cell and **a different factor** that affects the current it can deliver. [2]

States of matter (*Structure 1.1.2*)

Solids, liquids and gases

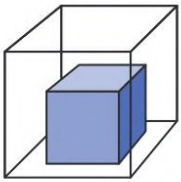
Matter is composed of particles. The types of interactions between these particles determine the state of matter of a substance: solid, liquid or gas. All substances can exist in these three states, depending on the temperature and pressure.

The states of matter of substances are shown by letters in brackets after the formula: (s) for solid, (l) for liquid and (g) for gas. For example:

- Water is a solid below 0 °C: $H_2O(s)$
- Water is a liquid between 0 and 100 °C: $H_2O(l)$
- Water is a gas above 100 °C: $H_2O(g)$.

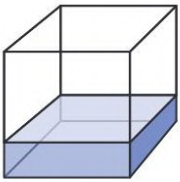
A special symbol, (aq), is used for molecules or other species in aqueous solutions. For example, the expression "NaCl(aq)" tells us that sodium chloride is dissolved in water while "NaCl(s)" refers to the pure compound (solid sodium chloride). The properties of the three states of matter are summarized in figure 11.

solid



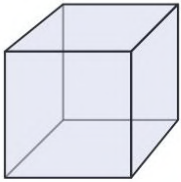
- fixed volume
- fixed shape
- cannot be compressed
- attractive forces between particles are strong
- particles vibrate in fixed positions but do not move around

liquid



- fixed volume
- no fixed shape
- cannot be compressed
- attractive forces between particles are weaker than those in solids
- particles vibrate, rotate, and move around

gas



- no fixed volume
- no fixed shape
- can be compressed
- attractive forces between particles are negligible
- particles vibrate, rotate, and move around faster than in a liquid



▲ Figure 11 Steam, liquid water and ice are the three states of water

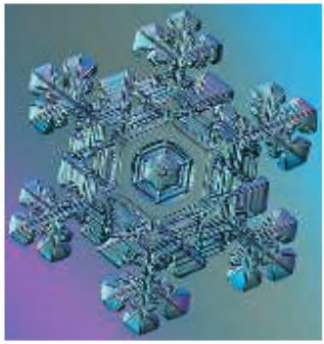
Changes of state

Substances change their states of matter as they absorb or release energy. Solid ice will absorb energy as it is heated. The particles continue to vibrate in fixed positions, but more violently, until a temperature known as the melting point is reached. At this point, the ice melts (changes its state from solid to liquid). A further increase in temperature accelerates the movement of particles, and eventually the water vaporizes and becomes a gas. The decrease in temperature reverses these changes of state.

Under certain conditions, solid substances can turn into gases directly, without melting. This change of state, known as **sublimation**, is typical for dry ice (solid carbon dioxide, $CO_2(s)$, figure 12), which is commonly used for refrigerating ice cream and biological samples.



▲ Figure 12 Sublimation of dry ice



▲ Figure 13 A snowflake, the product of deposition of water

The process opposite to sublimation is called *deposition*. At low temperatures, water vapour in the air solidifies and forms snowflakes of various shapes and sizes (figure 13).

When a substance changes from a more condensed state to a less condensed state, energy is absorbed by the particles from the surroundings. This happens when a solid becomes a liquid or a gas, and when a liquid becomes a gas. These are **endothermic** processes.

When a substance changes from a less condensed state to a more condensed state, the particles lose energy to the surroundings and, for a molecular substance, the intermolecular forces become stronger. This happens when a gas becomes a liquid or a solid, and when a liquid becomes a solid. The process of releasing energy to the surroundings is an **exothermic** process.

The changes of state occurring in these transformations are shown in figure 14.

Non-Newtonian fluids

Some substances, known as **non-Newtonian fluids**, do not behave like typical liquids. The **viscosity** of non-Newtonian fluids varies depending on the force applied to them. You will make a non-Newtonian fluid commonly known as maize starch slime or "oobleck", and explore its properties.

Relevant skills

- Inquiry 1: Identify dependent and independent variables
- Inquiry 2: Identify and record relevant qualitative observations

Safety

Wear eye protection.

Materials

- Spoon or large spatula
- 250 cm^3 beaker
- Powdered maize starch
- Water

Method

1. Add three or four heaped spoons of maize starch to the beaker. Note its appearance and consistency.
2. Slowly add water to the maize starch and mix. Continue adding water until the mixture achieves a thick consistency. Adjust by adding more maize starch or more water, as needed.
3. Spend some time exploring the properties of your mixture. It should harden if tapped, and flow smoothly if stirred slowly.

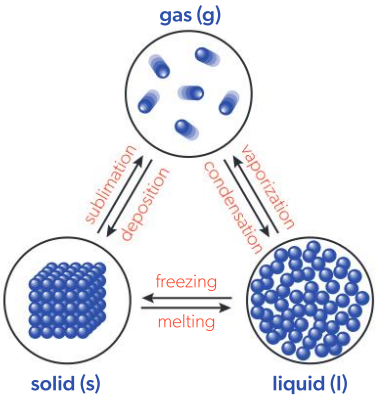
Questions

1. Describe the properties and identify the state of matter of each of the following:
   - powdered maize starch
   - water
   - the maize starch–water mixture.
2. Suppose you were asked to develop a research question relating to a maize starch–water mixture. Consider possible independent and dependent variables.
3. Research non-Newtonian fluids and identify other examples of these substances.
4. How has this experience changed the way you think about states of matter and their properties? Reflect on this, completing the following sentence starters:
   - I used to think…
   - Now, I think…

Linking questions

Why are some substances solid while others are fluid under standard conditions? (Structure 2.4)

Why are some changes of state endothermic and some exothermic? (Structure 2, Reactivity 1.2)



▲ Figure 14 Endothermic and exothermic changes of state

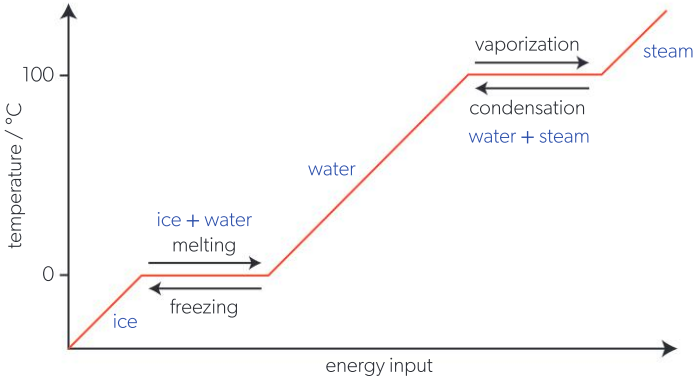


▲ Figure 15 Orange growers spray their fruit with water on cold nights. Freezing of water is an exothermic process that releases energy (in the form of heat) to the fruit, protecting it against cold

Kelvin temperature scale (*Structure 1.1.3*)

As temperature rises, the energies of particles increase. **Temperature** is a measure of the average kinetic energy of particles. As substances absorb energy, particles of a solid vibrate in the lattice more, particles in a liquid vibrate more and move faster, while in a gas they move faster.

When water is heated, there is no temperature change during the periods when a solid changes to a liquid and when a liquid changes to a gas (figure 16). The added energy is used to disrupt the solid lattice and overcome the intermolecular forces between molecules in the liquid.



▲ Figure 16 Graph of the heating curve for water

There were many attempts to measure relative temperature, but the first widely accepted temperature scale was introduced by the Polish-born Dutch physicist Daniel Gabriel Fahrenheit.

The kelvin is the base unit of temperature measurement in the International System of Units (SI). There are seven base units, and all other units of measurements can be derived from these (figure 17).



▲ Figure 17 The seven base SI units are kilogram (kg) for mass, meter (m) for length, second (s) for time, ampere (A) for electric current, kelvin (K) for temperature, mole (mol) for amount of substance, and candela (cd) for luminous intensity. All units of measurement can be derived from these seven base units

You will learn more about the mole in *Structure 1.4*.

Reactivity 1.1 Measuring enthalpy changes

What can be deduced from the temperature change that accompanies chemical or physical change?

Chemistry involves the study of chemical reactions and physical changes of state of the elements and their compounds. Conservation of energy is a fundamental principle of science, which is examined through observation and experimentation. The use of models, empirical or experimental data, the language of mathematics and scientific terminology, all contribute to our understanding of energy changes associated with chemical reactions.

An understanding of the relationships that exist between chemistry and energy involves understanding how energy is transferred between a chemical system and the surroundings. This information can in turn be used to develop an understanding of the relative stability of reactants and products, leading to better control over the progress of the reaction being studied.

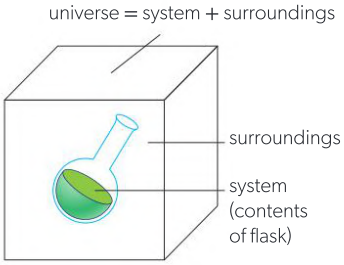
Understandings

Reactivity 1.1.1—Chemical reactions involve a transfer of energy between the system and the surroundings, while total energy is conserved.

Reactivity 1.1.2—Reactions are described as endothermic or exothermic, depending on the direction of energy transfer between the system and the surroundings.

Reactivity 1.1.3—The relative stability of reactants and products determines whether reactions are endothermic or exothermic.

Reactivity 1.1.4—The standard enthalpy change for a chemical reaction, $\Delta H^{\ominus}$, refers to the heat transferred at constant pressure under standard conditions and states. It can be determined from the change in temperature of a pure substance.



▲ Figure 1 The universe is the combination of the system and its surroundings

Energy transfer in chemical reactions (*Reactivity 1.1.1*)

In a chemical reaction, total energy is conserved. Chemical potential energy is stored in the chemical bonds of **reactants** and **products**, while the temperature of the reaction mixture is a function of the kinetic energy of the atoms, ions and molecules present.

All chemical reactions involve energy changes. Energy may be released into the **surroundings** from the **reaction system** or it may be absorbed by the reaction system from the surroundings. Most commonly, the energy is transferred in the form of heat, but it may also be in the form of electricity, sound or light.

In an **open system**, the transfer of matter and energy is possible across its boundary (for example, matter can be added to a beaker, and energy can be transferred through its sides). A **closed system** allows no transfer of matter, though energy may be transferred across the boundary. In an **isolated system**, matter and energy can neither enter nor exit the system.

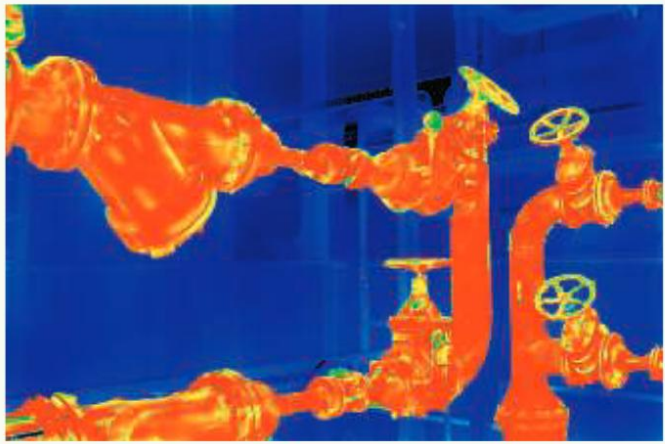




▲ Figure 2 In each of the above scenarios, energy is transferred. In hot springs, energy is transferred as heat, and in fireworks, as heat, sound and light. Both scenarios are open systems

Models

Most industrial processes take place in open or closed systems. The loss of heat during an industrial process not only affects the efficiency of the chemical reaction, but also contributes to a loss of useful energy, an increase in thermal pollution. Thermography can be used to model heat flow and loss from structures in chemical industries as a heat map, where red is hot and purple is cold.



▲ Figure 3 Thermograph of industrial engineering system

What are the advantages of modelling heat distribution and transfer? How can chemical engineers use the data collected to improve the efficiency of industrial processes? How does this help our environment?

What is the difference between heat and temperature?

Temperature, T, is an example of a **state function**. For a state function, any change in value is independent of the pathway between the initial and final measurements.

For example, if you take the temperature of the water in a swimming pool early in the morning (the initial value) and then again in the afternoon (final value), it will not tell you the complete story of any temperature fluctuations that may have occurred throughout the day. The calculation of the temperature change is simple:

$$\Delta T = T_{final} - T_{initial}$$



▲ Figure 4 If you record the temperature of a pool at the beginning and the end of a day, it will not give you an indication of the heating and cooling that has occurred throughout the day, only the overall temperature change

Other examples of state functions include volume, enthalpy and pressure.

Heat, q, is a form of energy that is transferred from a warmer body to a cooler body, as a result of the **temperature gradient**. Heat is sometimes referred to as **thermal energy**. It can be transferred by the processes of conduction, convection and radiation.

Heat has the ability to do **work**. When heat is transferred to an object, the result is an increase in the average **kinetic energy** of its particles. This causes an increase in temperature and potentially a phase change, for example, a change of state from liquid to gas.

At **absolute zero**, 0 K (−273.15 °C), all motion of the particles theoretically stops and the **entropy**, S, of a system reaches its minimum possible value. The absolute temperature (in kelvin) is proportional to the average kinetic energy of the particles of matter. As the temperature increases, the kinetic energy of the particles also increases.

Entropy is defined and explored in *Reactivity 1.4*.

ATL Communication skills

Your communication skills will develop incrementally throughout the entire chemistry programme. Communication skills include consistent and accurate application of scientific terminology. Your use of the terms "heat" and "temperature" in explanations will demonstrate your understanding of these concepts.

For example, consider the simple reaction between magnesium metal and hydrochloric acid.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

In this reaction, heat is released from the reaction system into the surroundings, and the temperature of the aqueous solution rises. When you think about heat, you are considering the transfer of thermal energy in the system. When you refer to the temperature of a system, you are describing the average kinetic energy of the particles within that system.



▲ Figure 5 Magnesium ribbon reacting with hydrochloric acid

Thermochemistry is the study of heat changes that occur during chemical reactions. Heat changes are often described in terms of **enthalpy**. At constant pressure, the **enthalpy change**, ΔH, is defined as the heat transferred from a closed system to the surroundings during a chemical reaction. The terms "enthalpy change" and "heat of reaction" are commonly used when describing the thermodynamics of a reaction. The most common unit of enthalpy change is kJ.

Activity

Imagine a glass of water containing ice cubes sitting in the summer sun. It will undergo a change in enthalpy.

- Is the glass of water an open, closed or isolated system?
- Identify the system and the surroundings.
- Explain the movement of energy in the form of heat, between the system and the surroundings.
- What would you observe on the outside of the glass? Explain this observation in terms of a change of state and transfer of energy.

Exothermic and endothermic reactions (*Reactivity 1.1.2*)

A chemical reaction in which heat is transferred from the system to the surroundings is defined as an **exothermic** reaction. In contrast, chemical reactions in which heat is absorbed into the system from the surroundings are defined as **endothermic** reactions.

When a chemical reaction takes place, the atoms of the reactants are rearranged to create new products. Chemical bonds in the reactants are broken, and new chemical bonds are made to form products. Energy is absorbed by the reaction system to break the chemical bonds, and therefore bond breaking is an endothermic process. This energy is termed the **bond dissociation energy** and it can be quantified for each type of bond. Energy is released into the surroundings when new chemical bonds are made, and therefore bond making is an **exothermic** process. The **transfer of energy** between the surroundings and the system is an important part of your understanding of the energy changes in a reaction.

Exothermic reactions have a negative enthalpy change, and endothermic reactions have a positive enthalpy change. The sign of the enthalpy change is defined from the perspective of the system and not the surroundings. For example, in an exothermic reaction, heat is being lost by the system and so the enthalpy change is negative. For an endothermic reaction, heat is absorbed by the system, so the enthalpy change is positive.

To determine whether a reaction is endothermic or exothermic, we can use a calorimeter. A calorimeter is any apparatus used to measure the amount of heat being exchanged between the system and the surroundings. In the school laboratory, experiments focus on the change in temperature, ΔT, of the reaction solvent, which in most cases is water.

Observations

In the laboratory, observations can be made using human senses, or with the aid of instruments such as data-logging equipment. The application of digital technology to collect data is one of the essential skills in the study of chemistry.

Energy profiles (*Reactivity 1.1.3*)

Energy profiles are a visual representation of the enthalpy change during a reaction. From an energy profile, you can determine the enthalpy of the reactants and the products, the **activation energy** (E_a), and the enthalpy change for the reaction.

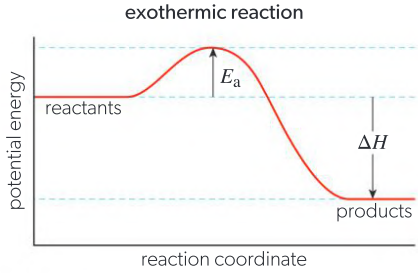
Many chemical reactions are exothermic. In these reactions, energy is released from the system to the surroundings. The reactants of this reaction are at a higher energy level and considered to be lower in stability. Products for exothermic reactions are at a lower energy level and considered to be more energetically favourable.

Activation energy, E_a, is the minimum energy required for the reaction to take place. You will study activation energy in *Reactivity 2.2*.

Consider the reaction between zinc and copper(II) sulfate solution. It is a single displacement reaction involving the displacement of the copper(II) ion by zinc:

$$Zn(s) + CuSO_4(aq) \rightarrow Cu(s) + ZnSO_4(aq)$$

Measured quantities of copper(II) sulfate solution and zinc are mixed in a calorimeter. The mixture is stirred, and the change in temperature of the solution is measured using a thermometer or data-logging equipment. In this reaction, heat is generated by the reaction system. As a result, the temperature of the solution increases, and the heat is transferred to the surroundings. The reaction is therefore exothermic.



▲ Figure 6 Using a thermometer or a temperature probe, you would observe an increase in the temperature of the reaction mixture in an exothermic reaction. The enthalpy of the products is lower than that of the reactants. You would describe the products as being energetically more stable than the reactants

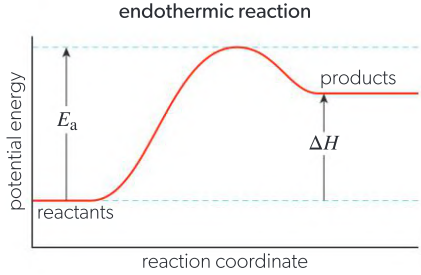
The graph in figure 6 is an example of an **energy profile**.

If you consider an endothermic reaction, the products of the reaction are at a higher energy level and therefore are less stable than the reactants.

Ammonium nitrate, NH_4NO_3, is an important component of fertilizers. When the solid dissolves in water to form aqueous ammonium and nitrate ions, the temperature of the solution decreases.

$$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$$

This heat is absorbed by the reaction system from the surroundings. The apparatus containing the reaction mixture will feel cold to touch. This is an example of an endothermic reaction (figure 7).



▲ Figure 7 Using a thermometer or a temperature probe, you would observe a decrease in the temperature of the reaction mixture in an endothermic reaction. The enthalpy of the products is greater than that of the reactants. The products are described as being energetically less stable than the reactants

Global impact of science

Developments in science may have ethical, environmental, political, social, cultural and economic consequences, which must be considered during decision making. The pursuit of science may have unintended consequences. German chemist Fritz Haber was awarded the Nobel Prize in Chemistry in 1918 for developing a method to chemically extract nitrogen from the air by reacting it with hydrogen. Haber's discovery allowed for the large-scale production of fertilizers that began during the green revolution and continues today. However, his process also provided Germany with a source of ammonia that was used for the production of explosives during the First World War. The global impact of science is evident in Haber's research.

ATL Communication skills

Communication skills cover a wide range of skills and forms of communication. Your ability to effectively communicate verbally and in written form most often comes to mind when you are thinking about improving communication skills. However, communication also involves your ability to read and write different forms of texts intended for different audiences. In science, you need to be able to write formal laboratory reports using specific terminology and accepted writing styles. Another form of communication you would utilize in writing reports and answering examination questions, is your ability to sketch graphs and extract data and meaningful information from graphs. Can you read and analyse the energy profiles that represent exothermic and endothermic reactions? Try to accurately sketch these diagrams, including all of the components.

Practice question

1. Barium hydroxide, $Ba(OH)_2$, reacts with ammonium chloride, NH_4Cl:

$Ba(OH)_2(s) + 2NH_4Cl(s) \rightarrow BaCl_2(aq) + 2NH_3(g) + 2H_2O(l) \quad \Delta H = +164\ kJ\ mol^{-1}$

Which of the following is correct for this reaction?

	Temperature	Enthalpy	Stability
A	increases	products have lower enthalpy than the reactants	products are less stable than the reactants
B	decreases	products have lower enthalpy than the reactants	products are more stable than the reactants
C	decreases	products have higher enthalpy than the reactants	products are less stable than the reactants
D	increases	products have higher enthalpy than the reactants	products are more stable than the reactants

Standard temperature and pressure (STP) conditions are denoted by the symbol ⊖. STP is a temperature of 273.15 K and a pressure of 100 kPa. Standard ambient temperature and pressure (SATP) refer to more practical reaction conditions of 298.15 K and 100 kPa. STP and SATP conditions are given in the section 2 of the data booklet.

Substance	Specific heat capacity / kJ kg^{-1} K^{-1}
water	4.18
ethanol	2.44
copper	0.385

▲ Table 1 The specific heat capacities of water, ethanol and copper

Standard enthalpy change, $\Delta H^{\ominus}$ (*Reactivity 1.1.4*)

The **standard enthalpy change for a reaction**, $\Delta H^{\ominus}$, refers to the heat transferred at constant pressure under standard conditions and states. It can be determined from the change in temperature of a pure substance. The units of $\Delta H^{\ominus}$ are kJ mol^{-1}.

To calculate $\Delta H^{\ominus}$ for a reaction, you therefore need to determine the amount of heat released or absorbed in the course of that reaction. This can be done by measuring the change in temperature of a pure substance to or from which this heat is transferred. When calculating the amount of heat lost or gained by a pure substance such as water, you need to know the **specific heat capacity**, c, of that substance.

The specific heat capacity of a pure substance is defined as the amount of heat needed to raise the temperature of 1 kg of that substance by 1 °C or 1 K. For example, the specific heat capacity of ethanol is 2.44 kJ kg^{-1} K^{-1}, so it takes 2.44 kJ to raise the temperature of 1 kg of ethanol by 1 K. The lower the specific heat capacity of a given substance, the higher the rise in temperature when the same amount of heat is transferred to the sample.

Specific heat capacity is an **intensive property** that does not vary in magnitude with the size of the system being described. For example, a 10 cm^3 sample of copper has the same specific heat capacity as a 1 ton block of copper. When you heat up a pure substance, the rise in temperature depends on:

- the identity of that substance
- the mass of that substance
- the amount of heat supplied.

Practice questions

2. Using table 1, calculate how much energy is required to raise the temperature of the following by 1 K.
   a. 1 kg of water
   b. 1000 kg of copper
3. When equal masses of two different substances, X and Y, absorb the same amount of energy, their temperatures rise by 5 °C and 10 °C, respectively. Which of the following is correct?
   a. The specific heat capacity of X is twice that of Y.
   b. The specific heat capacity of X is half that of Y.
   c. The specific heat capacity of X is one fifth that of Y.
   d. The specific heat capacity of X is the same as that of Y.
4. Using table 1, state which of the following statements is correct.
   a. More heat is needed to increase the temperature of 50 g of water by 50 °C than 50 g of ethanol by 50 °C.
   b. If the same heat is supplied to equal masses of ethanol and water, the temperature of the water will increase more.
   c. If equal masses of water at 20 °C and ethanol at 50 °C are mixed together, the final temperature will be 35 °C.
   d. If equal masses of water and ethanol at 50 °C cool down to room temperature, ethanol gives out more heat.

Specific heat capacity, c, is used to calculate the amount of heat, Q, absorbed by a pure substance using the relationship:

$$Q = mc\Delta T$$

where m is mass of the pure substance in kg and ΔT is the change in temperature of that substance in K.

Heat, Q, is related to enthalpy change, ΔH, by the following equation:

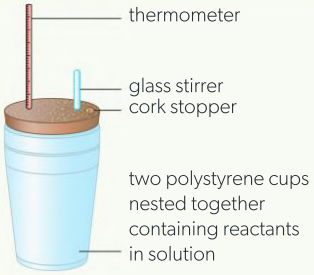
$$\Delta H = -\frac{Q}{n}$$

where n is the number of moles of the **limiting reactant**. In a reaction, the limiting reactant is the reacting substance with the least stoichiometric amount present, which therefore limits how much product can be formed. In contrast, the other reacting substances are said to be in **excess**.

Measurement

Performing reactions in a polystyrene coffee cup to measure the enthalpy change is a convenient experimental procedure. This method introduces systematic errors that can be analysed and the effect of their directionality assessed.

Systematic errors are a consequence of the experimental procedure. Their effect on empirical data is constant and always in the same direction. With the coffee-cup calorimeter, the measured change in enthalpy for a reaction will always be lower in magnitude than the actual value, as some heat will be transferred between the contents and the surroundings in every experiment.



▲ Figure 8 A coffee-cup calorimeter

Worked example 1

1. When a 1.15 g sample of anhydrous lithium chloride, LiCl, was added to 25.0 g of water in a coffee-cup calorimeter, a temperature rise of 3.80 K was recorded. Calculate the enthalpy change of dissolution for 1 mol of lithium chloride. Assume that the heat capacity of lithium chloride itself is negligible.

2. 180.0 J of heat is transferred to a 100.0 g sample of iron, resulting in a temperature rise from 22.0 °C to 26.0 °C. Calculate the specific heat capacity of iron.

Solution

1. $Q = mc\Delta T$

   $= 0.025\ \text{kg} \times 4.18\ \text{kJ kg}^{-1}\ \text{K}^{-1} \times 3.80\ \text{K}$

   $= 0.397\ \text{kJ}$

   Now you need to determine the energy gained for 1 mol of LiCl.

   $$n(\text{LiCl}) = \frac{1.15\ \text{g}}{42.39\ \text{g mol}^{-1}} = 0.0271\ \text{mol}$$

   $$\Delta H = -\frac{Q}{n} = \frac{-0.397\ \text{kJ}}{0.0271\ \text{mol}} = -14.6\ \text{kJ mol}^{-1}$$

2. First, determine the change in temperature, ΔT:

   $\Delta T = (299 - 295)\ \text{K} = 4\ \text{K}$.

   Substitute the values into $Q = mc\Delta T$:

   $0.180\ \text{kJ} = 0.100\ \text{kg} \times c \times 4\ \text{K}$

   Make c the subject of the equation and solve:

   $$c = \frac{0.180\ \text{kJ}}{0.100\ \text{kg} \times 4\ \text{K}} = 0.450\ \text{kJ kg}^{-1}\ \text{K}^{-1}$$

Practice questions

5. Calculate the energy absorbed by water when the temperature of 30 g of water is raised by 30 °C. The specific heat capacity of water is 4.18 J g^{-1} K^{-1}.
6. 0.675 kJ of heat is transferred to 125 g of copper metal. Copper metal has a specific heat capacity of 385 J kg^{-1} K^{-1}. Calculate the change in temperature of the copper metal.

Investigation to find the enthalpy change for a reaction

In this skills task, we will look at the method used to calculate the enthalpy change for the exothermic metal displacement reaction between zinc and copper(II) sulfate:

$$Zn(s) + CuSO_4(aq) \rightarrow Cu(s) + ZnSO_4(aq)$$

Relevant skills

- Tool 1: Measuring variables
- Tool 1: Applying techniques
- Tool 2: Applying technology to process data
- Tool 3: Processing uncertainties
- Tool 3: Graphing
- Inquiry 1: Controlling variables
- Inquiry 2: Processing data
- Inquiry 3: Evaluating

Materials

- electronic balance
- coffee-cup calorimeter
- measuring cylinder
- thermometer or temperature probe
- 1.0 mol dm^{-3} copper(II) sulfate solution
- zinc powder

Method

1. Using an electronic balance, accurately measure the mass of 25 cm^3 of 1.0 mol dm^{-3} $CuSO_4$ solution. Transfer the solution to the coffee-cup calorimeter.
2. Using a thermometer or a temperature probe, record the temperature of the solution every 30 seconds for up to three minutes, or until a constant temperature is achieved.
3. At three minutes, introduce between 1.3 g and 1.4 g of zinc powder, record the exact mass of zinc and commence stirring.
4. Continue to take temperature readings for up to five minutes after the maximum temperature has been reached.
5. Produce a temperature versus time graph to determine the change in temperature.
6. Use your value of ΔT to calculate the heat released, Q, and the enthalpy change for the reaction, ΔH.

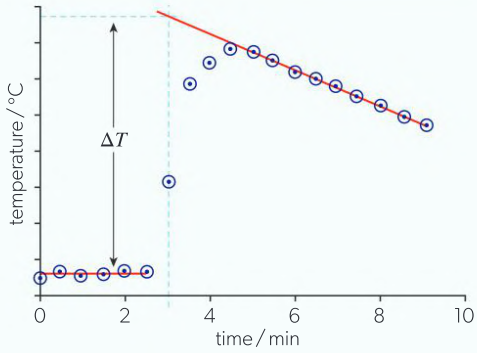
Assumptions and errors

A number of assumptions are made when using this method:

- The heat released from the reaction is completely transferred to the water.
- The coffee cup acts as an insulator against heat loss to the surroundings. However, the coffee cup also has a heat capacity and heat is transferred to it from the water, but it is much smaller than that of water.
- The maximum temperature reached is an accurate representation of the heat evolved during the reaction.
- The specific heat capacity of an aqueous solution is the same as that of water.

Loss of heat from the system to the surroundings is the main source of error in this experiment and one that is difficult to quantify. The change in temperature, ΔT, calculated from a graph will include a systematic error. This loss of heat means that the maximum temperature recorded will be lower than the theoretical value, making the calculated value of Q lower than the actual value. The effect of errors on the result of subsequent calculations is important in considering improvements in experimental procedures.

An accepted method of calculating the maximum temperature to compensate for systematic errors in data is to look at the cooling section of the curve after the reaction is complete, and extrapolate this back to the moment when zinc is introduced at 3 minutes, as shown in figure 9. A more accurate value for ΔT can then be determined.



▲ Figure 9 Example of a temperature vs time graph for a calorimetry experiment

Worked example 2

A coffee-cup calorimeter was used to measure the temperature change for the reaction between zinc powder and a 1.0 mol dm^{-3} solution of copper(II) sulfate. The following results were recorded:

Mass of copper(II) sulfate solution / g	28.8
Mass of zinc / g	1.37
ΔT / °C	39.0

Determine the amount of heat released and the enthalpy change for this reaction.

Solution

First, use $Q = mc\Delta T$ to determine the amount of heat released:

$$Q = 0.0288 \text{ kg} \times 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1} \times 39.0 \text{ K}$$

$$= 4.69 \text{ kJ}$$

Then, determine the limiting reactant for the reaction.

$$\text{Number of moles of zinc, } n(\text{Zn}) = \frac{m}{M_r}$$

$$= \frac{1.37 \text{ g}}{65.38 \text{ g mol}^{-1}}$$

$$= 0.0210 \text{ mol}$$

$$\text{Number of moles of copper(II) sulfate, } n(\text{CuSO}_4) = c \times v$$

$$= 1.00 \text{ mol dm}^{-3} \times 0.0288 \text{ dm}^3$$

$$= 0.0288 \text{ mol}$$

Zinc is present in a smaller amount, so it is the limiting reactant. You can calculate the enthalpy change of reaction from $\Delta H = -\frac{Q}{n}$:

$$\Delta H = -\frac{4.69 \text{ kJ}}{0.0210 \text{ mol}} = -223 \text{ kJ mol}^{-1}$$

TOK

In theory of knowledge, there are 12 concepts in focus. These are: evidence, certainty, truth, interpretation, power, justification, explanation, objectivity, perspective, culture, values and responsibility. Scientists perform experiments and process the raw data to enable us to draw conclusions. We compare experimental and theoretical values. What concepts do we utilize when justifying our conclusions? How do we use evidence? Are our judgments subjective or objective? When analysing and appraising experimental limitations, how do assumptions have an impact on our perceptions?

Combustion of primary alcohols

You can determine the enthalpy change of combustion of common alcohols in a school laboratory. After repeating the experiment several times with a homologous series of alcohols, you can subsequently analyse this data and identify patterns.

Relevant skills

- Tool 1: Recognise and address the relevant safety, ethical or environmental issues in an investigation
- Tool 1: Measuring temperature and mass
- Tool 1: Calorimetry
- Inquiry 1: Appreciate when and how to insulate against heat loss or gain
- Inquiry 2: Identify and record relevant qualitative observations and sufficient relevant quantitative data

Materials

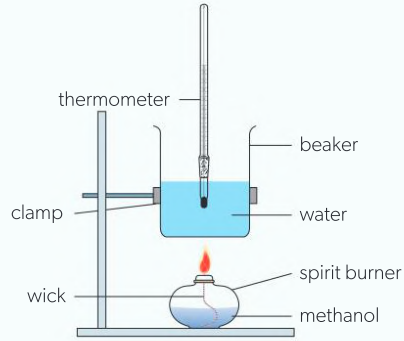
- five spirit burners, each containing one of the following alcohols: methanol, ethanol, propan-1-ol, butan-1-ol and pentan-1-ol
- electronic balance
- beaker or metal calorimeter
- tripod
- temperature probe or thermometer

Safety

Alcohols should be handled and disposed of with care because they are generally flammable, hazardous and volatile.

Instructions

1. Using suitable sources, identify the hazards and complete a risk assessment for this experiment. In your risk assessment, you should:
   - identify the **hazards**
   - assess the level of **risk**
   - determine relevant **control measures**
   - identify suitable **disposal methods** aligned with your school's health and safety policies.
2. Determine the initial mass of the spirit burners using an electronic balance.
3. Accurately determine the mass of 30 cm^3 of water contained in a 250 cm^3 beaker or metal calorimeter.
4. Using either a temperature probe or a thermometer, determine and record the initial temperature of the water.
5. Ignite a spirit burner under the beaker or calorimeter and allow the alcohol to burn to heat the water. The period over which it burns can be set in one of two different ways:
   a. allow each alcohol to burn until a temperature change of 30 °C is reached
   b. allow each alcohol to burn for a period of two minutes.
6. Determine the final mass of each spirit burner immediately after the flame is extinguished. Take extra care because the burner will be hot.
7. Use your values of ΔT of the water and Δm of the burner to calculate the amount of heat released, Q, and the enthalpy change of combustion, ΔH, for each alcohol.



▲ Figure 10 A typical arrangement of experimental apparatus for an enthalpy of combustion determination

ATL Research skills

Cite your sources fully, according to your school's citing and referencing system.

Worked example 3

A metal calorimeter was used to measure the amount of heat released by the combustion of methanol. The following results were recorded:

Mass of water / g	31.2
Change in mass of methanol / g	0.348
ΔT / °C	30.0

Determine the amount of heat released and the enthalpy change of combustion for methanol.

Solution

First, use $Q = mc\Delta T$ to determine the amount of heat released:

$$Q = 0.0312 \text{ kg} \times 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1} \times 30.0 \text{ K}$$

$$= 3.91 \text{ kJ}$$

Methanol reacts with oxygen in a combustion reaction. Methanol is the limiting reactant for this reaction because oxygen is present in excess in the air.

Number of moles of methanol, $n(CH_3OH) = \frac{m}{M_r}$

$$= \frac{0.348 \text{ g}}{32.05 \text{ g mol}^{-1}}$$

$$= 0.0109 \text{ mol}$$

You can calculate the enthalpy change of reaction from $\Delta H = -\frac{Q}{n}$:

$$\Delta H = -\frac{3.91 \text{ kJ}}{0.0109 \text{ mol}} = -359 \text{ kJ mol}^{-1}$$

Thermochemistry experiments provide a useful set of raw data, and involve experimental procedures that can be evaluated for random and systematic errors. The identification of the systematic errors and examination of their directionality are essential aspects of the analysis of experimental results. Calorimetry experiments typically give a smaller change in temperature than is predicted from theoretical values. This is the result of heat loss from the system, which is difficult to measure. Scientists usually make the assumption that the heat lost to the environment is negligible. TOK helps us to understand our judgments of discrepancies between experimental and theoretical values.

Thermometric titration

The neutralization reaction between an acid and a base is exothermic. In this skills task, you will determine the unknown concentration of hydrochloric acid by measuring the change in temperature while sodium hydroxide is added to the acid. The temperature will reach a maximum when the acid and base are mixed together in stoichiometric amounts.

Relevant skills

- Tool 1: Calorimetry and acid–base titration
- Tool 2: Use sensors
- Tool 3: Calculate and interpret percentage error
- Tool 3: Understand the significance of uncertainties in raw and processed data
- Tool 3: Propagate uncertainties and state them to an appropriate level of precision
- Tool 3: Extrapolate graphs
- Inquiry 1: Appreciate when and how to insulate against heat loss or gain
- Inquiry 3: Identify and discuss sources of systematic and random error

Safety

- Wear eye protection.
- Sodium hydroxide solution is corrosive.
- Hydrochloric acid is corrosive.

Materials

- two 250 cm^3 polystyrene cups
- thermometer or temperature probe
- graduated pipette and filler
- burette
- ~50.0 cm^3 sodium hydroxide solution of known concentration
- 30.0 cm^3 hydrochloric acid of unknown concentration

Method

1. Read through the safety, materials and method. Use this information, and relevant safety data, to complete a risk assessment for this practical work and show it to your teacher.
2. Review the titration, percentage error and uncertainties sections in the *Skills* chapter.
3. Rinse and fill the burette with sodium hydroxide solution. Record its concentration.
4. Add 25 cm^3 of acid solution to the cup and place it under the burette. Nest it inside a second cup, for additional thermal insulation. For safety, these cups should be placed inside a beaker to avoid tipping over.
5. Position the temperature probe in the acid and record the initial temperature of the acid in the cup.
6. Add a small volume (~5 cm^3) of sodium hydroxide solution to the acid, stirring gently. Record the highest temperature reached with this addition.
7. Continue adding small volumes of sodium hydroxide solution and recording the temperature until the temperature decreases over several consecutive readings.
8. Clear up as instructed by your teacher.

Questions

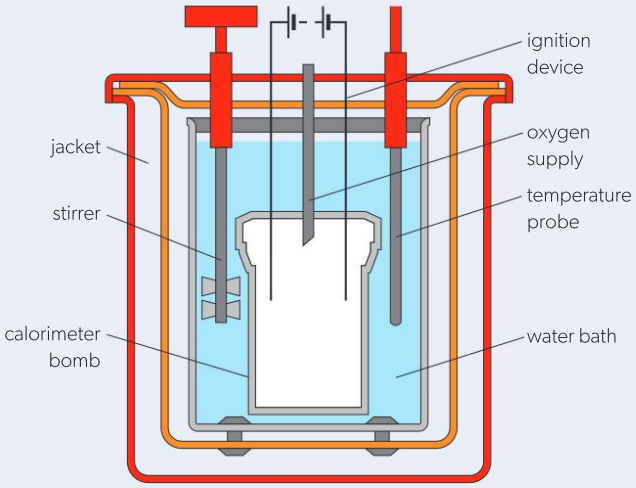
1. Plot a graph showing temperature vs volume of sodium hydroxide solution added.
2. Extrapolate the two sections of the graph to find the maximum temperature reached during the titration.
3. Determine the concentration of the acid, along with absolute and percentage uncertainties. Make sure you state all values to an appropriate level of precision.
4. Calculate the percentage error of your experimental acid concentration.
5. Determine the enthalpy of neutralization, along with absolute and percentage uncertainties. State all values to an appropriate level of precision.
6. Calculate the percentage error of your experimental enthalpy of neutralization.
7. Comment on the relative impacts of systematic and random error on the values obtained for the acid concentration and enthalpy of neutralization.
8. Suggest and explain two improvements that could be made to this method.

Measurement

Experimental enthalpy values can be assessed in terms of their accuracy and precision. Random errors in measurement lead to imprecision, whereas systematic errors cause inaccuracy. What are some of the sources of random and systematic errors in an enthalpy of neutralization experiment? To what extent are these errors quantifiable?

ATL Thinking skills

Calorimetry experiments conducted in research laboratories utilize the same principles as the calorimetry experiments described in this chapter. The instrument used is called a **bomb calorimeter** (figure 11). A sample is burned inside a chamber (called a "bomb"), and the resulting temperature change of the surrounding water is measured.



▲ Figure 11 Diagram of a bomb calorimeter used in research laboratories

1. Study the diagram carefully and list all the features that are labelled.
2. Deduce the purpose of each feature.
3. Consider why the measurements obtained with a bomb calorimeter are highly accurate and precise.
4. What properties of water make it suitable for calorimetry experiments?

End of topic questions

Topic review

1. Using your knowledge from the *Reactivity 1.1* topic, answer the guiding question as fully as possible:

   What can be deduced from the temperature change that accompanies chemical or physical change?

Exam-style questions

Multiple-choice questions

2. Which is correct for the following reaction?

   $2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$
   $\Delta H = -1049 \text{ kJ mol}^{-1}$

   A Reactants are less stable than products and the reaction is endothermic.
   B Reactants are more stable than products and the reaction is endothermic.
   C Reactants are more stable than products and the reaction is exothermic.
   D Reactants are less stable than products and the reaction is exothermic.

3. Which statement is correct?

   A In an exothermic reaction, the products have more energy than the reactants.
   B In an exothermic reversible reaction, the activation energy of the forward reaction is greater than that of the reverse reaction.
   C In an endothermic reaction, the products are more stable than the reactants.
   D In an endothermic reversible reaction, the activation energy of the forward reaction is greater than that of the reverse reaction.

4. Which statement is correct for this reaction?

   $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$
   $\Delta H = -26.6 \text{ kJ mol}^{-1}$

   A 13.3 kJ are released for every mole of Fe produced.
   B 26.6 kJ are absorbed for every mole of Fe produced.
   C 53.2 kJ are released for every mole of Fe produced.
   D 26.6 kJ are released for every mole of Fe produced.

5. In which reaction do the reactants have a lower energy than the products?

   A $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
   B $HBr(g) \rightarrow H(g) + Br(g)$
   C $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$
   D $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

Extended-response questions

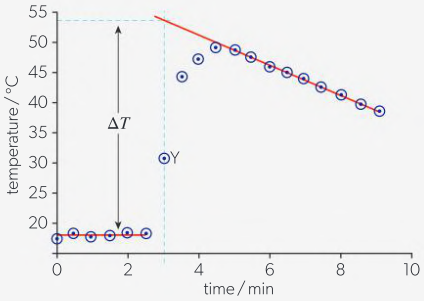
6. Nitrogen dioxide and carbon monoxide react according to the following equation:

$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$
$\Delta H = -226 \text{ kJ mol}^{-1}$

a. Calculate the enthalpy change for the reverse reaction. [1]

b. State the equation for the reaction of NO_2 in the atmosphere to produce acid deposition. [1]

7. Powdered zinc was reacted with 25.00 cm^3 of 1.000 mol dm^{-3} copper(II) sulfate solution in an insulated beaker. Temperature was plotted against time.



a. Estimate the time at which the powdered zinc was placed in the beaker. [1]

b. State what point **Y** on the graph represents. [1]

The maximum temperature used to calculate the enthalpy change of reaction was chosen at a point on the extrapolated (red) line.

c. State the maximum temperature that should be used, and outline **one** assumption made in choosing this temperature on the extrapolated line. [1]

d. To determine the enthalpy of reaction, the experiment was carried out five times. The same volume and concentration of copper(II) sulfate was used but the mass of zinc was different each time. Suggest, with a reason, if zinc or copper(II) sulfate should be in excess for each trial. [1]

The formula $q = mc\Delta T$ was used to calculate the amount of energy released.

The values used in the calculation were $m = 25.00$ g and $c = 4.18$ J g^{-1} K^{-1}.

e. State an assumption made when using these values for m and c. [1]

f. Predict, giving a reason, how the final enthalpy of reaction calculated from this experiment would compare with the theoretical value. [1]

8. A potato chip (crisp) was ignited, and the flame was used to heat a test tube containing water.

Mass of water/g	7.8
Mass of chip/g	1.2
Initial temperature/°C	21.3
Final temperature/°C	22.6

a. Calculate the heat required, in kJ, to raise the temperature of the water, using data in the table above and from section 2 of the data booklet. [1]

b. Determine the enthalpy of combustion of the potato chip, in kJ g^{-1}. [1]

Hess's law (*Reactivity 1.2.2*)

If you have travelled to New York City, Shanghai, Paris, Seoul, London, Madrid or Delhi, you have experienced the subways that criss-cross these enormous cities and transport millions of people every day. In any transport network there is more than one way to travel between point A and B. If you are an adventurous traveller, half the fun is often working out the fastest route.



▲ Figure 3 The extensive metro system in Seoul provides many alternative pathways for navigating the urban sprawl

The same idea can be true for chemistry. A chemical equation usually shows the net reaction – it can be a summary of a number of different reactions, which, when added together, result in an overall reaction.

For example, the overall reaction for the oxidation of elemental sulfur, S(s), to sulfur trioxide, $SO_3(l)$, is as follows:

$$S(s) + \frac{3}{2}O_2(g) \rightarrow SO_3(l)$$

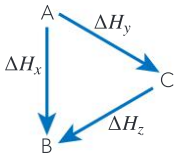
However, this process is difficult to carry out in one step. Typically, sulfur is oxidized first to sulfur dioxide, $SO_2(g)$, and then sulfur dioxide is oxidized further to sulfur trioxide:

Step 1: $S(s) + O_2(g) \rightarrow SO_2(g)$

Step 2: $SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(l)$

If you add up the enthalpy changes that occur in each of these individual steps, the total would be the same as the enthalpy change of the overall reaction. This is the basis of Hess's law:

Regardless of the route by which a chemical reaction proceeds, the enthalpy change will always be the same, as long as the initial and final states of the system are the same.



▲ Figure 4 Hess's law

Hess's law is an application of the conservation of energy law. Figure 4 illustrates Hess's law: the enthalpy change for reaction A → B (ΔH_x) is equal to the sum of the enthalpy changes for the reactions A → C and C → B ($\Delta H_y + \Delta H_z$).

Hess's law can be applied to find the unknown enthalpy change for a given reaction by combining other reactions with known enthalpy changes.

Worked example 2

The reaction for the formation of methanol is shown below:

$C(s) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3OH(l)$

Use Hess's law and equations 1–3 to calculate the enthalpy change for the reaction above.

1	$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$	$\Delta H = -726\,kJ\,mol^{-1}$
2	$C(s) + O_2(g) \rightarrow CO_2(g)$	$\Delta H = -394\,kJ\,mol^{-1}$
3	$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	$\Delta H = -286\,kJ\,mol^{-1}$

Solution

There are two methods for using Hess's law for a question like this: the **summation of equations method**, and the **enthalpy cycle diagram method**.

Summation of equations method

First, inspect the formation of methanol reaction scheme. In your calculation, you need to use enthalpy values corresponding to reactions that involve carbon, hydrogen and oxygen as reactants, and methanol as a product.

Start with the reactants. Carbon is a reactant in equation 2, so this equation can be used as written:

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -394\,kJ\,mol^{-1}$$

You require two moles of hydrogen as a reactant. Therefore, equation 3 can be used in the direction as written, but with doubled stoichiometric coefficients. This means that the enthalpy value must also be doubled:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l) \qquad \Delta H = -572\,kJ\,mol^{-1}$$

We also need half a mole of oxygen. Oxygen is present as a reactant in two of the above equations, so ignore this for now.

For the product methanol, you need to use equation 1. However, the equation must be reversed to make methanol a product, not a reactant. When reversing the chemical equation, the sign of the enthalpy change must also be changed:

$$CO_2(g) + 2H_2O(l) \rightarrow CH_3OH(l) + \frac{3}{2}O_2(g) \qquad \Delta H = +726\,kJ\,mol^{-1}$$

Now add the three equations together, cancelling the species common to both sides of the reaction, and add the enthalpy values:

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -394\,kJ\,mol^{-1}$$

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l) \qquad \Delta H = -572\,kJ\,mol^{-1}$$

$$CO_2(g) + 2H_2O(l) \rightarrow CH_3OH(l) + \frac{3}{2}O_2(g) \qquad \Delta H = +726\,kJ\,mol^{-1}$$

Total $C + \cancel{2O_2} \frac{1}{2}O_2 + 2H_2 + \cancel{CO_2} + \cancel{2H_2O} \rightarrow \cancel{CO_2} + \cancel{2H_2O} + CH_3OH + \frac{3}{2}\cancel{O_2}$ $\qquad \Delta H = -240\,kJ\,mol^{-1}$

Cancelling the common species gives the overall equation $C + 2H_2 + \frac{1}{2}O_2 \rightarrow CH_3OH$, which is identical to the reaction in the question. Therefore, the total enthalpy change for the reaction is equal to –240 kJ mol^{-1}. Note that here and below the states are omitted to save space.

Enthalpy cycle diagram method

You can represent the calculations needed to determine the enthalpy change of reaction in the form of an enthalpy cycle diagram.

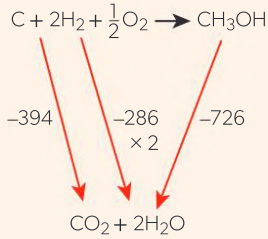
First, write the equation you are trying to find the enthalpy change for: $C + 2H_2 + \frac{1}{2}O_2 \rightarrow CH_3OH$. Below this, write the other species from equations 1, 2 and 3 not included in the original reaction: $CO_2 + 2H_2O$. Oxygen, O_2, is usually not included in enthalpy cycle diagrams.

Then, draw two arrows from the reactants to the species at the bottom to represent equations 2 and 3. Draw an arrow from the products to the species at the bottom to represent equation 1.

Finally, label the arrows with the enthalpy change values. As with the summation of equations method, the enthalpy change for equation 3 needs to be doubled to give the value for two moles of hydrogen. The resulting enthalpy cycle diagram is shown in figure 5.

Then, calculate the sum of the enthalpy change values, following the alternative pathway from the reactants to the products, via the CO_2 and $2H_2O$ intermediates. If the pathway you are following is in the opposite direction of an arrow (in this case, the last arrow), reverse the sign of the enthalpy change. This calculation is summarized below:

$$(-394) + (-286 \times 2) + (+726) = -240\,\text{kJ mol}^{-1}$$



▲ Figure 5 Enthalpy cycle diagram for the formation of methanol

ATL Communication skills

In the previous worked example, you have seen two methods for representing and solving Hess's law calculations: the summation of equations method and the enthalpy cycle diagram method. In what situations would you use an enthalpy cycle? In what situations would you use the summation of equations method? What are the strengths and limitations of each method? To what extent are the methods different if they represent the same concept?

Whichever method you prefer, make sure you show your reasoning clearly in assessments. Often candidates make arithmetical errors when calculating the enthalpy of reaction. It is advisable to clearly show your full working rather than simply recording the final answer. This gives the examiner the opportunity to assign partial marks where applicable.

Practice questions

4. Calculate the enthalpy change for the following reaction, using the summation of equations method, and equations 1–3 below:

$$3H_2(g) + 2C(s) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$$

1	$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$	$\Delta H = -1367\,\text{kJ mol}^{-1}$
2	$C(s) + O_2(g) \rightarrow CO_2(g)$	$\Delta H = -394\,\text{kJ mol}^{-1}$
3	$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	$\Delta H = -286\,\text{kJ mol}^{-1}$

5. Determine the enthalpy change for the following reaction, using the enthalpy cycle diagram method and equations 1–3 below:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(l)$$

1	$N_2(g) + O_2(g) \rightarrow 2NO(g)$	$\Delta H = +66\,\text{kJ mol}^{-1}$
2	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	$\Delta H = -92\,\text{kJ mol}^{-1}$
3	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	$\Delta H = -572\,\text{kJ mol}^{-1}$

Using Hess's law to find enthalpy change

When heated, potassium hydrogencarbonate decomposes to produce potassium carbonate, carbon dioxide and water. The enthalpy change for this decomposition is difficult to determine directly. You will determine the enthalpy change for two other reactions, and then apply Hess's law to find the enthalpy change for the decomposition of potassium hydrogencarbonate.

Relevant skills

- Tool 1: Calorimetry
- Tool 3: Record, propagate and express uncertainties
- Inquiry 2: Carry out relevant and accurate data processing
- Inquiry 3: Evaluate the implications of methodological weaknesses, limitations and assumptions on the conclusions

Safety

- Wear eye protection.
- Hydrochloric acid is an irritant.
- Anhydrous potassium carbonate is an irritant.
- The reaction between carbonates and acids effervesces. Cover the reaction vessel with a lid to prevent loss of reagents and spillage.

Materials

- 2 mol dm^{-3} hydrochloric acid
- anhydrous potassium carbonate (solid)
- potassium hydrogencarbonate (solid)

Method

1. Read through the method and write a list of equipment required for the practical. Show it to your teacher for approval before starting.
2. Place approximately 3 g of potassium carbonate into a weighing boat. Record the exact mass.
3. Use a measuring cylinder to collect 30 cm^3 of hydrochloric acid and transfer it into a coffee-cup calorimeter. Record the temperature of the acid.
4. Add the potassium carbonate to the acid solution and stir the mixture, monitoring the temperature. Record the maximum temperature reached.
5. Dispose of the reaction mixture appropriately.
6. Repeat the procedure, using potassium hydrogencarbonate instead of potassium carbonate. (Note that this time, record the lowest temperature reached because the reaction is endothermic.)

Analysis

Part 1: Known enthalpy changes of reaction

1. Write a balanced equation, including state symbols, for the reaction between:
   a. potassium carbonate and hydrochloric acid
   b. potassium hydrogencarbonate and hydrochloric acid
2. Show that the acid is in excess in both cases.
3. Process your experimental data to determine the enthalpy change for the reaction between:
   a. potassium carbonate and hydrochloric acid
   b. potassium hydrogencarbonate and hydrochloric acid

Part 2: Applying Hess's law

4. Write a balanced equation, including state symbols, for the decomposition of potassium hydrogencarbonate to produce potassium carbonate and other products.
5. Draw an enthalpy cycle connecting the three reactions:
   - potassium carbonate and hydrochloric acid
   - potassium hydrogencarbonate and hydrochloric acid
   - thermal decomposition of potassium hydrogencarbonate
6. Apply Hess's law, and your processed data from part 1, to determine the enthalpy change for the thermal decomposition of potassium hydrogencarbonate.
7. Propagate uncertainties to determine the absolute uncertainty of the enthalpy values you have obtained.
8. Look up the theoretical value for the enthalpy change for the decomposition of potassium hydrogencarbonate online. Use this value to determine the percentage error of your result.
9. Identify two assumptions made throughout this investigation and discuss their validity.
10. Identify and explain two major sources of systematic error in this investigation.

AHL

Standard enthalpy changes of combustion, $\Delta H_c^\ominus$, and formation, $\Delta H_f^\ominus$ (*Reactivity 1.2.3 and Reactivity 1.2.4*)

When describing a standard enthalpy change, there are a number of steps that must be completed.

1. Determine the type of reaction.
2. Write an equation to describe the reaction.
3. Include state symbols in the equation.

The **standard enthalpy change of combustion**, $\Delta H_c^\ominus$, is the enthalpy change that occurs when one mole of a substance in its standard state is burned completely in oxygen. The standard state of a pure substance is the form that it takes under standard conditions: 25.00 °C (298.15 K), which is taken as being room temperature, and a pressure of 1.00×10^5 Pa.

When a **hydrocarbon** undergoes combustion, it reacts with oxygen to produce carbon dioxide, CO_2, and water, H_2O. Butane, C_4H_{10}, is a highly flammable gas and a component of liquefied petroleum gas (LPG). The equation for its combustion is as follows:

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$$

The standard enthalpy of combustion for butane, $\Delta H_c^\ominus$, is $-2878\ \text{kJ mol}^{-1}$. This value, and the standard enthalpies of combustion for other substances, are given in section 14 of the data booklet, along with their standard states.

Exothermic and endothermic reactions are introduced in *Reactivity 1.1*.

The chemical equation for the combustion of butane can also be written with the enthalpy of combustion value included in the equation. The negative enthalpy change indicates an **exothermic** reaction, so energy is released into the surroundings. Therefore, the value is included on the product side:

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l) + 2878\ \text{kJ mol}^{-1}$$

Activity

Write equations to describe the standard enthalpy change of combustion for the following compounds, and state the enthalpy change values by referring to section 14 of the data booklet. Ensure that you include state symbols and that the equations are balanced:

a. octane, C_8H_{18}
b. cyclohexanol, $C_6H_{12}O$
c. methanoic acid, CH_2O_2
d. glucose, $C_6H_{12}O_6$
e. chloroethane, C_2H_5Cl (hint: a strong, corrosive acid is one of the products)

Global impact of science

Much of the world's energy supply is produced by the combustion of hydrocarbons. Increasingly, the scientific community is researching alternative energy sources to reduce harmful emissions of greenhouse gases. This will enable governments to reach emission targets set by international agreements, such as COP21 in Paris and COP26 in Glasgow.

The combustion of ammonia is being investigated as a possible carbon-neutral fuel source. However, one problem with this new technology is the high level of NO_x emissions. Industrial processes are rarely totally "green": there are environmental costs associated with every method of energy production. How do we measure the environmental impact of different fuel sources, and which factors are most important for a fuel to be sustainable? Who should make decisions about what fuel sources are used to produce energy?

The **standard enthalpy change of formation**, $\Delta H_f^\ominus$, of a substance is the energy change that occurs when one mole of a substance is formed from its constituent elements in their standard states. Most elements are solids in their standard state, but there are 11 elements that are gases under standard conditions, for example argon, Ar, and fluorine, F_2. Two elements are liquid under standard conditions: mercury, Hg, and bromine, Br_2.

The constituent elements of butane are carbon, C, and hydrogen, H_2. Therefore, the formation of butane is described by the following equation:

$$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g)$$

The enthalpy change of formation of butane, $\Delta H_f^\ominus$, is $-126\ kJ\ mol^{-1}$. This value, and other values for the standard enthalpy change of formation for many common compounds, can be found in section 13 of the data booklet. The standard states of these compounds are given in the same section.

Activity

Write equations to describe the standard enthalpy change of formation for the following compounds, and state the enthalpy change values by referring to section 13 of the data booklet:

a. propane, C_3H_8
b. chloromethane, CH_3Cl
c. ethanol, C_2H_6O
d. benzoic acid, C_6H_5COOH
e. carbon monoxide, CO
f. methylamine, CH_3NH_2

You will notice that when writing chemical equations for standard enthalpy changes of combustion and formation, fractional stoichiometric coefficients may be necessary to balance the equation. This is one of the few cases when it is correct to use fractions to balance the final chemical equation.

AHL

The allotropes of carbon and their structures are discussed in *Structure 2.2*.

Under standard conditions, the element carbon exists as several allotropes, for example, diamond, graphite and buckminsterfullerene. Allotropes of carbon have different $\Delta H_f^\ominus$ values. If an element has several allotropes, the normal convention is that the most stable allotrope is the standard state. The enthalpy of formation is 0 kJ mol^{-1} for all elemental substances in their standard states. Graphite is taken as the standard for the carbon allotropes, and so it has a standard enthalpy of formation of 0 kJ mol^{-1}.

ATL Thinking skills

Reflect on the rationale for averaging bond enthalpy values. Evaluate the strengths and limitations of this approach.

Linking question

Would you expect allotropes of an element, such as diamond and graphite, to have different $\Delta H_f^\ominus$ values? (Structure 2.2)

Applying Hess's law to enthalpy changes of combustion

As discussed earlier in this topic, enthalpy is a state function, so the enthalpy change of a reaction is the difference between the initial and final values of the enthalpy of the system. The pathway between this initial and final state of a chemical reaction does not affect the overall enthalpy change.

The overall standard enthalpy change for a reaction, $\Delta H_r^\ominus$, can be calculated using enthalpy of combustion data. The difference between the sum of the enthalpy changes of combustion of the reactants and the sum of the enthalpy changes of combustion of the products is equal to the overall enthalpy change of the reaction:

$$\Delta H_r^\ominus = \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})$$

This is a direct application of Hess's law.

Practice question

6. Using Hess's law and enthalpy of combustion data, the enthalpy change of formation of pentane is determined to be −177 kJ mol^{-1}, but the value given in the data booklet is −173 kJ mol^{-1}. Suggest why the values are different.

Applying Hess's law to enthalpy changes of formation

The standard enthalpy change for a reaction can also be calculated using enthalpy of formation data. The difference between the sum of the enthalpy changes of formation of the products and the sum of the enthalpy changes of formation of the reactants is equal to the overall enthalpy change of the reaction:

$$\Delta H_r^\ominus = \sum(\Delta H_f^\ominus \text{ products}) - \sum(\Delta H_f^\ominus \text{ reactants})$$

Worked example 3

Calculate the standard enthalpy change of formation of pentane, C_5H_{12}, using the enthalpy of combustion data from section 14 of the data booklet.

Solution

Step 1: Write a balanced chemical equation for the formation of 1 mol of pentane: $5C(s) + 6H_2(g) \rightarrow C_5H_{12}(l)$

Step 2: Write equations for the combustion of carbon, hydrogen and pentane and find their standard enthalpy change values from section 14 of the data booklet.

1	$C(s) + O_2(g) \rightarrow CO_2(g)$	$\Delta H^\ominus = -394\,kJ\,mol^{-1}$
2	$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	$\Delta H^\ominus = -286\,kJ\,mol^{-1}$
3	$C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$	$\Delta H^\ominus = -3509\,kJ\,mol^{-1}$

Step 3: Use these equations with either the summation of equations method or enthalpy cycle diagram method to determine the enthalpy change for the reaction. These methods were introduced earlier in this topic.

Summation of equations method

Inspect the formation of pentane reaction scheme. In your calculation, you need to use enthalpy values corresponding to the combustion reactions that involve carbon, hydrogen and pentane. Carbon is a reactant in equation 1, but five moles are required for the overall equation. Therefore you should multiply the enthalpy of combustion of carbon by five:

$$5C(s) + 5O_2(g) \rightarrow 5CO_2(g) \qquad \Delta H = -1970\,kJ\,mol^{-1}$$

You also require six moles of hydrogen as a reactant. Equation 2 should be used, with the enthalpy change multiplied by six:

$$6H_2(g) + 3O_2(g) \rightarrow 6H_2O(l) \qquad \Delta H_c^\ominus = -1716\,kJ\,mol^{-1}$$

For the product pentane, you should use equation 1, but the equation must be reversed to make pentane a product, not a reactant. As a result, the enthalpy change sign changes from negative to positive:

$$5CO_2(g) + 6H_2O(l) \rightarrow C_5H_{12}(l) + 8O_2(g) \qquad \Delta H^\ominus = +3509\,kJ\,mol^{-1}$$

Now add the three equations together, cancelling the species common to both sides of the reaction, and add the enthalpy values:

	$5C(s) + 5O_2(g)$	$\rightarrow$	$5CO_2(g)$	$\Delta H_c^\ominus = -1970\,kJ\,mol^{-1}$
	$6H_2(g) + 3O_2(g)$	$\rightarrow$	$6H_2O(l)$	$\Delta H_c^\ominus = -1716\,kJ\,mol^{-1}$
	$5CO_2(g) + 6H_2O(l)$	$\rightarrow$	$C_5H_{12}(l) + 8O_2(g)$	$\Delta H^\ominus = +3509\,kJ\,mol^{-1}$
Total	$5C + \cancel{5O_2} + 6H_2 + \cancel{3O_2} + \cancel{5CO_2} + \cancel{6H_2O}$	$\rightarrow$	$\cancel{5CO_2} + \cancel{6H_2O} + C_5H_{12} + \cancel{8O_2}$	$\Delta H^\ominus = -177\,kJ\,mol^{-1}$

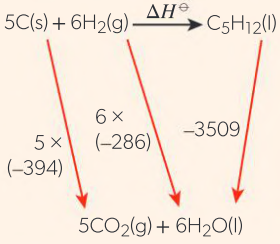
Cancelling the common species gives the overall equation $5C(s) + 6H_2(g) \rightarrow C_5H_{12}(l)$. Therefore, the total enthalpy change for the reaction is $-177\,kJ\,mol^{-1}$.

Enthalpy cycle diagram method

The enthalpy cycle diagram is on the right. Oxygen, $O_2(g)$, is not included in enthalpy cycle diagrams.

Calculate the sum of the enthalpy change values, following the arrows from the reactants to the products, via the $CO_2(g)$ and $H_2O(l)$ intermediates. If you are going against the direction of an arrow, reverse the sign of the enthalpy change.
This calculation is summarized below:

$$\Delta H^\ominus = 5 \times (-394) + 6 \times (-286) + 3509 = -177\,kJ\,mol^{-1}$$



AHL

Worked example 4

Calculate the enthalpy of combustion of pentane, $C_5H_{12}(l)$, using the enthalpy of formation data from section 13 of the data booklet.

Solution

Step 1: Write a balanced chemical equation for the combustion of 1 mol of pentane:

$$C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$$

Step 2: Write equations for the formation of pentane, carbon dioxide, and water from their elements and find their standard enthalpy change values from section 12 of the data booklet.

1	$5C(s) + 6H_2(g) \rightarrow C_5H_{12}(l)$	$\Delta H_f^\ominus = -173\ kJ\ mol^{-1}$
2	$C(s) + O_2(g) \rightarrow CO_2(g)$	$\Delta H_f^\ominus = -394\ kJ\ mol^{-1}$
3	$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	$\Delta H_f^\ominus = -286\ kJ\ mol^{-1}$

Step 3: Use these equations with either the summation of equations method or enthalpy cycle diagram method to determine the enthalpy change for the reaction.

Summation of equations method

In your calculation, you need to use enthalpy values corresponding to the formation reactions that involve pentane, carbon dioxide and water. Pentane is a reactant in the overall equation, but a product in equation 1. Therefore, equation 1 must be reversed. As a result, the enthalpy change sign changes from negative to positive:

$$C_5H_{12}(l) \rightarrow 5C(s) + 6H_2(g) \qquad \Delta H^\ominus = +173\ kJ\ mol^{-1}$$

Carbon dioxide is a product in an equation 2, but five moles are required for the overall equation. Therefore, you should multiply the enthalpy of formation of carbon dioxide by five:

$$5C(s) + 5O_2(g) \rightarrow 5CO_2(g) \qquad \Delta H_f^\ominus = -1970\ kJ\ mol^{-1}$$

You require six moles of water as a product. Equation 3 should be used, with the enthalpy change multiplied by six:

$$6H_2(g) + 3O_2(g) \rightarrow 6H_2O(l) \qquad \Delta H_f^\ominus = -1716\ kJ\ mol^{-1}$$

Now add the three equations together, cancelling the species common to both sides of the reaction, and add the enthalpy values:

	$C_5H_{12}(l)$	→	$5C(s) + 6H_2(g)$	$\Delta H^\ominus = +173\ kJ\ mol^{-1}$
	$5C(s) + 5O_2(g)$	→	$5CO_2(g)$	$\Delta H_f^\ominus = -1970\ kJ\ mol^{-1}$
	$6H_2(g) + 3O_2(g)$	→	$6H_2O(l)$	$\Delta H_f^\ominus = -1716\ kJ\ mol^{-1}$
Total	$C_5H_{12} + 5C + 5O_2 + 6H_2 + 3O_2$	→	$5C + 6H_2 + 5CO_2 + 6H_2O$	$\Delta H^\ominus = -3513\ kJ\ mol^{-1}$

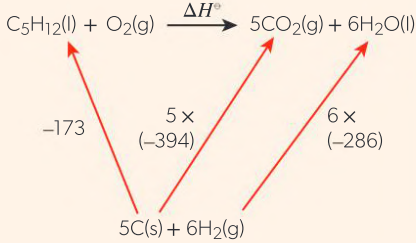
Cancelling the common species gives the overall equation $C_5H_{12}(l) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(l)$. Therefore, the total enthalpy change for the reaction is $-3513\ kJ\ mol^{-1}$.

Enthalpy cycle diagram method

The enthalpy cycle diagram is on the right.

Calculate the sum of the enthalpy change values, following the arrows from the reactants to the products, via the C(s) and $H_2(g)$ intermediates. If you are going against the direction of an arrow, reverse the sign of the enthalpy change. This calculation is summarized below:

$$\Delta H^\ominus = (+173) + 5 \times (-394) + 6 \times (-286) = -3513\ kJ\ mol^{-1}$$



Practice questions

7. Calculate the enthalpy change of formation of propanone, $CH_3COCH_3(l)$, using enthalpy change of combustion data from section 14 of the data booklet.

   Use the summation of equations method and the enthalpy cycle diagram method to support your answer.

8. Hydrogen gas can be made using the following reaction:

   $CH_4(g) + H_2O(g) \rightarrow 3H_2(g) + CO(g)$

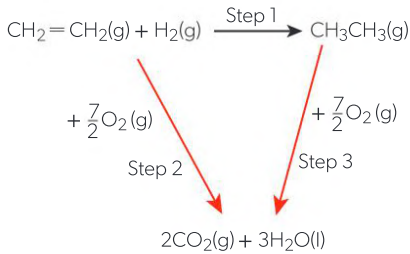
   Section 13 of the data booklet lists the standard enthalpies of formation, $\Delta H_f^\ominus$, for some of the species in the reaction above.

   a. Outline why no value is listed for $H_2(g)$.

   b. Determine the value of $\Delta H^\ominus$, in kJ mol^{-1}, for the reaction using the standard enthalpy change of formation values.

   c. Outline why the value of enthalpy change of this reaction calculated from bond enthalpies is less accurate.

9. Consider the following enthalpy cycle diagram:



   a. Calculate the standard enthalpy change, $\Delta H^\ominus$, of step 2 using section 14 of the data booklet.

   b. Determine the standard enthalpy change, $\Delta H^\ominus$, of step 1.

   c. Suggest one reason why the calculated value of $\Delta H^\ominus$ using Hess's law in part **b** can be considered accurate and one reason why it can be considered approximate.

Comparing ΔH values

The standard enthalpy of combustion of propan-1-ol is −2021 kJ mol^{-1}. In this task, you will compare this value to those obtained with some of the methods described in this topic.

Relevant skills

- Tool 3: Use arithmetic and algebraic calculations to solve problems
- Tool 3: Calculate and interpret percentage error
- Inquiry 3: Compare the outcomes of an investigation to the accepted scientific context
- Inquiry 3: Identify and discuss sources and impacts of systematic and random errors

Method

1. Determine the enthalpy of combustion of propan-1-ol by each of the following methods:

   a. Refer to average bond enthalpy values.

   b. Conduct a calorimetry experiment. This could be real (as described in *Reactivity 1.1*), or simulated, if you have access to a suitable simulation software. Record the relevant measurement uncertainties and propagate them.

   c. Refer to standard enthalpies of formation.

2. Determine the percentage error of each of the values obtained in methods **a**, **b** and **c**.

3. Discuss and explain the differences between the standard enthalpy of combustion of propan-1-ol and each of your calculated values obtained by methods **a**, **b** and **c**.

Reactivity 1.2 Energy cycles in reactions

How does application of the law of conservation of energy help us to predict energy changes during reactions?

The utilization of energy is central to our lives. Agricultural, industrial, and domestic activities all consume vast amounts of energy daily. **Thermodynamics** is the study of energy and its interconversions. The first law of thermodynamics states that energy can be converted from one form to another and that the total amount of energy for a given system remains constant.

This law is often called the **law of conservation of energy**. According to this law, energy can be neither created nor destroyed; it can only be converted from one form to another. This means that for a given reaction we can account for and quantify all the energy changes. This is one of the most fundamental principles of science.

Understandings

Reactivity 1.2.1—Bond-breaking absorbs and bond-forming releases energy.

Reactivity 1.2.2—Hess's law states that the enthalpy change for a reaction is independent of the pathway between the initial and final states.

AHL

Reactivity 1.2.3—Standard enthalpy changes of combustion, $\Delta H_c^\ominus$, and formation, $\Delta H_f^\ominus$, data are used in thermodynamic calculations.

Reactivity 1.2.4—An application of Hess's law uses enthalpy of formation data or enthalpy of combustion data to calculate the enthalpy change of a reaction

Reactivity 1.2.5—A Born–Haber cycle is an application of Hess's law, used to show energy changes in the formation of an ionic compound.

Bond-breaking and bond-forming (*Reactivity 1.2.1*)

When a chemical reaction takes place, the atoms of the reactants are rearranged to create products. Chemical bonds in the reactants are broken and new chemical bonds are made to form the products. Energy is required to break the chemical bonds: bond-breaking is an **endothermic** process. Energy is released when new chemical bonds form: bond-making is an **exothermic** process. This transfer of energy between the system and the surroundings is an essential part of understanding the energy changes involved in a chemical reaction (*Reactivity 1.1*).

Laws and theories

The **law of conservation of energy** states that energy can be neither created nor destroyed; it can only be converted between different forms. Laws allow predictions to be made but, unlike theories, laws do not have explanatory power. What other laws have you come across in your study of chemistry?

Bond enthalpy

Imagine a simple hydrogen molecule, H_2. The breaking of the hydrogen molecule into individual hydrogen atoms requires energy. The **bond enthalpy (*BE*)** is defined as the energy required to break one mole of bonds by **homolytic fission** in one mole of gaseous covalent molecules under standard conditions. The process of homolytic fission distributes the electrons from the bond equally between the two new species. This results in the formation of radicals, indicated by the • symbol.

For example, the bond enthalpy of the H–H bond is equal to the enthalpy change of the following reaction:

$$H_2(g) \rightarrow 2H\bullet(g) \qquad \Delta H = +436\,kJ\,mol^{-1}$$

Bond-breaking is an endothermic process, and therefore it has a **positive enthalpy value**.

Selected *BE* values are provided in table 1 (on the next page) and section 12 of the data booklet. Bond enthalpy values are derived from experimental data on the breaking of the same bond in a wide variety of compounds. For example, the C–H bond enthalpy will vary through the alkane series because the chemical environment of the individual bonds changes.

If a molecule of methane underwent a series of steps in which one hydrogen atom was removed at a time, the bond enthalpy for each removal would be different. This is because the chemical environment changes upon the removal of successive hydrogen atoms.

$$CH_4(g) \rightarrow \bullet CH_3(g) + \bullet H(g) \qquad BE_1(C\text{–}H) = +439\,kJ\,mol^{-1}$$

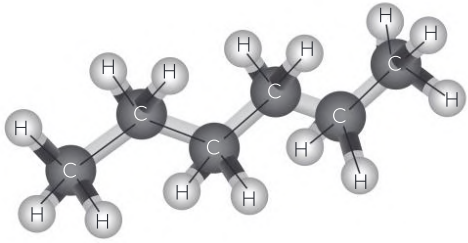
$$\bullet CH_3(g) \rightarrow \bullet CH_2(g) + \bullet H(g) \qquad BE_2(C\text{–}H) = +462\,kJ\,mol^{-1}$$

$$\bullet CH_2(g) \rightarrow \bullet CH(g) + \bullet H(g) \qquad BE_3(C\text{–}H) = +424\,kJ\,mol^{-1}$$

$$\bullet CH(g) \rightarrow \bullet C(g) + \bullet H(g) \qquad BE_4(C\text{–}H) = +338\,kJ\,mol^{-1}$$

Can you recognize the differences in the products of each equation and explain how the chemical environment changes for each successive step?

Bond enthalpies are **average values** and are therefore only an approximation. The average bond enthalpy value of the C–H bond is $+414\,kJ\,mol^{-1}$.



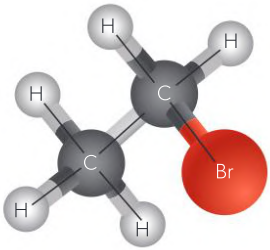
▲ Figure 1 When you examine the structure of this hexane molecule, can you see how the chemical environment of the C–H bonds differs throughout the molecule?

Bond	Average bond enthalpy / kJ mol^{-1}	Bond	Average bond enthalpy / kJ mol^{-1}
H–H	436	C=O	804
O–O	144	C=N	615
O=O	498	C≡N	890
O–H	463	N–N	158
C–H	414	N=N	470
C–C	346	N≡N	945
C=C	614	Cl–Cl	242
C≡C	839	Br–Br	193
C–O	358	I–I	151

▲ Table 1 Average bond enthalpies at 298 K

TOK

Measured energy changes in chemical reactions can be explained by the model of bonds broken and bonds formed. These explanations are based on empirical data and further data can be obtained to modify and refine the model. What role do models play in the acquisition of knowledge in the natural sciences? How do we determine whether a model is useful? How are models used in other areas of knowledge?



▲ Figure 2 A molecule of bromoethane, C_2H_5Br, has one C–C bond, one C–Br bond and five C–H bonds

You can use bond enthalpy data to calculate the enthalpy change of reaction, ΔH:

$$\Delta H = \Sigma(BE \text{ of bonds broken}) - \Sigma(BE \text{ of bonds formed})$$

The enthalpy change of a reaction calculated from bond enthalpies will differ from its actual value, as bond enthalpies are average values. Additionally, bond enthalpy data do not take intermolecular forces into account, which is particularly important when substances in a reaction are solid or liquid.

When performing bond enthalpy calculations, it is useful to draw out the full structural formulas of the reactants and products involved in the reaction. From this, you can determine the type and number of each chemical bond present.

Models

The model of bond breaking and formation is used to represent the energy changes taking place within a reaction system. We can assess the extent to which this model is in agreement with empirical data, by comparing the theoretical values generated from bond energy data with experimental evidence. Using empirical data to replace or modify proposed models is a central methodology in science.

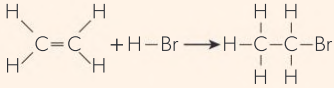
Worked example 1

Using bond enthalpy data, determine the enthalpy change of reaction for the following process:

$$C_2H_4(g) + HBr(g) \rightarrow C_2H_5Br(g)$$

Solution

Start by drawing the full structural formulas of all of the reactants and products:



In the reactants, there are four C–H bonds, one C=C bond and one H–Br bond. In the product, there are five C–H bonds, one C–C bond, and one C–Br bond. Therefore, the enthalpy change for the reaction is as follows:

$$\Delta H = \Sigma(BE \text{ of bonds broken}) - \Sigma(BE \text{ of bonds formed})$$

$$= [4BE(\text{C–H}) + BE(\text{C=C}) + BE(\text{H–Br})] - [5BE(\text{C–H}) + BE(\text{C–C}) + BE(\text{C–Br})]$$

Then, substitute in the BE values from the data booklet:

$$= [(4 \times 414) + 614 + 366] - [(5 \times 414) + 346 + 285]$$

$$= 2636 - 2701$$

$$= -65 \text{ kJ mol}^{-1}$$

Practice questions

1. The chloralkali process is an industrial process used to make chlorine gas and sodium hydroxide. Much of the chlorine gas is combined with hydrogen gas to produce hydrogen chloride:

   $$Cl_2(g) + H_2(g) \rightarrow 2HCl(g)$$

   The enthalpy change for this reaction is -185 kJ mol^{-1}. Using section 12 of the data booklet, calculate the bond enthalpy, in kJ mol^{-1}, for the H–Cl bond.
2. Enthalpy changes depend on the number and type of bonds broken and formed. Hydrogen gas can be formed industrially by the reaction of natural gas with steam:

   $$CH_4(g) + H_2O(g) \rightarrow 3H_2(g) + CO(g)$$

   Determine the enthalpy change, ΔH, for this reaction, in kJ mol^{-1}, using section 12 of the data booklet.
3. Methane and chlorine react to produce chloromethane and hydrogen chloride.
   a. Write the balanced chemical equation for this reaction.
   b. Using bond enthalpy values from the data booklet, determine the enthalpy change for this reaction.
   c. Deduce and explain whether this reaction is exothermic or endothermic.
   d. State which are more energetically stable, the reactants or the products.

Linking question

How would you expect bond enthalpy data to relate to bond length and polarity? (*Structure 2.2*)

How does the strength of a carbon–halogen bond affect the rate of a nucleophilic substitution reaction? (*Reactivity 3.4*)

Lattice enthalpy

Lattice enthalpy values tell us how strong the ionic bonds are in particular ionic lattice. Lattice enthalpy, $\Delta H^{\ominus}_{\text{lattice}}$, is the standard enthalpy change that occurs on the formation of gaseous ions from one mole of the solid lattice. It is a measure of the strength of an ionic bond because, in order for the ions to become gaseous, all the electrostatic forces of attraction between cations and anions in the lattice need to be overcome. A general equation for the lattice dissociation process is shown below:

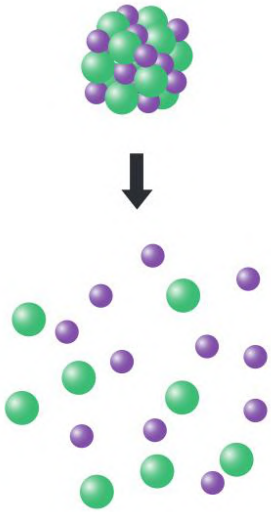
$$MX(s) \longrightarrow M^+(g) + X^-(g) \qquad \Delta H^{\ominus}_{\text{lattice}} > 0$$

The process is **endothermic**. Experimental values of lattice enthalpy at 298 K for some compounds can be found in the data booklet. Lattice enthalpies are often quoted as negative values that represent the exothermic formation of the lattice from gaseous ions—the opposite process to that shown in figure 14. However, in this book we shall consider only the endothermic formation of gaseous ions from a lattice, which is consistent with the definition given in the data booklet.

Lattice enthalpy increases as the energy required to overcome the electrostatic forces of attraction between ions increases. Two factors affecting lattice enthalpy are **ionic radius** and **ionic charge**. The strength of the electrostatic attraction between oppositely charged ions:

- increases with increasing ionic charge
- decreases with increasing ionic radius.

Look at the variations in lattice enthalpy of the compounds in table 6.



▲ Figure 14 Lattice enthalpy is the energy required to overcome the electrostatic forces of attraction holding ions together in the lattice

Ionic compound	$\Delta H^{\ominus}_{\text{lattice}}$/kJ mol^{-1}	Ionic radius/10^{-12} m		Ionic charge	
		Cation	Anion	Cation	Anion
KF	829	138	133	+1	–1
NaF	930	102	133	+1	–1
CaF_2	2651	100	133	+2	–1

▲ Table 6 Lattice enthalpies of selected compounds

NaF has a greater lattice enthalpy than KF because the cations in NaF are smaller, and therefore the electrostatic attraction between Na^+ ions and F^- ions is greater. The lattice enthalpy of CaF_2 is considerably larger than that of KF. This is in part due to the smaller ionic radius of Ca^{2+} compared to K^+. However, it is mainly due to the greater charge on the Ca^{2+} cation, which results in greater electrostatic attraction between the cations and anions in CaF_2.

Factors affecting the lattice enthalpy of the group 1 chlorides

Charge density is a term used to describe charge per unit volume. In this task, you will explore the charge density of the group 1 cations and relate this to the trend in lattice enthalpies of the group 1 chlorides.

Relevant skills

- Tool 2: Use spreadsheets to manipulate data
- Tool 3: General mathematics
- Inquiry 1: State and explain predictions
- Inquiry 2: Identify, describe and explain patterns, trends and relationships

Instructions

Part 1: Prediction

1. For the group 1 cations, predict the relationship between:
   a. ionic radius and lattice enthalpy of their chlorides
   b. charge density and lattice enthalpy of their chlorides.
2. Sketch the graphs you expect to obtain for the relationships above.

Part 2: Data collection

3. Collect the following data for the group 1 chlorides: ionic charge, ionic radius, lattice enthalpy. Possible sources of information include the data booklet and online databases. Cite each source appropriately.
4. Input your data into a spreadsheet and organize it into a suitable table.

Part 3: Analysis

5. Calculate the volume of each cation.
6. Calculate the charge density of each cation.
7. Plot two graphs: one graph showing the relationship between ionic radius and lattice enthalpy, and the other, between charge density and lattice enthalpy.
8. Describe and explain the trends shown in the graphs.
9. Discuss the differences between the two graphs.
10. Evaluate your prediction, including a comparison of the graphs you obtained in 7 and the sketched graphs you obtained in 2.
11. Consider possible extensions to this investigation: what other aspects of ionic radius, charge density and lattice enthalpy could you explore?

Practice questions

1. Write equations, including state symbols, that represent the lattice enthalpies of KBr, CaO and $MgCl_2$.
2. State and explain whether you expect KF or K_2O to have a lower lattice enthalpy value.
3. State and explain which of the following ionic compounds you expect to have the greatest lattice enthalpy value: NaCl, $MgCl_2$, Na_2O or MgO.
4. Describe and explain the trend in lattice enthalpy of the group 1 chlorides down the group from LiCl to CsCl.

Properties of ionic compounds

The properties of ionic compounds are due to their structural features: they contain cations and anions held together by strong electrostatic attractive forces in a lattice.

AHL

Born–Haber cycles (*Reactivity 1.2.5*)

Born–Haber cycles are another application of Hess's law. These cycles combine the enthalpy changes associated with several steps involved in the formation of an ionic compound. The steps in the cycle include ionization energy, enthalpy of atomization, electron affinity, lattice enthalpy, and enthalpy of formation.

Ionization energy, IE, is the standard enthalpy change that occurs on the removal of one mole of electrons from one mole of atoms or positively charged ions in the gaseous state. For metal ions with multiple valence electrons, the first, second, and sometimes further ionization energies are defined:

IE_1: $M(g) \rightarrow M^+(g) + e^-$ $\quad IE > 0$

IE_2: $M^+(g) \rightarrow M^{2+}(g) + e^-$ $\quad IE > 0$

The process of ionization is endothermic. The values of first ionization energy can be found in section 9 of the data booklet.

The **enthalpy of atomization**, $\Delta H^{\ominus}_{at}$, is the standard enthalpy change that occurs when one mole of gaseous atoms of an element is formed. This process is endothermic. For a solid monatomic species, the value of $\Delta H^{\ominus}_{at}$ is equal to the enthalpy of sublimation:

$M(s) \rightarrow M(g)$ $\quad \Delta H^{\ominus}_{at} > 0$

For a gaseous diatomic species, $\Delta H^{\ominus}_{at}$ is equal to half of the bond enthalpy (section 12 of the data booklet):

$\frac{1}{2}X_2(g) \rightarrow X(g)$ $\quad \Delta H_{at}{}^{\ominus} > 0$

The **electron affinity**, EA, is the standard enthalpy change on the addition of one mole of electrons to one mole of atoms in the gaseous state:

$X(g) + e^- \rightarrow X^-(g)$ $\quad EA < 0$

Electron affinity of non-metals is typically negative, but there are exceptions, such as the electron affinity of helium or nitrogen. Values of first and second electron affinity can be found in section 9 of the data booklet.

The **lattice enthalpy,** $\Delta H^{\ominus}_{lattice}$, is defined as the standard enthalpy change that occurs when gaseous ions are formed from one mole of structural units of a solid ionic lattice:

$MX(s) \rightarrow M^+(g) + X^-(g)$ $\quad \Delta H^{\ominus}_{lattice} > 0$

The process is endothermic. Experimental values of lattice enthalpy at 298.15 K can be found in section 16 of the data booklet.

Lattice enthalpies are often quoted as negative values that represent the exothermic formation of the solid lattice from gaseous ions. However, in DP chemistry, the lattice enthalpy always refers to the endothermic formation of gaseous ions from the solid lattice.

We have already defined the standard enthalpy of formation, $\Delta H_f^\ominus$, of a substance as the enthalpy change that occurs when one mole of a substance is formed under standard conditions from its constituent elements in their standard states. Standard enthalpy of formation of an ionic substance, such as NaCl(s), can be represented by a single equation:

$$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s) \qquad \Delta H_f^\ominus = -411\ kJ\ mol^{-1}$$

ATL Self-management skills

The correct use of subject-specific terminology is essential to your ability to demonstrate your knowledge and understanding of new concepts in chemistry.

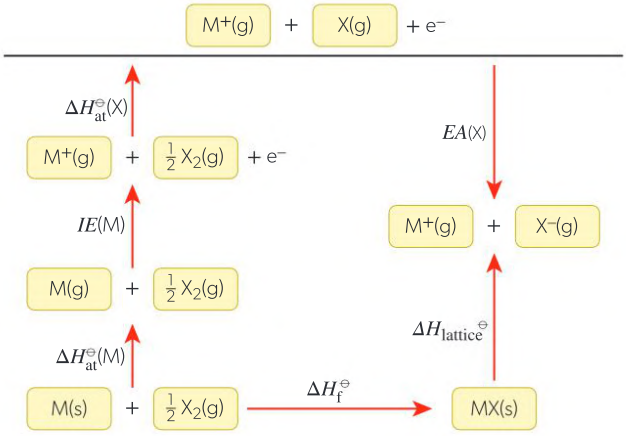
You will encounter several key terms in this section: lattice enthalpy, standard enthalpy of formation, standard state, enthalpy of atomization, ionization energy and electron affinity. Make a glossary that has an entry for each term, detailing the following:

Term	Symbol	Definition	Equation
lattice enthalpy	$\Delta H_{lattice}^\ominus$	Standard enthalpy change that occurs on the formation of gaseous ions from one mole of a solid lattice	$MX(s) \rightarrow M^+(g) + X^-(g)$

You can continue to add terms to your chemistry glossary over time.

Using the Born–Haber cycle

The lattice enthalpy, enthalpy of atomization, ionization energy, electron affinity and enthalpy of formation can be combined to construct a Born–Haber cycle. A generalized Born–Haber cycle is shown in figure 6.



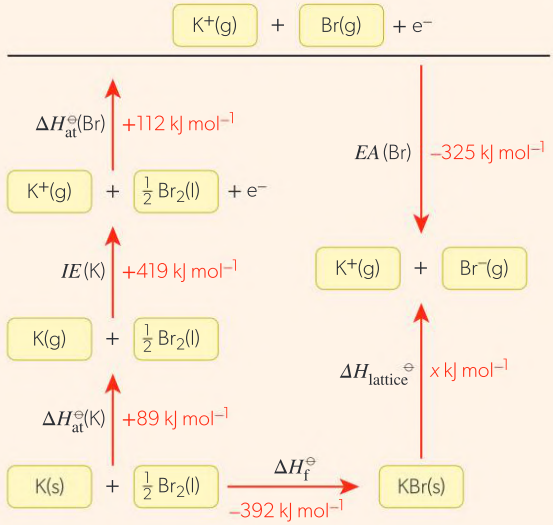
▲ Figure 6 A generalized Born–Haber cycle

AHL

The unknown value of enthalpy change for any step in a Born–Haber cycle can be determined by following the opposite pathway in the cycle. For example, if you know the values of enthalpy of atomization, ionization energies, electron affinities and lattice enthalpy, you can find the standard enthalpy of formation of an ionic compound. The enthalpy of formation is equal to the sum of the enthalpy changes for each of the other steps. Much like with the enthalpy cycle diagrams you constructed before, you should reverse the sign of any enthalpy changes where the pathway goes against the direction of the arrow in the cycle.

Worked example 5

Determine the lattice enthalpy of potassium bromide using the Born–Haber cycle in figure 7.



▲ Figure 7 Born–Haber cycle to calculate the lattice enthalpy of potassium bromide

Solution

The lattice enthalpy is the enthalpy change for the reaction $KBr(s) \rightarrow K^+(g) + Br^-(g)$. In figure 7, this step is shown from the bottom right of the cycle, to the middle right, labelled $\Delta H_{lattice}^{\ominus}$. To find the lattice enthalpy, follow the alternative pathway, going from KBr(s) clockwise around to $K^+(g) + Br^-(g)$. Add the values of the enthalpy change for each step:

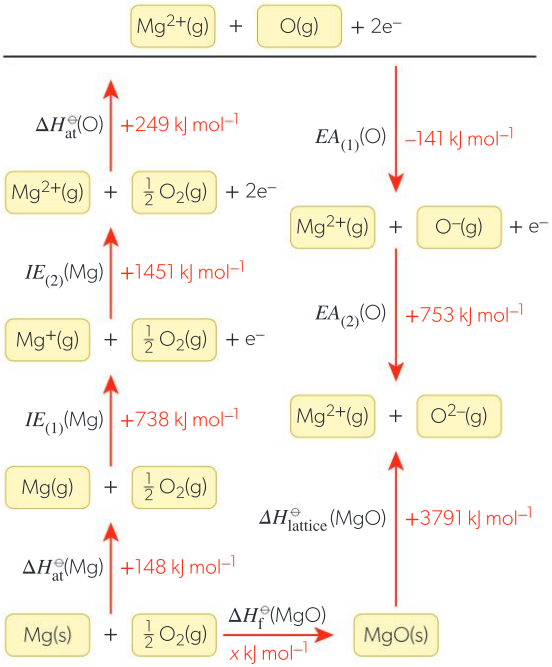
$$\Delta H_{lattice}^{\ominus}(KBr) = -\Delta H_f^{\ominus} + \Delta H_{at}^{\ominus}(K) + IE(K) + \Delta H_{at}^{\ominus}(Br) + EA(Br)$$
$$= -(-392) + (+89) + (+419) + (+112) + (-325)$$
$$= +687\ kJ\ mol^{-1}$$

Note that you followed the reverse of the enthalpy of formation reaction arrow, so the sign was reversed: +392 instead of −392. Remember to express your answer for the lattice enthalpy of an ionic compound as a positive value, because it is an endothermic process.

AHL

Practice question

10. Interpret and utilize the values from the Born–Haber cycle in figure 8 to calculate the enthalpy of formation for the ionic compound magnesium oxide, MgO. MgO has doubly charged ions.



▲ Figure 8 Born–Haber cycle to calculate the enthalpy of formation for magnesium oxide



▲ Figure 9 The lattice structure of crystalline magnesium oxide: grey = Mg^{2+}, red = O^{2-}

Linking question

What are the factors that influence the strength of lattice enthalpy in an ionic compound? (*Structure 2.1*)

Factors affecting the value of lattice enthalpy are detailed in *Structure 2.1*.

End of topic questions

Topic review

1. Using your knowledge from the *Reactivity 1.2* topic, answer the guiding question as fully as possible:

   How does application of the law of conservation of energy help us to predict energy changes during reactions?

Exam-style questions

Multiple-choice questions

2. Which equation represents the N–H bond enthalpy in NH_3?

   A $NH_3(g) \rightarrow N(g) + 3H(g)$

   B $\frac{1}{3}NH_3(g) \rightarrow \frac{1}{3}N(g) + H(g)$

   C $NH_3(g) \rightarrow \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$

   D $NH_3(g) \rightarrow \bullet NH_2(g) + \bullet H(g)$

3. Bond enthalpies can be used to calculate the enthalpy change of the following reaction, in $kJ\,mol^{-1}$.

   $$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

   Referring to section 12 of the data booklet, deduce which of the following represents the calculation of the enthalpy change of reaction from bond enthalpies.

   A $(6 \times 391) - [(3 \times 436) + 945]$

   B $(3 \times 391) - (436 + 945)$

   C $-[(3 \times 436) + 945] + (3 \times 391)$

   D $-(6 \times 391) + [(3 \times 436) + 945]$

AHL

4. Which equation represents lattice enthalpy?

   A $NaCl(g) \rightarrow Na^+(g) + Cl^-(g)$

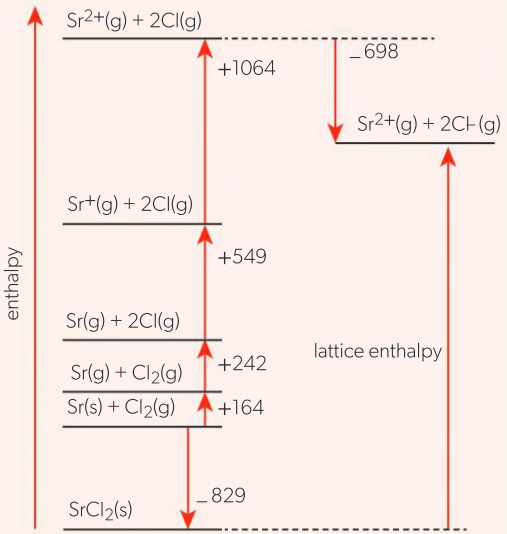
   B $NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$

   C $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

   D $NaCl(s) \rightarrow Na^+(s) + Cl^-(s)$

AHL

5. Using the Born–Haber cycle below, determine which calculation represents the lattice enthalpy of strontium chloride, $SrCl_2$, in $kJ\,mol^{-1}$.



   A $-(-829) + 164 + 242 + 549 + 1064 - (-698)$

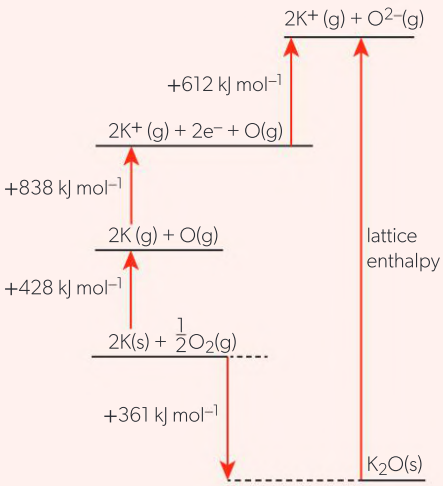
   B $-829 + 164 + 242 + 549 + 1064 - 698$

   C $-(-829) + 164 + 242 + 549 + 1064 - 698$

   D $-829 + 164 + 242 + 549 + 1064 - (-698)$

AHL

6. The Born–Haber cycle for potassium oxide, K_2O, is shown below:



Which calculation represents the lattice enthalpy in kJ mol^{-1}?

A $-361 + 428 + 838 + 612$
B $-(-361) + 428 + 838 + 612$
C $-361 + 428 + 838 - 612$
D $-(-361) + 428 + 838 - 612$

Extended-response questions

7. The following equation describes the industrial production of ammonia using the Haber process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

a. Calculate the enthalpy change of reaction for this process using bond enthalpy data. [3]

b. State and explain whether the reaction is exothermic or endothermic. [2]

8. Ammonia reacts with oxygen to form nitrogen and steam.

a. Write a balanced chemical equation for this reaction, showing the states of all species. [1]

b. Draw full structural formulas for all reactants and products to determine the nature of the covalent bonds present. [4]

c. Using bond enthalpy values from section 12 of the data booklet, determine the enthalpy change for this reaction. [3]

9. Octane, C_8H_{18}, is a component of liquid gasoline. When gasoline is burned as fuel, octane is combusted in oxygen via the following reaction:

$$C_8H_{18}(l) + \frac{25}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(g)$$

a. Using bond enthalpy data, calculate the enthalpy change for this reaction. [3]

b. The enthalpy change for this reaction was determined experimentally to be -5470 kJ mol^{-1}. Give **two** reasons why the enthalpy change value you obtained from bond enthalpy data differs from the experimental value. [2]